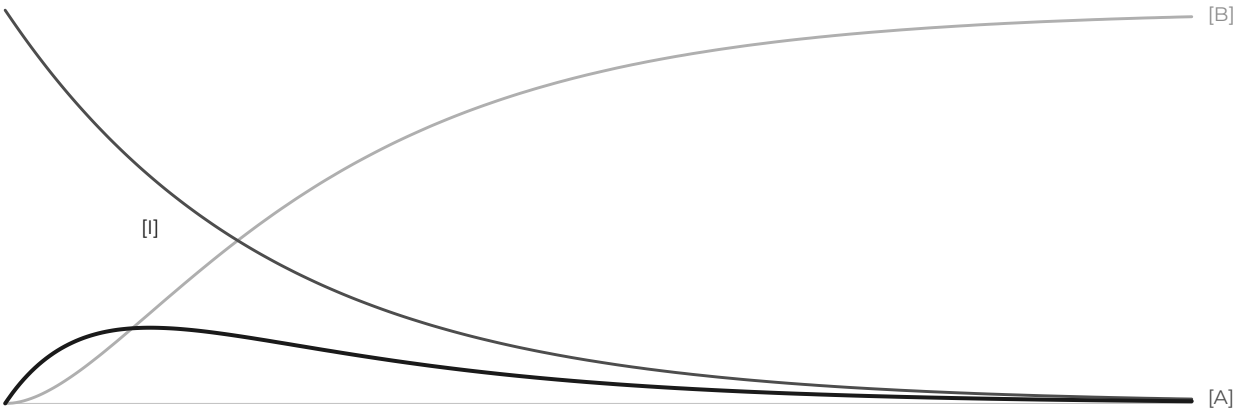
MONOGRAPH

The Steady-State Hypothesis

When it holds and how much it misses:
from ozone photolysis to ammonia synthesis

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Supervisor: Prof. Juliano Alves Bonacin



Escola de Talentos do Instituto Principia

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Abstract

In this work, I explain the steady-state hypothesis (SSH) from scratch, the approximation used in chemical kinetics to simplify the rate law of mechanisms with reactive intermediates, and I ask when it is good, how much it misses and whether data can show that error. I test a time-scale criterion (the intermediate must last a short time compared with how fast the reactants change) against the exact solution of known mechanisms, by numerical simulation. In the mechanism $A \rightarrow I \rightarrow B$, the largest error in [B] is 7.7 % of $[A]_0$ when the step that consumes the intermediate is 10 times faster than the one that forms it. In the mechanism $A \rightleftharpoons I \rightarrow P$, the SSH requires $k_1 \ll k_{-1} + k_2$, and the pre-equilibrium, $k_2 \ll k_1 + k_{-1}$. The two criteria mirror each other: swapping k_1 and k_2 leaves the exact product curve unchanged and turns one approximation into the other. In a band around $k_1 = k_2$, neither works, and a total law, which does not neglect the intermediate in the balance, is off by at most 14 % of $[A]_0$. In the stratosphere, with the Chapman mechanism, the relative error the SSH introduces in $[O_3]$ at steady state is about 1.4×10^{-10} at 25 km and 1.4×10^{-9} at 30 km. The 13 ozone data points from a short article I wrote in December 2025 follow the exponential decay the SSH predicts, with $k = 0.497$/s (standard error 0.008/s) and $R^2 = 0.998$ on the linear scale of the data, but they do not test the hypothesis: at their times, the exact curve and the SSH curve coincide. In ammonia synthesis, I simulated the start-up of a reactor: the gas composition reaches 95 % of its final value in about 11.5 s. With synthetic data, model curves with randomly drawn noise, I show which assay would test each law and that a curve of a single product can have R^2 above 0.99 and, read through the SSH formula, give a constant off by a factor of 10: testing the SSH requires measuring the reactant too, or closing the mass balance. In a sample from the public SABIO-RK database, with 1968 pairs of Michaelis constant and enzyme concentration taken from published assays, the Segel and Slemrod condition, computed at the largest $[S]_0$ of the assay, holds in about 98 % of the pairs that record $[S]_0$, but about 9 in every 100 pairs (8 to 10, with 95 % confidence) have an enzyme concentration above one tenth of the Michaelis constant, the range in which the error of the fitted value of that constant stops being small. These figures hold only for the entries that give the enzyme concentration in mol/L, about 14 % of the sample. Finally, I propose a bench test, not yet done, with invertase and a glucose meter, in which the glucose the strip does not read yet plays the role of the intermediate and can be measured. What I say about its result is a model prediction, and the program that will analyze the readings has only been tested with synthetic data. This is the revised and expanded version, dated October 16, 2026, of the monograph I wrote in July 2025 at the Escola de Talentos do Instituto Principia.

Keywords: chemical kinetics; steady-state hypothesis (SSH); pre-equilibrium; ozone; ammonia synthesis; enzyme kinetics; numerical simulation.

Resumo

Neste trabalho, explico do zero a hipótese do estado estacionário (HEE), a aproximação usada em cinética química para simplificar a lei de velocidade de mecanismos com intermediários reativos, e pergunto quando ela é boa, quanto ela erra e se os dados conseguem mostrar esse erro. Testo um critério de escalas de tempo (o intermediário precisa durar pouco perto do tempo em que os reagentes mudam) contra a solução exata de mecanismos conhecidos, por simulação numérica. No mecanismo $\mathrm{A} \rightarrow \mathrm{I} \rightarrow \mathrm{B}$, o maior erro em [B] é 7,7 % de $[\mathrm{A}]_0$ quando a etapa que consome o intermediário é 10 vezes mais rápida que a que o forma. No mecanismo $\mathrm{A} \rightleftharpoons \mathrm{I} \rightarrow \mathrm{P}$, a HEE pede $k_1 \ll k_{-1} + k_2$, e o pré-equilíbrio, $k_2 \ll k_1 + k_{-1}$. Os dois critérios são espelhos um do outro: trocar k_1 por k_2 não muda a curva exata do produto e leva uma aproximação na outra. Numa faixa em volta de $k_1 = k_2$, nenhuma das duas serve, e uma lei total, que não despreza o intermediário no balanço, erra no máximo 14 % de $[\mathrm{A}]_0$. Na estratosfera, com o mecanismo de Chapman, o erro relativo da HEE em $[\mathrm{O_3}]$ no estado estacionário é de cerca de $1{,}4 \times 10^{-10}$ a 25 km e $1{,}4 \times 10^{-9}$ a 30 km. Os 13 pontos de dados do ozônio de um artigo curto que escrevi em dezembro de 2025 seguem a queda exponencial que a HEE prevê, com $k = 0{,}497/\mathrm{s}$ (erro padrão de $0{,}008/\mathrm{s}$) e $R^2 = 0{,}998$ na escala linear dos dados, mas não testam a hipótese: nos tempos deles, a curva exata e a da HEE coincidem. Na síntese de amônia, simulei a partida de um reator: a composição do gás chega a 95 % do valor final em cerca de 11,5 s. Com dados sintéticos, curvas de modelo com ruído sorteado, mostro que ensaio testaria cada lei e que uma curva de um só produto pode ter R^2 acima de 0,99 e, lida pela fórmula da HEE, dar uma constante errada por um fator 10: para testar a HEE, é preciso medir também o reagente ou fechar o balanço. Numa amostra do banco público SABIO-RK, com 1968 pares de constante de Michaelis e concentração de enzima tirados de ensaios publicados, a condição de Segel e Slemrod, calculada no maior $[\mathrm{S}]_0$ do ensaio, vale em cerca de 98 % dos pares que registram $[\mathrm{S}]_0$, mas cerca de 9 em cada 100 pares (de 8 a 10, com 95 % de confiança) têm a concentração de enzima acima de um décimo da constante de Michaelis, faixa em que o erro do valor ajustado dessa constante deixa de ser pequeno. Essas contas valem só para as entradas que trazem a concentração de enzima em mol/L, cerca de 14 % da amostra. Por fim, proponho um teste de bancada, ainda não feito, com invertase e um glicosímetro, em que a glicose que a fita ainda não lê faz o papel do intermediário e pode ser medida. O que digo sobre o resultado é previsão do modelo, e o programa que vai analisar as leituras só foi testado com dados sintéticos. Esta é a versão revisada e ampliada, de 16 de outubro de 2026, da monografia que escrevi em julho de 2025 na Escola de Talentos do Instituto Principia.

Palavras-chave: cinética química; hipótese do estado estacionário; pré-equilíbrio; ozônio; síntese de amônia; cinética enzimática; simulação numérica.

List of Figures

Figure 1 – How Wilhelmy followed the hydrolysis of sucrose 16
Figure 2 – Average rate measured with two secants 20
Figure 3 – Instantaneous rate and tangent line at $t = 2$ s 21
Figure 4 – Zero-order reaction and autocatalytic reaction 23
Figure 5 – Effect of order on [A], ln[A] and $1/[A]$ vs. t 26
Figure 6 – Linearization grid for orders 0, 1 and 2 27
Figure 7 – Effect of the activation energy on $k(T)$ and Arrhenius lines 28
Figure 8 – The constant k does not change with time; the rate v falls 29
Figure 9 – Maxwell-Boltzmann distribution at two temperatures 29
Figure 10 – Real factor for 10 K against temperature and the factor-of-2 rule 30
Figure 11 – Energy profile of an exothermic and an endothermic step 31
Figure 12 – Gibbs energy profile of the S_N2 substitution 32
Figure 13 – Energy profile with and without a catalyst 32
Figure 14 – The biscuit factory: the line in front of the oven grows 34
Figure 15 – Energy profiles of an irreversible and a reversible step 35
Figure 16 – Energy profile of the two steps: slow step and pre-equilibrium 38
Figure 17 – The inverted biscuit factory, a picture of the SSH 40
Figure 18 – Exact and SSH solutions for $A \rightarrow I \rightarrow B$ 41
Figure 19 – Full simulation of the two-step mechanism 42
Figure 20 – The two ozone mechanisms 46
Figure 21 – Photolysis of O_3 in a cell: full solution and SSH 47
Figure 22 – O_3 decay with and without the $O + O_2 + M$ reaction 48
Figure 23 – Exponential fit to the ozone data from the December 2025 article 48
Figure 24 – Synthetic data: k_{obs} against J in O_3 photolysis 50
Figure 25 – Synthetic data: initial rate of O_3 loss with O_2 51
Figure 26 – The O atom right after the light is switched on 53
Figure 27 – Full Chapman mechanism and with the SSH at 25 and 30 km 54
Figure 28 – Relative error of the SSH in stratospheric ozone 54
Figure 29 – Chapman mechanism with and without the chlorine cycle 55
Figure 30 – Equilibrium ammonia against temperature, at four pressures 58
Figure 31 – N_2 conversion per pass against temperature 59
Figure 32 – N_2 conversion per pass against pressure 60
Figure 33 – Steps of ammonia synthesis on the iron surface 61
Figure 34 – Exact N coverage against the SSH 63
Figure 35 – Fits to the ammonia data from the December 2025 article 65
Figure 36 – Synthetic data: does AICc find the startup model? 67

Figure 37 – Synthetic data: reactor startup at four flow rates 68
Figure 38 – Reactor startup and time scales 69
Figure 39 – Michaelis-Menten curve: initial rate against [S] 72
Figure 40 – Induction period in the mechanism A $\rightarrow$ I $\rightarrow$ P 76
Figure 41 – SSH error in the mechanism A $\rightarrow$ I $\rightarrow$ B 77
Figure 42 – Validity map of the SSH, the pre-equilibrium and the total law 79
Figure 43 – Error of the SSH, the pre-equilibrium and the total law 80
Figure 44 – Distribution of ε_{K_M} in SABIO-RK 84
Figure 45 – Fraction of pairs with large ε_{K_M}, by EC class 85
Figure 46 – K_M and k_{cat} errors caused by the enzyme 86
Figure 47 – A case of high R^2 and a wrong constant 88
Figure 48 – When the points reveal the SSH error in A $\rightleftharpoons$ I $\rightarrow$ P 89
Figure 49 – Model prediction for invertase at three doses 93
Figure 50 – Test of the analysis program with synthetic data 94
Figure 51 – Map of per-pass N_2 conversion in temperature and pressure 141
Figure 52 – Sweep of J, absorbance and O_2 fraction in the illuminated cell 142
Figure 53 – Chapman ozone and lifetimes against altitude 143
Figure 54 – Bootstrap histograms of k (ozone) and τ (ammonia) 145
Figure 55 – Largest SSH error in [B] against k_2/k_1, 100 ratios 146

List of Tables

Table 1 – Unit of the rate constant by overall order 24
Table 2 – Integrated law, straight-line plot, slope and half-life 25
Table 3 – Initial rates of NO consumption 25
Table 4 – Activation energy for a factor of 2 and real factor at 25–35 °C 30
Table 5 – Molecularity and rate law of five reactions 34
Table 6 – SSH errors in the mechanism A → I → B 41
Table 7 – The three ways to treat an intermediate 43
Table 8 – Constants used in the ozone chapter 44
Table 9 – Fits to the ozone data from the December 2025 article 49
Table 10 – Time scales of the Chapman mechanism at 25 and 30 km 54
Table 11 – Temperature and pressure: effect on rate and on equilibrium 59
Table 12 – Industrial conditions of ammonia synthesis, from four sources 60
Table 13 – Parameters of the Dyson and Simon law and equilibrium constant 64
Table 14 – Five models for the ammonia data from the December 2025 article 66
Table 15 – Mechanism A ⇌ I → P: exact solution, SSH and pre-equilibrium 78
Table 16 – Error of the SSH, the pre-equilibrium and the total law at points of the map . . . 81
Table 17 – Six steps to decide whether the SSH can be used 82
Table 18 – What was already known and what this work adds 99
Table 19 – Unit of k according to the reaction order 122
Table 20 – Scripts and what they generate, part 1 of 5: chapters 1, 2, 3, 6, and 7 125
Table 21 – Scripts and what they generate, part 2 of 5: chapter 4 126
Table 22 – Scripts and what they generate, part 3 of 5: chapter 5 127
Table 23 – Scripts and what they generate, part 4 of 5: chapter 8 128
Table 24 – Scripts and what they generate, part 5 of 5: Appendices A and C 129
Table 25 – Ozone data from the December 2025 article 137
Table 26 – Ammonia data from the December 2025 article 137
Table 27 – Fundamental physical constants 138
Table 28 – Literature constants for ozone kinetics 138
Table 29 – Literature constants for the ammonia chapter 139
Table 30 – Per-pass N_2 conversion over a temperature and pressure grid 140
Table 31 – Sweep of the illuminated-cell model: effective k_{obs} and SSH error 142
Table 32 – Chapman mechanism as a function of altitude, 20 to 36 km 144
Table 33 – Bootstrap of the ozone and ammonia fits (95% CI) 144
Table 34 – Sweep of the SSH error in the A → I → B mechanism 145
Table 35 – K_M error caused by the enzyme in the SABIO-RK pairs 147
Table 36 – General data table 147

List of symbols

$[\mathrm{A}]$, $[\mathrm{A}]_0$, $[\mathrm{I}]_{\mathrm{ss}}$	concentration of A; initial concentration; concentration of I at steady state (mol m^{-3}; in practice, mol L^{-1}, or molecule cm^{-3} in the gas phase)
c°	standard concentration, 1 mol L^{-1} (used inside ln)
t, $t_{1/2}$, $t_{1/n}$	time; half-life; time for $1/n$ of the initial concentration to remain (s)
v, v_0	reaction rate (always positive); initial rate (mol L^{-1} s^{-1})
v_{X}	rate of consumption of the reactant or of formation of the product X, always positive (mol L^{-1} s^{-1})
$\dot{\xi}$, n_{X}	rate of conversion of the reaction; amount of substance of species X (mol s^{-1}; mol)
a, b, p, q	stoichiometric coefficients in the reaction $a\,\mathrm{A} + b\,\mathrm{B} \rightarrow p\,\mathrm{P} + q\,\mathrm{Q}$
n, m	order of the rate law with respect to A; order with respect to the second reactant
k, k_1, k_{-1}, k_{obs}	rate constant; of one step (forward); of the reverse step; observed constant (unit depends on the order)
E_{a}, E_0, A	Arrhenius activation energy; minimum collision energy needed to react; pre-exponential factor (J mol^{-1}, in practice kJ mol^{-1}; same unit as k)
R, T, P	gas constant; temperature; pressure (J mol^{-1} K^{-1}; K; Pa, in practice atm)
Q_{10}	ratio $k(T + 10\,\mathrm{K})/k(T)$, used in biochemistry
k_{B}, N_{A}, h	Boltzmann constant; Avogadro constant; Planck constant (J K^{-1}; mol^{-1}; J s)
σ_{col}, d_{AB}	collision cross section; sum of the radii of the two molecules (m^2; m)
μ, $\bar{u}_{\mathrm{rel}}$	reduced mass of the A-B pair; mean relative speed between the molecules (kg; m s^{-1})
$\mathcal{N}_{\mathrm{A}}$, Z_{AB}	number of molecules of A per volume; number of A-B collisions per volume and per time (m^{-3}; m^{-3} s^{-1})
$\mathcal{P}$	steric factor (fraction of collisions with the right orientation to react)
‡, $\Delta G^\ddagger$, $\Delta H^\ddagger$, $\Delta S^\ddagger$, κ	transition state; Gibbs energy, enthalpy, and entropy of activation; transmission coefficient of the Eyring equation (chapter 2), close to 1 (J mol^{-1}; J mol^{-1} K^{-1})
$\Delta_{\mathrm{r}}H^\circ$	standard enthalpy of reaction (kJ mol^{-1})
K_{eq}, K	equilibrium constant; in a reversible elementary step, $K = k_1/k_{-1}$, as in the mechanism $\mathrm{A} \rightleftharpoons \mathrm{I} \rightarrow \mathrm{P}$ of chapters 7 and 8

I; τ_{I}, τ_{reaction}, τ	reactive intermediate; lifetime of the intermediate; characteristic time over which the reactants change; characteristic time, in general (s)
α, α_0, α_∞	(chapter 1) angle by which the solution turns the plane of polarized light in the polarimeter; its value at the start; its value at the end
J, J_1, J_3	photolysis frequency; of the step $O_2 + h\nu \longrightarrow 2\,O$; of the step $O_3 + h\nu \longrightarrow O_2 + O$, with $J \equiv J_3$ in chapter 4 (s^{-1})
k_2, k_4 (chapter 4)	constant of the step $O + O_2 + M \longrightarrow O_3 + M$; constant of the step $O + O_3 \longrightarrow 2\,O_2$ (cm^6 $molecule^{-2}$ s^{-1}; cm^3 $molecule^{-1}$ s^{-1})
[M], [O_x]	concentration of the third body (the air); odd oxygen, [O] + [O_3] (molecule cm^{-3})
σ, Φ, F	absorption cross section of O_3 for light; quantum yield; photon flux (cm^2; dimensionless; photons cm^{-2} s^{-1})
Abs, ℓ	absorbance; optical path length of the light in the cell (dimensionless; cm)
x, x_0	(chapter 4) ratio $k_4[O_3]/(k_2[O_2][M])$ between the two fates of the O atom, destroying an O_3 or remaking one; its value at the start (dimensionless)
J_{ref}, b, $k_{\mathrm{obs},0}$	(chapter 4, assay with synthetic data) reference value of J, equal to $k_{\mathrm{obs}}/2$ from the 13 points; slope fitted in $\exp(-bJt)$, which the mechanism predicts to be 2; observed constant at the start of the reaction (s^{-1}; dimensionless; s^{-1})
a, b	(chapter 5) constants of the simple surface model: $a = 2k_{\mathrm{ads}}P_{N_2}$ forms N* and $b = k_{\mathrm{hid}}$ consumes it (s^{-1})
θ_i, θ_*	surface coverage by species i; fraction of free sites (dimensionless, between 0 and 1)
k_{ads}, k_{hid}, K_{N}	dissociative adsorption constant of N_2; constant of the hydrogenation steps of adsorbed N; equilibrium constant for N adsorption (s^{-1} atm^{-1}; s^{-1}; $atm^{1/2}$)
β, α	departure from equilibrium in the Temkin-Pyzhev law (0 with no ammonia, 1 at equilibrium); Temkin exponent, between 0.5 and 0.75 (dimensionless)
k_+, k_-	forward and reverse rate constants in the Temkin-Pyzhev law and in the Dyson and Simon law (chapter 5)
K_{a}, a_i, y_i, φ_i	equilibrium constant in activities for ammonia synthesis; activity of species i; mole fraction; fugacity coefficient (atm^{-1}; atm; dimensionless)
η	effectiveness factor of the catalyst pellet: the ratio between the real rate, with the gas's resistance to reaching the interior of the pores, and the rate there would be without that resistance (dimensionless, between 0 and 1)
ζ	(closed vessel, chapter 5) ratio between the number of surface sites and the number of N_2 molecules at the start (dimensionless)
$[E]_0$, $[S]_0$, [ES]	total enzyme concentration; initial substrate concentration; concentration of the enzyme-substrate complex (mol L^{-1})

K_M, v_{max}, k_{cat}, ε	Michaelis-Menten constant; maximum rate; catalytic constant, equal to k_2 with the enzyme saturated; ratio $[E]_0/([S]_0 + K_M)$ that defines when the SSH holds (mol L^{-1}; mol L^{-1} s^{-1}; s^{-1}; dimensionless)
ε_{K_M}, K_{VC}	(chapter 8) ratio $[E]_0/K_M$, an upper bound for ε; Van Slyke–Cullen constant, k_2/k_1 (dimensionless; mol L^{-1})
[C], $K_{M,ap}$, r	(chapter 8) concentration of the enzyme-substrate complex in the simulation, the same as [ES]; apparent K_M returned by the fit; ratio $k_2/(k_{-1} + k_2)$ (mol L^{-1}; mol L^{-1}; dimensionless)
θ_{ox}, k_{red}, k_{ox}	fraction of oxidized catalyst sites in the oxidation of SO_2; reduction and reoxidation constants of the surface (dimensionless; s^{-1} atm^{-1})
R^2, MAE, RMSE	coefficient of determination; mean absolute error; root mean square error of the fits
AICc, BIC, DW	corrected Akaike information criterion for small samples; Bayesian information criterion, of Schwarz; Durbin-Watson statistic of the residuals
CI	confidence interval, usually 95 %
τ, t_0, k'_a, k'_b	(ammonia fits, chapter 5) characteristic time of the simple exponential; dead time; constants of the steps in series A $\longrightarrow$ I $\longrightarrow$ P (min; min^{-1})
τ_W, γ	(ammonia fits, chapter 5) scale and shape of the Weibull curve (min; dimensionless)
F, F_{ref}, α, β	(chapter 5, assay with synthetic data) gas flow rate in the reactor; flow rate at which the constants equal those of the 13 points; exponents of $k'_a \propto F^\alpha$ and $k'_b \propto F^\beta$ (only the ratio F/F_{ref} enters the calculations; α and β are dimensionless)
t_{ind}	(chapter 7) time at which the induction period ends (s)
λ_+, λ_-	(chapter 7) the two roots that govern the exact solution of the mechanism A $\rightleftharpoons$ I $\rightarrow$ P: the larger one, of the fast start, and the smaller one, of the slow part (s^{-1})
q	(chapter 7) $4k_1k_2/(k_1 + k_{-1} + k_2)^2$, from 0 to 1: measures how close the two time scales of the exact solution are; the error of the total law depends only on it (dimensionless)
ρ	(chapter 8) ratio k_{fast}/k_{slow} in the mechanism A $\rightarrow$ I $\rightarrow$ B (dimensionless)
σ	(chapters 7 and 8) standard deviation of the noise in the points, as a fraction of the initial concentration (dimensionless)
κ, k_1^{SSH}	(chapter 8) ratio $k_1/(k_{-1} + k_2)$ of the SSH criterion in the mechanism A $\rightleftharpoons$ I $\rightarrow$ P; the k_1 computed through the SSH formula, close to $k_1/(1 + \kappa)$ (dimensionless; s^{-1})
k_{fast}, k_{slow}	(chapter 8) the larger and the smaller of the two constants of the mechanism A $\rightarrow$ I $\rightarrow$ B (s^{-1})

D, N	(chapter 8) deviation between the exact curve and the best SSH curve, without noise, at the times of an experimental design; number of points (fraction of the initial concentration; dimensionless)
$[\alpha]$, $[\beta]$, G	(chapter 8, bench test) concentrations of alpha- and beta-glucose; total glucose, $[\alpha] + [\beta]$ (mol L^{-1}; in practice, mmol L^{-1} or mg dL^{-1})
R, J, J_{SSH}	(chapter 8, bench test) glucose meter reading, $[\beta]/f_\beta$; glucose the strip does not see yet, $G - R$; the SSH value of J, v/k_m (same unit as G)
k_m, $k_{\alpha\beta}$, $k_{\beta\alpha}$, f_β	(chapter 8, bench test) mutarotation constant, $k_{\alpha\beta} + k_{\beta\alpha}$; constants of the $\alpha \rightarrow \beta$ interconversion and of its reverse; fraction of β at equilibrium (min^{-1}; dimensionless)
k_{ap}, K_F, K_G	(chapter 8, bench test) initial invertase rate divided by $[S]_0$, $v_{max}/(K_M + [S]_0)$; constants with which fructose and glucose slow down invertase (min^{-1}; mol L^{-1})

Contents

1	**Introduction**	**16**
1.1	Why study reaction rates	16
1.2	Stepwise reactions and the problem of intermediates	17
1.3	The question this work asks	17
1.4	How this work was done	18
1.5	What each chapter does	18
2	**Fundamentals of chemical kinetics**	**20**
2.1	Reaction rate	20
2.1.1	Average rate	20
2.1.2	Instantaneous rate	20
2.1.3	Stoichiometry and reaction rate	21
2.1.4	Units of the rate	21
2.2	Rate law and order	22
2.2.1	Definition	22
2.2.2	Order and stoichiometric coefficient	22
2.2.3	Zero order and autocatalysis	22
2.2.4	Orders that change	23
2.2.5	Units of the rate constant	24
2.3	How concentration changes with time	24
2.4	How to find the rate law	25
2.4.1	Initial rates	25
2.4.2	Isolation and pseudo-first order	25
2.4.3	Graphical method	26
2.5	Temperature and the Arrhenius equation	27
2.5.1	Arrhenius and the activation energy	27
2.5.2	The straight-line form	28
2.5.3	Rule of thumb: the factor of 2	29
2.6	Why molecules react	30
2.6.1	Collision theory	30
2.6.2	The Eyring equation	31
2.7	Mechanisms	33
2.7.1	Elementary step and molecularity	33
2.7.2	Rate-determining step	33
2.7.3	Rate law of an elementary step	34
2.7.4	Elementary equilibrium	35
3	**The steady-state hypothesis**	**36**
3.1	A bit of history	36
3.2	The idea and the definition	36
3.3	Three ways to treat the same intermediate	37
3.3.1	The rate-determining step	38
3.3.2	Pre-equilibrium	38
3.3.3	The steady-state hypothesis	39
3.3.4	The SSH contains the other two	39

3.3.5 The biscuit factory, inverted . . . 39
3.4 Example: A → I → B . . . 40
3.5 The two-step mechanism by computer . . . 42
3.6 Summary . . . 43

4 Case study 1: ozone . . . 44
4.1 Ozone and ultraviolet light . . . 44
4.2 Two versions of the mechanism . . . 45
4.2.1 The cell: photolysis of ozone . . . 45
4.3 The SSH in the two-step mechanism . . . 46
4.3.1 If O_2 is present: the general form . . . 47
4.4 The ozone data from the December 2025 article . . . 48
4.4.1 An assay with synthetic data . . . 50
4.5 The SSH in the stratosphere: the Chapman mechanism . . . 52
4.5.1 Three time scales . . . 54
4.5.2 The chlorine catalytic cycle . . . 54
4.6 What ozone teaches us about the SSH . . . 55

5 Case study 2: ammonia synthesis . . . 57
5.1 The Haber-Bosch process . . . 57
5.2 Thermodynamics versus kinetics . . . 57
5.3 What happens on the iron surface . . . 60
5.4 The SSH in the adsorbed species . . . 62
5.5 The ammonia data from the December 2025 article . . . 65
5.5.1 An assay with synthetic data for the startup . . . 66
5.6 Simulating a reactor . . . 69
5.7 What ammonia teaches about the SSH . . . 70

6 Other applications . . . 71
6.1 Enzymes: the Michaelis-Menten equation . . . 71
6.2 The production of sulfur trioxide . . . 73
6.3 Hydrogen and bromine: the rate law without a simple order . . . 74

7 When the steady-state hypothesis fails . . . 75
7.1 The time-scale criterion . . . 75
7.2 Failure 1: the start of the reaction . . . 75
7.3 Failure 2: an intermediate that is too slow . . . 76
7.4 Failure 3: too much intermediate . . . 77
7.5 Failure 4: fast reversible steps . . . 78
7.5.1 Swapping k_1 and k_2 . . . 78
7.5.2 The validity map . . . 79
7.5.3 One law that covers both cases . . . 80
7.6 Branched chains . . . 81
7.7 What to do when the SSH fails . . . 81
7.8 Practical checklist . . . 82

8 The hypothesis against data . . . 83
8.1 How large is ε in published assays? . . . 83
8.1.1 The sample . . . 83
8.1.2 What I computed . . . 84
8.1.3 Result . . . 84

8.1.4 What a large ε does to the measured K_M . . . 85
8.2 When the data show that the SSH went wrong . . . 87
8.2.1 What the product points do not tell . . . 87
8.2.2 When the fast phase shows . . . 89
8.2.3 The article data and enzymes . . . 90
8.3 A proposed bench test . . . 91
8.3.1 The glucose the strip does not see yet . . . 91
8.3.2 What the model predicts . . . 92
8.3.3 The protocol . . . 95
8.3.4 Commitments before the test . . . 96

9 Conclusion . . . 97
9.1 Answer to the question . . . 97
9.2 What was already known and what this work adds . . . 99
9.3 Next steps . . . 100

BIBLIOGRAPHY . . . 101

Glossary . . . 109

APPENDIX 114

APPENDIX A – Full derivations . . . 115
A.1 Integrated rate laws of order 0, 1, 2 and general, half-life, $1/n$-life time and fraction reacted . . . 115
A.2 A + B with different initial concentrations . . . 116
A.3 First-order reversible reaction . . . 116
A.4 The exact solution of A $\rightarrow$ I $\rightarrow$ B, and the case $k_1 = k_2$ . . . 117
A.5 The algebra of the two-step mechanism . . . 118
A.6 Pre-equilibrium with the second step also reversible . . . 118
A.7 Michaelis-Menten and Briggs-Haldane . . . 119
A.8 The ammonia surface . . . 119
A.9 Mars and van Krevelen . . . 120
A.10 $H_2 + Br_2$ . . . 120
A.11 Collision theory, the Eyring equation, and Arrhenius from two points . . . 121
A.12 Units of the rate constant . . . 122
A.13 Chapman and the general form with O_2 . . . 122
A.14 The total law: why the error depends only on q . . . 123

APPENDIX B – Code . . . 125
B.1 Figures and tables generated by script . . . 125
B.2 Environment . . . 129
B.3 What each script does . . . 130
B.4 Key excerpts . . . 135
B.4.1 Integration of the full system . . . 135
B.4.2 Curve fitting . . . 135
B.5 How to reproduce everything . . . 136

APPENDIX C – Tables of constants and data points . . . 137
C.1 Simulation sweeps . . . 140

C.2 General data table 147

CHAPTER 1

Introduction

1.1 Why study reaction rates

Chemical kinetics is the branch of chemistry that studies reaction rates and the step-by-step path that takes reactants to products. Unlike thermodynamics, which tells us whether a reaction can happen, kinetics tells us how, and how fast, it happens. Controlling an industrial process, setting the dose of a drug, or knowing how long a pollutant lasts in the air all depend on that second question.

The first person to study that rate systematically was the German chemist Ludwig Ferdinand Wilhelmy, in 1850 (WILHELMY, 1850). He analyzed the acid hydrolysis of sucrose, the breakdown of table sugar into glucose and fructose in the presence of acid:

$$\underset{\text{sucrose}}{C_{12}H_{22}O_{11}} + \underset{\text{water}}{H_2O} \xrightarrow{H^+} \underset{\text{glucose}}{C_6H_{12}O_6} + \underset{\text{fructose}}{C_6H_{12}O_6} \tag{1.1}$$

The acid (H^+, above the arrow) speeds up the reaction and comes out of it without being consumed. To follow the reaction, Wilhelmy used a polarimeter, an instrument that measures how much a substance turns the plane of polarized light, that is, light filtered so that its wave vibrates in a single plane (Figure 1a). The light goes through a polarizer, crosses a tube with the solution, and reaches a second polarizer, the analyzer. The observer turns the analyzer until it matches the turn the solution gave the light, and reads that angle, α.

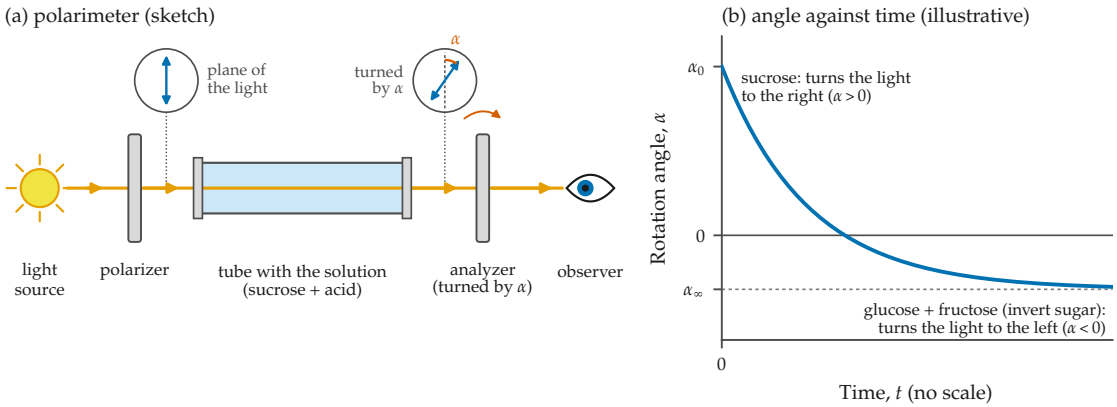


Figure 1. How Wilhelmy followed the hydrolysis of sucrose. (a) Sketch of a polarimeter: the light goes through the polarizer, the tube with the solution, and the analyzer, which the observer turns by an angle α. The circles show the plane of the light seen face-on, before and after the tube. (b) Rotation angle against time: it starts positive, with sucrose, falls as an exponential, crosses zero, and ends negative, with invert sugar. Curve without scale.

Source: author's own drawing; illustrative sketch and curve of the reaction studied by Wilhelmy (1850).

Sucrose turns the light to the right ($\alpha > 0$), and the mixture of glucose and fructose turns it to the left. That is why this mixture is called invert sugar (WILHELMY, 1850). Since sucrose and its products turn that light by different amounts, Wilhelmy was able to measure the progress of the reaction over time and proposed a differential equation in which the rate was proportional to the concentration of the reactant. In modern

notation, with S for sucrose, it reads

$$v = -\frac{\mathrm{d[S]}}{\mathrm{d}t} = k[\mathrm{S}] \tag{1.2}$$

where v is the reaction rate, t is time, and k is a constant that grows with the amount of acid. Water is in large excess and barely changes, so only [S] appears in the law. With $[\mathrm{S}] = [\mathrm{S}]_0$ at $t = 0$, the solution is

$$[\mathrm{S}] = [\mathrm{S}]_0\, \mathrm{e}^{-kt} \tag{1.3}$$

Each sugar turns the light in proportion to its concentration, and α is the sum of these parts. So the angle falls from its initial value α_0 to its final value α_∞ following the same exponential as Equation (1.3), $\alpha = \alpha_\infty + (\alpha_0 - \alpha_\infty)\,\mathrm{e}^{-kt}$, and crosses zero on the way (Figure 1b). This assumes that the glucose formed already turns the light by its final amount. Glucose has two forms, which turn into each other (mutarotation, Section 8.3); if that exchange is slow compared with the hydrolysis, the curve of α lags behind. A lag of this kind is what I propose to measure in Chapter 8. Wilhelmy was the first to describe a chemical reaction with a rate law.

1.2 Stepwise reactions and the problem of intermediates

Many reactions do not happen in a single collision. They go through a sequence of elementary steps, each with its own rate constant (Chapter 2). These steps usually form and consume species called **intermediates**, that is, atoms, ions, or molecules that really exist, but rarely last long enough to be isolated or even detected directly.

When a mechanism has an intermediate, the rate law of the overall reaction becomes hard to write using only what we can measure, which is the concentrations of reactants, products, and catalysts. In the early twentieth century, Max Bodenstein (BODENSTEIN, 1913) and, independently, David Chapman and Leo Underhill (CHAPMAN; UNDERHILL, 1913) solved this problem while studying the reaction between hydrogen and chlorine under light. Their idea was to assume that the intermediate's concentration changes little with time, compared with the rates at which it forms and is consumed, $\mathrm{d[I]}/\mathrm{d}t \approx 0$. This is the **steady-state hypothesis** (SSH), also called the steady-state approximation, the central subject of this work. The hypothesis does not mean the intermediate's concentration stays constant. In general it is small and follows, with a delay, the fall of the reactants themselves (IUPAC, 2025b). Chapter 3 gives the full definition, along with the criterion that says when the hypothesis holds.

The SSH shows up in very different settings. In catalytic reactions, the catalyst (a substance that speeds up a reaction without being consumed) often acts by forming intermediates on its own surface. That is the case of the iron catalyst in ammonia synthesis (Chapter 5) and the vanadium catalyst in one step of sulfuric acid manufacture (Chapter 6). In biochemical processes, the SSH gives the Michaelis-Menten equation, for enzymes (Chapter 6). In the light-driven decomposition of ozone, it gives the concentration of the free oxygen atom, both in a closed cell and in the mechanism Sydney Chapman proposed, in 1930, for the ozone layer of the stratosphere (CHAPMAN, 1930) (Chapter 4). Chapter 7 gathers the cases where the hypothesis stops holding, and Chapter 8 sets it against data.

1.3 The question this work asks

The question is: when is the steady-state hypothesis a good approximation, how much error does it carry, and can data show that error? I study two cases in detail, the light-driven decomposition of ozone and ammonia synthesis, and take the same question to general mechanisms and to the enzyme assays already published.

It splits into four parts:

(a) What simple criterion, comparing time scales, tells us whether the hypothesis holds?

(b) How large is the hypothesis's error, compared with the full solution of the same mechanism, for ozone (photolysis in a cell and in the stratosphere) and for ammonia (species held on the catalyst surface, with stated illustrative parameters, and reactor start-up)?

(c) Do the data from the December 2025 article (RIBEIRO, 2025b) follow the rate laws the

hypothesis gives?

(d) Can the data show when the hypothesis is wrong? And does it hold in the enzyme assays already published?

1.4 How this work was done

I wrote the first version of this monograph in July 2025, at the Escola de Talentos do Instituto Principia (RIBEIRO, 2025a), and a short article on the SSH in December 2025 (RIBEIRO, 2025b). This text, dated October 16, 2026, is the revised and expanded version of that monograph.

Every number used in this work comes from one of five sources, and each figure and table says which. The first is verified literature, always with a citation. The second is the data from the December 2025 article (RIBEIRO, 2025b), 13 points for ozone and 13 for ammonia. The third is the data from SABIO-RK (WITTIG *et al.*, 2018), a public database of enzyme kinetics, from which I took a sample of entries. The fourth is simulation, that is, Python code, listed in Appendix B, that solves the full system of rate equations and compares it with the SSH, or fits curves to the data. The fifth is synthetic data: model curves with noise drawn by the code, which show what an assay could reveal if the model and the noise are right. They are never measurements. The bench test of Chapter 8 is a proposal: I have not done it yet, and what I say about its result is a model prediction.

Many of the ideas I use are already in textbooks and papers. Each chapter says where each one comes from, and Section 9.2 separates what was already known from what this work adds.

Chapter 2 follows a general chemistry textbook, Flowers *et al.* (2019); to go further, I recommend Engel & Reid (2013), Chang & Goldsby (2013), McQuarrie & Simon (1997), and Maron & Lando (1974). The statistical basis of the activation energy, also from that chapter, is in Hill (1986), and the Hammond postulate (Section 2.6) comes from Hammond (1955). The conceptual basis of the SSH, the subject of Chapter 3, is discussed by Laidler (1987), and Frost & Pearson (1961) gathers several examples of it. Catalytic mechanisms, like those in Chapter 6, appear in Espenson (1995). The kinetics used in process engineering, which goes into the simulation of the ammonia reactor (Section 5.6), is in Hougen & Watson (1947). The relationship between thermodynamics and kinetics, the theme of Chapter 5, is treated by Atkins, Paula & Keeler (2022) and Levine (2009).

1.5 What each chapter does

Chapter 2 gathers the fundamentals the following chapters use: how to measure and write the rate of a reaction, how to find its rate law, the effect of temperature through the Arrhenius equation, and why multistep mechanisms have a rate-determining step. Chapter 3 defines the SSH, tells its history, and compares it, in the same two-step mechanism, with the other two ways of treating an intermediate (the rate-determining step and the pre-equilibrium), comparing the law each one gives, and tests the SSH against the exact solution of two examples, one of them solved on the computer.

The next two chapters each apply the SSH to a real case and compare the rate law it gives with the data from the December 2025 article. The two cases sit at opposite ends: in ozone, the intermediate is a free atom in the gas, which reaches steady state in milliseconds; in ammonia, the intermediates are held on the surface of a solid catalyst. Chapter 4 deals with ozone photolysis in a cell and with Chapman's mechanism in the stratosphere. Chapter 5 deals with the species held on the surface of the iron catalyst in ammonia synthesis and includes a simulation of reactor start-up. In both, I use synthetic data to show which assay would test each law. Chapter 6 closes the applications with the Michaelis-Menten equation, one step of sulfuric acid manufacture, and the rate law of hydrogen with bromine.

Chapter 7 gathers, with numbers, the cases where the SSH fails. For the mechanism $A \rightleftharpoons I \rightarrow P$, it gives the right criterion for the SSH and for the pre-equilibrium, shows that swapping the constants of the steps $A \rightarrow I$ and $I \rightarrow P$ leaves the product curve unchanged, maps where each one holds and the band where neither works, and presents a third law, the total

law, that contains both. It closes with a six-step checklist. Chapter 8 takes the SSH to data. In a sample of SABIO-RK, it computes, in published enzyme assays, how large the Segel and Slemrod parameter is, which tells whether there is too much enzyme for the SSH to hold. It then shows with synthetic data when a set of points can reveal the SSH error, and proposes a bench test, with invertase and a glucose meter. Chapter 9 answers each part of this introduction's question, using those numbers, and separates what was already known from what is new. The appendices hold the longer derivations (Appendix A), the code (Appendix B), and the tables of constants and of the data from the December 2025 article (Appendix C).

2 CHAPTER Fundamentals of chemical kinetics

This chapter gathers the ideas from chemical kinetics that the following chapters use: how to measure the rate of a reaction, how to write and find its rate law, and how the concentration of a reactant changes with time. The basis is a general chemistry textbook (FLOWERS *et al.*, 2019). I worked out the derivations myself, and the longer calculations are in Appendix A.

2.1 Reaction rate

2.1.1 Average rate

Consider some species X whose concentration falls with time in a hypothetical reaction (Figure 2). I use square brackets for concentration: [X] is the concentration of X, in moles per liter (mol/L) or, in SI units, moles per cubic meter (mol/m^3).

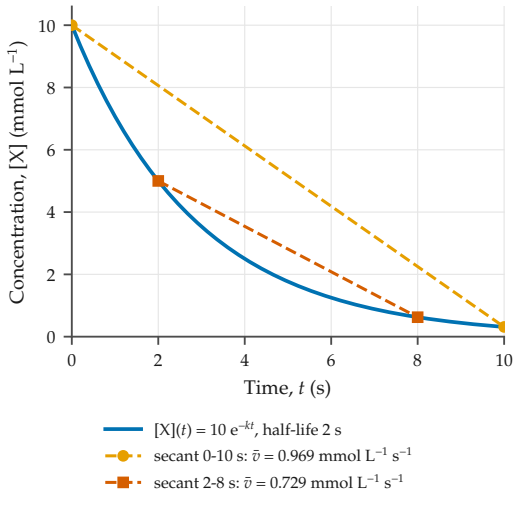


Figure 2. Average rate of a hypothetical decay reaction, measured with two secants: from 0 to 10 s and from 2 to 8 s. The magnitude of the secant's slope is the average rate over the interval.

Source: simulation (`codigo/figuras_fundamentos.py`), illustrative values.

Between two instants, the **average rate** of change of [X] is the ratio of the change in concentration to the length of the interval, that is, the slope of the secant line joining the two points on the curve. On the curve in Figure 2, between $t = 0$ and $t = 10\,\text{s}$,

$$\left.\frac{\Delta[\text{X}]}{\Delta t}\right|_{0\to10} = \frac{(0.3125 - 10.0)\,\text{mmol/L}}{10\,\text{s}} \approx -0.969\,\text{mmol/(L s)}, \tag{2.1}$$

and between $t = 2\,\text{s}$ and $t = 8\,\text{s}$,

$$\left.\frac{\Delta[\text{X}]}{\Delta t}\right|_{2\to8} = \frac{(0.625 - 5.00)\,\text{mmol/L}}{6\,\text{s}} \approx -0.729\,\text{mmol/(L s)}. \tag{2.2}$$

The concentration of X falls with time, so this slope is negative. Since X is consumed (it is a reactant), we call the positive number $-\Delta[\text{X}]/\Delta t$ the **average rate of consumption of X**: about 0.969 mmol/(L s) between 0 and 10 s, and 0.729 mmol/(L s) between 2 and 8 s. The first value is larger, so the reaction was faster in the first stretch, when there was more X, than in the second. This comparison comes back, in a different sense, in Section 2.2.

2.1.2 Instantaneous rate

The average rate depends on the interval chosen, and the reaction keeps slowing down as X runs out. If the interval Δt gets smaller and smaller, the secant line rotates until it touches the curve at a single point and becomes the **tangent** line. The slope of that line is the **instantaneous rate**, that is, how fast [X] is changing at that exact instant:

$$\frac{\text{d}[\text{X}]}{\text{d}t} = \lim_{\Delta t\to0}\frac{\Delta[\text{X}]}{\Delta t}. \tag{2.3}$$

The symbol $\text{d}[\text{X}]/\text{d}t$ reads "derivative of [X] with respect to time." For a reactant it is negative, and the instantaneous rate of consumption is $-\text{d}[\text{X}]/\text{d}t$, always a positive number.

On the curve in Figure 3, $[\text{X}](t) = [\text{X}]_0\,\text{e}^{-kt}$, with $[\text{X}]_0 = 10\,\text{mmol/L}$ and $k = 0.3466/\text{s}$ (the concentration falls to half every 2 s). At $t = 2\,\text{s}$,

[X] = 5.00 mmol/L, and the instantaneous rate is

$$\left.\frac{\mathrm{d[X]}}{\mathrm{d}t}\right|_{t=2\,\mathrm{s}} = -k\,[\mathrm{X}](2\,\mathrm{s}) = -(0.3466/\mathrm{s})(5.00\,\mathrm{mmol/L}) \approx -1.73\,\mathrm{mmol/(L\,s)}. \tag{2.4}$$

The instantaneous rate of consumption of X at that instant is $-\mathrm{d[X]}/\mathrm{d}t \approx +1.73\,\mathrm{mmol/(L\,s)}$. The tangent line at $t = 2\,\mathrm{s}$ crosses the time axis at 4.885 s and the concentration axis at 8.466 mmol/L (Figure 3).

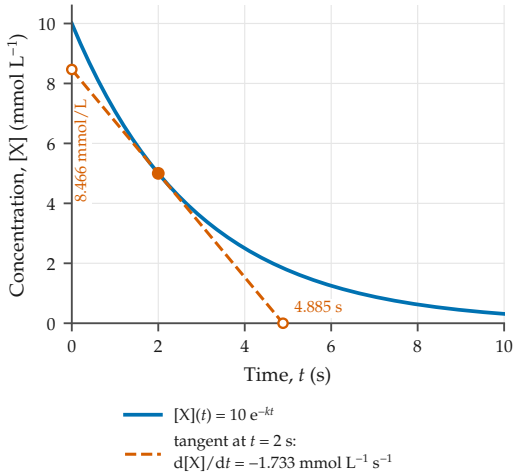


Figure 3. Instantaneous rate at $t = 2$ s on the same decay curve: the exact tangent at that point crosses the vertical axis at 8.466 mmol/L and the time axis at 4.885 s.

Source: simulation (`codigo/figuras_fundamentos.py`), illustrative values.

2.1.3 Stoichiometry and reaction rate

In a reaction, not every species changes concentration at the same rate: if a product's coefficient is twice a reactant's, it forms twice as fast. To separate this stoichiometric difference from the rate of the reaction itself, IUPAC (the International Union of Pure and Applied Chemistry, which standardizes names and definitions in chemistry) defines the **rate of conversion**, $\dot{\xi}$ (mol/s). If n_X is the amount of substance of X (mol) and ν_X is the stoichiometric coefficient of X in the reaction (negative for a reactant, positive for a product), then

$$\dot{\xi} = \frac{1}{\nu_X}\frac{\mathrm{d}n_X}{\mathrm{d}t}, \tag{2.5}$$

and this value is the same for every species in the reaction (IUPAC, 2025d).

Example 2.1. The amount of H_2O_2 in the decomposition $2\,H_2O_2 \rightarrow 2\,H_2O + O_2$ falls by 10 mol/min: $\mathrm{d}n_{H_2O_2}/\mathrm{d}t = -10\,\mathrm{mol/min}$. The rate of conversion is $\dot{\xi} = -(1/2)\,\mathrm{d}n_{H_2O_2}/\mathrm{d}t = 5\,\mathrm{mol/min}$. Since the coefficient of O_2 is 1 and that of H_2O is 2, O_2 forms at +5 mol/min and H_2O, at +10 mol/min.

Example 2.2. In the decomposition $2\,O_3 \rightarrow 3\,O_2$, if $\mathrm{d}n_{O_3}/\mathrm{d}t = -20\,\mathrm{mol/min}$, the rate of conversion is $\dot{\xi} = 10\,\mathrm{mol/min}$, and O_2 forms at $3\dot{\xi} = +30\,\mathrm{mol/min}$. This reaction comes back, with a different mechanism (photolysis by ultraviolet light), in Chapter 4.

In a solution of constant volume, it is more practical to divide everything by the volume V and work with concentrations. For the reaction $a\,A + b\,B \rightarrow p\,P + q\,Q$, I define the **reaction rate** v (always positive) as

$$v = -\frac{1}{a}\frac{\mathrm{d[A]}}{\mathrm{d}t} = -\frac{1}{b}\frac{\mathrm{d[B]}}{\mathrm{d}t} = \frac{1}{p}\frac{\mathrm{d[P]}}{\mathrm{d}t} = \frac{1}{q}\frac{\mathrm{d[Q]}}{\mathrm{d}t}, \tag{2.6}$$

valid at constant volume and with no appreciable buildup of intermediates (IUPAC, 2025d). In the integrated rate laws of Section 2.3 I write reactant A with coefficient 1, so that $\mathrm{d[A]}/\mathrm{d}t = -k[\mathrm{A}]^n$. If the coefficient of A in the actual reaction is a, just replace k with ak.

2.1.4 Units of the rate

The unit of v depends on the system:

- homogeneous reaction in solution: mol/(L s) (or mol/(m^3 s) in SI);
- gas-phase reaction: mol/(L s), or, in terms of partial pressure, atm/s;
- atmospheric reaction: molecule/(cm^3 s) (concentration in molecules per cubic centimeter instead of moles per liter, and I return to this unit in Chapter 4);
- heterogeneous reaction, on a surface: mol/(m^2 s) (per unit area instead of volume).

2.2 Rate law and order

2.2.1 Definition

The **rate law** is the equation that relates the reaction rate to the concentrations of the species we can measure and control: reactants, products, and catalysts. For $a\,\mathrm{A} + b\,\mathrm{B} \rightarrow c\,\mathrm{C} + d\,\mathrm{D}$, a common form is

$$v = k\,[\mathrm{A}]^n\,[\mathrm{B}]^m, \tag{2.7}$$

where k is the **rate constant**, n is the **order** of the reaction with respect to A (the exponent of [A] in the law), and m is the order with respect to B. The constant k depends on temperature but not on concentrations, and its unit changes with $n + m$ (Section 2.2.5). Orders are usually 0, 1, or 2, but they can also be negative or fractional (Section 2.2.4). They almost never go above 3, and traffic helps explain why:

> **Analogy: cars at the intersection**
>
> *Picture a busy intersection. Two cars crashing there is common. Now picture three cars crashing at the same time, at the same spot and in the same thousandth of a second: that is extremely rare. Four or five, practically never, and if it ever happens it makes the evening news. It is the same with molecules. An elementary step in which three molecules collide together is rare: the chance of three particles colliding at the same time is less than one thousandth of the chance of two (FLOWERS et al., 2019). With four or more, it is smaller still. In an elementary step, the overall order equals the number of molecules that collide in it, its molecularity (Section 2.7). That is why orders above 3 almost never show up.*

An intermediate forms and is consumed during the reaction itself, and it is usually hard to measure directly. That is why the rate law measured in the lab is written only with species we can control, and an intermediate predicted by a mechanism must be rewritten in terms of the reactants before it can enter that law. This is done through the pre-equilibrium or the steady-state hypothesis (SSH), also called the steady-state approximation, the two subjects of Chapter 3.

2.2.2 Order and stoichiometric coefficient

The orders in a rate law are not, in general, the coefficients of the balanced equation. They only match when the reaction is **elementary**, meaning it happens in a single event (Section 2.7). If the reaction is not elementary, the orders depend on the mechanism. When one step is much slower than the others, the rate of the reaction is that step's rate. If that step involves an intermediate, it must be written in terms of the reactants using the faster steps before it, and that is why the orders can differ from the coefficients of the rate-determining step.

For example, for $2\,\mathrm{NO} + \mathrm{Cl_2} \rightarrow 2\,\mathrm{NOCl}$ a two-step mechanism was proposed, the first step fast and reversible and the second slow:

$$\begin{aligned} \mathrm{NO} + \mathrm{Cl_2} &\rightleftharpoons \mathrm{NOCl_2}, \\ \mathrm{NOCl_2} + \mathrm{NO} &\longrightarrow 2\,\mathrm{NOCl}. \end{aligned} \tag{2.8}$$

The rate-determining step has coefficient 1 for NO, but the rate law the mechanism predicts is $v = k[\mathrm{NO}]^2[\mathrm{Cl_2}]$ (FLOWERS *et al.*, 2019). According to IUPAC, in a reaction with more than one step there is no general relationship between the overall coefficients and the orders (IUPAC, 2025a). The reverse is not true either. Orders matching the coefficients only raise the chance that a reaction is elementary; to be sure, you need to know the mechanism or have direct evidence.

2.2.3 Zero order and autocatalysis

In a **zero-order** reaction, the rate does not depend on the reactant's concentration: $v = k$, and $\mathrm{d}[\mathrm{A}]/\mathrm{d}t = -k$. As long as the reaction stays in this regime, adding more reactant does not speed it up. It sounds strange, but a room full of chairs makes it clear:

> **Analogy: chairs in a room**
>
> *Picture a room with 100 chairs. Each person who arrives has to sit down to fill out a form, which takes a fixed amount of time, then gets up and leaves. The "reaction" is this whole cycle: sit down, fill out the form, leave. The rate is the number of people who leave with a com-*

pleted form, per minute. With few people in the room, almost every chair is free, and doubling the number of people doubles the rate: that is order 1 in the number of people. Now imagine the room gets crowded, say with 1000 people waiting to get in. All 100 chairs are occupied, and no matter how many more people arrive, the rate stays locked at the maximum pace at which chairs become free. That is order zero, and it only changes if we change the number of chairs or how long each person stays seated.

In catalysis (Sections 5.3 and 6.1), the chairs are the catalyst's **active sites**, the people are the reactant, and the time seated is the time the site takes to convert one molecule. That is why a catalyzed reaction can be zero order in the reactant (when every site is occupied) and first order in the catalyst (when doubling the number of sites doubles the rate).

Autocatalysis. At the start of the reaction $\mathrm{A} + \mathrm{P} \longrightarrow 2\,\mathrm{P}$, the rate is low, because there is little product P to catalyze it. As P builds up, the reaction speeds up to a maximum. Toward the end it slows down, because reactant A is running out. A rate law that gives this behavior is

$$v = k\,[\mathrm{A}]^{r}\,[\mathrm{P}]^{p}, \tag{2.9}$$

provided some product is already present at the start ($[\mathrm{P}]_0 > 0$; without it, the reaction never begins) (IUPAC, 2025e). In the simplest case, $r = p = 1$. If the reaction also has a slow uncatalyzed path, the rate law gains one more term, $v = k_0[\mathrm{A}]^{r} + k[\mathrm{A}]^{r}[\mathrm{P}]^{p}$. Figure 4 shows both curves: the zero-order one, a straight line until the reactant runs out, and the autocatalytic one, shaped like an S.

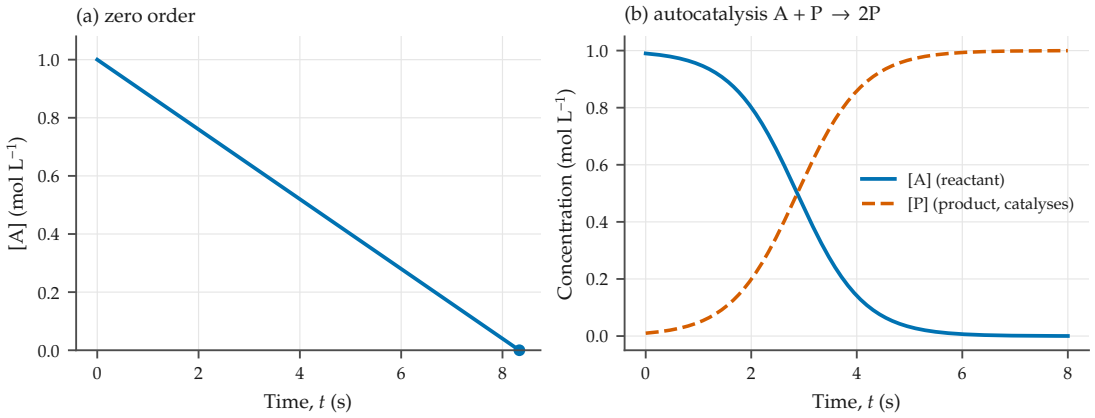


Figure 4. (a) Zero-order reaction: the concentration falls in a straight line until the reactant runs out. (b) Autocatalysis $\mathrm{A} + \mathrm{P} \longrightarrow 2\,\mathrm{P}$: the product speeds up its own formation, giving an S-shaped curve.

Source: simulation (`codigo/figuras_fundamentos.py`), illustrative values.

2.2.4 Orders that change

Not every rate law has fixed orders under all conditions, and not every law has an overall order.

Example 2.3. A multistep reaction, with an intermediate, can have the rate law

$$v = \frac{k_1[\mathrm{A}][\mathrm{B}]}{K + [\mathrm{A}]}, \tag{2.10}$$

where K is a constant with units of concentration. If $[\mathrm{A}] \ll K$, $v \approx (k_1/K)[\mathrm{A}][\mathrm{B}]$, that is, apparent order 1 in A and 1 in B. If $[\mathrm{A}] \gg K$, $v \approx k_1[\mathrm{B}]$, and the apparent order in A drops to 0 (IUPAC, 2025a). This form shows up again in the chairs of Section 2.2.3 and in Michaelis-Menten kinetics (Chapter 6), and I come back to it in Chapter 3.

Example 2.4. For $\mathrm{H_2(g)} + \mathrm{Br_2(g)} \longrightarrow 2\,\mathrm{HBr(g)}$, the measured rate law is

$$\frac{\mathrm{d}[\mathrm{HBr}]}{\mathrm{d}t} = \frac{k[\mathrm{H_2}][\mathrm{Br_2}]^{3/2}}{[\mathrm{Br_2}] + k'[\mathrm{HBr}]}. \tag{2.11}$$

Since $v = (1/2)\,\mathrm{d}[\mathrm{HBr}]/\mathrm{d}t$, the constant for v is $k/2$. This rate law does not have the form $k[\mathrm{H_2}]^{\alpha}[\mathrm{Br_2}]^{\beta}$, so it has no single overall order. At the start, with almost no HBr,

$v \approx (k/2)[H_2][Br_2]^{1/2}$, that is, order 1 in H_2 and 1/2 in Br_2. With a lot of HBr, $v \approx \big(k/(2k')\big)[H_2][Br_2]^{3/2}[HBr]^{-1}$, and the order in HBr is −1, meaning the product itself slows the reaction down. The mechanism behind this rate law, with the H and Br radicals (atoms or molecules with an unpaired electron, and so very reactive) as intermediates, is in Section 6.3.

2.2.5 Units of the rate constant

Since the rate always has the same unit (mol/(L s)), the unit of k changes with the overall order $n + m$ in Equation (2.7), so that both sides of the equation match.

Example 2.5. For $v = k[A]^\alpha[B]^\beta$,

$$[k] = \frac{[v]}{[A]^\alpha[B]^\beta} = \frac{\text{mol/(L s)}}{(\text{mol/L})^{\alpha+\beta}} = (\text{mol/L})^{1-(\alpha+\beta)}\,1/\text{s}. \tag{2.12}$$

In terms of partial pressures, $v = k\,P_A^\alpha P_B^\beta$ gives $[k] = \text{atm}^{1-(\alpha+\beta)}\,1/\text{s}$.

In the H_2+Br_2 rate law (Equation (2.11)), k' can only be added to $[Br_2]$ if both terms of the denominator have the same unit; so k' is dimensionless.

Isolating $[k]$ the same way gives $[k] = (\text{mol/L})^{-1/2}\,1/\text{s}$, or, in SI units, $(\text{mol/m}^3)^{-1/2}/\text{s}$. The step-by-step calculation is in Appendix A.12.

Table 1 gathers the units for the most common orders.

In atmospheric chemistry, concentration is usually given in molecules per cubic centimeter instead of moles per liter. A second-order k in $\text{cm}^3/(\text{molecule s})$ converts to L/(mol s) by multiplying by $N_A/1000$ (I return to this calculation in Chapter 4).

Table 1. Unit of the rate constant k for a rate law of overall order n.

Overall order n	Practical unit	SI unit
0	mol/(L s)	1×10^3 mol/(m^3 s)
1	1/s	1/s
2	L/(mol s)	1×10^{-3} m^3/(mol s)
3	L^2/(mol^2 s)	1×10^{-6} m^6/(mol^2 s)

Source: author's own work, based on Flowers *et al.* (2019), Table 12.1.

2.3 How concentration changes with time

The rate laws of Section 2.2 are differential equations, that is, they give the slope of the curve $[A](t)$ at each instant, and the curve itself still has to be found. For irreversible reactions, with the rate law written in terms of a single reactant, $d[A]/dt = -k[A]^n$, we can separate variables and integrate both sides. The full derivation, with the limits of each integral, is in Appendix A.1. For example, for zero order,

$$\int_{[A]_0}^{[A]} d[A]' = -k\int_0^t dt' \quad\Longrightarrow\quad [A] = [A]_0 - kt, \tag{2.13}$$

valid for $0 \le t \le [A]_0/k$. This domain restriction exists because the concentration cannot go negative: the reaction ends when the reactant runs out. Repeating the same procedure for the other orders leads to Table 2, which also shows which plot gives a straight line in each case, the subject of Section 2.4.

The laws in Table 2 hold for irreversible reactions whose rate law depends on a single reactant. They stop holding when two reactants are not in the same ratio as their coefficients (then the isolation trick is needed, Section 2.4.2), or when the reaction also has an appreciable reverse path. In that second case, for a reversible reaction that is first order in both directions, $A \rightleftharpoons B$, with rate constant k_1 forward and k_{-1} backward, the concentration of A does not fall to zero. It tends toward the equilibrium value $k_{-1}[A]_{tot}/(k_1 + k_{-1})$, with characteristic time $\tau = 1/(k_1 + k_{-1})$; the derivation is in Appendix A.3. Also, these equations only exist because the rate law has a simple form. For mechanisms with several steps, in general the only way forward is to solve the system on a computer, or to use an approximation such as the steady-state hypothesis (Chapter 3).

Table 2. Integrated rate law, plot that gives a straight line, that line's slope, and half-life for each order. In ln[A], read $\ln([A]/c^\circ)$, with $c^\circ = 1\,\text{mol/L}$; the slope does not change with this choice.

Order	Integrated law	Plot that gives a line	Slope	Half-life $t_{1/2}$
0	$[A] = [A]_0 - kt$	[A] vs. t	$-k$	$[A]_0/(2k)$
1	$\ln[A] = \ln[A]_0 - kt$	ln[A] vs. t	$-k$	$\ln 2/k$
2	$1/[A] = 1/[A]_0 + kt$	$1/[A]$ vs. t	$+k$	$1/(k[A]_0)$
$n \neq 1$	$[A]^{1-n} = [A]_0^{1-n} + (n-1)kt$	$[A]^{1-n}$ vs. t	$(n-1)k$	Appendix A.1

Source: author's own work.

2.4 How to find the rate law

2.4.1 Initial rates

One way to find the rate law is to run several trials at the same temperature, so k does not change, and measure the rate right at the start of each one, from the slope of the tangent at $t = 0$ (Section 2.1.2). At that instant the concentrations are whatever we put in the flask, and there is not yet enough product to react back. Between one trial and the next, only one reactant's concentration is changed at a time, and the ratio between the initial rates shows the order in that reactant.

Example 2.6. The general chemistry textbook by Flowers *et al.* (2019) gives this exercise for the reaction $2\,NO(g) + Cl_2(g) \rightarrow 2\,NOCl(g)$: three trials at the same temperature, with the initial rates of NO consumption in Table 3.

Between trials 1 and 2, $[NO]_0$ stays the same and $[Cl_2]_0$ rises by a factor of 1.5; the rate rises by the same factor of 1.5, which gives order 1 in Cl_2. Between trials 1 and 3, $[Cl_2]_0$ stays the same and $[NO]_0$ rises by a factor of 1.5; the rate rises by a factor of $2.25 = 1.5^2$, which gives order 2 in NO. The rate law is $-\mathrm{d}[NO]/\mathrm{d}t = k_{NO}[NO]^2[Cl_2]$, with

$$\begin{aligned} k_{NO} &= \frac{0.00300\,\text{mol/(L s)}}{(0.10\,\text{mol/L})^2(0.10\,\text{mol/L})} \\ &= 3.0\,\text{L}^2/(\text{mol}^2\,\text{s}) \\ &= 3.0 \times 10^{-6}\,\text{m}^6/(\text{mol}^2\,\text{s}). \end{aligned} \tag{2.14}$$

Since $v = (1/2)(-\mathrm{d}[NO]/\mathrm{d}t)$, the rate constant for the reaction (Equation (2.6)) is $k = k_{NO}/2 = 1.5\,\text{L}^2/(\text{mol}^2\,\text{s})$. This is the same reaction as in Equation (2.8): the rate law that comes out of the table has the order 2 in NO that the two-step mechanism predicts.

Table 3. Initial rates of NO consumption in $2\,NO(g) + Cl_2(g) \rightarrow 2\,NOCl(g)$, three trials at the same temperature.

Trial	$[NO]_0$ (mol/L)	$[Cl_2]_0$ (mol/L)	$-\mathrm{d}[NO]/\mathrm{d}t$ at $t = 0$ (mol/(L s))
1	0.10	0.10	0.00300
2	0.10	0.15	0.00450
3	0.15	0.10	0.00675

Source: literature: Flowers *et al.* (2019), Example 12.5 (textbook example values).

In the first version of this work, I had used a different exercise here, with a reaction of order 10 and a concentration of 100 mol/L, nearly twice the concentration of pure water itself. Those numbers do not describe any real reaction, so I swapped in the textbook example.

2.4.2 Isolation and pseudo-first order

When the rate law has two reactants, $v = k[A][B]$, integrating it is harder (general solution for $[A]_0 \neq [B]_0$ in Appendix A.2). One trick simplifies things: if one reactant is in large excess, its concentration barely changes during the reaction.

Example 2.7. In $A + B \longrightarrow$ products, $v = k[A][B]$, with B in large excess ($[B]_0 \gg [A]_0$), the relative change in [B] over the whole reaction is small, and the approximation $[B] \approx [B]_0$ holds. The reaction then behaves as if it were first order

in A, with an observed constant

$$k_{\text{obs}} = k[\text{B}]_0, \qquad [k_{\text{obs}}] = 1/\text{s}, \tag{2.15}$$

and $[\text{A}] = [\text{A}]_0\, \text{e}^{-k_{\text{obs}} t}$. We call this **pseudo-first order** (IUPAC, 2025a). Repeating the trial with several values of $[\text{B}]_0$ and measuring k_{obs} in each one, Equation (2.15) shows that k is the slope of the line k_{obs} vs. $[\text{B}]_0$.

2.4.3 Graphical method

Another way is to measure [A] as a function of time in a single reaction and test, on one plot, which of the three forms in Table 2 gives a straight line, because only the right transformation does that.

Figure 5 compares [A], ln[A], and 1/[A] vs. t for orders $n = 0, 1, 2, 3$, all with the same $[\text{A}]_0$ and the same initial rate v_0. In the 1/[A] panel, the curvature is given by

$$\frac{\text{d}^2}{\text{d}t^2}\left(\frac{1}{[\text{A}]}\right) = (2 - n)\, k^2\, [\text{A}]^{2n-3}, \tag{2.16}$$

positive (the curve rises faster and faster) when $n < 2$, zero (a straight line) at $n = 2$, and negative (the curve rises slower and slower) when $n > 2$, because the slope of 1/[A] is $k[\text{A}]^{n-2}$. With $[\text{A}]_0 = 1\,\text{mol/L}$ and $v_0 = 0.1\,\text{mol/(L s)}$, at $t = 6\,\text{s}$ the value of 1/[A] is 2.500, 1.822, 1.600, and 1.483 L/mol for $n = 0, 1, 2, 3$. Only the $n = 2$ curve is a straight line. At $t = 6\,\text{s}$, the $n = 0$ and $n = 1$ curves are already above it and the $n = 3$ curve is below it, exactly as Equation (2.16) predicts.

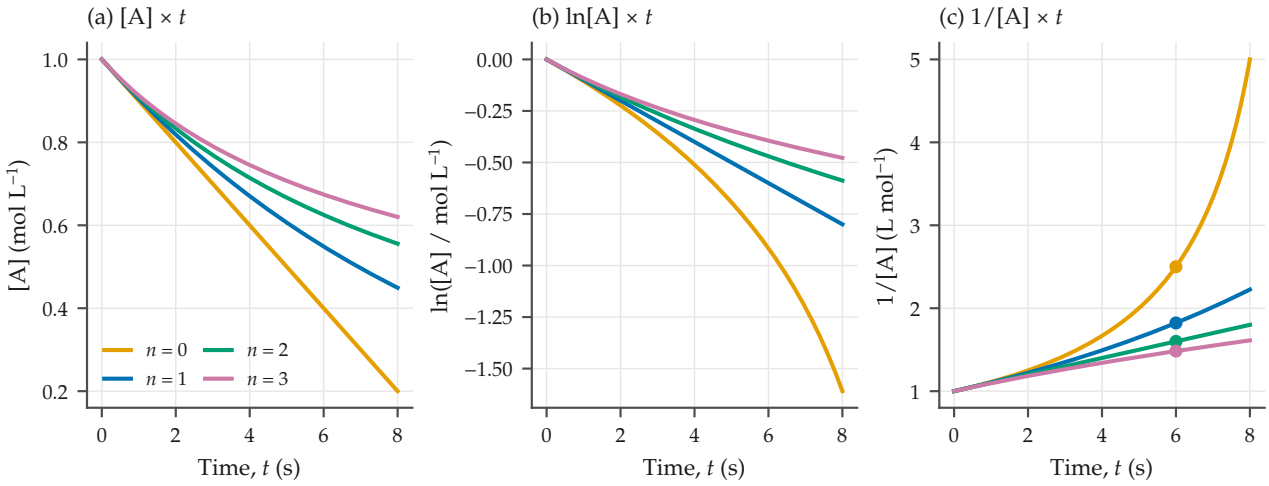


Figure 5. Effect of order $n = 0, 1, 2, 3$ on the shape of [A], ln[A], and 1/[A] vs. time, with the same $[\text{A}]_0$ and the same initial rate v_0. The points at $t = 6$ s in panel (c) are the 1/[A] values used in Equation (2.16).

Source: simulation (`codigo/figuras_fundamentos.py`), illustrative values.

Figure 6 organizes the same idea into a grid: each row is an order (0, 1, or 2), each column is a way of plotting [A] against t, and the highlighted panel, in each row, is the only one that gives a straight line.

The product appears only in the zero-order row, so its curve is not confused with the reactant's curve in the other two.

In practice, two precautions matter. First, the points need to cover a good part of the reaction (at least two or three half-lives), because near the start any curve looks almost straight, even with the wrong transformation. For an exact first-order decomposition, with 11 points equally spaced in time, the (wrong) plot of [A] against t gives $R^2 = 0.9990$ if the points cover only 20% of the reaction, but that falls to 0.9645 at 75% and 0.9104 at 90% (simulation: `codigo/fund_metodo_grafico.py`, in `dados/simulacao/fund_metodo_grafico.json`).

Second, it is better to look at the residuals, the difference between each point and the fitted line, than at the R^2 value alone. If the residuals bulge to one side, the order being tested is wrong, even if R^2 looks high.

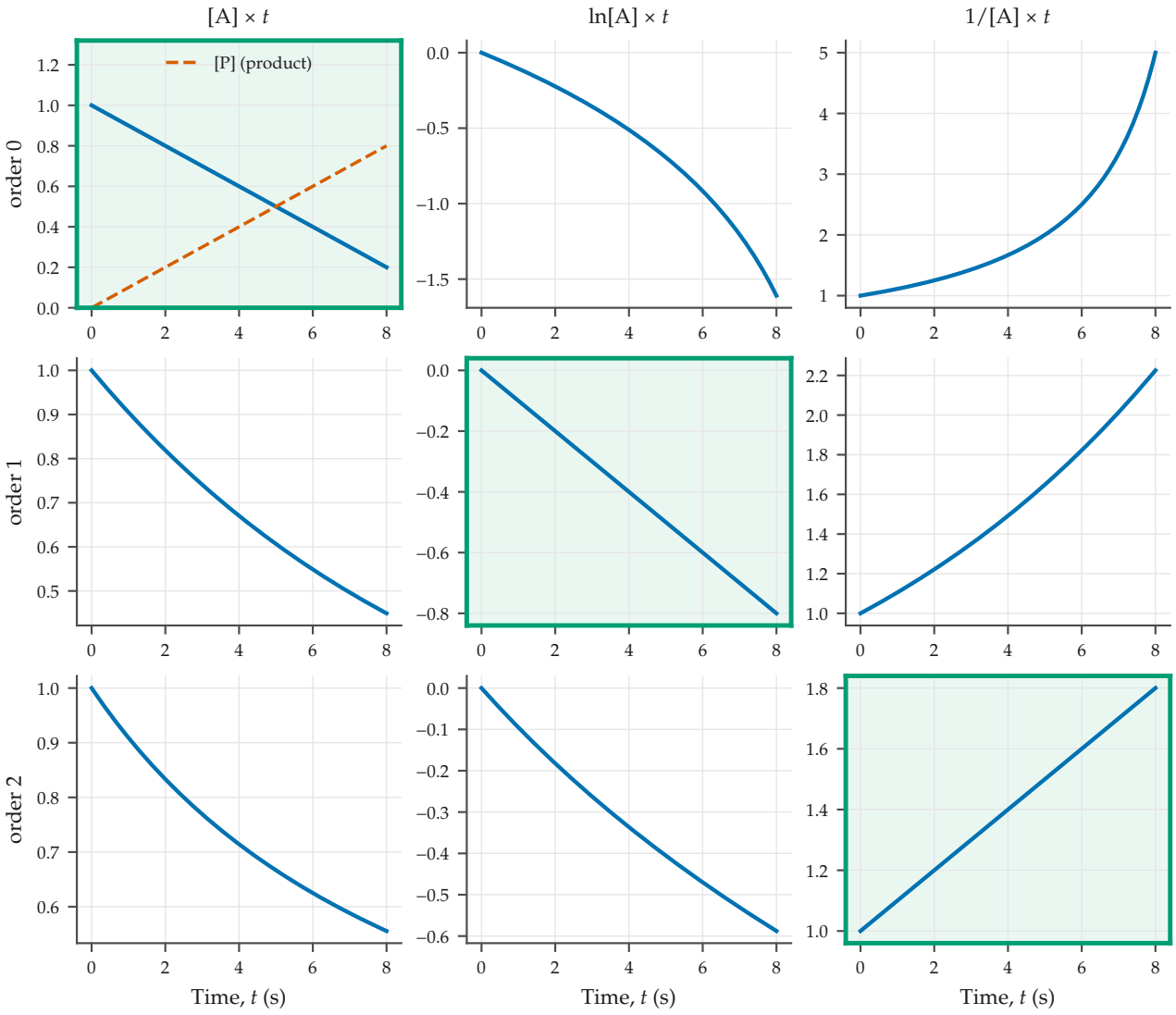


Figure 6. Linearization grid: each row is an order (0, 1, or 2); each column, a way of plotting concentration against time. The highlighted panel is, in each row, the only one that gives a straight line, and it is the one that shows the correct order. The product appears only in the zero-order row.

Source: simulation (`codigo/figuras_fundamentos.py`), illustrative values.

2.5 Temperature and the Arrhenius equation

As early as the nineteenth century it was known, by observation, that heat speeds up reactions, but an equation for that effect was still missing.

2.5.1 Arrhenius and the activation energy

Svante Arrhenius was born in 1859, in Sweden, studied at Uppsala, and, starting in 1881, worked in Stockholm with Erik Edlund, at the Academy of Sciences (NOBEL PRIZE OUTREACH, 2026b). In 1884, in his doctoral thesis, he argued that salts dissociate into ions in water even without an electric current, an idea accepted today but contested at the time, and it was for this electrolytic dissociation that he received the 1903 Nobel Prize in Chemistry (NOBEL PRIZE OUTREACH, 2026a; NOBEL PRIZE OUTREACH, 2026b).

In 1889, Arrhenius returned to an older problem, the hydrolysis of sucrose that Wilhelmy (1850) had already studied (Chapter 1). He proposed that only an "active" form of sucrose reacts with the acid. This active form is in fast equilibrium with the ordinary form, and going from one to the other absorbs heat. Applying the van 't Hoff equation to this equilibrium, Arrhenius

arrived at the relationship between the rate constant k and the absolute temperature T (in kelvin) (ARRHENIUS, 1889; LOGAN, 1982):

$$k = A\,\mathrm{e}^{-E_\mathrm{a}/(RT)}, \tag{2.17}$$

where A, the **pre-exponential factor**, has the same unit as k. E_a is the **activation energy** (J/mol, or kJ/mol in practice), the energy barrier the reactants need to overcome, and $R = 8.314\,\mathrm{J/(mol\,K)}$ is the gas constant. The term $\mathrm{e}^{-E_\mathrm{a}/(RT)}$ is, to a good approximation, the fraction of *collisions* between molecules with enough energy to react. This is different from the fraction of *molecules* with that energy, which is another calculation and comes back in Section 2.6.

Equation (2.17) is empirical: it describes the data well, but does not say where A and E_a come from. Giving physical meaning to these two parameters is the job of collision theory, created by Trautz (1916) and Lewis (1918), and of transition state theory, from Eyring (1935) and Evans & Polanyi (1935) (Section 2.6). Neither of them replaces the Arrhenius equation, and current tables of rate constants are still published in the Arrhenius form (BURKHOLDER *et al.*, 2019).

2.5.2 The straight-line form

Taking the natural logarithm of both sides of Equation (2.17),

$$\ln k = \ln A - \frac{E_\mathrm{a}}{R}\cdot\frac{1}{T}, \tag{2.18}$$

a straight line of $\ln k$ against $1/T$: the slope equals $-E_\mathrm{a}/R$, and the point where the line crosses the vertical axis (at $1/T = 0$) equals $\ln A$. In base-10 logarithm, the slope equals $-E_\mathrm{a}/(\ln 10 \cdot R)$, and $\ln 10 \approx 2.303$ is the "2.3" that shows up in this conversion, because $\ln x = (\ln 10)\log_{10} x$. This is how E_a is found from k measured at several temperatures. With just two, T_1 and T_2, the same line gives E_a without a plot (the calculation is in Appendix A.11). The larger E_a is, the steeper the line and the more k changes with temperature. A small E_a gives a reaction that is not very sensitive to it. Figure 7 shows this: panel (a) has $k(T)$ for two illustrative values of E_a, and panel (b) has the lines for three real atmospheric reactions, each drawn only over the temperature range in which it was measured.

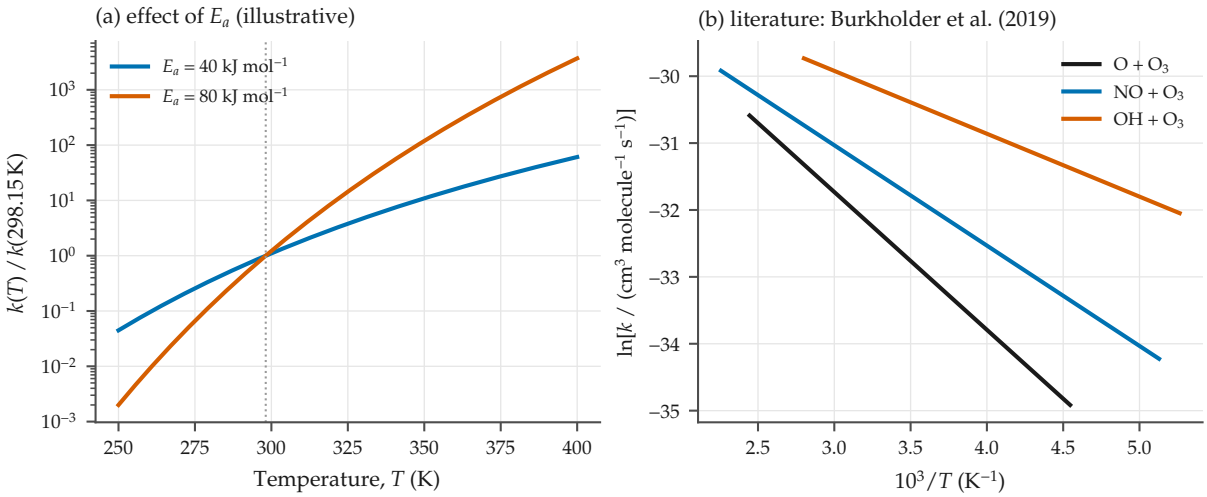


Figure 7. (a) Effect of activation energy on $k(T)$ (illustrative values): the larger E_a, the more k changes with temperature. (b) $\ln k$ against $1/T$ for three real atmospheric reactions, each line drawn only over the temperature range in which the constant was measured.

Source: (a) simulation (`codigo/figuras_fundamentos.py`), illustrative values; (b) literature: Burkholder *et al.* (2019) (JPL Publication 19-5, Tab. 1A p. 1-6, Tab. 1B p. 1-53, Tab. 1C p. 1-70); lines: simulation (`codigo/figuras_fundamentos.py`).

At a fixed temperature, k does not change with time. It depends only on temperature, and it does not matter how long the reaction has been running. What changes is the rate $v = k[\mathrm{A}]^n$, because the reactant keeps being consumed, as shown in Figure 8. In mechanisms with an in-

termediate, the rate of product formation can even start low and grow for a while, in what is called the induction period (Section 7.2). Even there, what changes is a concentration, and the constant k stays the same.

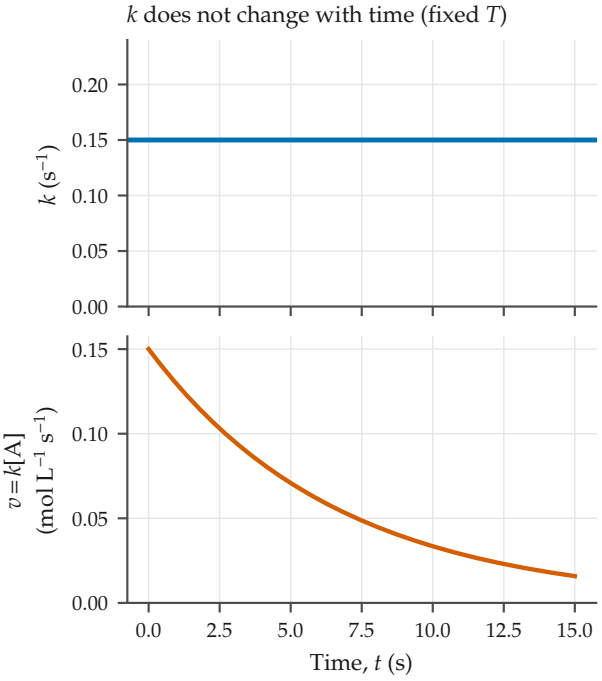


Figure 8. At a fixed temperature, the rate constant k does not change with time (top panel); what falls is the rate $v = k[\mathrm{A}]$, because the reactant keeps being consumed (bottom panel).

Source: simulation (`codigo/figuras_fundamentos.py`), illustrative values.

Since T sits in the exponent, k grows very fast with temperature, but not exponentially in T: the exponential is in k against $1/T$. Over a narrow temperature range, this growth looks like a simple exponential in T, and that resemblance is where the rule of thumb in the next section comes from. In an intuitive reading, what makes k grow is the rising fraction of molecules with enough energy to react, the "tail" of the Maxwell-Boltzmann distribution (Figure 9). The average energy of a gas grows little with temperature (from 25 °C to 35 °C, only about 3%), but the fraction of molecules above an energy of tens of kJ/mol nearly doubles over that interval. The exact difference between this fraction of molecules and the fraction of collisions in Equation (2.17) is in Section 2.6.

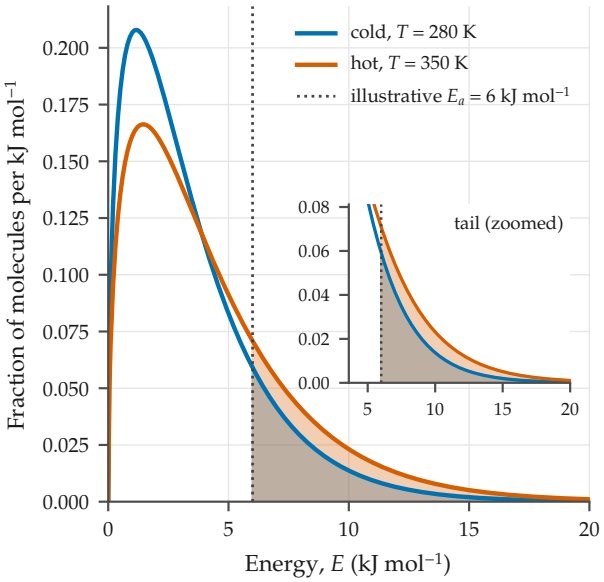


Figure 9. Maxwell-Boltzmann distribution of molecular energies at two illustrative temperatures (280 K, "cold," and 350 K, "hot"), each normalized to area 1. The shaded tails show the fraction of molecules with energy above an illustrative E_a of 6 kJ/mol; the inset zooms in on the tail.

Source: simulation (`codigo/figuras_fundamentos.py`; formula from Atkins, Paula & Keeler (2018)).

2.5.3 Rule of thumb: the factor of 2

Near room temperature, many reactions become roughly twice as fast when the temperature rises by 10 K. This is the **van 't Hoff rule** (VAN 'T HOFF, 1884). It is good for a quick mental calculation, but it is only an estimate:

$$k(T + 10\,\mathrm{K}) \approx k(T) \cdot 2. \tag{2.19}$$

By the Arrhenius equation, the exact factor for an interval of 10 K is

$$\begin{aligned} Q_{10} &= \frac{k(T + 10\,\mathrm{K})}{k(T)} \\ &= \exp\left[\frac{E_\mathrm{a}\,(10\,\mathrm{K})}{R\,T\,(T + 10\,\mathrm{K})}\right], \end{aligned} \tag{2.20}$$

and it equals exactly 2 only for one particular combination of E_a and T. Table 4 gives, over a 10 K range around five temperatures, the activation energy that gives exactly $Q_{10} = 2$. Near 25 °C, that value is about 53 kJ/mol. The same table shows the real factor between 25 and 35 °C for three activation energies, and Figure 10 plots that factor against temperature.

Table 4. (a) Activation energy that gives a factor of exactly 2 over five 10 K temperature ranges; (b) real factor $k(T + 10\,\mathrm{K})/k(T)$ for three activation energies, between 25 and 35 °C.

(a) Activation energy for an exact factor of 2		
Temperature range	T_1–T_2 (K)	E_a (kJ mol^{-1})
0–10 °C	273.15–283.15	44.6
25–35 °C	298.15–308.15	52.9
37–47 °C	310.15–320.15	57.2
100–110 °C	373.15–383.15	82.4
200–210 °C	473.15–483.15	131.7
(b) Real factor between 25 and 35 °C		
E_a (kJ mol^{-1})		Real factor
25		1.39
50		1.92
100		3.70

Source: simulation (`codigo/figuras_fundamentos.py` → `dados/simulacao/fund_q10.csv`).

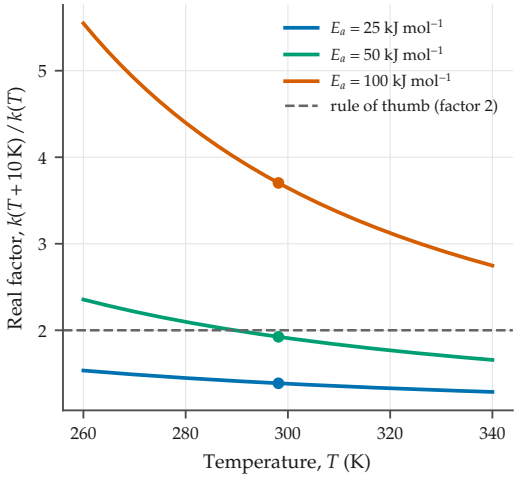


Figure 10. Real factor $k(T + 10\,\mathrm{K})/k(T)$ against temperature, for three activation energies, compared with the rule of thumb of the factor of 2. For each E_a, the rule is exact at only one temperature (Table 4).

Source: simulation (`codigo/figuras_fundamentos.py`).

The rule works best for E_a near 50 kJ/mol. Reactions with small E_a depart from it considerably. One example is $O + O_3 \longrightarrow 2\,O_2$, the first case study (Chapter 4): $E_a \approx 17.1$ kJ/mol (BURKHOLDER *et al.*, 2019) and the real factor between 25 and 35 °C is only 1.25. In $NO + O_3$ and $OH + O_3$, with E_a of 12.5 kJ/mol and 7.8 kJ/mol (BURKHOLDER *et al.*, 2019), the factors drop even further, to 1.18 and 1.11 (numbers in `dados/simulacao/fund_q10.csv`). The van 't Hoff rule deals with k. The van 't Hoff equation of chemical equilibrium, from Chapter 5, deals with K_{eq} and is a different thing.

2.6 Why molecules react

The Arrhenius equation describes how k changes with T, but does not explain the values of A and E_a. Two theories try to explain these values: collision theory, which sees the reaction as a crash between molecules, and transition state theory, which follows the energy path between reactants and products.

2.6.1 Collision theory

For two molecules to react, they must collide. That is the central postulate of collision theory. But not every collision produces a product. Two conditions must be met.

1. The collision must have energy equal to or greater than a minimum energy, E_0.
2. The molecules must collide with a spatial orientation favorable to forming the products.

Before the two conditions comes the number of collisions. In a gas mixture with $\mathcal{N}_A$ molecules of A and $\mathcal{N}_B$ of B per cubic meter, the number of collisions between A and B, per volume and per time, is

$$Z_{AB} = \sigma_{col}\,\bar{u}_{rel}\,\mathcal{N}_A\,\mathcal{N}_B, \qquad (2.21)$$

where σ_{col} (m^2), the **collision cross section**, is the effective meeting area between the molecules, and $\bar{u}_{rel}$ (m/s) is the average relative velocity between them. The link to concentration is $\mathcal{N}_A = N_A[A]$ (with [A] in mol/L, $\mathcal{N}_A = 1000\,N_A[A]$). Doubling [A] doubles the number of collisions. Raising the temperature from 25 °C to 35 °C increases $\bar{u}_{rel}$ by only about 2%, a small effect compared with what comes from the fraction with enough energy. In a reaction involving a solid, collisions only happen at the surface, which is why grinding the solid speeds up the reaction. This form holds for gases. In solution, the collision is replaced by the encounter between neighbors in the liquid, with a similar result.

The fraction of collisions with energy equal to or greater than E_0, which already appeared in Equation (2.17), is $e^{-E_0/(RT)}$. This is the exact result of the Maxwell-Boltzmann distribution for the energy along the line joining the centers of the two molecules. That fraction is different from the fraction of *molecules* with translational energy above the same value, which is the shaded area in Figure 9. Both calculations start from the same distribution, but answer different questions.

The second condition, orientation, is measured by the **steric factor**,

$$\mathcal{P} = \frac{k_{\text{experimental}}}{k_{\text{theoretical}}}, \tag{2.22}$$

where $k_{\text{theoretical}}$ is the k the theory predicts assuming every collision with enough energy reacts, whatever the orientation. When $\mathcal{P} \ll 1$, only a small fraction of orientations lead to products, a common case for large molecules with complicated geometry. When $\mathcal{P} \approx 1$, almost any orientation works, as between atoms or small molecules.

Combining the two conditions with the collision frequency, the rate constant of an elementary bimolecular reaction between A and B becomes

$$k = \mathcal{P}\, \sigma_{\text{col}}\, \bar{u}_{\text{rel}}\, N_{\text{A}}\, e^{-E_0/(RT)},$$
$$\bar{u}_{\text{rel}} = \sqrt{\frac{8k_{\text{B}}T}{\pi\mu}}, \tag{2.23}$$

in $\text{m}^3/(\text{mol s})$ ($\times 10^3$ gives $\text{L}/(\text{mol s})$). Here k_{B} is the Boltzmann constant, $\sigma_{\text{col}} = \pi d_{\text{AB}}^2$, with d_{AB} the sum of the molecules' radii, and $\mu = m_{\text{A}} m_{\text{B}}/(m_{\text{A}} + m_{\text{B}})$ is the **reduced mass** of the A-B pair. Comparing with the Arrhenius equation (Equation (2.17)), $A = \mathcal{P}\, \sigma_{\text{col}}\, \bar{u}_{\text{rel}}\, N_{\text{A}}$ and $E_{\text{a}} = E_0 + RT/2$, slightly larger than E_0 because of the dependence of $\bar{u}_{\text{rel}}$ on $\sqrt{T}$. The full derivation is in Appendix A.11 (ATKINS; PAULA; KEELER, 2018; LAIDLER, 1987; STEINFELD; FRANCISCO; HASE, 1999).

2.6.2 The Eyring equation

Transition state theory follows a different path. Instead of counting collisions, it looks at the **energy path** between reactants and products along the **reaction coordinate**, which is the axis that describes the progress of the bonds breaking and forming. Figure 11 shows this path for an exothermic and an endothermic step.

At the top of the barrier, marked with ‡ in Figure 11, sits the **activated complex**, or **transition state**: an unstable configuration in which old bonds are breaking while new ones form. It is the highest-energy point on the reaction path, with a lifetime on the order of 10^{-13} s or less, too short to be isolated or detected.

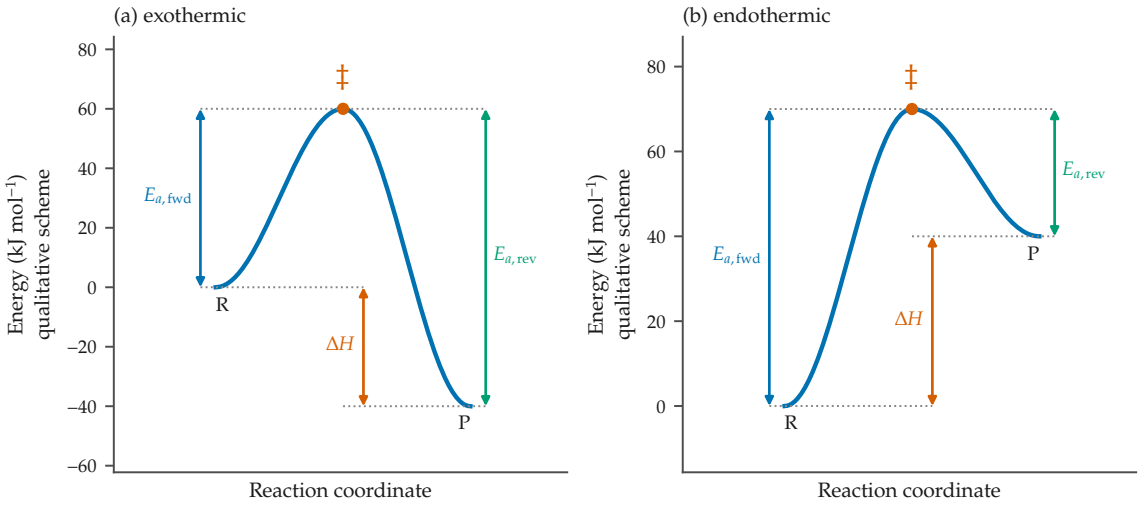


Figure 11. Schematic energy profile of an elementary step: (a) exothermic; (b) endothermic. The top of the barrier (‡) is the transition state; $E_{a,\text{forward}}$ and $E_{a,\text{reverse}}$ are measured from the reactants and the products; ΔH is, to a good approximation, the difference between them.

Source: author's own drawing.

An intermediate, by contrast, really exists, in a valley between two barriers; it is what the steady-state hypothesis deals with (Section 3.2).

I write the activated complex with ‡ over the species, R ⟶ [R···P]‡ ⟶ P (R for the reactants, P for the products). $E_{a,\text{forward}}$ is the energy the reactants need to reach the top, and $E_{a,\text{reverse}}$ is the same distance seen from the products.

The enthalpy difference between reactants and products, ΔH, is, to a good approximation, $E_{a,\text{forward}} - E_{a,\text{reverse}}$. The equality is exact in solution and when the number of gas molecules does not change; otherwise the two differ by a term of order RT.

Twenty years after Eyring (1935) and Evans & Polanyi (1935) formulated this theory, Hammond (1955) proposed a rule of thumb, the **Hammond postulate**: the transition state resembles more closely whichever species (reactant or product) has energy closest to its own.

Example 2.8 (A bimolecular nucleophilic substitution)**.** In the reaction $HO^- + CH_3Cl \longrightarrow [HO\cdots CH_3\cdots Cl]^{‡-} \longrightarrow CH_3OH + Cl^-$, the transition state carries the same negative charge as the reactants, distributed among the three central atoms.

Figure 12 shows this reaction's Gibbs energy profile, with the Gibbs energy of activation, $\Delta G^‡$, marked between the reactants and the top of the barrier.

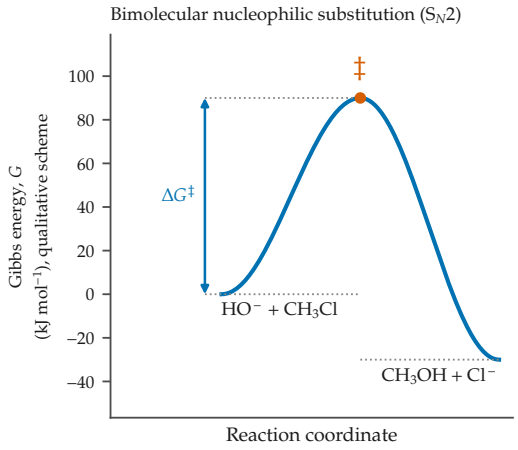


Figure 12. Gibbs energy profile of the bimolecular nucleophilic substitution (S_N2) between HO^- and CH_3Cl, with the Gibbs energy of activation $\Delta G^‡$ marked.

Source: author's own drawing.

A catalyst does not change the energy of the reactants or of the products. It opens a different path, with different steps and lower barriers (Figure 13). The catalyst takes part in those steps, but comes out whole at the end, and that is why the ΔH of the overall reaction does not change. In the Haber-Bosch process (Chapter 5), it is the iron surface that opens this lower path.

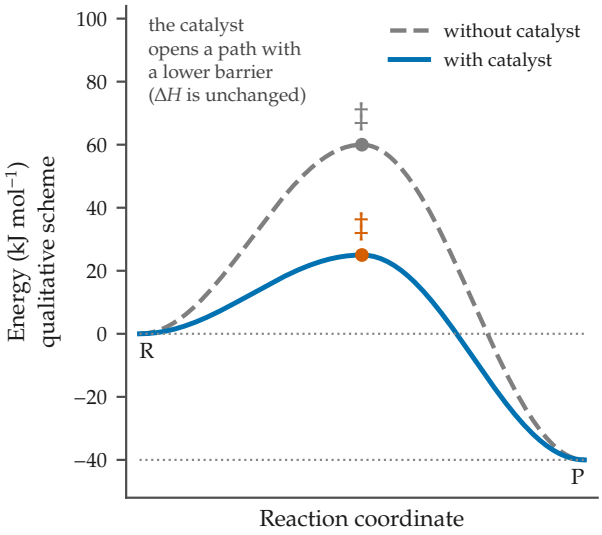


Figure 13. The catalyst opens a path with a lower barrier for the same reaction; the energy difference between reactants and products (ΔH) does not change.

Source: author's own drawing.

Transition state theory proposes, for k, the **Eyring equation**:

$$k = \kappa \frac{k_B T}{h} e^{-\Delta G^‡/(RT)}, \qquad \Delta G^‡ = \Delta H^‡ - T\Delta S^‡, \tag{2.24}$$

where h is Planck's constant, κ is the **transmission coefficient**, almost always close to 1, and $\Delta H^‡$ and $\Delta S^‡$ are the enthalpy and entropy of activation: the differences in enthalpy and in entropy between the transition state and the reactants. Entropy measures, in simple terms, how many arrangements the molecules can take; a transition state more ordered than the reactants has a negative $\Delta S^‡$. For a bimolecular step, the right-hand side is also divided by a standard concentration c° (1 mol/L), so that the units work out; the value of $\Delta G^‡$ depends on this choice. The term $k_B T/h$ is about 6.2×10^{12}/s at 25 °C. Its

inverse, 1.6×10^{-13} s, is the order of magnitude of the activated complex's lifetime mentioned above. Comparing with the Arrhenius equation, $E_a = \Delta H^{\ddagger} + RT$ for a unimolecular reaction or one in solution, and $E_a = \Delta H^{\ddagger} + 2RT$ for a bimolecular gas-phase reaction. Again, the difference is small, on the order of RT (derivation in Appendix A.11) (EYRING, 1935; EVANS; POLANYI, 1935; LAIDLER, 1987).

2.7 Mechanisms

Up to now I have treated the reaction as a single event. In practice, many reactions happen in several steps, each with its own rate constant. A **mechanism** is the sequence of steps that, added together, give the overall reaction.

2.7.1 Elementary step and molecularity

An **elementary step** is one where what the chemical equation shows happens in a single event: an encounter between particles, in steps with two or three molecules, or the breaking or rearrangement of a single molecule that received energy from an earlier collision or from light. The **molecularity** of an elementary step is the number of particles (molecules, atoms, or ions) that meet in it. A step with a single particle is unimolecular, with two it is bimolecular, and with three it is termolecular. Only in an elementary step are the orders in the rate law certain to equal the stoichiometric coefficients of the reactants.

Example 2.9 (Molecularity of five reactions)**.** Table 5 classifies five reactions from the atmosphere and the laboratory by their molecularity. Ozone photolysis and $O + O_3$ bring back the first case study (Chapter 4). The last two reactions have a 3rd-order rate law, consistent with a single collision among three particles, and that is why they are called termolecular. But the same law can also come from a two-step mechanism with a pre-equilibrium. A sign of this shows up in $2\,NO + O_2$, which gets slower as the temperature rises, with a negative apparent activation energy (TSUKAHARA; ISHIDA; MAYUMI, 1999). In other words, the same rate law can come from more than one mechanism. These two reactions come back as a bridge in Chapter 3.

2.7.2 Rate-determining step

When a mechanism has several steps, one of them usually controls the rate of the overall reaction. A factory helps to see which.

> **Analogy: the biscuit factory**
>
> *Picture a biscuit factory with two steps. In the first, a machine mixes the ingredients and shapes the raw biscuits. This machine is very fast, and turns out up to* 1000 kg/h *of raw dough. Next, the raw biscuits go into an oven, which bakes them. This oven, however, has limited capacity: it only bakes* 300 kg/h.
>
> *Even if the machine delivers* 1000 kg/h *of raw dough, the oven only bakes* 300 kg/h. *This means that the whole factory's output is* 300 kg/h, *the rate of the slowest step. Notice that* 700 kg/h *of raw dough is left over, and the line of trays in front of the oven never stops growing (great for anyone who likes raw biscuit dough, terrible for the factory owner). In chemistry, that line is an intermediate building up, as shown in Figure 14.*
>
> *This example illustrates the* ***rate-determining step****, or* ***slow step*** *(IUPAC, 2025c; MURDOCH, 1981): no matter how fast the other steps are, the slowest one (the oven, here) is the one that decides the rate of the whole process.*

In a multistep reaction, the same idea holds: the rate of the overall reaction is, to a good approximation, that of the slow step. While the line grows, 300 kg/h is the rate at which product comes out; raw dough goes in at 1000 kg/h. The two only match when the line stops growing, and that is why Equation (2.6) asks for intermediates that do not build up.

The factory also has an inverted version, with the oven faster than the machine. In that case, the line of trays always stays short, because each tray barely waits before going into the oven. It is this version that gives the right picture of the steady-state hypothesis (Chapter 3).

Table 5. Molecularity, rate law, and remark for five reactions.

Reaction	Molecularity	Rate law	Remark
$O_3 \xrightarrow{h\nu} O_2 + O$	unimolecular	depends on light	photolysis; Chapter 4
$O + O_3 \longrightarrow 2\,O_2$	bimolecular	$v = k[O][O_3]$	Chapter 4
$NO_2 + NO_2 \longrightarrow NO_3 + NO$	bimolecular	$v = k[NO_2]^2$	rate-determining step below 225 °C for the reaction in Example 2.10
$2\,NO + O_2 \longrightarrow 2\,NO_2$	termolecular (by the rate law)	$v = k[NO]^2[O_2]$	3rd-order rate law; a two-step mechanism is possible
$2\,NO + Cl_2 \longrightarrow 2\,NOCl$	termolecular (by the rate law)	$v = k[NO]^2[Cl_2]$	same

Source: literature: Flowers *et al.* (2019); Burkholder *et al.* (2019); Tsukahara, Ishida & Mayumi (1999).

2.7.3 Rate law of an elementary step

In a generic elementary step $a\,A + b\,B \rightarrow$ products, the rate law follows directly from the stoichiometric coefficients, with no experiment needed:

$$v = k[A]^a[B]^b. \tag{2.25}$$

This only holds because the step is elementary. In a reaction that is not elementary, the exponents come from data, as in Section 2.2.

One example will accompany the discussion of mechanisms into Chapter 3: the elementary step

$$1\,A + 2\,B \xrightarrow{k} 1\,C + 2\,D, \tag{2.26}$$

whose rate, by Equation (2.25), is

$$\begin{aligned} v &= -\frac{d[A]}{dt} = -\frac{1}{2}\frac{d[B]}{dt} \\ &= \frac{d[C]}{dt} = \frac{1}{2}\frac{d[D]}{dt} = k[A][B]^2. \end{aligned} \tag{2.27}$$

Elementary steps with more than three particles colliding at once almost never occur: $a + b \leq 3$.

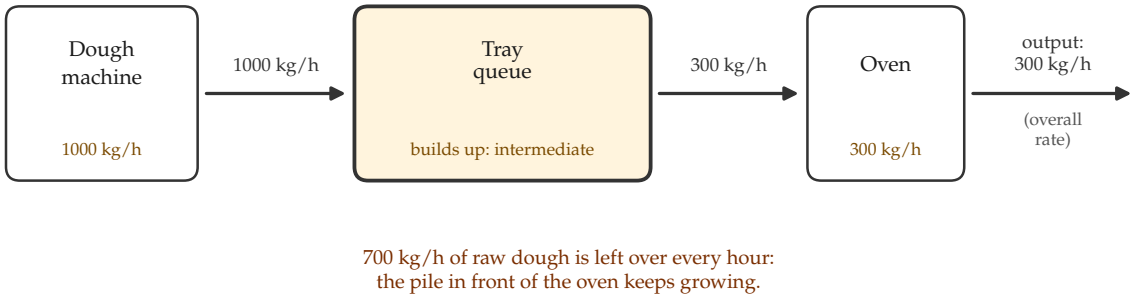


Figure 14. The biscuit factory: the dough machine produces 1000 kg/h, but the oven only processes 300 kg/h. The line of trays in front of the oven (the intermediate) keeps growing without stopping: 700 kg/h of raw dough is left over.

Source: author's own drawing.

Being elementary and, in Equation (2.26), practically irreversible, this reaction has the energy profile in Figure 15(a). Panel (b) shows the case where the reverse is also fast. The reverse of an elementary step is negligible when $E_{a,\text{reverse}}$ is much larger than $E_{a,\text{forward}}$ (the case in panel a) or when the products are removed from the system as soon as they form.

Example 2.10 (When a simple rate law is elementary)**.** The reaction $NO_2 + CO \longrightarrow NO + CO_2$ has, above about 225 °C, the rate law $v = k[NO_2][CO]$, which is that of a single collision between the two molecules. It is likely that, in that range, the reaction really is an elementary step. Below 225 °C, however, the measured rate law is $v = k[NO_2]^2$. CO does not even ap-

pear in it, and the mechanism now has two steps, $NO_2 + NO_2 \longrightarrow NO_3 + NO$ (slow) and $NO_3 + CO \longrightarrow NO_2 + CO_2$ (fast), the same rate-determining step from Table 5. In other words, it is the experiment that tells us whether a reaction is elementary, not the chemical equation alone (FLOWERS *et al.*, 2019).

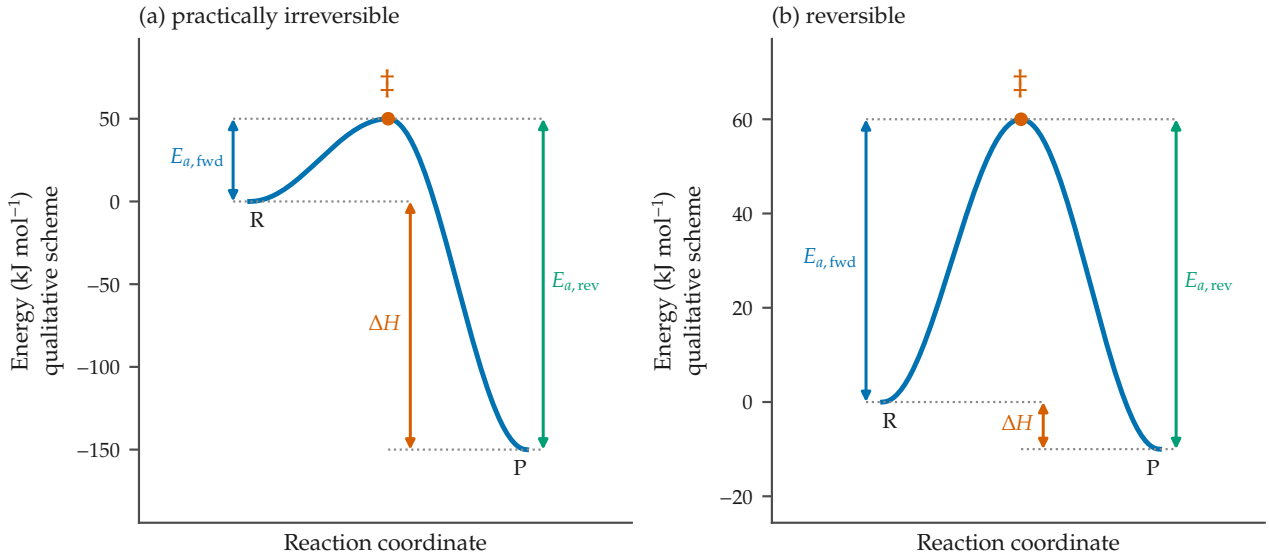


Figure 15. Energy profiles of an elementary step (illustrative values): (a) practically irreversible ($E_{a,\mathrm{reverse}} \gg E_{a,\mathrm{forward}}$); (b) reversible (similar forward and reverse barriers).

Source: author's own drawing.

2.7.4 Elementary equilibrium

When both directions of an elementary step are fast enough for the system to reach equilibrium, the same reaction from Equation (2.26) is written

$$1\,\mathrm{A} + 2\,\mathrm{B} \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} 1\,\mathrm{C} + 2\,\mathrm{D}. \tag{2.28}$$

Being elementary in both directions, the forward and reverse rates are $v_{\mathrm{forward}} = k_1[\mathrm{A}][\mathrm{B}]^2$ and $v_{\mathrm{reverse}} = k_{-1}[\mathrm{C}][\mathrm{D}]^2$. At equilibrium, the two are equal:

$$k_1[\mathrm{A}]_{\mathrm{eq}}[\mathrm{B}]^2_{\mathrm{eq}} = k_{-1}[\mathrm{C}]_{\mathrm{eq}}[\mathrm{D}]^2_{\mathrm{eq}}. \tag{2.29}$$

Isolating the ratio between the two rate constants gives the equilibrium constant of this reaction:

$$K_c = \frac{k_1}{k_{-1}} = \frac{[\mathrm{C}]_{\mathrm{eq}}[\mathrm{D}]^2_{\mathrm{eq}}}{[\mathrm{A}]_{\mathrm{eq}}[\mathrm{B}]^2_{\mathrm{eq}}}. \tag{2.30}$$

This result links kinetics to thermodynamics. In an elementary step, the equilibrium constant is always the ratio between the forward and reverse rate constants, $K = k_1/k_{-1}$.

3 CHAPTER The steady-state hypothesis

The steady-state hypothesis (SSH), also called the steady-state approximation, is the central subject of this work. In the previous chapter I dealt with multistep mechanisms and the idea of a rate-determining step. This chapter deals with the case where no step is clearly slower than the others, or where we do not know which one is: the case of the intermediate I, a species that the reaction both forms and consumes on the way from reactants to products. I start with the history of the hypothesis and give its definition, along with the criterion that tells us when it holds. Then I compare three ways of treating the same mechanism and test the SSH against the exact solution in two examples, one of them solved by computer.

3.1 A bit of history

The hypothesis was proposed in 1913, independently, by Max Bodenstein (BODENSTEIN, 1913) and by David L. Chapman and Leo K. Underhill (CHAPMAN; UNDERHILL, 1913). Both groups were studying the same reaction, hydrogen with chlorine under light, which forms HCl. Today we know it is a **chain reaction**: a chlorine atom, formed by light, reacts with H_2 and gives back a hydrogen atom, which reacts with Cl_2 and gives back a chlorine atom, and so on, many times before the chain breaks. While the chain runs, the formation and consumption of these free Cl and H atoms stay almost balanced. That is what the SSH needs in order to hold, and it was by analyzing the kinetics of this reaction that Bodenstein, Chapman, and Underhill arrived at the idea.

In 1925, George Briggs and J. B. S. Haldane used the same idea for enzymes (BRIGGS; HALDANE, 1925) (Section 6.1). In 1930, the geophysicist Sydney Chapman, not the Chapman of 1913, proposed the four-step mechanism that explains the ozone layer of the stratosphere (CHAPMAN, 1930) (Chapter 4).

At Oxford, Cyril Hinshelwood studied the reaction between hydrogen and oxygen, in particular its explosion limits (HINSHELWOOD, 1964), and also the decomposition of ozone dissolved in carbon tetrachloride (BOWEN; MOELWYN-HUGHES; HINSHELWOOD, 1931). In the Soviet Union, Nikolai Semenov explained why a gas mixture can suddenly switch from a slow reaction to an explosion. The reaction's intermediates are radicals that multiply at each step, in a branching chain, and the explosion happens when they stop being at steady state (SEMENOV, 1964, p. 491). I return to this idea in more detail in Section 7.6. In 1931, Semenov became director of the Institute of Chemical Physics of the USSR Academy of Sciences (NOBEL PRIZE OUTREACH, 2026d).

In 1956, Hinshelwood and Semenov shared the Nobel Prize in Chemistry "for their researches into the mechanism of chemical reactions" (NOBEL PRIZE OUTREACH, 2026c). One worked in England and the other in the Soviet Union, and even so the two reached similar results on chain reactions, each on his own.

3.2 The idea and the definition

An intermediate really exists, for a short or a long time, in a valley of the energy diagram, between two barriers (Section 2.6). Some can be isolated and kept in a flask. The most reactive ones, like the free oxygen atom in the ozone layer (Chapter 4), can only be detected, because they vanish almost as fast as they form, and those are the cases the SSH is for. The activated complex, on the other hand, the highest point on the reaction path, does not exist that way: it sits at the top of a barrier and does not last even one molecular vibration.

The SSH assumes that the intermediate's concentration changes with time much less than the rates at which it forms and is consumed:

$$\frac{\mathrm{d[I]}}{\mathrm{d}t} \approx 0, \qquad \text{that is,} \quad v_{\text{form}}(\mathrm{I}) \approx v_{\text{cons}}(\mathrm{I}). \tag{3.1}$$

The two rates in Equation (3.1) are each a positive number, because the formation and the consumption of I are never negative. What stays close to zero is the *difference* between the two; each of them, on its own, can be large. The SSH also does *not* assume that [I] stays constant. In general, [I] stays small and, after a fast initial rise, slowly tracks the fall of the reactants (IUPAC, 2025b). There are mechanisms in which the steady-state value does not depend on the reactants and [I] does stay constant while they fall: that is what happens with the oxygen atom in the photolysis of ozone under constant light, where $[\mathrm{O}]_{\text{ss}} = J/k_4$ (Chapter 4). But there the constancy is a result of *that* specific mechanism, and the hypothesis, by itself, does not require it. The phrase "steady state" is also used in another sense: a system in which no concentration changes with time any more, like a continuous-flow reactor (IUPAC, 2025b). That is a different subject, and Chapter 5 keeps the two apart.

The condition for Equation (3.1) to hold needs to be written with rate constants or with lifetimes, because just saying which of the two processes is "faster" is too vague. The intermediate's **lifetime**, τ_{I} (s), is the average time a molecule of I lasts before being consumed. If every step that consumes I is first order in I, with constant k'_j, then $\tau_{\mathrm{I}} = 1/\sum_j k'_j$. If one of those steps also depends on another reactant, that reactant's concentration is folded into k'_j. Calling τ_{reaction} (s) the time over which the reactant concentrations change appreciably, the criterion is

$$\tau_{\mathrm{I}} \ll \tau_{\text{reaction}}. \tag{3.2}$$

That is, the intermediate does not last long because the step that consumes I has a rate constant much larger than the step that forms it. In the A $\rightarrow$ I $\rightarrow$ B mechanism of Example 3.1, this means $k_2 \gg k_1$. That is why [I] stays small and the two rates in Equation (3.1) become nearly equal. This criterion is enough for the SSH to hold, but it is not required: for the O atom in ozone photolysis under constant light, it gets worse with time and the SSH stays exact (Sections 4.3 and 7.1).

The SSH holds when each molecule of I lasts a short time compared with the time over which the reactants change (Equation (3.2)), and because of that [I] stays much smaller than the reactant concentrations. Besides that, the initial stage, of a few τ_{I}, in which I is still building up, the induction period (Chapter 7), must already have passed. The SSH is useful when the rate law has a complicated form and when we do not know which step is the slow one. It does not require knowing this: the rate-determining-step and pre-equilibrium cases both follow from it as two limits (Section 3.3.4), as long as the intermediate stays at a small concentration. Example 3.1 puts this criterion into numbers, for a concrete mechanism.

3.3 Three ways to treat the same intermediate

I write the rate law using only species that can be measured and controlled (reactants, products, and catalysts), as the IUPAC recommends when defining the steady state (IUPAC, 2025b). An intermediate does not enter the rate law directly: it forms and is consumed within the reaction itself, and in general it cannot be measured directly. When the mechanism has an intermediate, there are three ways to reach a rate law that uses only measurable species: assume which step is the slow one, assume a pre-equilibrium, or use the SSH. All three work for the same two-step mechanism:

$$\mathrm{A} + \mathrm{B} \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} \mathrm{I} + \mathrm{D}, \tag{3.3}$$

$$\mathrm{I} + \mathrm{B} \xrightarrow{k_2} \mathrm{C} + \mathrm{D}. \tag{3.4}$$

Adding the two steps, the overall reaction is $1\,\mathrm{A} + 2\,\mathrm{B} \rightarrow 1\,\mathrm{C} + 2\,\mathrm{D}$, because the I that step (3.3) forms is used up in step (3.4) and does not appear in the final result. This is the same overall reaction as in the previous chapter (Equation (2.26)), which I treated there as a single elementary step.

Here it goes through two steps, with the intermediate I, and we do not know, from the start, which of the two is the slow one.

3.3.1 The rate-determining step

If step (3.3) is much slower than step (3.4), the rate of the overall reaction is the rate of step 1 itself:

$$v \approx k_1[\mathrm{A}][\mathrm{B}]. \tag{3.5}$$

It does not matter that step (3.3) is reversible: as long as it is the slower one, almost all of the I that forms goes straight on to step 2, and the reverse step (k_{-1}) barely counts.

Figure 16a shows the energy profile for this case, where the peak of step 1 is the higher of the two.

3.3.2 Pre-equilibrium

If, instead, step (3.4) is the slow one, step (3.3) goes back and forth many times before I moves on to products, and stays close to equilibrium, with the forward and reverse steps at nearly the same rate: $k_1[\mathrm{A}][\mathrm{B}] \approx k_{-1}[\mathrm{I}][\mathrm{D}]$, that is,

$$[\mathrm{I}] \approx \frac{k_1}{k_{-1}} \frac{[\mathrm{A}][\mathrm{B}]}{[\mathrm{D}]}. \tag{3.6}$$

Substituting Equation (3.6) into the rate of step 2,

$$\begin{aligned} v &\approx k_2[\mathrm{I}][\mathrm{B}] \\ &\approx \frac{k_1 k_2}{k_{-1}} \frac{[\mathrm{A}][\mathrm{B}]^2}{[\mathrm{D}]}. \end{aligned} \tag{3.7}$$

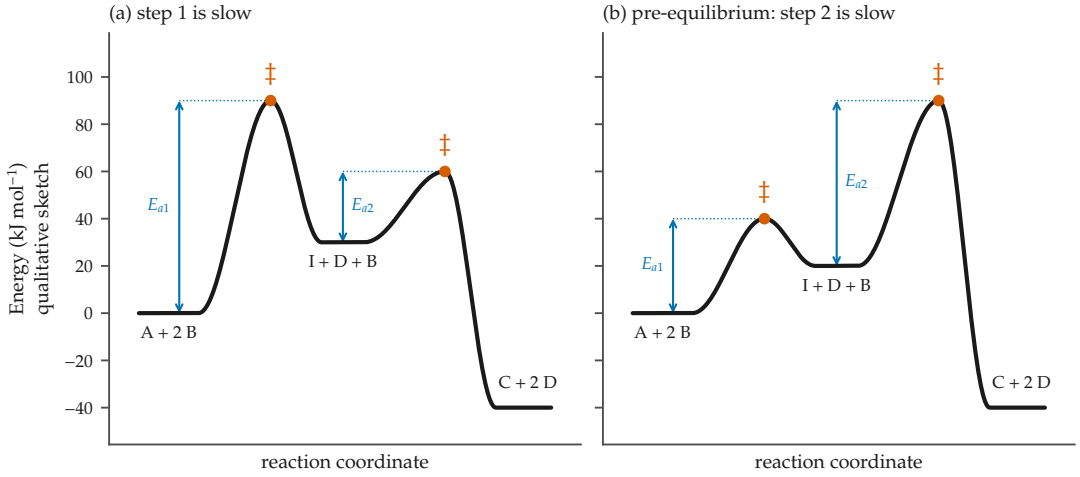


Figure 16. Qualitative energy profile (schematic, without a reaction-coordinate scale) of the mechanism in Equations (3.3) and (3.4). The numbers (kJ/mol) assume similar pre-exponential factors and the same [B] in both steps. What decides the regime is the fate of I: the barrier of the reverse of step 1, $E_{a,-1}$, against E_{a2}, both measured from the valley. (a) Step 1 is rate-determining: $E_{a,-1} = 60\,\mathrm{kJ/mol}$ is larger than $E_{a2} = 30\,\mathrm{kJ/mol}$, and almost all of I goes on to C. (b) Pre-equilibrium, with step 2 rate-determining: $E_{a,-1} = 20\,\mathrm{kJ/mol}$ is smaller than $E_{a2} = 70\,\mathrm{kJ/mol}$, and I goes back to A and B many times before moving on.

Source: author's own drawing.

The D in the denominator is **self-inhibition**: as product D builds up, it slows the reaction, because it pushes step (3.3) back toward A and B. This holds when the intermediate goes back to being a reactant much faster than it moves on to products,

$$k_{-1}[\mathrm{D}] \gg k_2[\mathrm{B}]; \tag{3.8}$$

without this condition, [I] does not get close to equilibrium. In the simpler mechanism $\mathrm{A} \rightleftharpoons \mathrm{I} \rightarrow \mathrm{P}$, this condition would read $k_{-1} \gg k_2$, but the right one is looser: $k_2 \ll k_1 + k_{-1}$ (Section 7.5). Right at the start of the reaction, when there is almost no D, Equation (3.8) fails, and Equation (3.7) would give an infinite rate as $[\mathrm{D}] \to 0$: a sign that this rate law only holds after a short initial period (Section 7.2). The observed activation energy is $E_{a,\mathrm{obs}} = E_{a1} + E_{a2} - E_{a,-1}$. It can even be negative, if step −1 has a larger barrier than the sum of the other two. Two third-order reactions, NO + Cl_2 (FLOWERS *et al.*, 2019) and NO + O_2 (TSUKAHARA; ISHIDA; MAYUMI, 1999), look

like triple collisions, but they can also come from a pre-equilibrium similar to this one. For NO + O_2, this apparent activation energy is negative, about −4 kJ/mol between 273 and 600 K (TSUKAHARA; ISHIDA; MAYUMI, 1999). The reaction gets slower as the temperature rises, something impossible for an ordinary elementary step. But here it is natural, because the constant that grows fastest with temperature is that of the reverse step, k_{-1}, which has the largest barrier.

3.3.3 The steady-state hypothesis

But the mechanism in Equations (3.3) and (3.4) does not, by itself, say which of the two steps is the slow one. We could solve the rate equations by computer, but that does not give a simple formula. This is exactly the case the SSH is for: instead of choosing a rate-determining step, it assumes directly that $\mathrm{d[I]}/\mathrm{d}t \approx 0$ (Equation (3.1)). Calling $r_1 = k_1[\mathrm{A}][\mathrm{B}] - k_{-1}[\mathrm{I}][\mathrm{D}]$ the net rate of step (3.3) and $r_2 = k_2[\mathrm{I}][\mathrm{B}]$ that of step (3.4),

$$\begin{aligned} \frac{\mathrm{d[I]}}{\mathrm{d}t} &= r_1 - r_2 \approx 0 \\ \Longrightarrow \quad [\mathrm{I}]_{\mathrm{SSH}} &= \frac{k_1[\mathrm{A}][\mathrm{B}]}{k_{-1}[\mathrm{D}] + k_2[\mathrm{B}]}. \end{aligned} \tag{3.9}$$

Product C only appears in step (3.4), so the reaction rate is simplest to compute as $v = \mathrm{d[C]}/\mathrm{d}t = k_2[\mathrm{I}][\mathrm{B}]$: it is the species that appears in the fewest steps (Section 3.5). Substituting Equation (3.9),

$$v = \frac{k_1 k_2 [\mathrm{A}][\mathrm{B}]^2}{k_{-1}[\mathrm{D}] + k_2[\mathrm{B}]}. \tag{3.10}$$

Equation (3.10) holds for any ratio between $k_{-1}[\mathrm{D}]$ and $k_2[\mathrm{B}]$, without requiring that one step be much slower than the other. In the simplest possible example of an intermediate, A → I → B (Example 3.1), the same ratio $\tau_{\mathrm{I}}/\tau_{\mathrm{reaction}}$ from Equation (3.2) is exactly the relative error left over in [I], compared with the exact [I], after the initial stage.

3.3.4 The SSH contains the other two

Equation (3.10) has two limits, and each of them is one of the earlier cases. If step (3.4) is much faster than the reverse of step (3.3) ($k_2[\mathrm{B}] \gg k_{-1}[\mathrm{D}]$), the term $k_2[\mathrm{B}]$ dominates the denominator and

$$v \approx k_1[\mathrm{A}][\mathrm{B}]: \tag{3.11}$$

step 1 is rate-determining, and Equation (3.11) is the same as Equation (3.5). If, instead, the reverse of step (3.3) is much faster than step (3.4) ($k_{-1}[\mathrm{D}] \gg k_2[\mathrm{B}]$), the term $k_{-1}[\mathrm{D}]$ dominates and

$$v \approx \frac{k_1 k_2}{k_{-1}} \frac{[\mathrm{A}][\mathrm{B}]^2}{[\mathrm{D}]}: \tag{3.12}$$

step 2 is rate-determining, step 1 stays in pre-equilibrium, and Equation (3.12) is the same as Equation (3.7). The example in the IUPAC entry on the rate-determining step uses this same form (IUPAC, 2025c). What decides which of the two cases holds is the more likely fate of the intermediate: going back to being a reactant (the term $k_{-1}[\mathrm{D}]$) or moving on to products (the term $k_2[\mathrm{B}]$). This equivalence only holds as long as the intermediate stays at a small concentration in both limits: if the pre-equilibrium lets I build up, the SSH stops holding, and the pre-equilibrium must then be used with I included in the mass balance (Chapter 7).

3.3.5 The biscuit factory, inverted

Analogy: the biscuit factory, inverted

Go back to the biscuit factory from the previous chapter (Section 2.7.2): the dough machine makes 1000 kg/h *and the oven can only bake* 300 kg/h, *so the raw dough, the intermediate, piles up on the waiting table, because it arrives much faster than it leaves. Now flip the two numbers: the machine now makes* 300 kg/h *(it is now the rate-determining step) and the oven can handle* 1000 kg/h. *Each tray that arrives goes almost straight into the oven. The waiting table stays nearly empty: what comes in each hour is almost the same as what goes out. If the machine slows down further, say because the flour is running low, the pile on the table shrinks with it, but it stays small and always "caught up" with the machine. This is the SSH: the intermediate, the dough waiting, is small in amount and tracks the reactants, because it is*

> *consumed almost as soon as it forms. The output is* 300 kg/h, *the rate of the first step. Figure 17 draws the inverted factory.*

The inverted factory is the case of Equation (3.11), with the machine (step 1, the slow one) playing the role of k_1 and the oven (step 2, the fast one) playing the role of k_2. It is the same factory as before, only now the bottleneck has moved. That is why it works as a picture of both cases: the rate-determining-step approach and the SSH give the same answer when the intermediate barely manages to build up.

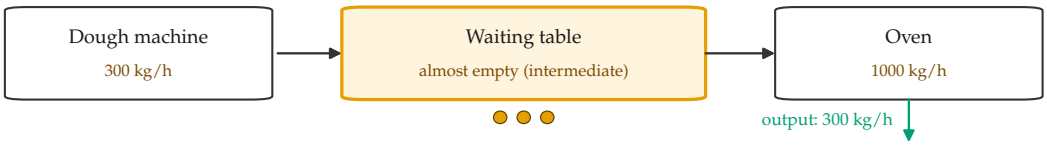


Figure 17. The biscuit factory from Section 2.7.2 (there with the dough machine at 1000 kg/h and the oven at 300 kg/h), inverted: now the dough machine is the slow one (300 kg/h) and the oven can handle far more than it receives (1000 kg/h). The waiting table, the intermediate in the analogy, stays small and always "caught up" with the machine: this is the picture of the steady-state hypothesis.

Source: author's own drawing.

3.4 Example: A → I → B

Example 3.1 (A → I → B: the simplest mechanism with an intermediate)**.** The simplest possible mechanism with an intermediate has two first-order steps, one after the other:

$$\mathrm{A} \xrightarrow{k_1} \mathrm{I} \xrightarrow{k_2} \mathrm{B}. \tag{3.13}$$

At the start there is only A: $[\mathrm{A}] = [\mathrm{A}]_0$ and $[\mathrm{I}] = [\mathrm{B}] = 0$ at $t = 0$. The rate equations are

$$\begin{aligned} \frac{\mathrm{d}[\mathrm{A}]}{\mathrm{d}t} &= -k_1[\mathrm{A}], \\ \frac{\mathrm{d}[\mathrm{I}]}{\mathrm{d}t} &= k_1[\mathrm{A}] - k_2[\mathrm{I}], \\ \frac{\mathrm{d}[\mathrm{B}]}{\mathrm{d}t} &= k_2[\mathrm{I}]. \end{aligned} \tag{3.14}$$

Adding all three, $[\mathrm{A}] + [\mathrm{I}] + [\mathrm{B}] = [\mathrm{A}]_0$ at every instant (mass balance).

By the SSH. Setting $\mathrm{d}[\mathrm{I}]/\mathrm{d}t \approx 0$ in Equation (3.14), $k_1[\mathrm{A}] \approx k_2[\mathrm{I}]$, and the reaction rate is

$$\begin{aligned} v = -\frac{\mathrm{d}[\mathrm{A}]}{\mathrm{d}t} &= \frac{\mathrm{d}[\mathrm{B}]}{\mathrm{d}t} + \frac{\mathrm{d}[\mathrm{I}]}{\mathrm{d}t} \\ &\approx \frac{\mathrm{d}[\mathrm{B}]}{\mathrm{d}t} = k_2[\mathrm{I}] \approx k_1[\mathrm{A}], \end{aligned} \tag{3.15}$$

with the last equality using the SSH. From Equation (3.15) itself, $[\mathrm{I}]_{\mathrm{SSH}} = (k_1/k_2)[\mathrm{A}]$; integrating $\mathrm{d}[\mathrm{B}]/\mathrm{d}t = k_1[\mathrm{A}]_0 \mathrm{e}^{-k_1 t}$ with $[\mathrm{B}] = 0$ at $t = 0$,

$$[\mathrm{B}]_{\mathrm{SSH}} = [\mathrm{A}]_0 \left(1 - \mathrm{e}^{-k_1 t}\right) = [\mathrm{A}]_0 - [\mathrm{A}]. \tag{3.16}$$

So the SSH counts all of the A that has already reacted as if it were already B, without waiting for it to pass through I. This is the same result as the rate-determining-step shortcut (Section 3.3.1) when step 1 is the slower one.

The exact solution. Solving Equation (3.14) without approximation (the full step-by-step derivation is in Appendix A.4), for $k_1 \neq k_2$,

$$\begin{aligned} [\mathrm{I}] &= \frac{k_1[\mathrm{A}]_0}{k_2 - k_1}\left(\mathrm{e}^{-k_1 t} - \mathrm{e}^{-k_2 t}\right), \\ [\mathrm{B}] &= [\mathrm{A}]_0 \left[1 - \frac{k_2\,\mathrm{e}^{-k_1 t} - k_1\,\mathrm{e}^{-k_2 t}}{k_2 - k_1}\right]. \end{aligned} \tag{3.17}$$

At the start, with $[\mathrm{I}] \approx 0$, the rate of formation of B is $k_2[\mathrm{I}] \approx 0$. That is, the product starts at zero rate, while the SSH says it already starts at $k_1[\mathrm{A}]_0$. The difference exists because I needs to build up first (Section 7.2). If $k_2 \gg k_1$ and $t \gg 1/k_2$, the term $\mathrm{e}^{-k_2 t}$ vanishes and $k_2 - k_1 \approx k_2$: Equation (3.17) becomes the SSH solution. This is why the criterion in Equation (3.2) requires $k_2 \gg k_1$ in this mechanism, since $\tau_{\mathrm{I}} = 1/k_2$ and the time over which A changes is $1/k_1$.

Figure 18 shows the exact curves and the SSH curves for three ratios k_2/k_1. With $k_2/k_1 = 100$ they are almost indistinguishable, except right at the start. With $k_2/k_1 = 2$ the difference stays large for much of the reaction and only vanishes once almost all of A has turned into B, because

the intermediate is not "fast" enough compared with the reactants' time scale.

The difference between the two solutions follows from Equations (3.16) and (3.17): $[\mathrm{B}]_{\mathrm{SSH}} - [\mathrm{B}] = [\mathrm{I}]$. The SSH gets ahead of [B] by exactly what is still held up in the intermediate. It is largest at the end of the initial stage and shrinks after that. Its maximum has an exact formula,

$$\frac{[\mathrm{I}]_{\max}}{[\mathrm{A}]_0} = \left(\frac{k_1}{k_2}\right)^{k_2/(k_2-k_1)}, \qquad (3.18)$$

which only approaches the simple ratio k_1/k_2 when $k_2 \gg k_1$: for $k_2/k_1 = 10$, k_1/k_2 lies above the exact value by 22.6 % of itself, and for $k_2/k_1 = 100$, by 4.5 %. Table 6 gathers these numbers. After the initial stage, the SSH misses [I] by k_1/k_2 of the exact value, or by $k_1/(k_2 - k_1)$ of the SSH value itself; both become the lifetime ratio of Equation (3.2) when $k_2 \gg k_1$.

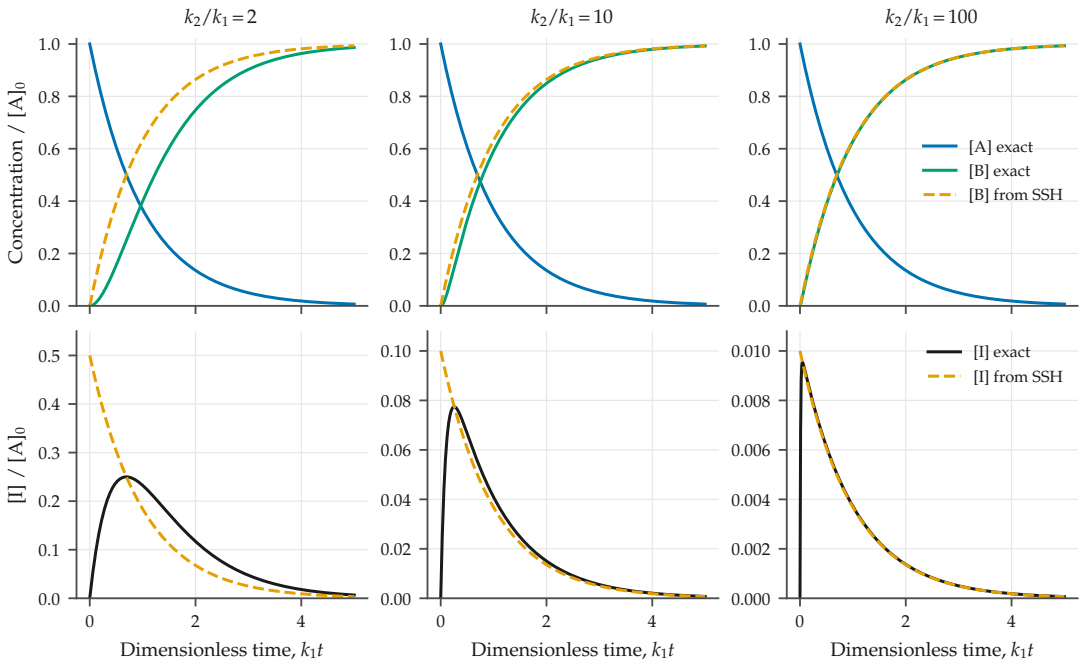


Figure 18. Mechanism A ⟶ I ⟶ B: exact solution (solid lines) and SSH solution (dashed) for $k_2/k_1 = 2$, 10, and 100. Top: [A] and [B]; bottom: [I], with its own scale in each panel. Time is in units of $1/k_1$ and concentrations are fractions of $[\mathrm{A}]_0$. The exact [I] curve meets the SSH curve at its highest point.

Source: simulation (`codigo/validade_ssh.py`).

Table 6. SSH errors in the mechanism A ⟶ I ⟶ B. Column 2: largest difference between the SSH's [B] and the exact [B], as a fraction of $[\mathrm{A}]_0$, which is the same number as $[\mathrm{I}]_{\max}/[\mathrm{A}]_0$ (Equation (3.18)), because $[\mathrm{B}]_{\mathrm{SSH}} - [\mathrm{B}] = [\mathrm{I}]$. Column 4: how far the simple ratio k_1/k_2 lies above the exact value in column 2, as a fraction of k_1/k_2 itself.

k_2/k_1	$\dfrac{\max([\mathrm{B}]_{\mathrm{SSH}} - [\mathrm{B}])}{[\mathrm{A}]_0} = \dfrac{[\mathrm{I}]_{\max}}{[\mathrm{A}]_0}$	k_1/k_2	error of the k_1/k_2 approximation
1	0.368	1	—
2	0.250	0.5	50 %
5	0.134	0.2	33 %
10	0.077	0.1	22.6 %
50	0.0185	0.02	7.7 %
100	0.0095	0.01	4.5 %
1000	0.0010	0.001	0.69 %

Source: simulation (`codigo/validade_ssh.py`); numbers in `dados/simulacao/validade_AIB_erro.csv`.

I evaluated Equation (3.18) at 100 log-spaced ratios k_2/k_1, from 1 to 1000, and checked ten of them by integrating the full system by computer. The largest difference was 7×10^{-14} of $[A]_0$. The maximum error falls off continuously between the points in Table 6 (Appendix C.1).

3.5 The two-step mechanism by computer

Example 3.2 (The mechanism in Equations (3.3) and (3.4), solved in full)**.** I return to the mechanism in Section 3.3, now without choosing which step is the slow one. I solve the five rate equations (one for each species) by computer, with *illustrative* rate constants and initial concentrations that I chose: $[A]_0 = 1$, $[B]_0 = 2$ mol L^{-1}, $k_1 = 1$, $k_{-1} = 10$, and $k_2 = 100$, all in L/(mol s), since all three steps are bimolecular. Figure 19 shows the result.

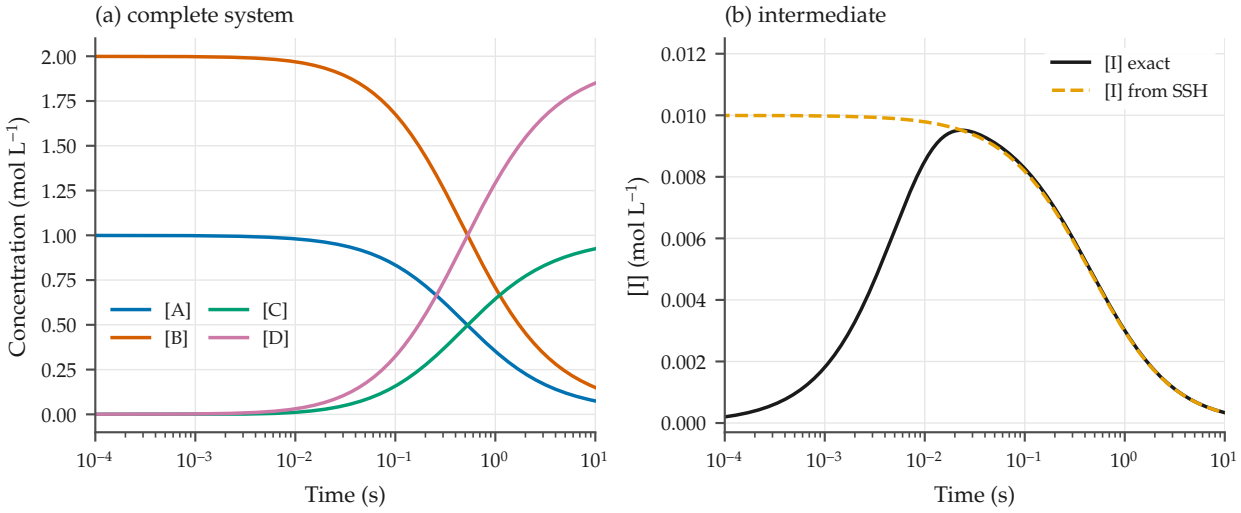


Figure 19. Simulation of the mechanism in Example 3.2, $A + B \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} I + D$ followed by $B + I \xrightarrow{k_2} C + D$, with $[A]_0 = 1$, $[B]_0 = 2$ mol L^{-1}, $k_1 = 1$, $k_{-1} = 10$, and $k_2 = 100$ (illustrative units). (a) Full solution: [A], [B], [C], and [D], time on a logarithmic scale. (b) The intermediate: exact [I] (solid line) and [I] from Equation (3.9) (dashed). After $t = 0.05$ s the two curves stay within 1.09 % of each other; the maximum of [I] is 9.516×10^{-3} mol L^{-1} at $t = 0.0232$ s.

Source: simulation (`codigo/figuras_ssh.py`); numbers in `dados/simulacao/ssh_exemplo20.csv` and `ssh_exemplo20.json`.

In the simulation, [I] rises quickly to a small value, much smaller than [A] and [B], and then closely tracks Equation (3.9), falling along with the reactants. The sum [A] + [I] + [C] stays equal to $[A]_0$ at every instant, with rounding error smaller than 10^{-14}, as it must in any integration. I checked the accuracy another way: repeating the integration with a second method (Radau), the largest point-by-point difference was 1.05×10^{-9} (`ssh_exemplo20.json`).

The rate of formation of I and its rate of consumption stay nearly equal, as Equation (3.1) requires, and in panel (b) of Figure 19 [I] rises and then falls, tracking [A] and [B].

A series mechanism can have many reversible steps and still behave, in the end, as if it were irreversible. If a mechanism has 1000 steps, 999 of them reversible and only one irreversible, but every path goes through that single irreversible step, that one step is enough to make the reverse rate of the overall reaction negligible too. This holds as long as the other 999 really are in series, one after another, with no parallel branches that could bypass the irreversible step.

The reaction rate, as in Section 3.3.3, can be calculated using the rate of formation or consumption of A, B, C, or D: they all give the same result, as long as the SSH holds. So it pays to

choose the species that appears in the fewest steps. Here, C appears only in step (3.4), and the calculation is most direct through it.

This freedom of choice assumes that the intermediate does not build up ($\mathrm{d[I]}/\mathrm{d}t \approx 0$). Without this assumption, the rate of consumption of A and the rate of formation of C can differ substantially. Only the relation between B and D, $-(1/2)\,\mathrm{d[B]}/\mathrm{d}t = (1/2)\,\mathrm{d[D]}/\mathrm{d}t$, always holds, because it follows directly from the stoichiometry of the two steps.

3.6 Summary

Table 7 gathers the three ways to treat the same intermediate. All three give the same law when one of the two fates of I dominates, moving on to products ($k_2[\mathrm{B}] \gg k_{-1}[\mathrm{D}]$) or going back to reactants (the opposite), and [I] stays small. The SSH, in addition, also covers the middle ground, where neither condition is clear. The rate-determining step and the pre-equilibrium are, each of them, a special case of the SSH (Section 3.3.4). The advantage of the SSH is that it does not require knowing beforehand which of the two cases holds. The disadvantage is that it too can fail, when the intermediate is not "fast" enough or when it builds up in too large an amount. Chapter 7 shows, with numbers, when this happens.

Table 7. The three ways to treat an intermediate in the mechanism of Equations (3.3) and (3.4).

Approximation	Condition	Resulting law	Typical error
Rate-determining step	$k_2[\mathrm{B}] \gg k_{-1}[\mathrm{D}]$: step 1 much slower	$v \approx k_1[\mathrm{A}][\mathrm{B}]$ (3.11)	of order k_1/k_2, as in Example 3.1
Pre-equilibrium	$k_{-1}[\mathrm{D}] \gg k_2[\mathrm{B}]$: step 1 goes back and forth fast	$v \approx \dfrac{k_1 k_2}{k_{-1}} \dfrac{[\mathrm{A}][\mathrm{B}]^2}{[\mathrm{D}]}$ (3.12)	small when [I] is small; grows if the pre-equilibrium lets I build up (Chapter 7)
SSH	$\tau_\mathrm{I} \ll \tau_\mathrm{reaction}$ (3.2), without knowing which step is the slow one	$v = \dfrac{k_1 k_2 [\mathrm{A}][\mathrm{B}]^2}{k_{-1}[\mathrm{D}] + k_2[\mathrm{B}]}$ (3.10)	of order $\tau_\mathrm{I}/\tau_\mathrm{reaction}$ (Chapter 7)

Source: author's own work.

4 CHAPTER Case study 1: ozone

Stratospheric ozone is a good place to see the steady-state hypothesis (SSH), also called the steady-state approximation, in action. The intermediate is a lone oxygen atom, O, which lives for thousandths of a second; ozone, O_3, lasts far longer. This difference in lifetimes is what makes the SSH work, both in the breakdown of O_3 by light in a closed cell (two steps) and in the mechanism that explains the ozone layer itself, proposed by Chapman in 1930 (CHAPMAN, 1930), which I use here in its four-step form.

This chapter uses the units of atmospheric chemistry: gas concentration in molecules per cubic centimeter (1 molecule cm^{-3} = 10^6 molecules per cubic meter), second-order rate constants in cm^3 molecule^{-1} s^{-1}, and third-order ones in cm^6 molecule^{-2} s^{-1}. The constants used are in Table 8.

Table 8. Constants used in the ozone chapter.

Quantity	Value		Source (page)
Rate constants (cm^3 molecule^{-1} s^{-1}; k_2 in cm^6 molecule^{-2} s^{-1})			
k_2: O + O_2 + M ⟶ O_3 + M (M = air)	$6.1 \times 10^{-34}\,(T/298)^{-2.4}$	6.1×10^{-34} at 298 K	JPL, Table 2-1, p. 2-5
k_4: O + O_3 ⟶ 2 O_2	$8.0 \times 10^{-12}\,e^{-2060/T}$	8.0×10^{-15} at 298 K	JPL, Table 1A, p. 1-6
k_5: Cl + O_3 ⟶ ClO + O_2	$2.3 \times 10^{-11}\,e^{-200/T}$	1.2×10^{-11} at 298 K	JPL, Table 1F, p. 1-228
k_6: ClO + O ⟶ Cl + O_2	$2.8 \times 10^{-11}\,e^{+85/T}$	3.7×10^{-11} at 298 K	JPL, Table 1F, p. 1-226
Photolysis of O_3 at the mercury line (253.65 nm)			
σ (cm^2), room temperature	$(1.1329 \pm 0.0035) \times 10^{-17}$ (JPL: 1.141×10^{-17})		Hodges *et al.*, p. 1; JPL, p. 4-18
Φ of O(^{1}D) and of O(^{3}P), 220 to 305 nm	0.90 and 0.10		IUPAC, p. 1497; JPL, p. 4-22
Stratosphere	at 25 km	at 30 km	
T (K)	221.552	226.509	NOAA, Table I, p. 61 and 63
[M] (molecule cm^{-3})	8.3341×10^{17}	3.8278×10^{17}	NOAA, Table II, p. 85 and 87
$[O_2]$ = 0.209476 [M]	1.7458×10^{17}	8.0183×10^{16}	NOAA, Table 3, p. 3
k_2 at these temperatures	1.24×10^{-33}	1.18×10^{-33}	Eqs. 4.6–4.7
k_4 at these temperatures	7.33×10^{-16}	8.98×10^{-16}	
k_5 at these temperatures	9.33×10^{-12}	9.51×10^{-12}	expressions for k_5 and k_6 above
k_6 at these temperatures	4.11×10^{-11}	4.08×10^{-11}	
J_1 (s^{-1})	$\sim 3.4 \times 10^{-12}$	$\sim 2.5 \times 10^{-11}$	Jacob, Fig. 10-5, p. 168 (reading)
J_3 (s^{-1})	$\sim 5.2 \times 10^{-4}$	$\sim 7.7 \times 10^{-4}$	Jacob, Fig. 10-5, p. 168 (reading)
$[O_3]$ observed (molecule cm^{-3})	$\sim 5 \times 10^{12}$	$\sim 3 \times 10^{12}$	Jacob, Fig. 10-5, p. 168 (reading)
Total chlorine entering the stratosphere	3240 parts per trillion (2020)		Salawitch *et al.*, Fig. Q6-1, p. 17
Fraction of chlorine as Cl or ClO	less than 0.1		Jacob, p. 181

Source: literature: Burkholder *et al.* (2019) (JPL); Hodges *et al.* (2019); Atkinson *et al.* (2004) (IUPAC); NOAA; NASA; USAF (1976); Jacob (1999); Salawitch *et al.* (2023). J_1, J_3, and the observed ozone come from a sketch for low latitudes (order of magnitude only): J_1 and J_3 from a reading of the book's vector coordinates; the observed ozone from a visual reading. For M = N_2, IUPAC gives a k_2 about 7 % lower (ATKINSON *et al.*, 2004, p. 1462), within the JPL uncertainty (factor 1.1); I use the JPL value in the cell as well.

4.1 Ozone and ultraviolet light

The Sun sends the Earth ultraviolet (UV) light of many wavelengths. The ozone layer absorbs all of the UV-C (100 nm to 280 nm, the most dangerous) and much of the UV-B (280 nm to 315 nm); the rest reaches the ground (SALAWITCH *et al.*, 2023, p. 7). This absorption is what protects life. On absorbing a UV photon, ozone breaks apart, $O_3 + h\nu \longrightarrow O_2 + O$ ($h\nu$ is the photon's energy, h the Planck constant and ν the frequency of

the light). This light-driven breakup is called **photolysis**. In the stratosphere, the O that comes out of it usually meets O_2 again within thousandths of a second and remakes ozone, which can then absorb light many more times. O_2 itself also absorbs UV, at a shorter wavelength, and splits into two O atoms.

Another reaction, however, truly destroys ozone: $O + O_3 \longrightarrow 2\,O_2$. Chapman combined this reaction, the photolysis of O_3 and of O_2, and the reaction that remakes ozone, into a mechanism that explains the origin of the layer (CHAPMAN, 1930). But this mechanism, on its own, predicts more ozone than is observed, because it is missing catalytic cycles involving oxides of nitrogen, hydrogen, and chlorine (JACOB, 1999, p. 169–177). Chlorofluorocarbons (CFCs) carry chlorine up to the stratosphere (MOLINA; ROWLAND, 1974), where chlorine and bromine destroy ozone through these cycles (SALAWITCH *et al.*, 2023, p. 24) (Section 4.5.2).

4.2 Two versions of the mechanism

Atmospheric oxygen takes part in four reactions (CHAPMAN, 1930), numbered here in the way that will hold for the rest of the chapter:

$$O_2 + h\nu \longrightarrow 2\,O \qquad (J_1), \tag{4.1}$$

$$O + O_2 + M \longrightarrow O_3 + M \qquad (k_2), \tag{4.2}$$

$$O_3 + h\nu \longrightarrow O_2 + O \qquad (J_3 \equiv J), \tag{4.3}$$

$$O + O_3 \longrightarrow 2\,O_2 \qquad (k_4). \tag{4.4}$$

That is four different quantities at once, so I will name each one before going on. J_1 is the frequency with which light breaks up O_2 (reaction 4.1). k_2 is the rate constant of the reaction that forms ozone (reaction 4.2). J_3, which I will just call J, is the frequency with which light breaks up O_3 (reaction 4.3). And k_4 is the rate constant of the reaction that destroys ozone (reaction 4.4). The values of all four are in Table 8.

The same four reactions describe two very different situations. In a closed cell, with ozone diluted in a non-reacting gas and lit by a lamp, only reactions (4.3) and (4.4) matter, that is, only J and k_4. This is the **two-step mechanism**. In the stratosphere, where ozone itself is also being produced, all four come into play, in what is called the **Chapman mechanism** (Section 4.5). That is where J_1 and k_2 come back into the picture.

4.2.1 The cell: photolysis of ozone

Picture a quartz cell with ozone diluted in a non-reacting gas (N_2 or Ar), lit by a mercury lamp whose strongest ultraviolet line, 253.65 nm (HODGES *et al.*, 2019) (the light known as "254 nm"), sits close to ozone's absorption maximum (BURKHOLDER *et al.*, 2019, p. 4-16). Adding reactions (4.3) and (4.4), the O atom disappears and what is left is $2\,O_3 \longrightarrow 3\,O_2$, the overall reaction from Example 2.2. The O atom is the intermediate: it is born in one step and consumed in the next.

The rate of photolysis (4.3) is $J[O_3]$, with J the **photolysis frequency** (the fraction of O_3 broken up per second, in s^{-1}):

$$J = \sigma\,\Phi\,F. \tag{4.5}$$

Three new quantities come in here. The **absorption cross section**, σ, is the target each molecule offers to photons (it is not the collision cross section σ_{col} from Chapter 2). The **quantum yield**, Φ, is the fraction of absorbed photons that breaks the molecule. F is the **photon flux**. At 253.65 nm, $\sigma = (1.1329 \pm 0.0035) \times 10^{-17}$ cm^2 (HODGES *et al.*, 2019, p. 1) and $\Phi = 1$ (90% of the atoms come out as O(^{1}D), 10% as O(^{3}P) (BURKHOLDER *et al.*, 2019, p. 4-22)).[1]

Equation (4.5) only holds if the cell is **optically thin** (with J the same throughout the volume), measured by the decadic absorbance $\text{Abs} = \sigma\ell[O_3]/\ln 10$ (ℓ is the optical path length). Absorbance measures how much light ozone removes from the beam: the higher it is, the less light comes out the other side. With $\ell = 1$ cm and Abs = 0.1, $[O_3] \approx 2.0 \times 10^{16}$ molecule cm^{-3} (about 830 ppm at 1 atm), and even then only 79% of the light gets through. In a thicker cell, the rate law would stop being first order.

With O_2 present, reaction (4.2) comes into play, remaking ozone: the third body M carries away

[1] In 1 atm of N_2, O(^{1}D) turns into O(^{3}P) in about 1 ns (BURKHOLDER *et al.*, 2019, p. 1-7), so I treat both as a single "O" when there is plenty of background gas. With little gas, part of the O(^{1}D) reacts directly with O_3 (Section 4.4.1).

the excess collision energy (BURKHOLDER *et al.*, 2019, p. 2-1). From JPL,

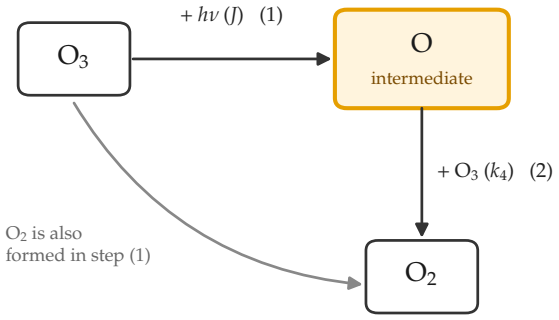
$$k_2(T) = 6.1 \times 10^{-34} \left(\frac{T}{298\,\mathrm{K}}\right)^{-2.4}, \qquad (4.6)$$

$$k_4(T) = 8.0 \times 10^{-12}\, \mathrm{e}^{-2060/T}, \qquad (4.7)$$

with T in kelvin (BURKHOLDER *et al.*, 2019, p. 1-6, 2-5). The unit of k_2 is $\mathrm{cm^6\,molecule^{-2}\,s^{-1}}$, and that of k_4 is $\mathrm{cm^3\,molecule^{-1}\,s^{-1}}$ (values at 298 K and 1 atm in Table 8).

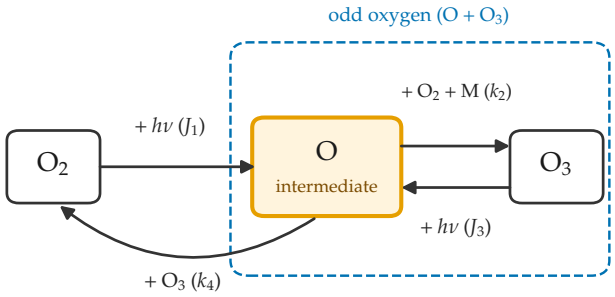
The two-step version thus describes ozone diluted in an inert gas, in a thin cell (the third condition, on O_2, comes in Section 4.3). The Chapman version explains the origin of the layer, but not a cell at 253.65 nm, where O_2 barely absorbs (cutoff at 242.3 nm (BURKHOLDER *et al.*, 2019, p. 4-7)) and where the pressure, temperature, and broadband sunlight needed are missing. Figure 20 sums up the two situations, and Table 8 gathers the chapter's constants.

(a) O_3 photolysis by UV light



(1) $O_3 + h\nu \rightarrow O_2 + O$
(2) $O + O_3 \rightarrow 2\,O_2$

(b) Chapman cycle (stratosphere)



(1) $O_2 + h\nu \rightarrow 2\,O$
(2) $O + O_2 + M \rightarrow O_3 + M$
(3) $O_3 + h\nu \rightarrow O_2 + O$
(4) $O + O_3 \rightarrow 2\,O_2$

Figure 20. The two mechanisms in this chapter. (a) Photolysis of O_3 in two steps: reactions (4.3) and (4.4). (b) The Chapman cycle, for the stratosphere: reactions (4.1) through (4.4). In both, the O atom is the intermediate (orange box); the dashed blue line groups odd oxygen, O and O_3.

Source: author's own drawing, with the mechanisms from Chapman (1930) and Burkholder *et al.* (2019).

4.3 The SSH in the two-step mechanism

Example 4.1 (Photolysis of ozone in an illuminated cell)**.** The cell has ozone diluted in an inert gas, with no O_2 at the start, and receives constant light. Only steps (4.3) and (4.4) hold. How does $[O_3]$ change with time t?

The O atom is born in step (4.3), at rate $J[O_3]$, and is consumed in step (4.4), at rate $k_4[O][O_3]$. Ozone is consumed in both, one molecule in each:

$$\frac{\mathrm{d}[O]}{\mathrm{d}t} = J[O_3] - k_4[O][O_3], \qquad (4.8)$$

$$\frac{\mathrm{d}[O_3]}{\mathrm{d}t} = -J[O_3] - k_4[O][O_3]. \qquad (4.9)$$

The O atom is the intermediate. By the SSH, its change is negligible compared with the rates of formation and consumption (Section 3.2):

$$\frac{\mathrm{d}[O]}{\mathrm{d}t} \approx 0 \quad \Longrightarrow \quad [O]_{ss} = \frac{J}{k_4}. \qquad (4.10)$$

Substituting this value into Equation (4.9), the two consumption terms become equal and the ozone rate law reduces to

$$\frac{\mathrm{d}[O_3]}{\mathrm{d}t} = -2J[O_3], \qquad k_{obs} = 2J. \qquad (4.11)$$

This is a first-order law in O_3. Integrating from $[O_3]_0$ to $[O_3]$,

$$[O_3] = [O_3]_0\, \mathrm{e}^{-2Jt}, \qquad t_{1/2} = \frac{\ln 2}{2J}. \qquad (4.12)$$

The complete derivation, in four steps and with every line of algebra, is in Appendix A.13.

Figure 21 shows that O reaches J/k_4 within milliseconds and stays there while ozone falls.

This constant plateau comes from the mechanism itself, because J/k_4 does not depend on $[O_3]$; the SSH on its own "assumes" nothing of the kind (Section 3.2). That is why it is only off at the start. After that, the exact solution and the SSH solution differ only by a factor on the order of $J\tau_O(0)$ (less than 2% in the figure), with $\tau_O(0) = 1/(k_4[O_3]_0)$. A practical consequence follows from this. Measuring k_{obs} and comparing it with $2J$ tests the *mechanism*, that is, whether the steps are the right ones (with J measured separately). Since the solution without the SSH gives practically the same curve, the comparison does not test the hypothesis.

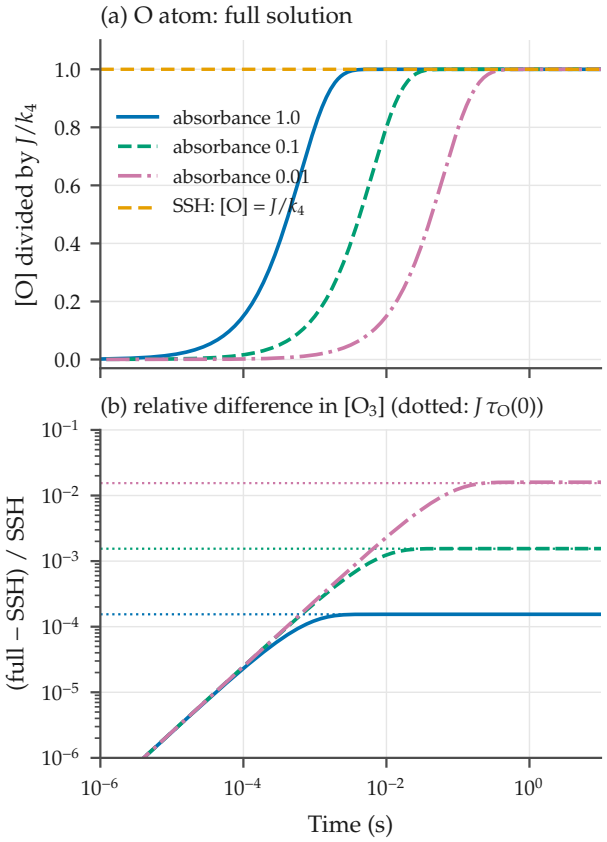


Figure 21. Two-step mechanism, full solution and SSH. (a) [O] divided by J/k_4. (b) Relative difference between the $[O_3]$ from the two solutions, constant at $J\tau_O(0)$: 0.015 %, 0.16 %, and 1.6 %, for initial absorbances of 1, 0.1, and 0.01 ($[O_3]_0$ of 2.0×10^{17}, 2.0×10^{16}, and 2.0×10^{15} molecule/cm^3). $J = 0.25$/s (illustrative), 298 K, no O_2.

Source: simulation (`codigo/ozonio.py`), with k_4 from Burkholder *et al.* (2019) and σ from Hodges *et al.* (2019).

The factor of 2 has a simple origin. Each photon breaks up one molecule of O_3 through step (4.3) and releases an O that, by the SSH, is consumed as soon as it forms, through step (4.4), destroying a second molecule of O_3. That is, every photon absorbed leads to the loss of two molecules of ozone, and the decomposition quantum yield is 2.

4.3.1 If O_2 is present: the general form

O_2 is a product of the reaction itself, so even starting without it, it builds up, and reaction (4.2) comes into play, competing with step (4.4) for the O atom. Reworking the SSH with the two paths out for O leads to a rate law that is still first order in O_3, but with a more complicated observed rate constant (the full algebra is in Appendix A.13):

$$\begin{aligned} \frac{d[O_3]}{dt} &= -k_{obs}[O_3], \\ k_{obs} &= \frac{2J\,k_4[O_3]}{k_4[O_3] + k_2[O_2][M]} \\ &= 2J\,\frac{x}{1+x}, \\ x &= \frac{k_4[O_3]}{k_2[O_2][M]}. \end{aligned} \tag{4.13}$$

The ratio x compares the two paths out for the oxygen atom: destroying an O_3 (numerator) or remaking one (denominator). Without O_2 ($x \gg 1$), $k_{obs} \to 2J$, and we are back to Equation (4.11). With a lot of O_2 ($x \ll 1$), the order changes:

$$\frac{d[O_3]}{dt} \approx -\frac{2J\,k_4}{k_2[O_2][M]}\,[O_3]^2, \tag{4.14}$$

second order in O_3 and slower, because almost every O atom goes back to forming ozone (Figure 22). At 298 K and 1 atm, k_{obs} stays within 10% of $2J$ only while $[O_3]/[O_2] \geq 17$. Starting without O_2, that lasts until 3.8% of the ozone has been used up.

This completes the list of conditions from Section 4.2. The formula $k_{obs} = 2J$ only holds for ozone diluted in an inert gas, in an optically thin cell, and while $[O_3]/[O_2] \gg k_2[M]/k_4$, that is, at the start of the reaction or at low total pressure.

[2] In *pure* ozone this changes: according to the abstract of Norrish & Wayne (1965), the quantum yield grew with O_3 pressure (up to 16.7 at 6.7 kPa), through a chain started by $O(^1D)$. With an inert gas, it falls to the limit of 2 from the two-step mechanism.

Adding more inert gas does not "push away" the O_2, because that gas is also M and speeds up the reaction that remakes ozone (4.2).[2]

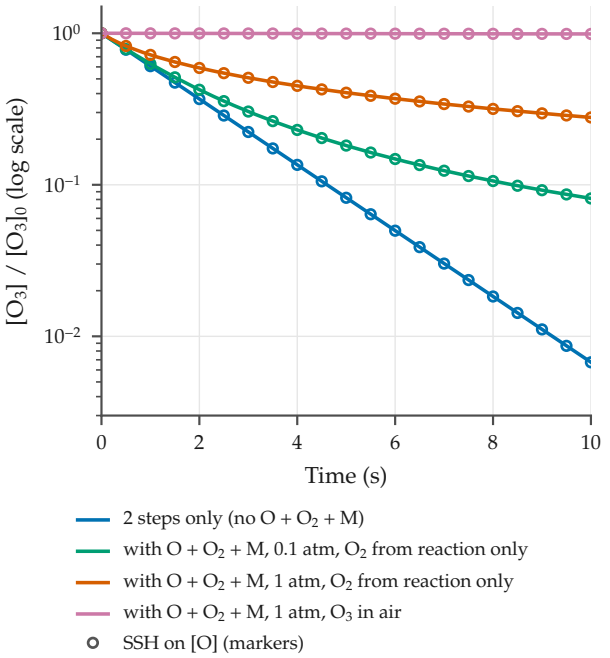


Figure 22. O_3 decay with the reaction O + O_2 + M ⟶ O_3 + M also present. On the logarithmic scale, a straight line means first order. With only the two steps (blue), $[O_3]$ falls as e^{-2Jt}; the higher the total background pressure, the more the O_2 formed remakes O_3, and the more the decay bends away from the line. Lines: full solution. Circles: SSH for the O atom. $[O_3]_0 = 2.0 \times 10^{16}$ molecule/cm^3, $J = 0.25$/s (illustrative), 298 K, background gas N_2.

Source: simulation (`codigo/ozonio.py`), with constants from Burkholder *et al.* (2019).

I ran the illuminated-cell model over a much wider range of J, O_2 fraction, and absorbance than the points shown here, and the SSH and the $k_{obs} = 2J$ formula only get worse once the O atom's lifetime stops being small compared with ozone's decay time. The scan puts numbers on what the calculation above already said: at 1 atm, the O_2 that photolysis itself forms slows the ozone's fall before any O_2 comes from outside (Appendix C.1).

4.4 The ozone data from the December 2025 article

The December 2025 article (RIBEIRO, 2025b) is a short text I wrote after the first version of this work. Here I go back to its ozone data with the tools of this chapter. There are 13 points of normalized $[O_3]$ against time, from $t = 0$ to $t = 8.6$ s, in steps of approximately 0.7 s (values in Table 25), along with a curve published with them; both are shown in Figure 23. The values have two decimal places, so each one carries a rounding error of up to ±0.005.

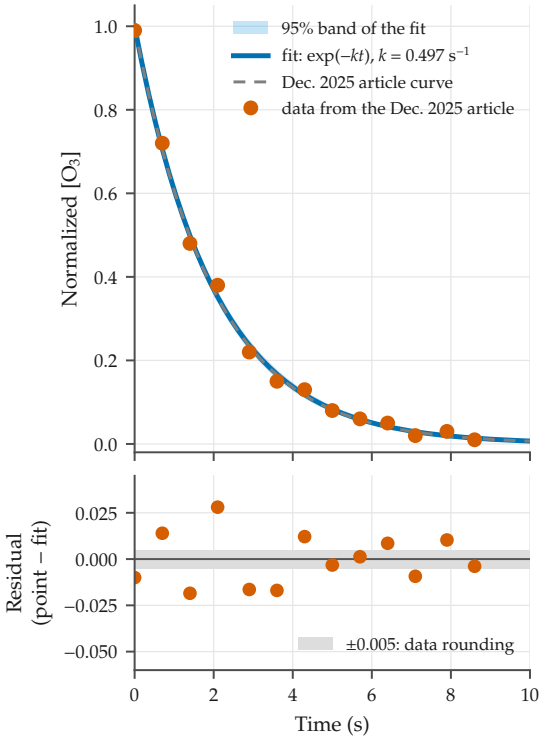


Figure 23. Fit of $\exp(-kt)$ to the 13 ozone data points from the December 2025 article. Top: points (circles), fit (blue, $k = 0.497$/s; the band spans the $\exp(-kt)$ curves with k at the ends of the 95 % interval of k), and the curve published in the article (gray dashed), which nearly coincides with the fit. Bottom: residuals (point minus fit); the gray band marks ±0.005, the rounding of the values. The vertical axis is normalized $[O_3]$, as in the article.

Source: data from the December 2025 article (RIBEIRO, 2025b); gray curve: the curve published in the December 2025 article; fit and band: simulation (`codigo/ajuste_ozonio.py`).

A simple exponential fits the 13 points well, as Figure 23 shows and Table 9 summarizes: $k_{obs} = 0.497$/s (standard error of 0.008/s; 95% confidence interval from 0.479 to 0.516/s), with $R^2 = 0.998$ on the linear scale of the data themselves. Letting the amplitude float ($A\,e^{-kt}$) does not improve the fit enough to make up for the extra parameter, and A comes out consistent with

1. On the line obtained by log-linearizing, $\ln[O_3]$ against t, the same k comes out equal within error, but the line's R^2 drops to 0.979, because the last three points, with values below 0.03, carry little weight on the original scale and a lot on the logarithmic one: the ±0.005 rounding turns into a huge relative error when the value is small.

The article's own curve describes the points as well as the fit does (sum of squared residuals of 0.00244 versus 0.00242). Its AICc comes out better only because I count zero parameters ($p = 0$), which assumes that $k = 0.5$/s was set without looking at the points. So I do not use that difference to choose between the curves.

Table 9. Three fits to the 13 ozone data points from the December 2025 article, and the curve published in the article itself. p: fitted parameters. CI: 95 % confidence interval (Student's t, $13 - p$ degrees of freedom). The models here are simple enough for this method to be adequate; in Chapter 5, with nonlinear models and more strongly correlated parameters, the CI comes from profile likelihood and bootstrap. MAE and RMSE: mean absolute error and root mean squared error. R^2 on the y scale, except for the line fit to $\ln y$, where it is the line's own R^2. The AICc (AKAIKE, 1974) only compares models on the same scale.

Model	p	k (1/s)	95 % CI of k	R^2	MAE	RMSE	AICc
$\exp(-kt)$	1	0.497	0.479 to 0.516	0.998	0.0117	0.0136	−106.5
$A\exp(-kt)$	2	0.496	0.472 to 0.519	0.998	0.0116	0.0136	−103.1
line fit to $\ln y$	2	0.502	0.454 to 0.551	0.979	0.0122	0.0143	—
article's curve	0	0.500	—	0.998	0.0117	0.0137	−109.2

Source: data from the December 2025 article (RIBEIRO, 2025b); fits: simulation (`codigo/ajuste_ozonio.py`).

R^2 only measures how closely a curve follows the shape of the points (SPIESS; NEUMEYER, 2010). Two curves from similar mechanisms can have the same R^2, and the value depends on the scale it is calculated on, as I have just shown (0.998 versus 0.979, for the same 13 points). That is why I use the AICc to compare models, and the chemistry of the steps to choose the mechanism. In Section 8.2 I show cases in which R^2 exceeds 0.99 while the constant is off by a factor of 10.

An exercise takes this a little further. If the decay follows the two-step mechanism, then Equation (4.11) gives $J = k_{obs}/2 \approx 0.25$/s, and from that comes an estimate of the photon flux that matches this J. The calculation assumes that reaction (4.2) does not matter. With background gas, the O_2 formed slows the decay, and the same k_{obs} would require a larger J; lowering the pressure does not fix this, because of $O(^1D)$ (Section 4.4.1).

With $\sigma = 1.1329 \times 10^{-17}$ cm^2 (HODGES *et al.*, 2019) and $\Phi = 1$, $F = J/(\sigma\Phi) \approx 2.2 \times 10^{16}$/(cm^2 s), about 17 mW/cm^2 at 253.65 nm. To get a sense of scale, I compared this value with published fluxes for low-pressure mercury lamps in laboratory reactors:

- about 2.6 times the top of the range in Peng *et al.* (2015);
- 6 times the maximum in Rowe, Lambe & Brune (2020);
- 112 to 165 times a collimated beam calculated from Rahn *et al.* (2003).

A decay this fast therefore calls for a UV source much more intense than the one in those setups. That does not make it impossible: the flux depends on the power, the number of lamps and the distance to the cell, and the article gives none of the three. In a thick cell or with O_2 present, the flux needed would be even larger. Comparing k_{obs} with $2J$ directly would take a cell with the photon flux measured separately; in Section 4.4.1 I show, with synthetic data, what that test would look like.

The 13 points follow an exponential, first-order decay, which is the shape the SSH predicts for the two-step mechanism. If the ozone in the cell had an absorbance above about 0.01 over 1 cm (more than 2×10^{15} molecules per cm^3), the O atom adjusts in under 0.07 s, far less than the 0.7 s between one point and the next, and the exact solution and the SSH solution give almost the same curve (a difference of up to 1.6%, Figure 21). The article does not give this concentration, so this is an assumption. What the points pin down, then, is the mechanism's constant k_{obs}. The rela-

tion $k_{obs} = 2J$ links that constant to the intensity of the light.

4.4.1 An assay with synthetic data

The 13 points are a single curve, and the article does not give the photon flux, the pressure, the background gas or the initial ozone concentration. So they pin down k_{obs}, but they do not test $k_{obs} = 2J$. To test it, one would have to change J and measure it separately, for example by chemical actinometry, which measures the light by the amount of product of a reaction with a known quantum yield (KUHN; BRASLAVSKY; SCHMIDT, 2004). To see what that test would look like, I generated *synthetic data*: curves from the cell model (Sections 4.3 and 4.3.1) plus noise drawn with a fixed seed (the draw comes out the same every time the code runs), made by `codigo/dados_sinteticos.py`, with the numbers in `dados/simulacao/dados_sinteticos.json`. None of these points is a measurement. They show what the assay could show if the model and the noise are right.

I took two things from the 13 points. The first is a reference value for J: $J_{ref} = k_{obs}/2 = 0.2486/s$, with the k_{obs} of Table 9. The second is the size of the noise: the standard deviation of the 13 residuals around $\exp(-kt)$, which is 0.0142. Each synthetic point is the model value plus a number drawn from a normal distribution with that standard deviation, independent from one point to the next.

The assay has two series. In the first, I change the light intensity: J is 0.25, 0.5, 1, 1.5 and 2 times J_{ref}. I use only steps (4.3) and (4.4), solved without the SSH, with initial absorbance 0.1. Each curve has 40 points, from $t = 0$ to $k_{obs}t \approx 4.3$, as in the 13 points, and each condition has 3 replicates. Figure 24 shows the curves and the k_{obs} fitted to each one.

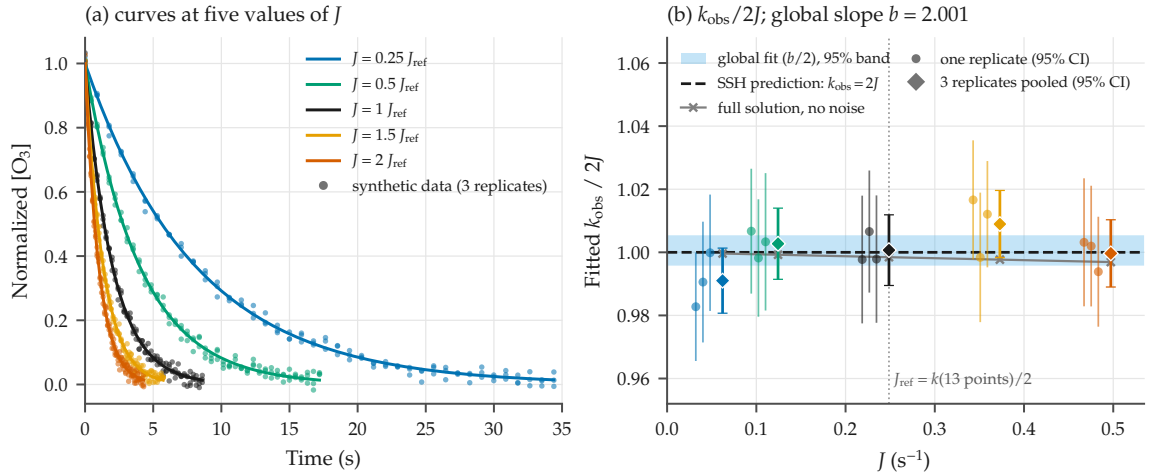


Figure 24. Synthetic data: an assay of O_3 photolysis at five values of J. These are model curves plus drawn noise, not measurements. (a) Normalized $[O_3]$ against time: 3 replicates of 40 points per condition (circles) and the exponential fitted to the 3 replicates pooled (lines). Two steps only, full solution without the SSH, initial absorbance 0.1, 298 K. Normal noise with standard deviation 0.0142, that of the 13 points of the December 2025 article around $\exp(-kt)$. (b) Fitted k_{obs} divided by $2J$, against J: one replicate (circles, drawn slightly to the left) and 3 replicates pooled (diamonds), with 95 % confidence interval. Dashed: the prediction $k_{obs} = 2J$ (Equation (4.11)). Blue band: $b/2$, with its 95 % interval, from the fit of $\exp(-bJt)$ to the 15 curves ($b = 2.001$). Crosses: the full solution without noise, 0.04 % to 0.3 % below 1, which is the SSH error. $J_{ref} = 0.2486/s$ is half the k fitted to the 13 points.

Source: synthetic data (`codigo/dados_sinteticos.py`), with the model of `codigo/ozonio.py`, k_4 from Burkholder *et al.* (2019) and σ from Hodges *et al.* (2019); J_{ref} and the noise taken from the data of the December 2025 article (RIBEIRO, 2025b).

The 15 values of $k_{obs}/2J$ lie between 0.98 and 1.02 (Figure 24b). Fitting the 15 curves at once with $\exp(-bJt)$ gives $b = 2.001$, with a 95 % confidence interval from 1.992 to 2.011. If each photon destroyed one O_3 molecule instead of two, b would be 1, and that value lies more than 200 standard errors away from the fit. A straight line of k_{obs} against J with a free intercept gives an

intercept of −0.0018/s (from −0.0034 to −0.0003). This interval excludes zero only narrowly because the fit's reduced chi-square (the scatter of the points divided by the expected scatter) is 0.62 and shrinks the error; without that reduction, it contains zero. At the lowest level of J, k_{obs} came out 1.7 standard errors below $2J$, and the SSH error would pull the other way. At the J_{ref} level alone, the 3 replicates of 40 points give k_{obs} an interval of ±0.0056/s, 3.3 times narrower than that of the 13 points.

This test is of the mechanism, not of the SSH. Without noise, the k_{obs} fitted to the full solution lies 0.04 % to 0.3 % below $2J$ (crosses in Figure 24b). That is the error of order $J\tau_O(0)$ from Section 4.3, smaller than the uncertainty of the slope, about 0.5 %. So, with the noise of the 13 points, this assay cannot see it.

In the second series, I fix $J = J_{ref}$, switch on reaction (4.2) at 0.1 atm and change the O_2 fraction in the gas at the start: 0, 1 %, 5 % and 20.9 % (that of air). Now k_{obs} changes with time, because the x of Equation (4.13) depends on $[O_3]$ and on $[O_2]$. Each O_3 used up turns into 3/2 of an O_2, so $[O_2] = [O_2]_0 + \frac{3}{2}([O_3]_0 - [O_3])$. With this, Equation (4.13) can be integrated:

$$2Jt = \left(1 - \frac{3k_2[M]}{2k_4}\right) \ln \frac{[O_3]_0}{[O_3]} + \frac{k_2[M]}{k_4}\left(\frac{[O_2]_0}{[O_3]_0} + \frac{3}{2}\right)\left(\frac{[O_3]_0}{[O_3]} - 1\right). \tag{4.15}$$

Without reaction (4.2), $k_2[M] \to 0$ and we are back to Equation (4.12). With it, the curve has a single free parameter, J, because k_2, k_4, [M] and the initial concentrations are known.

To measure k_{obs} at the start without assuming the form of the rate law, I use the method of initial rates (Section 2.4.1): 40 points per curve, from the start until half the O_3 is used up, and 3 replicates. I fit to $\ln[O_3]$ a degree-3 polynomial in t, $-k_{obs,0}\,t - c_2t^2/2 - c_3t^3/6$, whose linear coefficient is the constant at the start, $k_{obs,0}$; c_2 and c_3 only absorb the curvature. Without noise, this fit lies up to 1.5 % below the exact value. Figure 25 compares $k_{obs,0}$ with the prediction $2J\,x_0/(1 + x_0)$, where x_0 is the x at the start, with no fitted parameter.

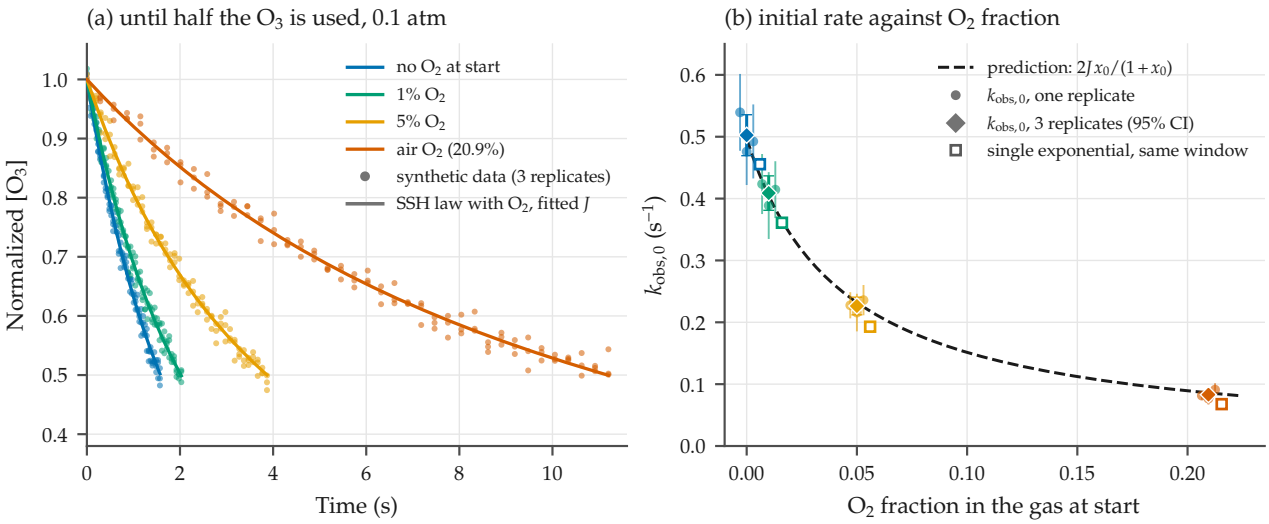


Figure 25. Synthetic data: initial rate of O_3 photolysis with O_2 in the gas. These are model curves plus drawn noise, not measurements. (a) Normalized $[O_3]$ until half the O_3 is used up, with the reaction $O + O_2 + M \longrightarrow O_3 + M$ switched on, 0.1 atm, $J = J_{ref}$ and four initial O_2 fractions: 3 replicates of 40 points (circles) and Equation (4.15) with fitted J (lines). (b) $k_{obs,0}$ fitted without the SSH law, with a degree-3 polynomial in $\ln[O_3]$, against the O_2 fraction: one replicate (circles) and 3 replicates pooled (diamonds, 95 % confidence interval). Dashed: the prediction $2J\,x_0/(1 + x_0)$ of Equation (4.13), with no fitted parameter. Open squares, in the color of each fraction: k of a single exponential over the same window, drawn slightly to the right so they do not hide the diamonds. Normal noise with standard deviation 0.0142.

Source: synthetic data (`codigo/dados_sinteticos.py`), with the model of `codigo/ozonio.py` and constants from Burkholder *et al.* (2019); J_{ref} and the noise taken from the data of the December 2025 article (RIBEIRO, 2025b).

At all four fractions, the prediction falls inside the 95 % interval of the fitted $k_{obs,0}$. The ratio between the two goes from 0.97 to 1.01, while the prediction falls from $2J$ to $0.17 \times 2J$. Fitting Equation (4.15) to each fraction, with J free, gives J between 0.993 and 1.004 times J_{ref}; with a single J for the 12 curves, it gives 0.2485/s (from 0.2471 to 0.2498). Without noise, the same law returns J within 0.3 % of J_{ref}: with O_2, the SSH also errs little in this model. A single exponential over the same window would get it wrong: it would give 0.79 to 0.92 times the predicted $k_{obs,0}$, and the AICc rejects it at all four fractions, by a difference of 16 to 123 in favor of Equation (4.15). That error does not come from the noise: without noise, the exponential gives the same 0.79 to 0.92.

The J series left reaction (4.2) switched off. With it switched on, the O_2 that the photolysis itself forms slows the ozone's fall. I fitted $\exp(-kt)$ to the full solution, without noise, over the window of the 13 points, with $J = J_{ref}$ and no O_2 at the start. The ratio $k/2J$ is 0.994 at a total pressure of 0.001 atm, 0.956 at 0.01 atm, 0.75 at 0.1 atm and 0.37 at 1 atm. At 1 atm, the curve is no longer even exponential ($R^2 = 0.83$). These numbers come from a model without $O(^1D)$, which only holds with plenty of background gas (footnote in Section 4.2.1). With little gas, part of the $O(^1D)$ reacts with O_3, and each photon destroys more than two molecules. Lowering the pressure, then, does not free the reading $J = k_{obs}/2$ from the O_2 formed; the way out is to fit J with a law that already accounts for that O_2.

The assay that would test this chapter's law would then have five parts: a thin cell (absorbance near 0.1); J measured separately at each intensity, by actinometry; background gas near 1 atm, so that $O(^1D)$ is deactivated, with J fitted by Equation (4.15), which already accounts for the O_2 formed; a lamp without the 185 nm mercury line (or a filter for it), because below 242 nm O_2 breaks apart and would make ozone inside the cell (PENG *et al.*, 2015; ROWE; LAMBE; BRUNE, 2020); and a series with O_2 in the gas from the start, to test Equation (4.13). Choosing conditions that tell rival models apart is a problem studied in statistics (BOX; HILL, 1967; ATKINSON; FEDOROV, 1975); here I tried only one simple design. The 13 points of the December 2025 article amount to a single condition of this assay ($J = J_{ref}$, no O_2), and J_{ref} and the noise come from them. That is why they cannot serve as a test.

Five caveats. The 13 residuals tend to alternate in sign (the Durbin-Watson statistic (DURBIN; WATSON, 1950) gives 3.2; with no pattern it would be near 2), and the independent noise I drew does not reproduce that. I treated J as known without error; in a real assay, actinometry has its own uncertainty, and the slope interval would come out wider. At low total pressure, the ozone itself becomes a larger share of the gas: with absorbance 0.1, $[O_3]_0 \approx 2.0 \times 10^{16}$ molecule/cm^3, about 0.8 % of the molecules at 0.1 atm and 8 % at 0.01 atm, and the model counts only the background gas as M. In the O_2 series, the window up to half the O_3 used up was computed with the law itself and J_{ref}, as if from a pilot run. And the model has neither the loss of O atoms at the wall nor a separate $O(^1D)$. The rate constant of $O(^1D)$ with O_3 is about 8 times that with N_2, and half of those reactions release two O atoms (BURKHOLDER *et al.*, 2019, p. 1-7); with little background gas, then, $J = k_{obs}/2$ comes out high. The chemistry of pure ozone (footnote in Section 4.3.1) is the extreme case of this.

4.5 The SSH in the stratosphere: the Chapman mechanism

In the stratosphere all four reactions (4.1) through (4.4) hold (JACOB, 1999, p. 163–164), with M the third body already introduced in Section 4.2.1 (stratospheric values in Table 8):

$$\frac{d[O]}{dt} = 2J_1[O_2] + J_3[O_3] - k_2[O][O_2][M] - k_4[O][O_3], \tag{4.16}$$

$$\frac{d[O_3]}{dt} = k_2[O][O_2][M] - J_3[O_3] - k_4[O][O_3]. \tag{4.17}$$

Reactions (4.2) and (4.3) only trade one O for one O_3 and do not change the sum of the two, the **odd oxygen** (the name comes from counting atoms, 1 in O and 3 in O_3). Adding the equations above, these terms cancel: what is left is only J_1,

which creates odd oxygen by breaking up O_2, and k_4, which destroys it in the reaction with O_3. Calling this sum $[O_x]$,

$$[O_x] = [O] + [O_3], \tag{4.18}$$

$$\frac{d[O_x]}{dt} = 2J_1[O_2] - 2k_4[O][O_3]: \tag{4.19}$$

odd oxygen is only born from the photolysis of O_2 and only dies in the O + O_3 reaction (JACOB, 1999, p. 166).

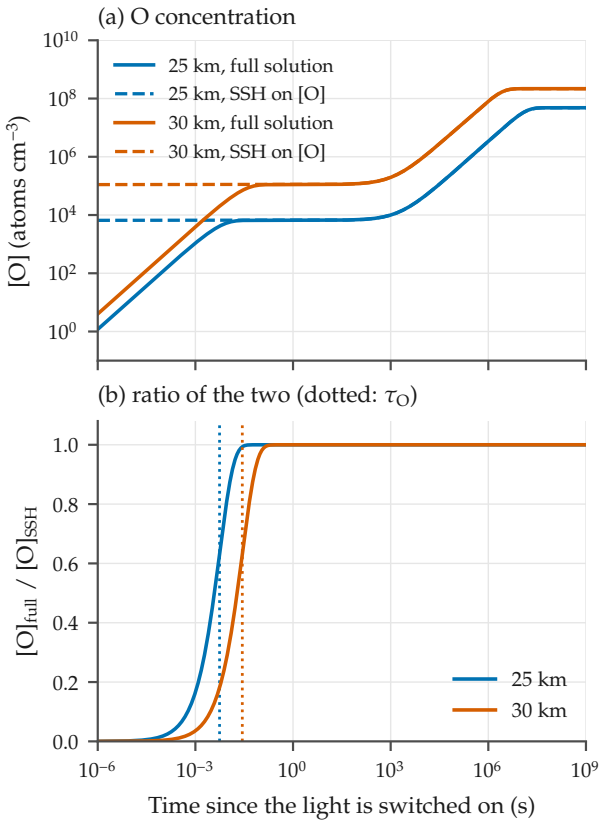


Figure 26. The O atom right after the light is switched on, at 25 and 30 km, on a log-log scale. (a) Full [O] (line) and SSH (dashed): the SSH starts right at the steady-state value; the full solution takes hundredths of a second to reach it. (b) Ratio between the two; dotted lines: τ_O (0.0055 s and 0.028 s), where the ratio reaches 1.

Source: simulation (`codigo/ozonio.py`), with the same constants as Figure 27.

The O atom is the shortest-lived intermediate. Two terms in Equation (4.16) are much smaller than the others, between 10^{-5} and 10^{-3} of them: production by O_2 photolysis and consumption in the reaction with O_3. Dropping both, the SSH gives

$$\frac{[O]_{ss}}{[O_3]} = \frac{J_3}{k_2[O_2][M]}, \tag{4.20}$$

which is 2.9×10^{-6} at 25 km and 2.1×10^{-5} at 30 km. There are tens to hundreds of thousands of O_3 molecules for every O, so $[O_x] \approx [O_3]$. Unlike in the cell, here $[O]_{ss}$ tracks J_3: by day the O exists, at night it turns back into O_3 within thousandths of a second (Figure 26), and the SSH keeps holding at every instant because O adjusts far faster than the light changes.

Substituting Equation (4.20) into the loss term of Equation (4.19), that loss becomes second order in O_3. If odd oxygen is also at steady state, production and loss become equal, and

$$[O_3]_{ss} = [O_2]\sqrt{\frac{J_1\, k_2[M]}{J_3\, k_4}}, \tag{4.21}$$

a second steady state, different from the first. Here ozone is almost all of the odd oxygen, and this equation only holds where its lifetime (Section 4.5.1) is short compared with the time the wind takes to bring in air from elsewhere (JACOB, 1999, p. 169).

With Table 8, this gives $[O_3]_{ss} \approx 1.7 \times 10^{13}$ and 1.0×10^{13} molecule/cm^3 (25 and 30 km), against an observed ozone of 5×10^{12} and 3×10^{12} in the same figure the J values came from, that is, about three times more. Figure 27 shows this simulated formation, starting from ozone-free air, up to the steady state. This agrees with Jacob (1999, p. 169): Chapman alone reproduces the shape of the layer, but overestimates ozone by a factor of 2 or more, because a sink is missing, the catalytic cycles with oxides of nitrogen (CRUTZEN, 1970), hydrogen, and chlorine (Section 4.5.2).

I ran the same Chapman model from 20 to 36 km, with J_1 and J_3 read from the same Jacob figure I already use at 25 and 30 km (at the other altitudes, from a new pixel-based reading of the same figure, which differs by up to 31% from the first). The steady-state $[O_3]$ peaks near 22-24 km, within the 20 to 30 km range of the observed peak of the ozone layer (JACOB, 1999, p. 168), though at values higher than the book's own Chapman curve (Appendix C.1).

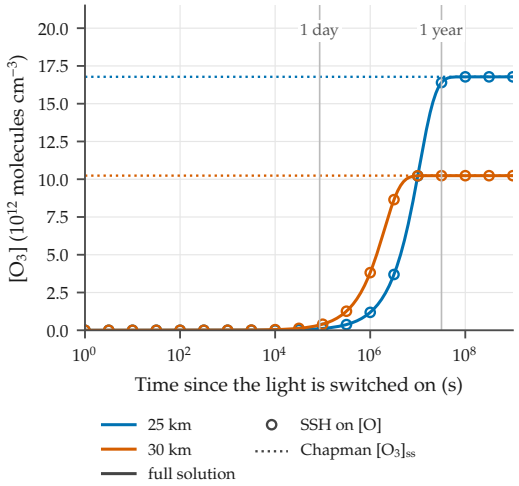


Figure 27. Formation of O_3 by the Chapman mechanism at 25 km (blue) and 30 km (orange), starting from ozone-free air, with midday light held constant. Lines: full solution. Circles: SSH for O (they coincide at the scale of the graph). Dotted: steady state from Equation (4.21), reached to 90 % in 241 days (25 km) and 44 days (30 km).

Source: simulation (`codigo/ozonio.py`), with constants from Burkholder *et al.* (2019), atmosphere from NOAA; NASA; USAF (1976), and J read from Figure 10-5 of Jacob (1999).

4.5.1 Three time scales

The O atom's lifetime, consumed almost entirely by reaction (4.2), is $\tau_O = 1/(k_2[O_2][M])$. A molecule of O_3 is broken up by light every $\tau_{O_3} = 1/J_3$, but the O from that breakup remakes ozone almost immediately, in τ_O. What changes the amount of ozone is the loss of odd oxygen, with $\tau_{O_x} \approx 1/(2k_4[O])$ (JACOB, 1999, p. 166) (Table 10).

Table 10. Time scales of the Chapman mechanism at the simulation's steady state, with midday light held constant throughout. τ_O: lifetime of the O atom. τ_{O_3}: lifetime of a molecule of O_3, soon remade. τ_{O_x}: lifetime of odd oxygen, $([O] + [O_3])/(2k_4[O][O_3])$.

Quantity	25 km	30 km
T (K)	221.6	226.5
[M] (molecule/cm^3)	8.33×10^{17}	3.83×10^{17}
τ_O (s)	0.0055	0.028
τ_{O_3} (min)	32	22
τ_{O_x} (days)	164	30
$[O]/[O_3]$	2.9×10^{-6}	2.1×10^{-5}
$[O_3]_{ss}$ (molecule/cm^3)	1.68×10^{13}	1.02×10^{13}

Source: simulation (`codigo/ozonio.py`); T and [M] from NOAA; NASA; USAF (1976); constants from Burkholder *et al.* (2019); J read from Figure 10-5 of Jacob (1999).

There are, then, three separate scales: milliseconds for O, tens of minutes for O_3, weeks to months for odd oxygen. In the thousandths of a second that O exists, the ozone around it barely changes (this is the inverted biscuit factory from Chapter 3), and that is why the SSH for O works very well, with the relative error in $[O_3]$ shown in Figure 28.

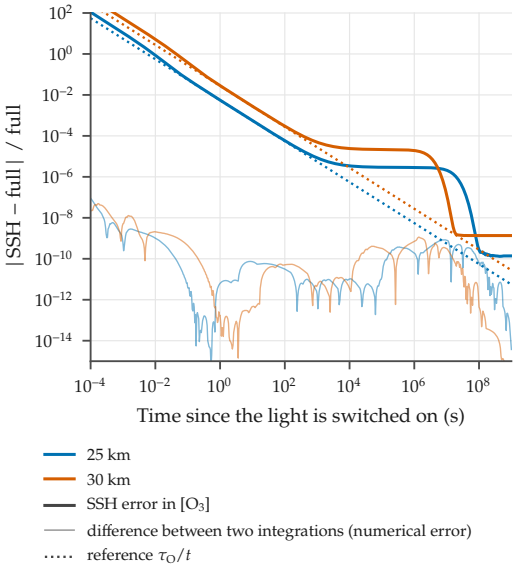


Figure 28. Relative error of the SSH in $[O_3]$, at 25 and 30 km. At the start, the error falls as τ_O/t (dotted lines); while O_3 is growing, it stays close to $[O]/[O_3]$ (3×10^{-6} and 2×10^{-5}); at steady state, it falls to about 1.4×10^{-10} at 25 km and 1.4×10^{-9} at 30 km. The thin lines give the order of magnitude of the numerical error (two independent integrations of the full solution).

Source: simulation (`codigo/ozonio.py`).

The steady state of odd oxygen is a cruder approximation. The reason is that τ_{O_x} is too long to ignore the exchange between day and night. In the lower stratosphere, it is also comparable to the time scale of transport by wind (JACOB, 1999, p. 169).

4.5.2 The chlorine catalytic cycle

CFCs rise intact to the stratosphere, where UV light breaks them up and releases chlorine atoms (MOLINA; ROWLAND, 1974), which enter a two-step cycle:

$$Cl + O_3 \longrightarrow ClO + O_2 \qquad (k_5), \qquad (4.22)$$

$$ClO + O \longrightarrow Cl + O_2 \qquad (k_6). \qquad (4.23)$$

Adding the two, Cl and ClO cancel out and what is left is $O + O_3 \longrightarrow 2\,O_2$, Chapman's reaction (4.4). Chlorine comes out the same as it went in, that is, it is a **catalyst** (STOLARSKI; CICERONE, 1974; MOLINA; ROWLAND, 1974). An analogous cycle swaps chlorine for an oxide of nitrogen, X = NO (CRUTZEN, 1970).

The SSH for Cl gives $k_5[\mathrm{Cl}][\mathrm{O_3}] = k_6[\mathrm{ClO}][\mathrm{O}]$. Since k_5 and k_6 are of the same order and there is far less O than O_3, almost all of the active chlorine sits as ClO, and that the rate-determining step is the reaction of ClO with O (JACOB, 1999, p. 177), whose rate competes with the loss of odd oxygen from pure Chapman in the ratio

$$\frac{v_{\mathrm{loss,\,Cl}}(\mathrm{O}_x)}{v_{\mathrm{loss}}(\mathrm{O}_x)} = \frac{k_6[\mathrm{ClO}]}{k_4[\mathrm{O_3}]}. \qquad (4.24)$$

$k_6/k_4 \approx 5.6 \times 10^4$ (25 km) and 4.5×10^4 (30 km): one molecule of ClO for every 50,000 of ozone is enough to match the loss from pure Chapman. I can run the calculation backward, with the observed ozone of Table 8: the losses become equal at a [ClO] of about 110 and 170 parts per trillion (3% to 5% of the total 2020 chlorine (SALAWITCH *et al.*, 2023, p. 17)). Since Cl and ClO together add up to less than 10% of the chlorine over most of the stratosphere (JACOB, 1999, p. 181), the ratio in Equation (4.24) reaches at most 3.0 (25 km) and 1.9 (30 km). In the simulation of Figure 29, which uses the higher ozone of pure Chapman, the same 10% makes chlorine account for 58% (25 km) and 42% (30 km) of the odd-oxygen loss. With far less ClO than ozone, the cycle can destroy odd oxygen as fast as pure Chapman, or faster, because every chlorine atom restarts the cycle until it becomes trapped, for a while, in a reservoir such as HCl (SALAWITCH *et al.*, 2023; JACOB, 1999, p. 24, 177) (Figure 29).

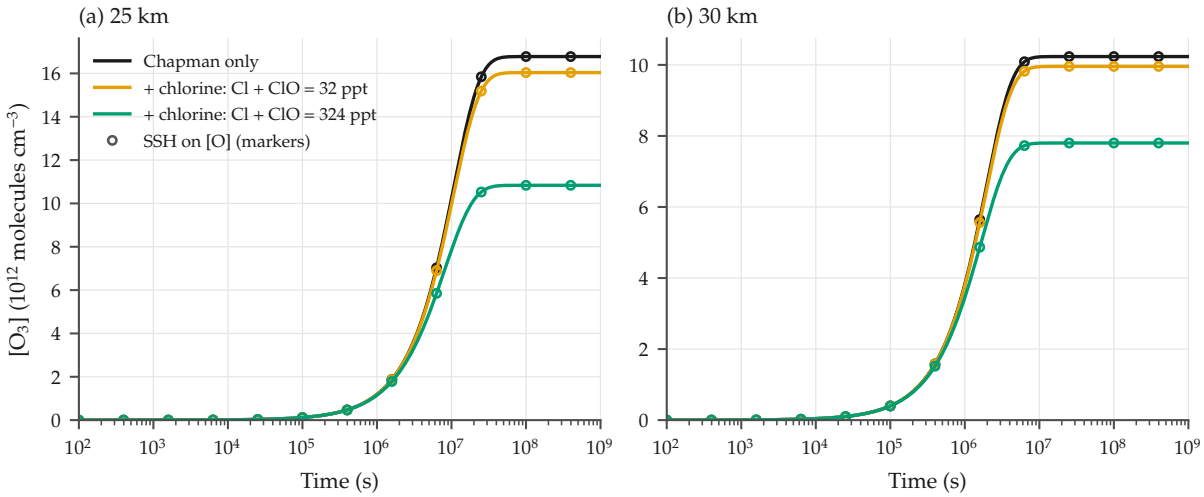


Figure 29. $[O_3]$ at 25 km (a) and 30 km (b), with Chapman alone (black) and with the chlorine cycle (reactions 4.22–4.23), for active chlorine (Cl + ClO) of 1 % and 10 % of the total 2020 chlorine (32 and 324 parts per trillion). At 10 %, the steady-state O_3 drops 35 % (25 km) and 24 % (30 km). Lines: full solution. Circles: SSH for O. The model has no oxides of nitrogen, OH, HO_2, bromine, or the HCl and $ClONO_2$ reservoirs.

Source: simulation (`codigo/ozonio.py`), with constants from Burkholder *et al.* (2019); total chlorine from Salawitch *et al.* (2023).

I ran the scenario of 10% of the total chlorine as ClO at more altitudes, and the drop in steady-state ozone is much larger at 20 km (two thirds) than at 25-30 km (a quarter to a third). The chemistry of chlorine itself is the same at every altitude. The difference comes from the scenario, which ties the amount of chlorine to the density of the air (Appendix C.1).

4.6 What ozone teaches us about the SSH

The SSH holds for the O atom. For ozone it does not hold, because O lives for milliseconds, but O_3 is the reactant itself, and the time over which it changes is the time scale of the reaction. In the cell, $[\mathrm{O}]_{\mathrm{ss}} = J/k_4$ does not depend on $[\mathrm{O_3}]$, so O stays constant while ozone falls. This is a

result of this particular mechanism; the SSH, in general, does not assume that the intermediate stays constant (Section 3.2). That is why the criterion $\tau_{\mathrm{O}} \ll \tau_{\mathrm{reaction}}$ (Equation (3.2)) is sufficient, but not required. Even as τ_{O} grows while ozone runs out, the SSH law for O holds exactly again after the first few milliseconds, because $\mathrm{d}[\mathrm{O}]/\mathrm{d}t$ vanishes at J/k_4 for any $[\mathrm{O_3}]$ (Section 7.1); in $[\mathrm{O_3}]$ only the fixed offset inherited from the start remains, up to 1.6% in Figure 21.

Comparing k_{obs} with $2J$ tests the mechanism. This comparison does not test the SSH, because the solution without the hypothesis gives almost the same curve, as long as O lives for much less time than the interval between readings. The same SSH shows how a reaction's order changes with conditions (first order without $\mathrm{O_2}$, second order and slower with a lot of $\mathrm{O_2}$).

In the stratosphere, O lives for milliseconds, tens to hundreds of thousands of times less than the interval between two breakups of an ozone molecule, so the SSH for O works very well, at every moment of the day. The steady state of odd oxygen, on the other hand, which gives Chapman's ozone, is an approximation on a much longer scale (weeks to months) and much cruder, one that agrees with observations only in order of magnitude and still needs wind transport and the catalytic cycles.

5 CHAPTER Case study 2: ammonia synthesis

The second case study of the steady-state hypothesis (SSH), also called the steady-state approximation, is the industrial synthesis of ammonia from nitrogen and hydrogen. In the ozone chapter, the intermediate was a loose atom in the gas. Here, the intermediates are atoms and pieces of molecules stuck to the surface of an iron catalyst. This chapter's SSH applies to that surface. The industrial reactor around it has a steady state of its own, which is a different thing.

5.1 The Haber-Bosch process

Until the early twentieth century, the nitrogen used in fertilizers and explosives came from limited natural reserves, such as Peruvian guano and Chilean saltpeter (ERISMAN *et al.*, 2008). Air has nitrogen to spare: more than 78% of the atmosphere is N_2. The trouble is that, in that form, nitrogen is of no use to plants or to the chemical industry (ERISMAN *et al.*, 2008). The two atoms of the N_2 molecule are held together by a very strong triple bond (ERTL, 2008), and breaking it was the challenge for more than a century.

Fritz Haber filed, on 13 October 1908, the patent for the "synthesis of ammonia from its elements": N_2 from the air reacts with hydrogen, at high pressure and high temperature, in the presence of a catalyst (ERISMAN *et al.*, 2008). In the laboratory, in 1909, Haber used osmium, and the whole world then had only about 80 kg of that metal (ERTL, 2008). Carl Bosch, at BASF, turned the process into an industrial one (ERISMAN *et al.*, 2008), and the first plant started running in 1913 (ERTL, 2008). The cheap catalyst came from Mittasch, who tested thousands of materials until he found one made from Swedish iron ore; that iron catalyst with promoters is still used in almost every plant today (ERTL, 2008). About 80% of the nitrogen produced goes into fertilizers, and the rest becomes other compounds (ERISMAN *et al.*, 2008), including explosives: ammonia is oxidized to nitric acid, which is used to make ammonium nitrate, nitroglycerin and TNT (ERISMAN *et al.*, 2008).

Erisman *et al.* (2008) estimate that, in 2008, about 48% of the world population was fed thanks to the nitrogen from the Haber-Bosch process. In 2000, the estimate was 44%. The authors themselves cite an even older estimate, from Smil, of about 40% at the end of the twentieth century. Howarth (2008) measures something else: it estimates that more than 80% of the nitrogen in the proteins of an average person came from the Haber-Bosch process, and links this to the Green Revolution, the expansion of agriculture in the twentieth century. The article does not show the arithmetic behind that estimate, and the two percentages neither add up nor compare, because one counts people fed and the other counts nitrogen atoms in proteins.

The process also has a cost. It uses a great deal of energy and accounts for about 1.2% of human CO_2 emissions (SMITH; HILL; TORRENTE-MURCIANO, 2020). And a large part of the nitrogen in fertilizers never reaches people: in 2005, about 100 Tg of nitrogen from the process were used in agriculture worldwide, but only 17 Tg turned into food (1 Tg = 10^{12} g, one million tonnes). The rest escapes into the air, the rivers and the sea (ERISMAN *et al.*, 2008).

5.2 Thermodynamics versus kinetics

In this section I show why equilibrium favors low temperatures and why the rate favors high temperatures. What happens on the catalyst surface, and the published laws I use to simulate the reactor, come in the following sections.

The overall reaction is

$$\begin{aligned} N_2(g) + 3\,H_2(g) &\rightleftharpoons 2\,NH_3(g), \\ \Delta_r H^\circ_{298} &= -91.8\,\text{kJ/mol}, \end{aligned} \tag{5.1}$$

where $\Delta_r H^\circ_{298}$ is the standard reaction enthalpy at 298 K, per mole of reaction, that is, for the 2 mol of NH_3 formed (CHASE JR., 1998). It is the heat exchanged, at constant pressure, with each gas in its standard state (the "°" indicates 1 bar). At 700 K it becomes more negative, about −105 kJ/mol (CHASE JR., 1998).

The negative sign says the reaction releases heat. A handy way to remember it is to write $N_2 + 3\,H_2 \rightleftharpoons 2\,NH_3 + \text{heat}$. By Le Châtelier's principle (ATKINS; PAULA; KEELER, 2018), a system at equilibrium reacts to a disturbance so as to reduce it. Heating is like adding "heat" to the products; that is why the equilibrium shifts to the left and the ammonia yield drops (cooling does the opposite). Pressure matters too. There are 4 mol of gas on the left and only 2 on the right, and raising the pressure pushes the equilibrium toward the side with fewer molecules, the ammonia side. Doubling the pressure multiplies by 4 the ratio required by the law of mass action at equilibrium.

Temperature, in turn, changes the equilibrium constant K_{eq} itself, through the van 't Hoff equation (JENKINS, 2008),

$$\frac{\mathrm{d}\ln K_{eq}}{\mathrm{d}T} = \frac{\Delta_r H^\circ}{R\,T^2}, \tag{5.2}$$

with T in kelvin and $R = 8.314\,\mathrm{J/(mol\,K)}$. Since $\Delta_r H^\circ < 0$, K_{eq} decreases as T increases. Do not confuse this with the rule of thumb of a factor of 2 every 10 °C (Section 2.5.3): that rule is about the *rate constant*, and here the subject is the equilibrium constant. Gillespie & Beattie (1930) fitted equilibrium measurements to a formula for K_a, the constant for the reaction written as $1/2\,N_2 + 3/2\,H_2 \rightleftharpoons NH_3$, in atm^{-1}. It is the same constant the rate law in Section 5.4 uses. It falls 13-fold between 600 and 800 K (values 6 to 9% higher in the NIST-JANAF tables (CHASE JR., 1998)).

Figure 30 shows the size of this effect. For a mixture of 3 mol of H_2 to 1 of N_2, at 200 atm, the equilibrium mole fraction of NH_3 drops from 0.386 at 400 °C to 0.190 at 500 °C; at 450 °C, it rises from 0.165 at 100 atm to 0.357 at 300 atm.

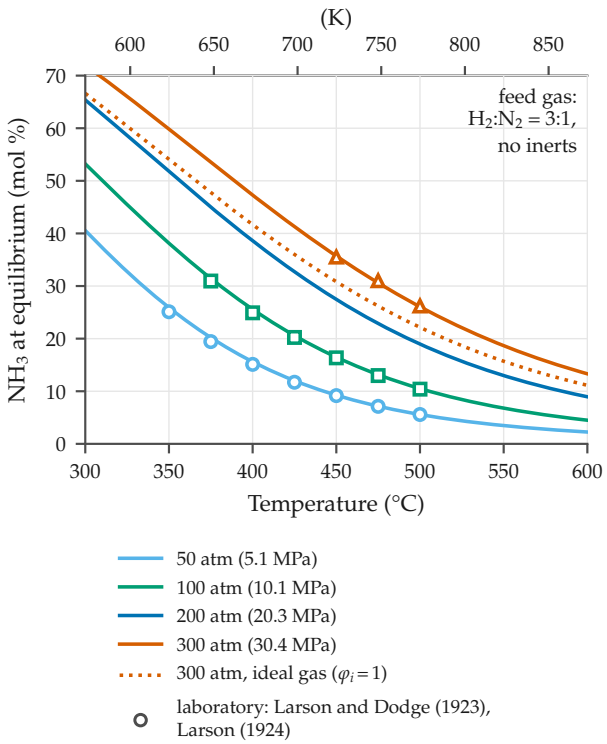


Figure 30. NH_3 at equilibrium, in mol %, against temperature, at four pressures (H_2:N_2 = 3:1, no inerts). Solid lines: with fugacity coefficients; dotted line (300 atm): ideal gas. Symbols: values measured in the laboratory.

Source: curves: simulation (`codigo/haber.py`; K_a from Gillespie & Beattie (1930); fugacity coefficients in the form of Cheema & Krewer (2018)); symbols: literature: Larson & Dodge (1923) and Larson (1924), values reproduced in Table 6 of Mahmoud El-Gharbawy, Khalifa Gad & Mahmoud Shehata (2021).

So far, this is only thermodynamics, which says how far the reaction can go. How long it takes is a question for kinetics, and the kinetic obstacle is the same triple bond of N_2. According to Gerhard Ertl's Nobel lecture, splitting $1/2\,N_2 + 3/2\,H_2$ into free atoms (N + 3 H) costs 1129 kJ per mole of NH_3 (ERTL, 2008). At 700 K, the typical thermal energy, RT, is only 5.8 kJ/mol, and that path is out of reach. That is why N_2 does *not* break into loose atoms in the gas. On the iron surface, it sticks and breaks along a path with much lower barriers, and that is the slowest step of the process (Section 5.3). The catalyst does not shift the equilibrium. It only opens another path, with lower barriers (ERTL, 2008), and comes out unchanged at the end. Temperature speeds up both directions of the reaction, each at its own pace. In the rate law I use to simulate the reac-

tor, the reverse constant grows faster than the forward one between 400 and 450 °C (8.2-fold versus 2.3-fold).

That leaves a standoff. At low temperature, equilibrium favors ammonia, but the reaction is too slow; at high temperature, the reaction is fast, but the equilibrium shifts toward the reactants. Table 11 summarizes the trade-off. Figures 31 and 32 show how it plays out in a simulated reactor using that rate law: conversion rises with temperature only up to a point, because after that the equilibrium is already close and the curve flattens out.

Table 11. Temperature and pressure in ammonia synthesis: what happens to the rate and to the equilibrium yield.

Condition	Kinetics	Equilibrium
High temperature ($T \uparrow$)	high rate ($k \uparrow$)	low yield ($K_{eq} \downarrow$)
Low temperature ($T \downarrow$)	low rate ($k \downarrow$)	high yield ($K_{eq} \uparrow$)
Intermediate temperature	trade-off	trade-off
High pressure ($P \uparrow$)	higher rate	higher yield

Source: author's own work.

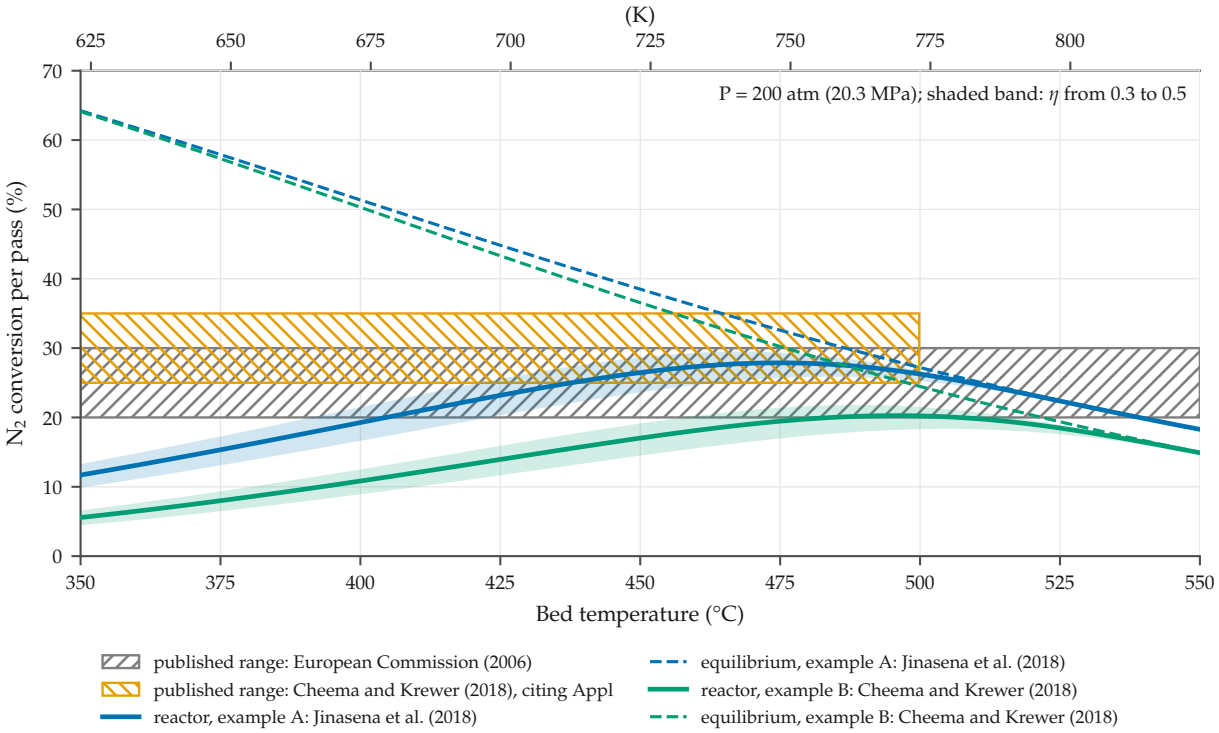


Figure 31. Fraction of the incoming N_2 that reacts in a single pass through the reactor, against the bed temperature, at 20.3 MPa (200 atm). Solid lines: isothermal bed with the rate law of Section 5.4, range of η (pellet effectiveness factor) from 0.3 to 0.5; dashed: equilibrium. Each example uses the flow rate and inlet composition of a published source. Rectangles: published industrial ranges.

Source: curves: simulation (`codigo/haber.py`; law of Dyson & Simon (1968); η range from Yancy-Caballero, Biegler & Guirardello (2015); flow rate, volume and composition from Jinasena, Lie & Glemmestad (2018) and Cheema & Krewer (2018)); rectangles: literature: European Commission (2006), p. 43, and Cheema & Krewer (2018), citing Appl.

In practice, plants choose an intermediate temperature, work at high pressure, and each pass through the reactor converts only part of the gas: what did not react is separated from the ammonia and sent back to the reactor (SMITH; HILL; TORRENTE-MURCIANO, 2020; EUROPEAN COMMISSION, 2006). Table 12 gathers the published ranges. Conversion per pass runs from about 15% (SMITH; HILL; TORRENTE-MURCIANO, 2020) to 20–30% (EUROPEAN COMMISSION, 2006) or 25–35% (Cheema & Krewer (2018), citing the Ullmann's encyclopedia). As Figures 31 and 32 show, the simulated model itself falls within this range when it uses the flow rate per catalyst volume from these sources, at 450 °C and 200 atm.

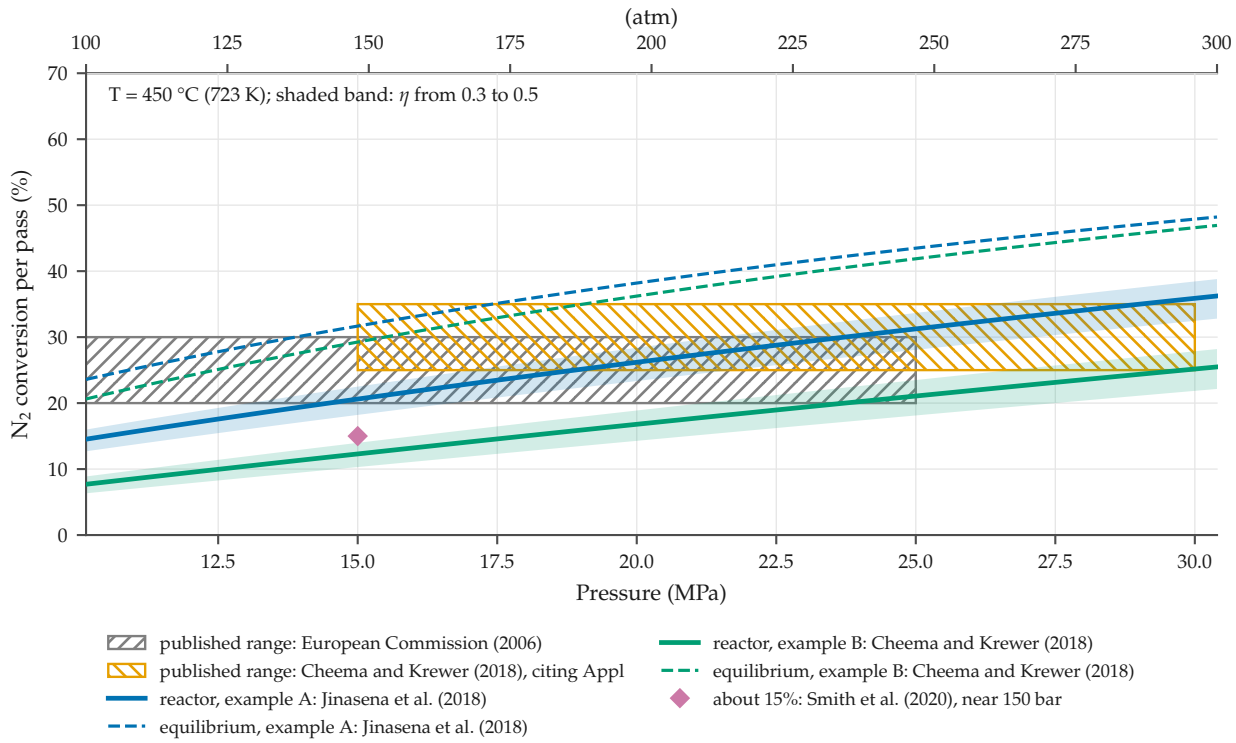


Figure 32. Fraction of the incoming N_2 that reacts in a single pass through the reactor, against pressure, at 450 °C (723 K). Lines and ranges as in Figure 31. The diamond marks the conversion of about 15 % cited by Smith et al. for plants near 150 bar.

Source: curves: simulation (`codigo/haber.py`; law of Dyson & Simon (1968); η range from Yancy-Caballero, Biegler & Guirardello (2015); flow rate, volume and composition from Jinasena, Lie & Glemmestad (2018) and Cheema & Krewer (2018)); rectangles and diamond: literature: European Commission (2006), p. 43; Cheema & Krewer (2018), citing Appl; Smith, Hill & Torrente-Murciano (2020).

Table 12. Operating conditions for the industrial synthesis of ammonia over iron, according to four sources.

Source	Temperature	Pressure	Conversion per pass	Note
Stoltze & Nørskov (1985)	375 °C to 500 °C	15.2 MPa to 30.4 MPa (150–300 atm)	—	conditions of the microkinetic calculation
Smith, Hill & Torrente-Murciano (2020)	400 °C to 450 °C	15 MPa to 25 MPa	about 15%; less than 20% near 150 bar	iron (magnetite or wustite) with several promoters
European Commission (2006)	350 °C to 550 °C	10 MPa to 25 MPa (100–250 bar)	20–30%	iron catalyst
Cheema & Krewer (2018), citing Appl	623 K to 773 K	15 MPa to 30 MPa (150–300 bar)	25–35%	H_2:N_2 from 2:1 to 3:1; inerts from 0 to 15% (molar)

Source: literature: Stoltze and Nørskov (1985); Smith et al. (2020); European Commission (2006), section 2.2.3.8; Cheema and Krewer (2018), section 1.1.

5.3 What happens on the iron surface

The iron is not consumed in the reaction. It offers places on its surface where molecules can stick: this is **adsorption**. The idea of picturing a catalyst's surface as a checkerboard, with squares where molecules can sit, comes from Irving Langmuir (ERTL, 2008).

Definition 5.1. A **site** is a place on the surface where a species can stick; the symbol * denotes an empty site and N* denotes a

nitrogen atom stuck to a site. The **coverage** θ_i of species i is the fraction of sites occupied by it, a number from 0 to 1; θ_* is the fraction of free sites. A species leaving the surface back into the gas is called **desorption**.

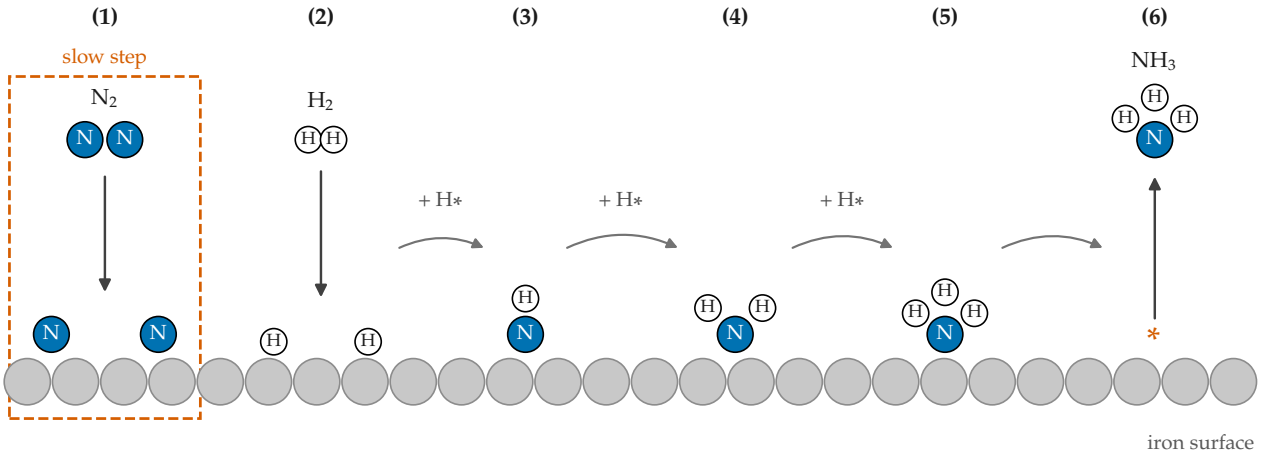
Since each site is either free or occupied by a single species, the coverages add up to 1:

$$\theta_* + \theta_N + \theta_H + \theta_{NH} + \theta_{NH_2} + \theta_{NH_3} = 1. \quad (5.3)$$

The accepted mechanism for ammonia synthesis over iron has six steps (ERTL, 2008; BOUDART, 1981), numbered from top to bottom:

$$N_2 + 2* \rightleftharpoons 2\,N* \quad (5.4)$$
$$H_2 + 2* \rightleftharpoons 2\,H* \quad (5.5)$$
$$N* + H* \rightleftharpoons NH* + * \quad (5.6)$$
$$NH* + H* \rightleftharpoons NH_2* + * \quad (5.7)$$
$$NH_2* + H* \rightleftharpoons NH_3* + * \quad (5.8)$$
$$NH_3* \rightleftharpoons NH_3 + * \quad (5.9)$$

The first two are dissociative adsorptions, in which the molecule sticks and breaks into two stuck atoms. Steps 3 through 5 are successive hydrogenations: the stuck N gains one H at a time, with both reactants side by side on the surface (a Langmuir-Hinshelwood-type mechanism). Step 6 is the desorption of ammonia. Adding step 1 once, step 2 three times and steps 3 through 6 twice each, the sites and the intermediates cancel out, leaving Equation (5.1). Figure 33 shows the cycle. Step 1, the adsorption of N_2, is the rate-determining step (ERTL, 2008; STOLTZE; NØRSKOV, 1985): on the most reactive face of an iron crystal in vacuum, only about one in a million N_2 molecules that hit the surface sticks and breaks apart (on other faces, far fewer still). That is why industrial iron carries promoters. Potassium helps break the stuck N_2, and oxides such as alumina and lime keep the iron grains from clumping together (ERTL, 2008). A gram of this catalyst has about 20 m^2 of surface (ERTL, 2008).



(1) $N_2(g) + 2* \rightarrow 2\,N*$

(2) $H_2(g) + 2* \rightleftharpoons 2\,H*$

(3) $N* + H* \rightleftharpoons NH* + *$

(4) $NH* + H* \rightleftharpoons NH_2* + *$

(5) $NH_2* + H* \rightleftharpoons NH_3* + *$

(6) $NH_3* \rightleftharpoons NH_3(g) + *$

$*$ = free site; X$*$ = species held on a site (intermediate). The SSH is applied to the coverages θ_N, θ_H, ...
In step (1), Ertl first shows adsorption of the whole N_2 ($N_2 + * \rightleftharpoons N_2*$); here both appear together.

Figure 33. Steps of ammonia synthesis on the iron surface. In step 1, the slowest one, N_2 sticks and breaks into two stuck N atoms; afterward, H atoms join the N one at a time until NH_3 forms and leaves. The asterisk is a free site; X* is a species stuck to a site.

Source: author's own drawing (mechanism from Ertl (2008)).

Analogy: chairs in a room

Recall the room with 100 chairs from Section 2.2. Each person who arrives needs to sit to fill out a form; then they get up and leave. On the surface, the chairs are the sites, the people are the gas molecules, and sitting down is adsorbing: the fraction of chairs occupied is the coverage θ. With few people in the room, almost every chair is free, and doubling the number of people doubles the number seated; with the room packed, say 1000 people, the 100 chairs are already full, and more people coming in does not seat anyone else. The N_2 molecule is more demanding than an ordinary person: it breaks into two atoms and needs two free chairs at once, like a couple who will only sit if the two spots are side by side.

Consider a gas A that adsorbs without breaking apart, A + * $\rightleftharpoons$ A*, with partial pressure P_A. Adsorption only happens if the molecule strikes a free site, and desorption only depends on how much is already occupied. When these two rates become equal, $\theta = K_A P_A/(1 + K_A P_A)$, with $K_A = k_{ads}/k_{des}$. This is the Langmuir isotherm (CHORKENDORFF; NIEMANTSVERDRIET, 2017) ("isotherm" because the curve $\theta(P)$ is drawn at fixed temperature). At low pressure, θ grows in proportion to P_A, like the room with few people. At high pressure, $\theta \to 1$: this is the packed room, and the curve stops rising (the same shape shows up in enzymes, Section 6.1). When the molecule breaks apart as it sticks, $A_2 + 2\,* \rightleftharpoons 2\,A*$, it needs two free sites at once. The same calculation, now with $(1-\theta)^2$ and θ^2, gives

$$\theta = \frac{\sqrt{K_{A_2} P_{A_2}}}{1 + \sqrt{K_{A_2} P_{A_2}}}. \qquad (5.10)$$

At low pressure, $\theta \approx \sqrt{K_{A_2} P_{A_2}}$, that is, the coverage grows with the square root of the pressure. The exponent 1/2 is the mark of a molecule that breaks apart as it sticks.

5.4 The SSH in the adsorbed species

Each coverage changes according to the steps that form and consume that species: step 1 forms two N* and step 3 consumes one; step 2 forms two H* and steps 3 through 5 consume one each; and so on. Calling v_j the net rate (forward minus reverse) of step j, per site and per second, the SSH says that the change in each coverage is negligible compared with the rates at which the species forms and is consumed (Equation (3.1)). It does not say the coverages stay constant: if the gas changes, they follow along. Setting all five changes to zero, the balance equations reduce to

$$v_2 = 3v_1, \qquad v_3 = v_4 = v_5 = v_6 = 2v_1. \qquad (5.11)$$

Under the SSH, every step moves at the pace of the slow step, each one multiplied by the number of times it appears in the cycle: no stuck species piles up, just as no dough piles up between the machines in the biscuit factory (Sections 2.7.2 and 3.3). The rate of the reaction is that of step 1, and ammonia leaves through step 6: $v = v_1$ and $v_{NH_3} = v_6 = 2v$, where v (per mole of N_2) and v_{NH_3} are, here, reaction frequencies per site.

Equation (5.11) is still not a usable rate law, because v_1 depends on coverages that are not measured in a reactor. In a reactor at steady state, the composition of the gas at each point of the bed does not change with time, and there $d\theta/dt = 0$ holds exactly. The SSH becomes an approximation when the gas changes with time, as in a reactor startup (Section 5.6). In that case it only requires that the number of atoms stuck to the surface be small compared with the amount of reactant in the gas. It is the same idea that, in enzymes, asks for little enzyme compared with the amount of substrate (Equation (6.9), in Chapter 6).

Example 5.1 (The coverage of N* in a reduced model)**.** Suppose three things: that N* is nearly the only species stuck to the surface ($\theta_* + \theta_N \approx 1$); that there is little ammonia in the gas, so the reverse of step 1 can be neglected; and that the hydrogenations and the desorption (steps 3 to 6) behave as a single first-order step in θ_N, with constant k_{hid}, which already absorbs the effect of the H_2 pressure. Then N* forms at rate $a(1-\theta_N)^2$, with $a = 2k_{ads}P_{N_2}$, and is consumed at rate $b\theta_N$, with $b = k_{hid}$, and the SSH gives $a(1-\theta_N)^2 = b\theta_N$:

a quadratic equation, whose physically meaningful root is

$$\theta_N = \frac{(2a + b) - \sqrt{b^2 + 4ab}}{2a}, \qquad v_{NH_3} = k_{hid}\,\theta_N \tag{5.12}$$

(the other root gives $\theta_N > 1$ and does not apply; the full algebra is in Appendix A.8). The limits show what the model says. If $a \ll b$, N* is consumed as soon as it forms, $\theta_N \approx a/b$ and $v_{NH_3} \approx a$; the N_2 bond-breaking step is in charge, and the rate is first order in it. If $a \gg b$, the surface fills up with N*, $\theta_N \to 1$ and $v_{NH_3} \to k_{hid}$: the rate stops growing with pressure, like the packed room. With illustrative values, $a/b = 0.1$; 1 and 10 give $\theta_N = 0.084$; 0.382 and 0.730.

The same model tells how long N* takes to reach that value: near it, a small deviation of θ_N dies away with the characteristic time $\tau_N = 1/\sqrt{b^2 + 4ab}$ (Appendix A.8), the intermediate's lifetime from Equation (3.2). Figure 34 compares $\theta_N(t)$ solved in full with the SSH value, for the same a/b, with illustrative parameters. Since time is in units of $1/k_{hid}$, the curves depend only on ratios, such as a/b. In panel (b), the gas sits in a closed vessel, and I call ζ the ratio between the number of surface sites and the number of N_2 molecules at the start. With few sites for many molecules ($\zeta = 0.01$), the N_2 in the gas runs out slowly, and the coverage θ_N changes with time instead of standing still. With gas of fixed composition, the SSH's relative error in θ_N is 30 %, 9 % and 0.08 % at time $1/k_{hid}$ (for $a/b = 0.1$, 1 and 10) and falls to 2.9 %, 0.1 % and 2×10^{-9} at time $3/k_{hid}$. In the closed vessel, after $5\tau_N$ and while $\theta_N > 0.01$, the largest relative difference is 0.6 %, 0.6 % and 5.5 % (simulation, `codigo/amonia_superficie.py`).

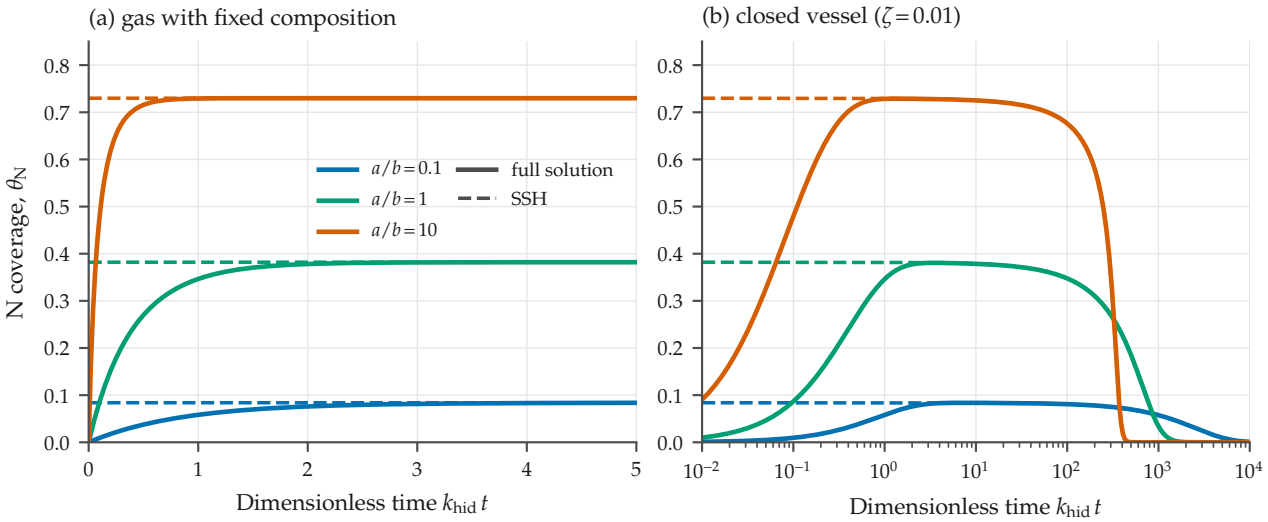


Figure 34. Fraction of the surface covered by N (θ_N) in the simple surface model, solved in full (solid lines) and with the SSH (dashed lines), for $a/b = 0.1$; 1 and 10. Time is in units of $1/k_{hid}$. (a) Gas of fixed composition: the SSH only misses at the start, while the surface fills up. (b) Closed vessel, with few sites for many N_2 molecules ($\zeta = 0.01$): θ_N changes with time, and the SSH tracks the change. The parameters are illustrative.

Source: simulation (`codigo/amonia_superficie.py`), illustrative parameters.

Example 5.1 dropped the reverse of step 1 and the effect of ammonia in the gas. A second path recovers both: steps 2 through 6 have constants much larger than step 1, the SSH pins the difference between their forward and reverse rates to a small value compared with them, and each one sits in *near-equilibrium*, which is the idea of pre-equilibrium (Section 3.3.2); the SSH, on its own, only says that no stuck species piles up. This gives $\theta_N/\theta_* = K_N P_{NH_3}/P_{H_2}^{3/2}$, with K_N measuring how much N* the surface would carry in equilibrium with the NH_3 and H_2 in the gas. Using this in the rate of step 1 (away from equilibrium)

and in the site balance with N* dominant,

$$\begin{aligned} v &= \frac{k_{\text{ads}}\, P_{\text{N}_2}\,(1-\beta)}{\left(1 + K_{\text{N}}\, P_{\text{NH}_3}/P_{\text{H}_2}^{3/2}\right)^2}, \\ \beta &= \frac{P_{\text{NH}_3}^2}{K_{\text{eq}}\, P_{\text{N}_2}\, P_{\text{H}_2}^3}, \\ v_{\text{NH}_3} &= 2v, \end{aligned} \tag{5.13}$$

with K_{eq} the equilibrium constant of Equation (5.1). The number β measures how far the system is from equilibrium ($\beta = 0$ with no ammonia; $\beta = 1$ at equilibrium, where $v = 0$, as it should be). With little ammonia, Equation (5.13) reduces to the result of Example 5.1 with $a \ll b$. With the surface nearly full of N*, it reduces to $v \approx (k_{\text{ads}}/K_{\text{N}}^2)(P_{\text{N}_2} P_{\text{H}_2}^3/P_{\text{NH}_3}^2) - k_{\text{des}}$, which is the Temkin and Pyzhev law (Equation (5.14)) with $\alpha = 1$, not the α of 0.5 to 0.75 used in practice.

In 1940, Mikhail Temkin and Vladimir Pyzhev published a rate law for this synthesis, assuming, as here, that the adsorption of N_2 controls the rate (TEMKIN; PYZHEV, 1940 apud NADIRI *et al.*, 2024). With partial pressures,

$$v = k_+\, P_{\text{N}_2} \left(\frac{P_{\text{H}_2}^3}{P_{\text{NH}_3}^2}\right)^{\alpha} - k_- \left(\frac{P_{\text{NH}_3}^2}{P_{\text{H}_2}^3}\right)^{1-\alpha}, \tag{5.14}$$

where k_+ and k_- are the forward and reverse rate constants, and α is a number between 0 and 1, used between 0.5 and 0.75 (NADIRI *et al.*, 2024; YANCY-CABALLERO; BIEGLER; GUIRARDELLO, 2015) (Table 13). The law is consistent with thermodynamics for any α: at equilibrium, $v = 0$ gives $k_+/k_- = K_{\text{eq}}$, no matter the exponent (worked out in Appendix A.8), but it diverges as $P_{\text{NH}_3} \to 0$.

Table 13. Parameters of the Dyson and Simon rate law and of the equilibrium constant.

Quantity	Value	Source
k_0 (pre-exponential factor of k_-)	8.849×10^{14} kmol/(m^3 h)	Dyson & Simon (1968), via Cheema & Krewer (2018)
E (activation energy of k_-)	170.56 kJ/mol (40,765 cal mol^{-1})	idem
α	0.5 (used: 0.5 to 0.75)	Cheema & Krewer (2018); Yancy-Caballero, Biegler & Guirardello (2015)
$\log_{10} K_{\text{a}}(T)$	polynomial in T and $\log_{10} T$ (`codigo/constantes.py`)	Gillespie & Beattie (1930), via Xie, Xia & Wang (2022)
K_{a} at 600, 700 and 800 K	3.83×10^{-2}; 8.81×10^{-3}; 2.84×10^{-3} atm^{-1}	calculated with the formula above
φ_{N_2}, φ_{H_2}, φ_{NH_3} at 700 K and 200 atm	1.097; 1.058; 0.925	correlations in Cheema & Krewer (2018)
Range of the fitted data	149 to 309 atm; 603 to 768 K; H_2:N_2 from 1.15 to 6.23	Nadiri *et al.* (2024)
Pellet effectiveness factor	0.3 to 0.5	Yancy-Caballero, Biegler & Guirardello (2015)

Source: literature: Dyson and Simon (1968), via Cheema and Krewer (2018); Gillespie and Beattie (1930), via Xie et al. (2022); Nadiri et al. (2024). Values of K_{a} and φ_i calculated with the formulas from the sources (`codigo/constantes.py`).

The form that reactor models use today replaces partial pressures with activities, which removes the constants' dependence on total pressure (DYSON; SIMON, 1968; NADIRI *et al.*, 2024):

$$v = k_- \left[K_{\text{a}}^2\, a_{\text{N}_2} \left(\frac{a_{\text{H}_2}^3}{a_{\text{NH}_3}^2}\right)^{\alpha} - \left(\frac{a_{\text{NH}_3}^2}{a_{\text{H}_2}^3}\right)^{1-\alpha} \right], \tag{5.15}$$

where $v_{\text{NH}_3} = 2v$ and the activity $a_i = y_i \varphi_i P$ is the partial pressure ($y_i P$, with y_i the mole fraction) corrected by the fugacity coefficient φ_i (1 in an ideal gas). At 200 atm, the synthesis gases are no longer ideal, and φ_i corrects for that. With P in atm, a_i and K_{a} enter as pure numbers (divided by 1 atm), so the bracket in Equation (5.15) has no unit, and k_0 has the unit of the rate. K_{a} is the Gillespie and Beattie equilibrium constant and $k_- = k_0 \exp(-E/RT)$ is the reverse rate constant. This is the law of Dyson & Simon (1968), used in industrial reactors and in the reactor model of Section 5.6; the parameters are in Table 13. It is inaccurate below about 600 K (NADIRI *et al.*, 2024). The rate it gives is that of the catalyst itself: in an actual pellet, it is lower by an effectiveness factor η between 0.3 and 0.5,

because the gas takes time to reach the inside of the pores (YANCY-CABALLERO; BIEGLER; GUIRARDELLO, 2015).

5.5 The ammonia data from the December 2025 article

The December 2025 article has a second data set, with 13 points of the "reactor efficiency" over the first minutes of a startup (RIBEIRO, 2025b). As with the ozone data (Section 4.4), the values are in Table 26 (Appendix C). They have two decimal places, so each one carries a rounding of up to ±0.005. The efficiency is a normalized quantity.

The points rise from 0 to nearly 1 between 0 and 8.6 min, with a slower start than a simple exponential: the values rise by 0.20 from 0 to 0.7 min and by 0.25 from 0.7 to 1.4 min. A curve $1 - e^{-t/\tau}$ always rises less in each equal interval, so none of them passes through the first two points. Figure 35 and Table 14 compare five descriptions of these points: the curve published in the article itself (M0, $1 - e^{-t/\tau}$ with $\tau = 1.667$ min); a simple exponential fitted to the points (M1); the same exponential with a dead time at the start (M2); two steps in series, A ⟶ I ⟶ P (M3); and a Weibull curve (WEIBULL, 1951), $1 - e^{-(t/\tau_W)^\gamma}$, with scale τ_W and shape γ (M4).

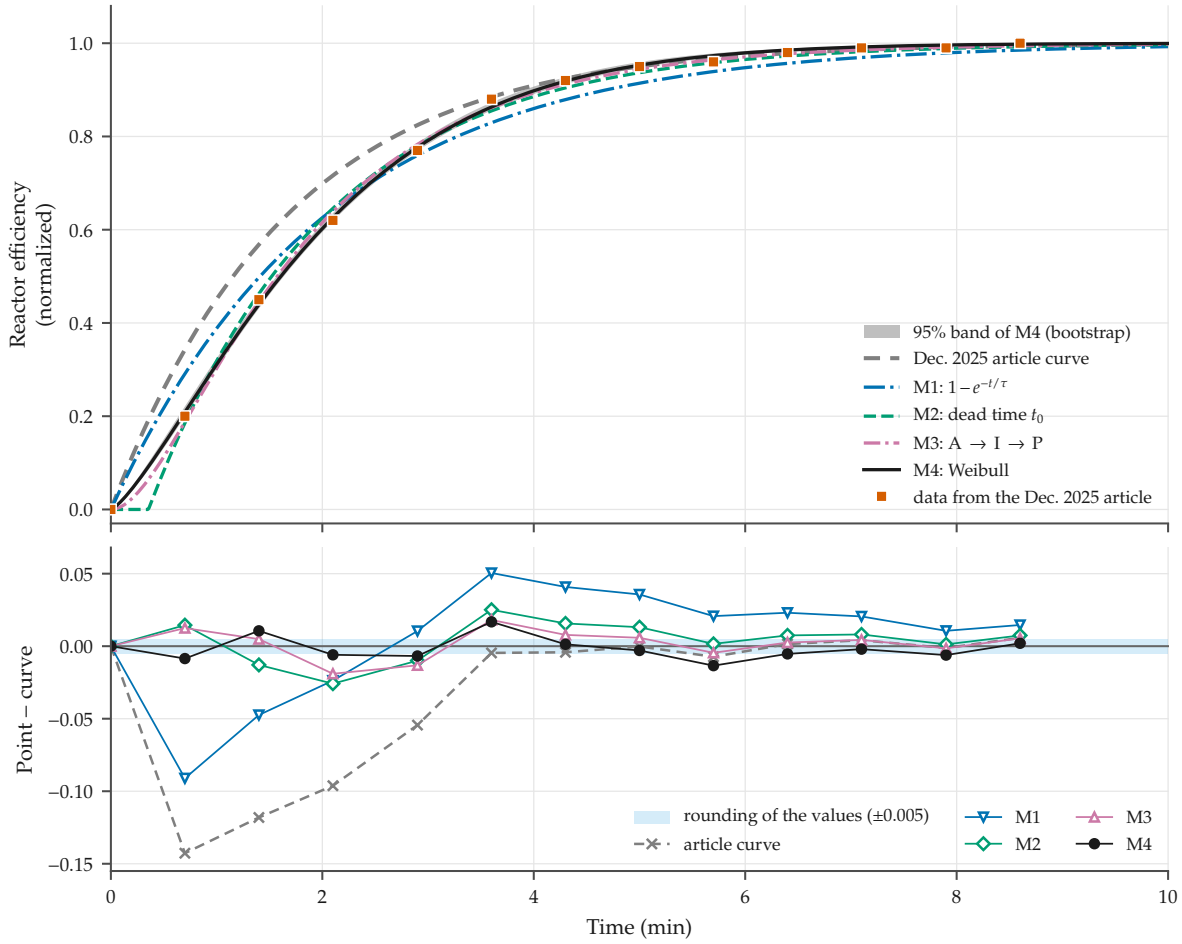


Figure 35. Top: the 13 ammonia data points from the December 2025 article (squares), the curve published in the article (dashed gray) and four models fitted to the points (M1 to M4); the gray band holds, point by point, 95 % of the curves of the model with the lowest AICc (M4) over 2000 replicas of the residual *bootstrap*. Bottom: point minus curve, for each curve; the light-blue band shows the rounding of the values (±0.005). The vertical axis is the article's "reactor efficiency," a normalized quantity.

Source: data from the December 2025 article (RIBEIRO, 2025b), with the curve published in the same article; fits and 95 % band: simulation (`codigo/ajuste_amonia.py`).

By AICc, which penalizes a model with more parameters, the order is M4, M3, M2, M1 and last the article's curve. The BIC gives the same order, and it does not change if I drop the point $(0; 0)$, which every model gets right only because the curve is forced to start at zero. M2, M3 and M4 have the same number of parameters and very similar R^2 (0.998; 0.999; 0.9994). All three describe the delay at the start well, and what separates them is the small residuals in the middle of the curve. Each one explains that delay in its own way: a dead time (M2), two steps in series (M3) or a sigmoid (M4). In M3, the curve is symmetric in the two constants (swapping one for the other gives the same curve), so the slow step can be either the first or the second (a known ambiguity of series reactions (ALCOCK; BENTON; MOORE, 1970)). With independent noise of the size seen here, these residuals already let AICc pick the right model most of the time, and changing one condition, such as the flow rate, would show which constant follows it (Section 5.5.1).

Table 14. Five models fitted to the 13 ammonia data points from the December 2025 article. In brackets, the 95 % confidence interval by profile likelihood.

Model (p)	Parameters [95 % CI]	R^2	AICc	ΔAICc	Weight	BIC	DW
M0: article curve (0)	none ($1/\tau = 0.6$/min, fixed)	0.9649	−70.8	46.4	0.000	−70.6	0.55
M1: $1 - e^{-t/\tau}$ (1)	$\tau = 2.03$ [1.84; 2.24] min	0.9862	−80.1	37.1	0.000	−80.1	0.76
M2: dead time (2)	$t_0 = 0.355$ [0.278; 0.424] min $\tau = 1.68$ [1.58; 1.79] min	0.9982	−103.3	13.9	0.001	−104.2	1.25
M3: A ⟶ I ⟶ P (2)	$k'_a = 0.648$ [0.601; 0.715] min^{-1} $k'_b = 2.17$ [1.71; 2.77] min^{-1}	0.9991	−112.0	5.2	0.069	−112.9	1.77
M4: Weibull (2)	$\tau_W = 2.127$ [2.085; 2.170] min $\gamma = 1.307$ [1.257; 1.360]	0.9994	−117.2	0.0	0.930	−118.2	2.21

Source: data from the December 2025 article (RIBEIRO, 2025b); fits: simulation (`codigo/ajuste_amonia.py`). p is the number of free parameters. Weight is the Akaike weight (AKAIKE, 1974): the relative probability that this is the best model among the five, given what the points show, and the weights add up to 1. DW is the Durbin-Watson statistic (DURBIN; WATSON, 1950) of the residuals, which runs from 0 to 4. Near 2, neighboring residuals do not resemble each other. Well below 2, they tend to share the same sign, which indicates that the model leaves a trend unexplained. Numbers in `dados/simulacao/amonia_ajuste.json`.

I also ran two bootstraps (Appendix C.1). In the first, I draw with replacement 13 pairs (t, y) from the 13 points and fit again: in 5000 replicates, M4 wins by AICc in 70.5%, M3 in 20% and M2 in 9%. In the second, I add to the M4 curve residuals drawn from M4 itself. Since these data come from M4, it wins in 97%, which does not test the choice. This second bootstrap gives the τ of M1 a narrower interval than the one in the table (1.99 to 2.06 min, against 1.84 to 2.24), because it uses the residuals of M4, about four times smaller than those of M1 (0.009 against 0.038).

Against the curve published in the article, the points sit more than 0.02 below it only up to 2.9 min. From 3.6 min on, the difference stays below 0.008, a little above the rounding of the values (up to ±0.005), and from 7.1 min on the points are already at 0.99 or higher. The delay is all in the first minutes of the startup.

Reaching a plateau is what IUPAC calls the steady state of a continuous-flow reactor: the outlet composition that no longer changes (IUPAC, 2025b). This reactor steady state is not this chapter's SSH, which is about the coverages of the species stuck to the surface (Section 5.4) and also holds during the rise, as long as the coverages adjust much faster than the gas changes and the surface holds little matter compared with the gas (Section 7.4).

5.5.1 An assay with synthetic data for the startup

The 13 points show the delay, but not its cause. To see what a fuller assay would show, I again generated *synthetic data*, as in Section 4.4.1: curves from models M1 to M4 of Table 14, with the parameters fitted to the 13 points, plus normal noise drawn with a fixed seed, made by `codigo/`

`dados_sinteticos.py`. The standard deviation of the noise, 0.0090, is that of the residuals of M4, the model with the lowest AICc, without the point (0; 0), through which every model passes. None of these points is a measurement. They show what the assay could show if the models and the noise are right.

The first question is whether AICc finds the right model. For each true model, I drew 300 data sets in two designs: the 13 times of the article and an assay of 3 replicates of 40 points, from 0 to 8.6 min. With 13 points, AICc gets the model right in 84 % to 98 % of the sets; with 120 points, it always gets it right when the truth is M2, M3 or M4 (Figure 36). The exception is M1. When the truth is the simple exponential, a model with one extra parameter wins in about 16 % of the sets, in both designs (first row of Figure 36). With 120 points, this is the known behavior of the AIC when one model is a special case of the other: an extra parameter that does not exist gets through whenever the improvement in the fit is worth more than the penalty, of about 2. That happens 16 % of the time (the chance that the square of a number drawn from the standard normal distribution exceeds 2), and that chance does not fall with more points. With 13 points, the AICc penalty is larger, and the similar fraction comes from other small-sample effects. The BIC (SCHWARZ, 1978), whose penalty grows with the number of points, lowers this false alarm to 2 % with 120 points.

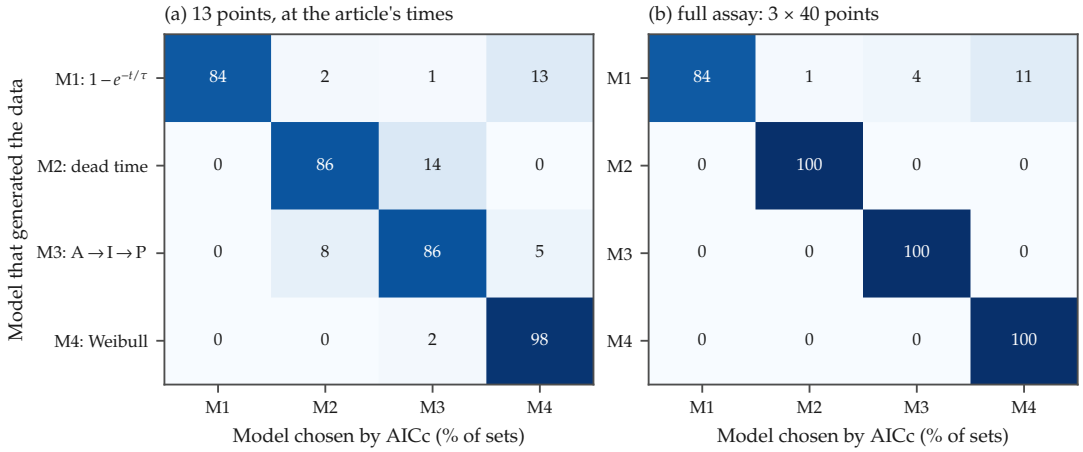


Figure 36. Synthetic data: how often AICc picks the right model for the reactor startup. These are model curves plus drawn noise, not measurements. Each row is the model that generated the data (M1 to M4 of Table 14, with the parameters fitted to the 13 points); each column, the model with the lowest AICc. The number is the percentage of 300 synthetic sets, rounded, so a row may not add up to 100. (a) The 13 times of the article, with the point at $t = 0$ fixed at zero. (b) 3 replicates of 40 points, from 0 to 8.6 min. Normal noise with standard deviation 0.0090, that of the M4 residuals on the 13 points.

Source: synthetic data (`codigo/dados_sinteticos.py`), with the models of `codigo/ajuste_amonia.py`; parameters and noise taken from the data of the December 2025 article (RIBEIRO, 2025b).

The same calculation says something about the 13 points of the article. On them, the AICc of M4 is 5.2 lower than that of M3. If the true curve were M3, a difference this large or larger would appear in 5 of the 300 synthetic sets (about 2 %; 95 % interval from 0.5 % to 3.8 %). With the noise of the M3 residuals, which is larger, it would be 7 of 300; with the values rounded to 0.01, like those of the article, 1 of 100. If the truth were M4, such a difference would appear in 263 of the 300 (88 %). With independent noise, then, the 13 points look more like the Weibull than like two first-order steps in series.

The second question is which constant follows a change of condition. In the reactor I simulate in Section 5.6, the time for the outlet to reach 95 % of its final value falls with the gas flow rate F as $F^{-0.97}$: from 22.3 s at half that reactor's flow rate to 5.8 s at twice it. It is the time to replace the gas in the bed. The fast step of M3, with constant k'_b (the slow one has k'_a), has $1/k'_b = 28$ s on the 13 points, 2.3 times the time the gas takes to cross

that bed. The article does not say which reactor it is or what the flow rate is, so this comparison only shows that the order of magnitude allows one to think of the gas exchange. I tested three hypotheses, chosen by me just for the assay:

(i) the fast step is the gas exchange: k'_b proportional to F, with k'_a fixed;

(ii) it is the slow step that follows the flow rate: k'_a proportional to F, with k'_b fixed;

(iii) control: the curve is the Weibull (M4), which only changes scale, with τ_W proportional to $1/F$.

Hypothesis (i) treats the gas exchange as a first-order step, as in a well-mixed volume. In the bed of Section 5.6, where the gas moves forward without mixing, the response has another shape: the 95 % time is 2.5 times the 63 % time, against 3.0 for an exponential. Under each hypothesis, I drew 3 replicates of 40 points at four flow rates, from 0.5 to 2 times the reference flow rate F_{ref}, the one at which the constants equal those of the 13 points. The noise draws are the same under all three hypotheses. Figure 37 shows hypothesis (i).

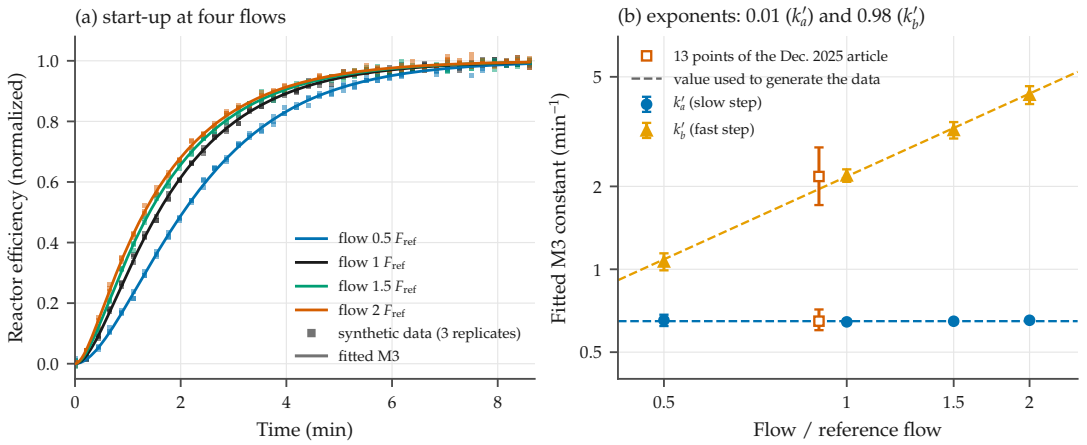


Figure 37. Synthetic data: reactor startup at four flow rates, under hypothesis (i): the fast step of M3 is the gas exchange (k'_b proportional to the flow rate; k'_a fixed). These are model curves plus drawn noise, not measurements. (a) Normalized efficiency: 3 replicates of 40 points per flow rate (squares) and M3 fitted to the 3 replicates pooled (lines). (b) Fitted k'_a and k'_b against the relative flow rate, on log scales, with 95 % confidence interval. Dashed: the values used to generate the data. Open squares: M3 fitted to the 13 points of the December 2025 article, with the profile interval of Table 14, drawn slightly left of flow 1 so they do not hide the others. The exponents in the title come from fitting $k'_a \propto F^{\alpha}$ and $k'_b \propto F^{\beta}$ to the 480 points. Normal noise with standard deviation 0.0090.

Source: synthetic data (`codigo/dados_sinteticos.py`), with M3 of `codigo/ajuste_amonia.py`; parameters and noise taken from the data of the December 2025 article (RIBEIRO, 2025b); open squares: fit to the article data, simulation (`codigo/ajuste_amonia.py`).

Under hypotheses (i) and (ii), M3 wins by AICc at all four flow rates and in each of the 12 curves on its own. I fitted M3 to all the curves together, with $k'_a \propto F^{\alpha}$ and $k'_b \propto F^{\beta}$. Hypothesis (i) gives $\alpha = 0.01$ (from -0.01 to 0.03) and $\beta = 0.98$ (from 0.94 to 1.02). Hypothesis (ii) gives $\alpha = 1.00$ (from 0.98 to 1.02) and $\beta = 0.02$ (from -0.02 to 0.07). Such an assay would tell which of the two constants follows the flow rate, that is, whether the gas exchange is the slow step or the fast one, which a single curve does not tell. Which step comes first stays unanswered: the M3 curve is symmetric in the two constants at any flow rate.

In the control (iii), M4 wins at every flow rate and in every curve, and both exponents of M3 come out near 1 ($\alpha = 1.05$, from 1.03 to 1.08; $\beta = 0.87$, from 0.79 to 0.95), as should happen when the whole curve changes scale. Neither 95 % interval contains 1 because M3 is not the right model here. In the models that did not generate the data, the parameters also change with the flow rate, but with no simple reading: in (i), the γ of M4 goes from 1.45 to 1.18, and the dead time of M2 from 0.56 to

0.19 min.

With the noise of the 13 points, more points per curve would already separate the two-step delay from the dead time and the Weibull. Only a change of condition ties one constant to a physical cause, such as the gas exchange. A real experiment would need several known flow rates, with replicates.

Two caveats. The three hypotheses are my choices: the article does not say what the "efficiency" measures, nor the temperature, the pressure or the flow rate, and a real startup also has the heating of the bed and the reduction of the catalyst (Section 5.6), which none of them includes. And the conclusion about the Weibull holds within the assumption of independent noise, which the residuals of M4 do not contradict (Durbin-Watson of 2.2 in Table 14).

5.6 Simulating a reactor

To see the Dyson and Simon law (Equation (5.15)) in action, I simulate a catalyst bed at uniform temperature (450 °C, 200 atm) with plug flow (all the gas moves forward at the same pace, without back-mixing), using the flow rate, volume and void fraction of a reactor described by Jinasena, Lie & Glemmestad (2018) (Figure 38).

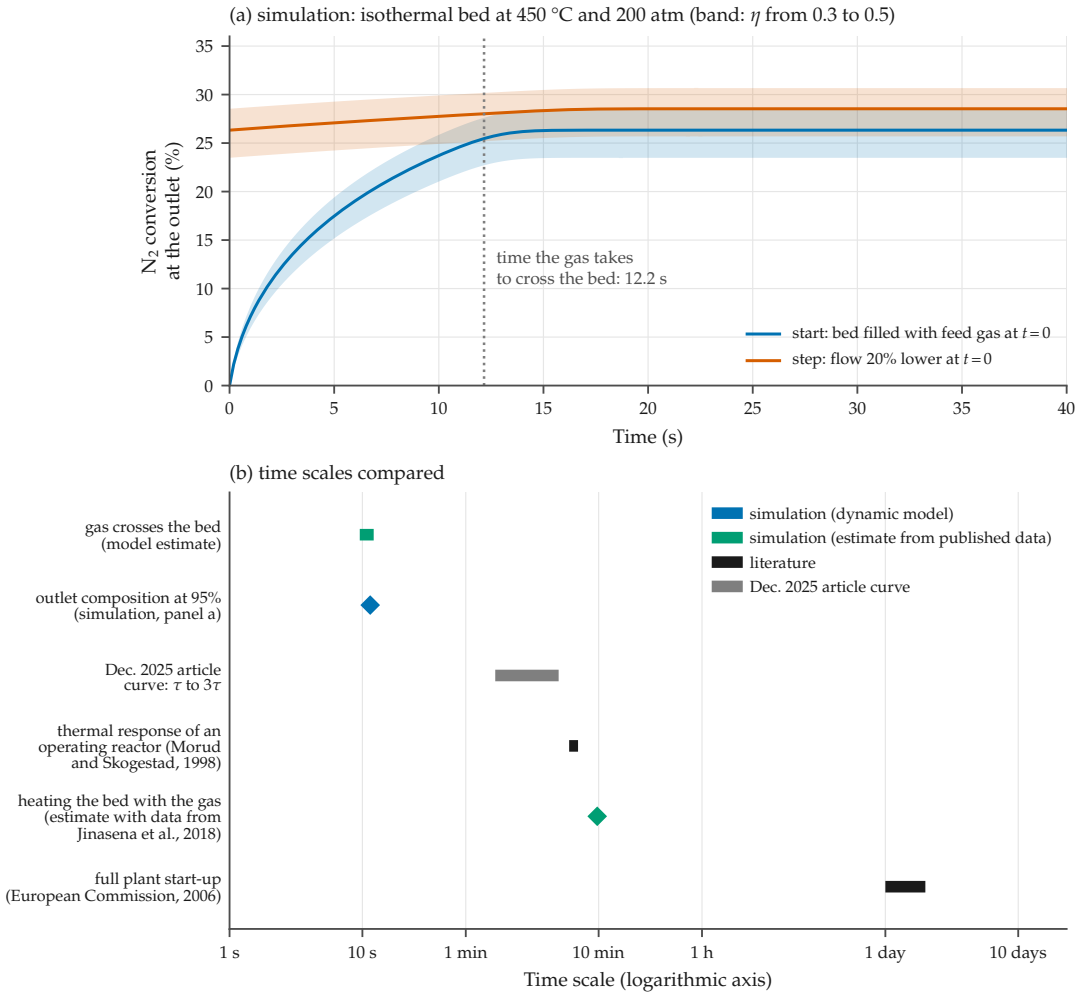


Figure 38. (a) N_2 conversion at the outlet of a bed at 450 °C and 200 atm, after the gas starts flowing (blue) and after a 20 % drop in flow rate (red); colored band from $\eta = 0.3$ to 0.5. The catalyst enters already active and hot, so only the time for the gas to cross the bed is visible. (b) Compared time scales, on a logarithmic axis.

Source: (a) and the first two rows of (b): simulation (`codigo/haber.py`; law of Dyson & Simon (1968); data from Jinasena, Lie & Glemmestad (2018)); thermal response: literature: Morud & Skogestad (1998); plant startup: literature: European Commission (2006), p. 47; article curve: the curve published in the December 2025 article (RIBEIRO, 2025b).

The model starts from a bed already reduced, hot and at operating pressure, and follows the composition of the gas after it starts flowing (startup) or after a 20% drop in flow rate (step change). It follows only the composition along the length of the bed, with the surface already at steady state at each instant, because that is where this chapter's SSH comes in. The heating of the reactor, the reduction of the catalyst, the heat of reaction, the differences between the center and the edge of the bed (radial profile) and the recycle are left for a larger model.

The gas takes about 12.2 s to cross the bed, and the outlet composition reaches 95% of its final value between 11.4 and 11.5 s after startup (for effectiveness factors η from 0.3 to 0.5), a little before one full pass: at $t = 0$ the bed is already full of feed gas, which reacts along the whole bed at once. This is far from other time scales of the same process. A calculation of mine, with the heat capacity of the bed and the specific heat of the gas given in the appendix of Jinasena, Lie & Glemmestad (2018) (`codigo/haber.py`), gives about 9.8 min to heat the bed with the gas alone. An industrial reactor already in operation responds to a thermal disturbance in about 7 min (MORUD; SKOGESTAD, 1998). And a full plant startup, with the heater and the catalyst reduction, takes 1 to 2 days (EUROPEAN COMMISSION, 2006, p. 47). The characteristic time of the article's curve, about 1.7 min, falls between the response of the composition (seconds) and the thermal response of a running reactor (minutes).

I ran the surface model over a grid of 40 ratios a/b and four ratios ζ, and this Dyson-Simon reactor over a temperature-pressure grid larger than the one shown here. Both still match the points already computed, and the new part is seeing where the SSH errs most (fast reaction, scarce gas) and where the reactor's rate law is being extrapolated outside the range with published data (Appendix C.1).

5.7 What ammonia teaches about the SSH

Ammonia shows the SSH in a different place from ozone. There, the SSH dealt with a molecule loose in the gas; here, it deals with the fraction of the surface occupied by each species stuck to the catalyst. The simple model of Example 5.1 and the near-equilibrium form (Equation (5.13)) use the same idea, the change in an intermediate being negligible compared with the rates that form and consume it, and both give a rate law for the coverages, just as one was given for the O atom and for the enzyme-substrate complex.

In the reactor the case is different. The Dyson and Simon law already has the SSH built into the coverages; what changes with time, in the simulation of Section 5.6, is the composition of the gas, and the hypothesis about the surface stays the same. The ammonia data from the December 2025 article rise with a delay to a plateau, and the model that describes them best, by AICc, is the Weibull curve, followed by the two steps in series; the plateau has the shape of the reactor's steady state, although the article does not say what the efficiency measures. This chapter's SSH sits in the part of the process that shows up least in a factory, that of the atoms and molecules stuck, for fractions of a second, to the surface of a piece of iron.

6 CHAPTER Other applications

Chapters 4 and 5 showed the steady-state hypothesis (SSH), also called the steady-state approximation, in two large cases: the atmosphere and the industrial synthesis of ammonia. This chapter closes the subject with three smaller examples. One is an enzyme reaction, of the kind that comes up whenever biochemistry or medicine is discussed. Another is a step in the manufacture of sulfuric acid, one more giant chemical process. The third picks up an account left open since Chapter 2, the rate law for the reaction between hydrogen and bromine.

6.1 Enzymes: the Michaelis-Menten equation

The Michaelis-Menten equation describes the rate of a reaction catalyzed by an enzyme as a function of the substrate concentration. The shape of this equation had already appeared in Henri (1902), and it was carefully tested by Michaelis & Menten (1913), who measured the initial rate of the breakdown of sucrose (ordinary table sugar) by invertase (JOHNSON; GOODY, 2011). They fixed two flaws in Henri's work: they controlled the pH with an acetate buffer and, before each reading, waited for the mutarotation of glucose (the slow interconversion between its two forms, which changes the angle read on the polarimeter) to finish (MICHAELIS; MENTEN, 1913; JOHNSON; GOODY, 2011). Only in 1925 did Briggs & Haldane show that the same equation also follows from a weaker hypothesis, the steady-state hypothesis, without needing to assume that the first step of the mechanism sits at equilibrium. It is that more general derivation that I carry out next.

Consider the two-step mechanism

$$\mathrm{E} + \mathrm{S} \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} \mathrm{ES} \xrightarrow{k_2} \mathrm{E} + \mathrm{P}, \tag{6.1}$$

where E is the free enzyme, S the substrate, ES the enzyme-substrate complex and P the product. k_1, k_{-1} and k_2 are the rate constants of each step.

The complex ES is the intermediate. By the SSH (Section 3.2), its change is negligible compared with the rates at which it forms and is consumed:

$$\frac{\mathrm{d}[\mathrm{ES}]}{\mathrm{d}t} \approx 0. \tag{6.2}$$

The rate of formation of ES is $k_1[\mathrm{E}][\mathrm{S}]$, and the rate of consumption (by the reverse to E + S or by forming the product) is $(k_{-1} + k_2)[\mathrm{ES}]$. By Equation (6.2), these two rates are equal:

$$k_1[\mathrm{E}][\mathrm{S}] = (k_{-1} + k_2)[\mathrm{ES}]. \tag{6.3}$$

Total enzyme is conserved, $[\mathrm{E}]_0 = [\mathrm{E}] + [\mathrm{ES}]$, that is, $[\mathrm{E}] = [\mathrm{E}]_0 - [\mathrm{ES}]$. Substituting into Equation (6.3) and distributing,

$$k_1[\mathrm{E}]_0[\mathrm{S}] - k_1[\mathrm{ES}][\mathrm{S}] = (k_{-1} + k_2)[\mathrm{ES}]. \tag{6.4}$$

Isolating [ES], that is, collecting on one side the terms that multiply it,

$$[\mathrm{ES}] = \frac{k_1[\mathrm{E}]_0[\mathrm{S}]}{k_1[\mathrm{S}] + k_{-1} + k_2} = \frac{[\mathrm{E}]_0[\mathrm{S}]}{[\mathrm{S}] + \dfrac{k_{-1} + k_2}{k_1}} \tag{6.5}$$

(dividing numerator and denominator by k_1). I define the **Michaelis constant**

$$K_M \equiv \frac{k_{-1} + k_2}{k_1}, \tag{6.6}$$

with the same unit as concentration (mol/L). Thus,

$$[\mathrm{ES}] = \frac{[\mathrm{E}]_0[\mathrm{S}]}{K_M + [\mathrm{S}]}. \tag{6.7}$$

The rate of the reaction is the rate of formation of the product, $v = \mathrm{d}[\mathrm{P}]/\mathrm{d}t = k_2[\mathrm{ES}]$. Substituting Equation (6.7) and calling $v_{\max} \equiv k_2[\mathrm{E}]_0$ (the rate when all the enzyme is stuck in ES), we arrive at the Michaelis-Menten equation:

$$v = \frac{v_{\max}[\mathrm{S}]}{K_M + [\mathrm{S}]}. \tag{6.8}$$

k_2 is also called k_{cat}: the number of substrate molecules that each enzyme converts per second, when it is saturated.

Michaelis & Menten (1913) had reached this same formula by a more restricted path. They assumed that the first step, $\text{E} + \text{S} \rightleftharpoons \text{ES}$, always sits at equilibrium: it is much faster, in both directions, than the second step ($k_2 \ll k_{-1}$). In this pre-equilibrium, the ratio [E][S]/[ES] is the dissociation constant of the complex, $K_S = k_{-1}/k_1$ (the full calculation is in Appendix A.7), and Equation (6.8) comes out the same, with K_S in place of K_M. K_S is the special case of Equation (6.6) when $k_2 \ll k_{-1}$: there K_M tends to $k_{-1}/k_1 = K_S$.

The derivation above uses two approximations, besides Equation (6.2). First, almost none of the substrate is stuck in ES, because there is far more substrate than enzyme, so the free substrate is almost all of the substrate. Second, while the complex ES forms, almost no substrate is used up, so $[\text{S}] \approx [\text{S}]_0$. That is why Equation (6.8) holds, strictly speaking, for the **initial rate**, $v_0 = v_{\max}[\text{S}]_0/(K_M + [\text{S}]_0)$, with $[\text{S}]_0$ in place of [S]. Both approximations are good when

$$\varepsilon = \frac{[\text{E}]_0}{[\text{S}]_0 + K_M} \ll 1 \tag{6.9}$$

(SEGEL; SLEMROD, 1989). An example built just to see the order of magnitude: with $[\text{E}]_0 = 1 \times 10^{-9}$ mol/L, $[\text{S}]_0 = 1 \times 10^{-3}$ mol/L and $K_M = 1 \times 10^{-4}$ mol/L, Equation (6.9) gives $\varepsilon \approx 9 \times 10^{-7}$, much smaller than 1. If the enzyme and the substrate were at the same concentration, and K_M were small compared with $[\text{S}]_0$, ε would come close to 1, and both approximations would stop holding. Chapter 7 returns to this criterion, due to Segel & Slemrod (1989), and shows what happens when it fails. In Section 8.1, I check whether the condition $[\text{E}]_0 \ll [\text{S}]_0 + K_M$ holds in a sample of published assays.

Equation (6.8) has three simple regimes. When $[\text{S}] \ll K_M$,

$$v_0 \approx \frac{v_{\max}}{K_M}[\text{S}], \tag{6.10}$$

first order in [S]. When $[\text{S}] = K_M$,

$$v_0 = \frac{v_{\max}}{2}: \tag{6.11}$$

K_M is the substrate concentration at which the rate reaches half its maximum. Only when $k_2 \ll k_{-1}$, and K_M approaches K_S, does this constant also measure the enzyme's affinity for the substrate: a smaller K_M, in that case, indicates that the enzyme holds on to the substrate more tightly. When $[\text{S}] \gg K_M$,

$$v_0 \approx v_{\max}: \tag{6.12}$$

the enzyme is saturated, and adding more substrate no longer speeds up the reaction.

Figure 39 shows the curve v_0 against [S], with the two reference points marked: the asymptote $v_{\max}$ and the point $[\text{S}] = K_M$.

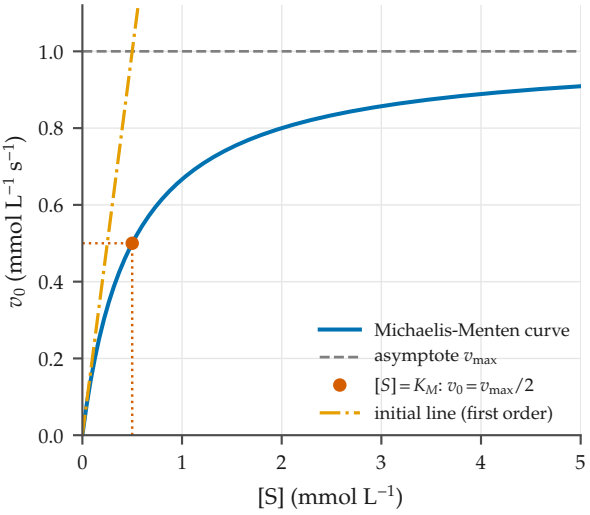


Figure 39. Initial rate v_0 against substrate concentration [S] from the Michaelis-Menten equation, with $v_{\max} = 1$ mmol/(L s) and $K_M = 0.5$ mmol/L (illustrative values, the same as in the first version of this work). The curve tends to the horizontal line $v_0 = v_{\max}$ (dashed). At $[\text{S}] = K_M$, $v_0 = v_{\max}/2$ (point marked). For $[\text{S}] \ll K_M$, $v_0 \approx (v_{\max}/K_M)[\text{S}]$ (dash-dot line).

Source: simulation (`codigo/figuras_outras.py`), illustrative values from the first version of this work.

Analogy: a Brazilian churrascaria

Picture a table at an all-you-can-eat barbecue, with several people seated: each person is an enzyme. The waiters bring meat and leave pieces on the table: that is the substrate, and the more meat, the larger [S]. *Each person chews one piece at a time; with a busy mouth, that person is the complex* ES. *Someone with a free mouth takes a piece when the waiter passes by (k_1). Someone who is chewing can put the piece back on the*

plate (k_{-1}) or finish chewing and swallow (k_2). With little meat on the table ($[S] \ll K_M$), almost everyone has a free mouth waiting for the waiter, and the rate of eating grows along with the meat served. With plenty of meat ($[S] \gg K_M$), everyone is chewing all the time; throwing more meat on the table does not make anyone chew faster. The rate reaches its maximum, $v_{max} = k_2[E]_0$, which only depends on how many people are at the table and how fast each one chews. What limits the rate is the time spent chewing (the busy mouth); how hungry the diners are does not come into it. K_M is, then, the amount of meat on the table at which half the people have a full mouth.

It is the same lesson as the chairs in a room (Section 2.2). In a full room, the rate of filling seats depends only on when some chair frees up, however many people are left standing. At the churrascaria, the waiter can bring all the meat he likes: v_{max} only changes with the number of mouths at the table and with the time each one takes to chew.

6.2 The production of sulfur trioxide

Sulfuric acid is manufactured, on an industrial scale, by the contact process: sulfur dioxide (SO_2) is oxidized to sulfur trioxide (SO_3) over a solid vanadium pentoxide (V_2O_5) catalyst, and the SO_3 then reacts with water to form the acid. It is another example of heterogeneous catalysis, in which the reaction happens on the surface of a solid, outside the gas phase. Here too there is an intermediate that the SSH resolves. What is new is that this intermediate is a state of the surface itself, instead of a molecule.

The catalyst surface has sites that can be occupied by an oxygen atom (oxidized site, O∗) or empty (reduced site, ∗). The redox cycle of Mars & van Krevelen (1954) has two steps:

$$SO_2 + O* \longrightarrow SO_3 + * \qquad (k_{red}), \tag{6.13}$$

$$O_2 + 2* \longrightarrow 2\,O* \qquad (k_{ox}). \tag{6.14}$$

In the first, SO_2 takes the oxygen from an oxidized site and leaves as SO_3, leaving the site empty. In the second, O_2 refills two empty sites at once. Adding step (6.13) twice with step (6.14) once, $2SO_2 + O_2 \rightarrow 2SO_3$: the catalyst appears on both sides and cancels out, that is, it comes out regenerated at each cycle.

Let θ_{ox} be the fraction of oxidized sites (coverage, between 0 and 1) and $1 - \theta_{ox}$ the fraction of empty sites. The rates of the two steps are

$$v_{red} = k_{red} P_{SO_2}\, \theta_{ox}, \tag{6.15}$$

$$v_{ox} = k_{ox} P_{O_2}\, (1 - \theta_{ox}), \tag{6.16}$$

where P_{SO_2} and P_{O_2} are the partial pressures of the two gases (in Pa; in practice, in atm) and k_{red}, k_{ox} are the rate constants of each step, in $1/(\text{s atm})$. First order in free sites is an assumption of the model, and nothing forces it to hold in general.

Here the SSH is applied to the coverage of the surface, in place of a concentration: the oxygen that SO_2 takes is replaced by O_2 at the same rate. Each event of step (6.13) consumes one oxidized site; each event of step (6.14) creates two. The steady-state condition, $d\theta_{ox}/dt \approx 0$, equates the sites consumed to the sites created per unit time:

$$k_{red} P_{SO_2} \theta_{ox} = 2\, k_{ox} P_{O_2} (1 - \theta_{ox}). \tag{6.17}$$

Isolating θ_{ox},

$$\theta_{ox} = \frac{2 k_{ox} P_{O_2}}{k_{red} P_{SO_2} + 2 k_{ox} P_{O_2}}. \tag{6.18}$$

The rate of formation of SO_3 is the rate of step (6.13), $v = v_{red} = k_{red} P_{SO_2} \theta_{ox}$. Substituting Equation (6.18),

$$v = \frac{2\, k_{red} k_{ox}\, P_{SO_2} P_{O_2}}{k_{red} P_{SO_2} + 2 k_{ox} P_{O_2}}. \tag{6.19}$$

There are two simple limits. If reoxidation is much faster than reduction ($k_{ox} P_{O_2} \gg k_{red} P_{SO_2}$), the surface stays almost entirely oxidized ($\theta_{ox} \approx 1$), and

$$v \approx k_{red} P_{SO_2}: \tag{6.20}$$

first order in SO_2 and zero order in O_2. It is a possible limit of the mechanism, and it does not contradict the stoichiometry of the overall reaction, which consumes both gases. At the other extreme, if reduction is much faster, the surface stays almost entirely reduced ($\theta_{ox} \approx 0$), and

$$v \approx 2\, k_{ox} P_{O_2}. \tag{6.21}$$

Mars & van Krevelen (1954) proposed this two-step scheme (the reactant reduces the catalyst, oxygen reoxidizes it) while studying the oxidation of aromatic hydrocarbons over vanadium oxide, and showed that the same formula also describes literature data on the oxidation of SO_2 over these catalysts. Vanadium pentoxide really is the catalyst of the contact process, still used today. But the real industrial catalyst is more complicated than this two-state model. It is a molten salt, with potassium promoters, and its surface chemistry has more steps than the two used here. The calculation with the site balance written out in full is in Appendix A.9.

6.3 Hydrogen and bromine: the rate law without a simple order

Section 2.2 presented, for the reaction

$$\mathrm{H_2(g) + Br_2(g) \longrightarrow 2\,HBr(g)}, \tag{6.22}$$

the measured rate law, without saying where it comes from. There it is written for $\mathrm{d[HBr]/d}t$. The rate of the reaction is

$$v = \frac{1}{2}\frac{\mathrm{d[HBr]}}{\mathrm{d}t}; \tag{6.23}$$

the same law, written for v, is

$$v = \frac{k[\mathrm{H_2}][\mathrm{Br_2}]^{3/2}}{[\mathrm{Br_2}] + k'[\mathrm{HBr}]}, \tag{6.24}$$

with k equal to half the constant that multiplies $\mathrm{d[HBr]/d}t$ in Section 2.2 (and k', the same constant as there). No order holds for the reaction as a whole, and the product (HBr) appears in the denominator. Bodenstein & Lind (1907) measured this law, and Herzfeld (1919) proposed the chain mechanism that explains it.

Example 6.1 (The chain mechanism for H_2 + Br_2)**.** The mechanism has five steps. The first starts the chain, breaking Br_2 into two atoms (k_1). Two propagate it, alternating between the H and Br atoms (k_2 and k_3). One slows it down, because it consumes the product (k_4), and the last one ends it (k_5):

$$\mathrm{Br_2 \longrightarrow 2\,Br} \qquad (k_1), \tag{6.25}$$
$$\mathrm{Br + H_2 \longrightarrow HBr + H} \qquad (k_2), \tag{6.26}$$
$$\mathrm{H + Br_2 \longrightarrow HBr + Br} \qquad (k_3), \tag{6.27}$$
$$\mathrm{H + HBr \longrightarrow H_2 + Br} \qquad (k_4), \tag{6.28}$$
$$\mathrm{2\,Br \longrightarrow Br_2} \qquad (k_5). \tag{6.29}$$

The H and Br atoms are the intermediates. I apply the SSH (Section 3.2) to the concentrations of both: $\mathrm{d[H]/d}t \approx 0$ and $\mathrm{d[Br]/d}t \approx 0$. Adding these two conditions, the three propagation terms (k_2, k_3 and k_4) appear with opposite signs and cancel. Only initiation and termination are left: $2k_1[\mathrm{Br_2}] = 2k_5[\mathrm{Br}]^2$, that is, $[\mathrm{Br}]_{\mathrm{ss}} = \sqrt{(k_1/k_5)[\mathrm{Br_2}]}$. From the SSH for H comes $[\mathrm{H}]_{\mathrm{ss}}$ as a function of $[\mathrm{Br}]_{\mathrm{ss}}$, and the rate of the reaction (Equation (6.23)) ends up as a function of only $[\mathrm{H_2}]$, $[\mathrm{Br_2}]$ and [HBr]. The full calculation, step by step, is in Appendix A.10. The result is Equation (6.24) again, with

$$k = k_2\sqrt{\frac{k_1}{k_5}}, \qquad k' = \frac{k_4}{k_3}. \tag{6.30}$$

It is the same reasoning as in Chapters 4 and 5. A short-lived intermediate, or two, like H and Br here, makes it possible to trade a system of differential equations for a piece of algebra.

CHAPTER

7 When the steady-state hypothesis fails

The steady-state hypothesis (SSH) assumes $\mathrm{d[I]}/\mathrm{d}t \approx 0$ for an intermediate I (Section 3.3.3). It is not always a good approximation, and the first version of this work, from July 2025, had already pointed out three ways it can fail (RIBEIRO, 2025a); I add a fourth (Section 7.4). In each one I compare the SSH with the full solution of the same mechanism, by simulation (`codigo/validade_ssh.py`).

7.1 The time-scale criterion

I have already shown, in the mechanism A $\rightarrow$ I $\rightarrow$ B (Chapter 3), that the SSH holds when $k_2 \gg k_1$: the intermediate lasts a short time compared with the time over which A changes. The general form of this criterion (Equation (3.2)) uses the intermediate's **lifetime**, τ_I (s), which is the average time a molecule of I lasts before being consumed. If every step that consumes I is first order in I, with constant k'_j (1/s),

$$\tau_\mathrm{I} = \frac{1}{\sum_j k'_j}. \tag{7.1}$$

When one of these steps also depends on another reactant, that reactant's concentration goes inside k'_j. Calling τ_reaction (s) the time over which the reactant concentrations change appreciably, the criterion is $\tau_\mathrm{I} \ll \tau_\mathrm{reaction}$. The smaller this ratio, the smaller the error of the SSH: in the mechanism A $\rightarrow$ I $\rightarrow$ B it measures the relative error in [I] itself (Section 3.3.3). For enzymes, Segel & Slemrod (1989) refine this criterion with their own small parameter, ε (Section 7.4).

The criterion is enough for the SSH to hold, but it is not required. There are mechanisms where it fails and the SSH remains exact. That is the case of the O atom in the photolysis of O_3 under constant light (Section 4.3). O_3 breaks apart under the light, with photolysis frequency J (1/s), and O is consumed by $\mathrm{O} + \mathrm{O_3} \longrightarrow 2\,\mathrm{O_2}$, with constant k_4 (numbering from Chapter 4). The change in [O] is $[\mathrm{O_3}](J - k_4[\mathrm{O}])$, which vanishes at $[\mathrm{O}] = J/k_4$ for any $[\mathrm{O_3}]$. The lifetime $\tau_\mathrm{O} = 1/(k_4[\mathrm{O_3}])$ grows as O_3 runs out, that is, the criterion, taken alone, gets worse over time. Even so, after a short transient at the start, the SSH for O remains exact to the end. Here [O] stays constant because of the mechanism itself, without the hypothesis having to assume it (Section 3.2).

In ammonia synthesis, the intermediates are the species stuck to the catalyst surface (Section 5.4), and the criterion asks that each one last a short time compared with the time over which the gas composition changes. During a reactor startup, that composition takes between 11.4 s and 11.5 s to reach 95% of its final value, depending on the effectiveness factor of the pellet, η, from 0.3 to 0.5 (Section 5.6; numbers in `dados/simulacao/amonia_reator.json`). The surface model (Section 5.4), with illustrative parameters, does not give in seconds how long each stuck species lasts, and the comparison remains open. Reaching that final value means reaching the reactor's steady state, which is different from the SSH: IUPAC calls the steady state, in a continuous-flow reactor, the regime in which every concentration becomes independent of time (IUPAC, 2025b), while the SSH is about the adsorbed intermediate.

7.2 Failure 1: the start of the reaction

The first failure on the July 2025 list is correct: in the early stages, [I] changes fast (RIBEIRO, 2025a). In the mechanism A $\rightarrow$ I $\rightarrow$ P (the same one from Chapter 3, here with the product called P), at $t = 0$ there is still no I. The rate of forma-

tion is $k_1[\mathrm{A}]_0$, the highest it ever reaches, and the rate of consumption is zero, which is far from $\mathrm{d}[\mathrm{I}]/\mathrm{d}t \approx 0$. The SSH, by contrast, starts right away from $[\mathrm{I}]_{\mathrm{SSH}} = (k_1/k_2)[\mathrm{A}]_0$ and gives the product an initial rate of $k_1[\mathrm{A}]_0$. In the exact solution (Equation (3.17)), the product starts with zero rate, because it must wait for I to build up.

This stretch is the **induction period**. At the maximum of [I], $\mathrm{d}[\mathrm{I}]/\mathrm{d}t = 0$: $k_1 \mathrm{e}^{-k_1 t} = k_2 \mathrm{e}^{-k_2 t}$. Taking the logarithm, the instant at which it ends is

$$t_{\mathrm{ind}} = \frac{\ln(k_2/k_1)}{k_2 - k_1}. \tag{7.2}$$

Figure 40 shows both curves for $k_2/k_1 = 3$, 10 and 100. In the [P] plot (panel a), the SSH curve, $[\mathrm{P}] = [\mathrm{A}]_0(1 - \mathrm{e}^{-k_1 t})$, does not depend on k_2. The exact curve starts flat and trails behind it, with a delay close to $1/k_2$. In the [I] plot (panel b), the exact curve rises from zero, crosses the SSH curve right at its highest point and, afterward, stays a little above it. The induction period lasts only a few $1/k_2$: $k_2 t_{\mathrm{ind}}$ is 1.65 in the first case, 2.56 in the second and 4.65 in the third (`dados/simulacao/validade_resumo.json`). Measured in units of $1/k_1$, it shrinks as k_2/k_1 grows. At $t = 10/k_2$, the exact [P] is still 2%, 6% and 10% below the SSH value in the three cases, in relative terms (as a fraction of $[\mathrm{A}]_0$, 0.018, 0.041 and 0.009).

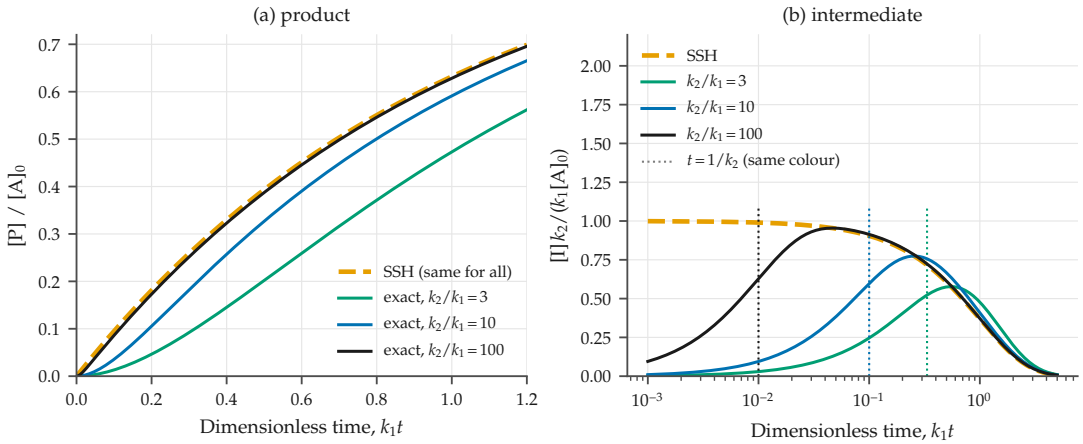


Figure 40. Induction period in the mechanism A → I → P, for $k_2/k_1 = 3$, 10 and 100. (a) [P] as a fraction of $[\mathrm{A}]_0$: exact solutions (solid lines) and the SSH solution (dashed), which does not depend on k_2. (b) [I] multiplied by $k_2/(k_1[\mathrm{A}]_0)$, so that the SSH curve is the same in the three cases; the vertical dotted lines mark $t = 1/k_2$. Time in units of $1/k_1$.

Source: simulation (`codigo/validade_ssh.py`).

The July 2025 version had a figure in which the rate constant k itself rose with time up to a maximum value (RIBEIRO, 2025a). At a fixed temperature, k does not change with time (Section 2.5). The correct idea behind that figure is that, with an intermediate, the *rate* of product formation can start at zero and only rise later. That is the induction period.

The ammonia data from the December 2025 article (RIBEIRO, 2025b) also rise with a delay at the start, in a shape similar to the exact [P] curve: they start more slowly than a simple exponential and then climb to nearly 1. The mechanism A → I → P itself is one of the models I fitted to them, and it describes that delay well (M3, Section 5.5). But the Weibull curve (M4) comes out ahead of it by the Akaike criterion (AICc).

7.3 Failure 2: an intermediate that is too slow

The second failure on the list is also correct: with long-lived intermediates, the SSH may not hold (RIBEIRO, 2025a). Under the SSH, every A that has reacted turns into the product B right away. In the exact solution, part of it stays held up as I. Since $[\mathrm{A}] + [\mathrm{I}] + [\mathrm{B}] = [\mathrm{A}]_0$ in both cases, and the SSH uses $[\mathrm{I}]_{\mathrm{SSH}} = 0$, the difference is exactly what is still retained in the intermediate:

$$[\mathrm{B}]_{\mathrm{SSH}} - [\mathrm{B}] = [\mathrm{I}]. \tag{7.3}$$

If the step that consumes I is not much faster than the one that forms it, I builds up, and the difference in Equation (7.3) grows. Table 6 gives the numbers: the largest error in [B], as a fraction of $[A]_0$, rises from 7.7 % with $k_2 = 10k_1$ to 37 % with $k_2 = k_1$, and Figure 41 shows how it evolves over time. When $k_2 \leq k_1$ there is not even a fixed error after the induction period, because the ratio between the exact [I] and the SSH [I] keeps growing without limit.

The biscuit factory (Section 2.7.2) is this case taken to the extreme. The dough machine makes 1000 kg/h and the oven only bakes 300 kg/h: 700 kg of raw dough pile up every hour, without end. The raw dough is the intermediate. In the reversed factory (Section 3.3.5), with the oven faster than the machine, the raw dough does not pile up, and that is the situation the SSH describes.

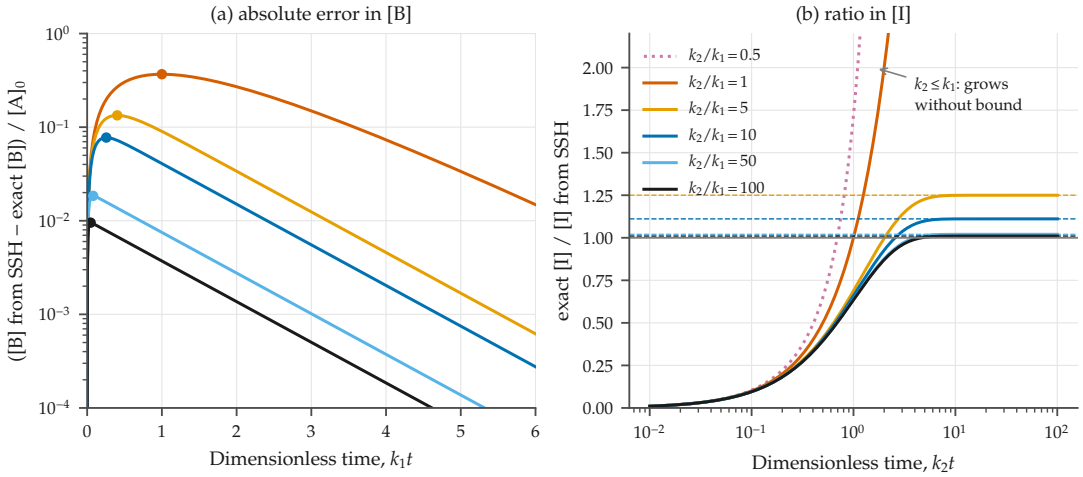


Figure 41. SSH error in the mechanism A ⟶ I ⟶ B. (a) Difference between the SSH [B] and the exact [B], as a fraction of $[A]_0$, against time in units of $1/k_1$; the point marks the maximum. (b) Ratio between the exact [I] and the SSH [I] against time in units of $1/k_2$; the thin dashed lines mark the final values $k_2/(k_2 - k_1)$. The colors are the same in both panels. With $k_2 \leq k_1$ the ratio keeps growing without limit.

Source: simulation (`codigo/validade_ssh.py`).

7.4 Failure 3: too much intermediate

This failure was not on the July 2025 list. It is not enough for the intermediate to form quickly: it also needs to stay in a small amount. That is what Segel & Slemrod (1989) show for the Michaelis-Menten mechanism (Section 6.1), with the parameter $\varepsilon = [E]_0/([S]_0 + K_M)$ (Equation (6.9)). The simulation shows this with illustrative values (`dados/simulacao/validade_segel.csv`). With $[E]_0 = 10^{-3}[S]_0$ and $K_M = 0.1[S]_0$, $\varepsilon = 9.1 \times 10^{-4}$ and the largest difference in [P] is 2.8×10^{-4} of $[S]_0$, a small value, as expected. Now imagine $[E]_0 = [S]_0$ and a small K_M, $0.01[S]_0$. The complex forms in a time $t_C = 1/[k_1([S]_0 + K_M)]$, only 0.005 of the time $t_S = ([S]_0 + K_M)/(k_2[E]_0)$ in which the substrate changes (SEGEL; SLEMROD, 1989), which on its own would look like a good sign. But $\varepsilon = 0.99$, close to 1 and far from small, and the difference reaches 0.36 of $[S]_0$. Almost all the substrate turns into intermediate right at the start. The SSH counts that substrate as if it were still free, and that is why it is badly wrong. In Section 8.1 I count how many published enzyme assays, in a sample from SABIO-RK (WITTIG *et al.*, 2018), fall outside this condition.

In surface catalysis (Section 5.4) the same condition holds. The criterion asks that each adsorbed species last a short time compared with the time over which the gas changes, *and* that the amount of matter stuck to the surface be small compared with what is in the gas. This second condition is the analogue of $\varepsilon \ll 1$. In the room with chairs (Section 2.2.3), the seated people (the stuck species) can occupy every chair (the sites) without breaking the SSH for the coverages. What matters is that they be few compared with

the line waiting outside (the gas).

7.5 Failure 4: fast reversible steps

The third item on the July list is also correct: with fast reversible reactions, it may be necessary to use the pre-equilibrium instead of the SSH (RIBEIRO, 2025a). Consider the mechanism

$$\mathrm{A} \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} \mathrm{I} \xrightarrow{k_2} \mathrm{P}, \tag{7.4}$$

with $K = k_1/k_{-1}$, the equilibrium constant of A $\rightleftharpoons$ I. Under the SSH,

$$v = \frac{k_1 k_2}{k_{-1} + k_2}[\mathrm{A}], \tag{7.5}$$

valid when $k_1 \ll k_{-1} + k_2$, that is, when [I] stays small compared with [A]. In this chapter the SSH uses $[\mathrm{A}] \approx [\mathrm{A}]_0 - [\mathrm{P}]$, neglecting what is held as I. Under the pre-equilibrium, if the pair A $\rightleftharpoons$ I reaches equilibrium well before I turns into P, $[\mathrm{I}] \approx K[\mathrm{A}]$ and, with the balance $[\mathrm{A}] + [\mathrm{I}] = [\mathrm{A}]_0 - [\mathrm{P}]$,

$$v = \frac{k_2 K}{1 + K}\left([\mathrm{A}]_0 - [\mathrm{P}]\right). \tag{7.6}$$

The pair A $\rightleftharpoons$ I takes a time of order $1/(k_1 + k_{-1})$ to equilibrate, and I takes a time of order $1/k_2$ to turn into P. That is why the pre-equilibrium condition is $k_2 \ll k_1 + k_{-1}$, and not just $k_2 \ll k_{-1}$. Rae & Berberan-Santos (2004) point out, building on the analysis of Pyun (1971), that this is the right condition and that requiring $k_{-1} \gg k_2$ is too restrictive. If $k_1 \gg k_2$, the pre-equilibrium holds even with a small k_{-1}: almost all of A turns into I right at the start, and then I turns into P slowly, with $v \approx k_2([\mathrm{A}] + [\mathrm{I}])$.

The two laws nearly agree when k_1 and k_2 are both much smaller than k_{-1}. Then both denominators are close to k_{-1}, [I] stays small, and both laws become $v \approx (k_1 k_2/k_{-1})[\mathrm{A}]$. Table 15 shows three regimes with $k_{-1} = 1/\mathrm{s}$ and $k_2 = 0.1/\mathrm{s}$. I chose the comparison times to see the three at a similar point of the reaction, when about 60% of A has turned into P. With $K = 0.01$ this takes longer, because the reaction is slower. In that regime, I is less than 1 % of the unreacted A and the SSH stays closer to the exact solution. With $K = 10$ and $K = 100$, on the other hand, most of the unreacted A sits as I. The SSH is badly wrong, with $[\mathrm{P}]_{\mathrm{SSH}}$ reaching 1.000 against 0.628 for the exact solution, and the pre-equilibrium gets it right.

Table 15. Mechanism A $\rightleftharpoons$ I $\rightarrow$ P with $k_{-1} = 1/\mathrm{s}$ and $k_2 = 0.1/\mathrm{s}$: [P] as a fraction of $[\mathrm{A}]_0$ at instant t, from the exact solution, the SSH (Equation (7.5)) and the pre-equilibrium (Equation (7.6)). The last column is the fraction of the unreacted A that sits as I, in the exact solution.

$K = k_1/k_{-1}$	t (s)	exact	SSH	pre-equilibrium	[I]/([A] + [I])
0.01	1000	0.5938	0.5971	0.6285	0.009
10	10	0.5935	0.9999	0.5971	0.908
100	10	0.6281	1.0000	0.6285	0.990

Source: simulation (`codigo/validade_ssh.py`); numbers in `dados/simulacao/validade_preeq_K.csv`.

7.5.1 Swapping k_1 and k_2

The mechanism in Equation (7.4) has an exact solution (PEREZ-BENITO, 2017). With $[\mathrm{I}]_0 = [\mathrm{P}]_0 = 0$, it gives for the product

$$[\mathrm{P}] = [\mathrm{A}]_0\left(1 - \frac{\lambda_+ \mathrm{e}^{-\lambda_- t} - \lambda_- \mathrm{e}^{-\lambda_+ t}}{\lambda_+ - \lambda_-}\right), \tag{7.7}$$

where λ_+ and λ_- (1/s) are the two roots of

$$\lambda^2 - (k_1 + k_{-1} + k_2)\,\lambda + k_1 k_2 = 0. \tag{7.8}$$

The smaller root, λ_-, governs the slow part of the reaction, and the larger one, λ_+, the fast start. Equation (7.8) contains only two combinations of the constants: the sum $k_1 + k_{-1} + k_2$ and the product $k_1 k_2$. Swapping k_1 and k_2 changes neither. So [P] is the same at every instant, whether step 1 or step 2 is the slow one. I checked this on 729 sets of constants, from 10^{-4} to 10^4 1/s, and the largest difference in [P] was 3×10^{-16} of $[\mathrm{A}]_0$ (`dados/simulacao/validade_simetria.json`). In the biscuit factory (Section 2.7.2), it is like swapping the speed of the dough machine with that of the oven: the number of finished bis-

cuits per hour does not change, only the amount of raw dough waiting in between. Someone who watches only the output cannot tell which of the two is the slow one. In data analysis this is called the slow-fast ambiguity (SCHRÖDER *et al.*, 2016).

The same swap takes the SSH constant, $k_1k_2/(k_{-1}+k_2)$, into the pre-equilibrium one, $k_1k_2/(k_1+k_{-1})$. So the pre-equilibrium error at a point of the $(k_2/k_{-1},\ K)$ plane is exactly the SSH error at the point mirrored across the diagonal $k_1 = k_2$. And the criterion of one is the mirror of the other's: $k_1 \ll k_{-1}+k_2$ for the SSH, $k_2 \ll k_1+k_{-1}$ for the pre-equilibrium. In the SSH, I must disappear (back to A or on to P) much faster than it forms. In the pre-equilibrium, A $\rightleftharpoons$ I must equilibrate well before I turns into P.

7.5.2 The validity map

Figure 42 extends the comparison to the $(k_2/k_{-1},\ K)$ plane, from 10^{-3} to 10^{3} on both axes. Panel (b) is (a) mirrored across the diagonal: over the 3721 grid points, the difference between the two was zero. Below the dashed line, where $k_1/(k_{-1}+k_2) < 0.1$, the largest SSH error in [P] stays below 8 % of $[A]_0$ throughout the reaction. On the line it goes from 3.5 %, when $k_2 \ll k_{-1}$, to 7.7 %, when $k_2 \gg k_{-1}$. It is the value for the irreversible mechanism with $k_2 = 10k_1$ (Table 6). To the left of the dotted line, where $k_2/(k_1+k_{-1}) < 0.1$, the pre-equilibrium error is, by the mirror, the same 7.7 % at most. In the bottom-left corner both hold.

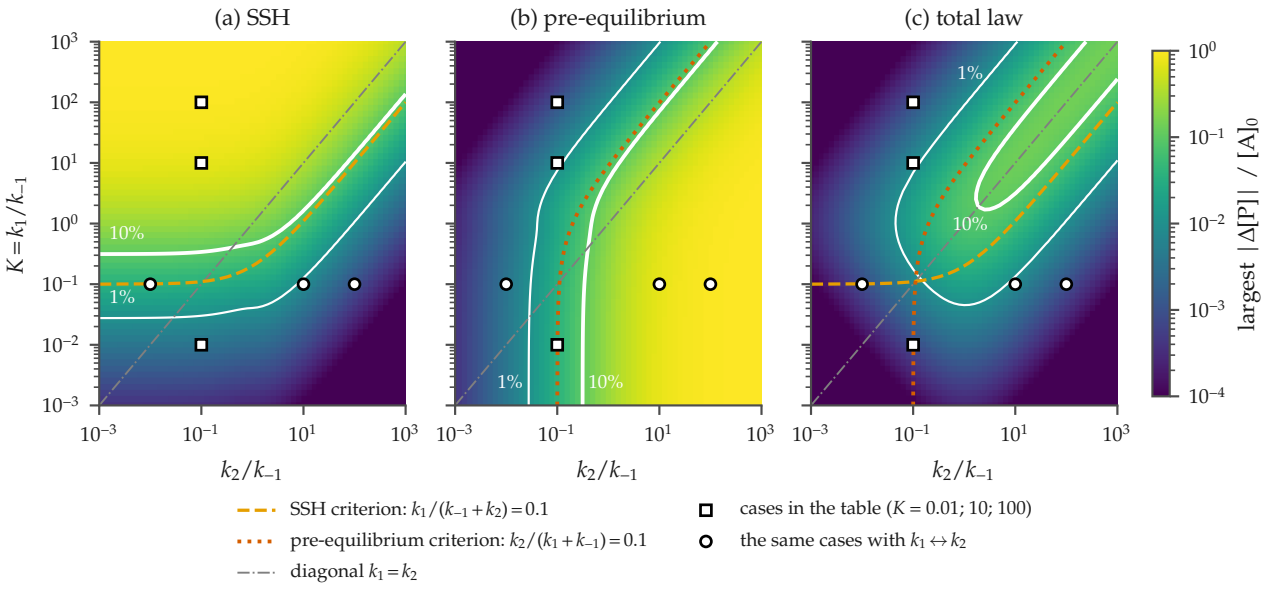


Figure 42. Validity map of the SSH (a), the pre-equilibrium (b) and the total law (c, Equation (7.9)) in the mechanism A $\rightleftharpoons$ I $\rightarrow$ P. Color is the largest difference between the approximation's [P] and the exact [P], as a fraction of $[A]_0$, throughout the reaction; the white curves mark 1 % and 10 %. Dashed: $k_1/(k_{-1}+k_2) = 0.1$ (SSH criterion). Dotted: $k_2/(k_1+k_{-1}) = 0.1$ (pre-equilibrium criterion). Dash-dot: the diagonal $k_1 = k_2$; panel (b) is panel (a) mirrored across it. Squares: the three cases from Table 15; circles: the same cases with k_1 and k_2 swapped.

Source: simulation (`codigo/validade_ssh.py`); numbers in `dados/simulacao/validade_mapa.csv` and `validade_mapa_lei_total.csv`.

Neither works in a band around the diagonal $k_1 = k_2$, which starts at $k_1 = k_2 \approx 0.4\,k_{-1}$. There neither step is much faster than the other, and the approximation has nothing to rest on. With $k_1 = k_2 = k_{-1}$, the SSH and the pre-equilibrium are off by 18.5 % of $[A]_0$; with $k_1 = k_2 = 10\,k_{-1}$, by 34 %. On the diagonal, when k_{-1} is much smaller than k_1 and k_2, the error reaches $1/e \approx 37$ %, the same as for the irreversible mechanism with $k_1 = k_2$ (Table 6). In that limit, both are off by more than 10 % as long as k_1/k_2 lies between 0.14 and 7.3. Outside the band, even with a

large K, one of them works. With $K = 1000$ and $k_2 = 10\,k_{-1}$, almost all of A turns into I, but the pre-equilibrium with the mass balance is off by only 0.95 % (grid sweep in Appendix C.1).

7.5.3 One law that covers both cases

Both laws are limiting cases of a third one: the SSH applied without neglecting I in the balance. With $\mathrm{d[I]}/\mathrm{d}t = 0$ and $[\mathrm{A}] = ([\mathrm{A}]_0 - [\mathrm{P}]) - [\mathrm{I}]$, one gets $[\mathrm{I}] = k_1([\mathrm{A}]_0 - [\mathrm{P}])/(k_1 + k_{-1} + k_2)$ and

$$v = \frac{k_1 k_2}{k_1 + k_{-1} + k_2}\,([\mathrm{A}]_0 - [\mathrm{P}])\,. \qquad (7.9)$$

I call Equation (7.9) the total law, because it uses the total that has not yet become product, [A] + [I]. For enzymes, the same change of variable is the basis of the total quasi-steady-state approximation (BORGHANS; BOER; SEGEL, 1996). If $k_1 \ll k_{-1} + k_2$, the k_1 drops out, [I] stays small and Equation (7.5) comes back. If $k_2 \ll k_1 + k_{-1}$, the k_2 drops out, and Equation (7.6) comes back. It is the same idea as in Section 3.3.4, where the SSH contained the rate-determining step and the pre-equilibrium. Here, the total law contains the SSH and the pre-equilibrium with the mass balance. In other words, what fails in Table 15 with large K is not $\mathrm{d[I]}/\mathrm{d}t \approx 0$, but neglecting I in the balance. Figure 43a compares the three laws on a cut through the map: in the gray band, where the other two are off by more than 10 %, the total law stays below 14 %.

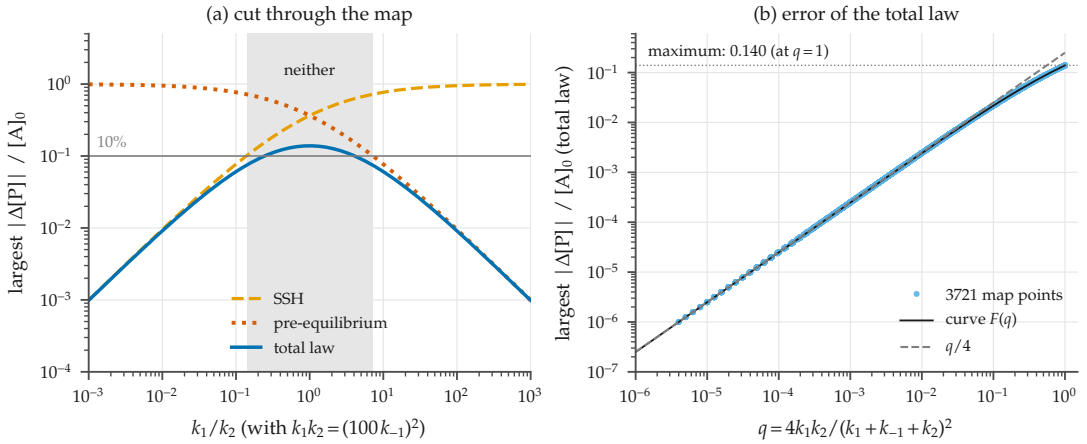


Figure 43. (a) Largest error in [P], as a fraction of $[\mathrm{A}]_0$, of the SSH, the pre-equilibrium and the total law (Equation (7.9)) along a cut through the map of Figure 42, with $k_1 k_2 = (100\,k_{-1})^2$, against k_1/k_2. In the gray band, the SSH and the pre-equilibrium are both off by more than 10 %. (b) Largest error of the total law against q (Equation (7.10)): the 3721 map points fall on a single curve, close to $q/4$ for small q and with a maximum of 0.140 at $q = 1$.

Source: simulation (`codigo/validade_simetria.py`); numbers in `dados/simulacao/validade_lei_total_curva.csv` and `validade_simetria.json`.

Panel (c) of Figure 42 shows the error of the total law. At the 3721 grid points it was never worse than the better of the other two, and it only exceeds 10 % in a narrow band around the diagonal, with k_1 and k_2 both above $1.7\,k_{-1}$ (on the diagonal, above $1.9\,k_{-1}$). When k_{-1} is much smaller than both, that band goes from $k_1/k_2 = 0.24$ to 4.1. The total law's error depends on a single combination of the constants,

$$q = \frac{4k_1 k_2}{(k_1 + k_{-1} + k_2)^2}, \qquad (7.10)$$

which goes from 0 to 1 and measures how close the two time scales of the exact solution, $1/\lambda_-$ and $1/\lambda_+$, are: from Equation (7.8), $q = 4\lambda_+\lambda_-/(\lambda_+ + \lambda_-)^2$. With $q \ll 1$, they are well separated, and the slow root is close to $k_1 k_2/(k_1 + k_{-1} + k_2)$, the constant in Equation (7.9). For small q, the largest error of the total law is about $q/4$ (Figure 43b; the derivation is in Appendix A.14). It passes 1 % at $q = 0.043$ and 10 % only at $q = 0.63$; the largest possible value is 0.140 of $[\mathrm{A}]_0$, with $k_{-1} \to 0$ and $k_1 = k_2$. Table 16 compares the three laws at points of the map.

Table 16. Largest error in [P], as a fraction of $[A]_0$, of the three laws at points of the map in Figure 42, all with $k_{-1} = 1/s$. The columns $k_1/(k_{-1} + k_2)$ and $k_2/(k_1 + k_{-1})$ are the SSH and pre-equilibrium criteria; q is that of Equation (7.10).

k_1 (1/s)	k_2 (1/s)	$k_1/(k_{-1} + k_2)$	$k_2/(k_1 + k_{-1})$	q	SSH	pre-equilibrium	total law
100	0.1	90.9	0.00099	0.0039	0.942	0.00097	0.00097
0.1	100	0.00099	90.9	0.0039	0.00097	0.942	0.00097
1000	10	90.9	0.010	0.039	0.950	0.0095	0.0091
0.1	0.1	0.091	0.091	0.028	0.032	0.032	0.0066
1	1	0.50	0.50	0.44	0.185	0.185	0.077
10	10	0.91	0.91	0.91	0.339	0.339	0.130
1000	1000	0.999	0.999	0.999	0.368	0.368	0.139

Source: simulation (`codigo/validade_simetria.py`); numbers in `dados/simulacao/validade_pontos_referencia.csv`.

On the diagonal with $k_1 = k_2 = 10\,k_{-1}$, where the SSH and the pre-equilibrium are off by 34 %, the total law is off by 13 %. Like the other two, it does not describe the induction period: it starts with rate $k_1 k_2 [A]_0/(k_1 + k_{-1} + k_2)$, not zero. With small q, its largest error shows up right after that period, while the exact [P] still lags behind.

There is a practical consequence. The curve of [P] against time only reveals the sum $k_1 + k_{-1} + k_2$ and the product $k_1 k_2$. The mechanisms with (k_1, k_{-1}, k_2) equal to (100; 1; 0.1), (0.1; 1; 100), (49.8; 51.1; 0.201) and (3.16; 94.8; 3.16), in 1/s, give the same [P], with a difference below 3×10^{-15}. The SSH is off by 94 %, 0.1 %, 24.5 % and 1.2 % in these four cases. The total law is off by 0.1 % in all of them, because it only uses the sum and the product. So following only the product does not tell which step is the slow one. To find out, one must also follow A or I (Section 8.2).

Much of this is already known. The comparison between the SSH and the pre-equilibrium in this mechanism has appeared for decades in chemical education journals (PYUN, 1971; GELLENE, 1995; PEREZ-BENITO, 2017), and the right pre-equilibrium criterion is in Rae & Berberan-Santos (2004). The swap of k_1 and k_2 that leaves the product curve unchanged is the slow-fast ambiguity (SCHRÖDER *et al.*, 2016). The slow root close to $k_1 k_2/(k_1 + k_{-1} + k_2)$ and the condition $q \ll 1$ are in Eilertsen, Schnell & Walcher (2024), for an enzyme model with the same equations. In this chapter I put these pieces together with simple calculations: the map of the error in [P] over the (k_2/k_{-1}, K) plane, the mirror between the SSH map and the pre-equilibrium map, the band where neither works, and the error of the total law written as a function of q alone.

7.6 Branched chains

In a chain reaction, the intermediates are radicals that regenerate with every turn. In a **branched** chain, one radical can generate more than one new radical. This is how Semenov explained the sharp jump from a nearly nil reaction to ignition in phosphorus vapor. In Oxford, Hinshelwood and his coworkers saw the same thing at the lower explosion limit of the hydrogen-oxygen mixture (HINSHELWOOD, 1964, p. 476). If the destruction of the radicals is faster than the branching, they can sit at steady state, and the reaction is slow (SEMENOV, 1964, p. 491). If branching is faster than destruction, the number of radicals grows without limit, and that is the explosion. There is no steady state for them, and the SSH has nothing to describe.

7.7 What to do when the SSH fails

The most direct way out is to solve the full system of rate equations on a computer, with no approximation. The two two-step mechanisms in this chapter have an exact formula, and I checked each one this way (the difference stayed below 10^{-12} of $[A]_0$); for Michaelis-Menten, the exact solution is the numerical one. There is a practical caution: with very different rate constants, the system is **stiff**. A simple integration

method would need steps the size of the fastest intermediate's lifetime, even after it reaches steady state, and would become too slow: slow enough to brew a pot of coffee while it runs. Methods for stiff systems, such as the Radau method I use here (HAIRER; WANNER, 1996; VIRTANEN *et al.*, 2020), take large steps over that stretch without losing accuracy. Pantea *et al.* (2014, p. 421) note that the full equations of chemical mechanisms tend to be stiff (TURÁNYI; TOMLIN; PILLING, 1993; TURÁNYI; TOMLIN, 2014), and that current software solves large systems of this kind accurately.

The second way out is to switch approximations: the pre-equilibrium, if the reversible pair equilibrates well before I turns into product ($k_2 \ll k_1 + k_{-1}$, Section 7.5), or the slow step, if it is the second step (Section 7.3). When neither step is much faster, the total law remains (Equation (7.9) and Section 7.5.3). In large mechanisms, the SSH may not even yield a formula. Pantea *et al.* (2014, p. 422) prove that the classical procedure, which isolates each intermediate in an algebraic expression, cannot be carried through to the end for a second-order reaction with only three intermediates.

7.8 Practical checklist

Table 17 sums up the chapter in six steps. If the first five answer "yes," the SSH tends to err little, by an amount set by the ratio $\tau_\mathrm{I}/\tau_\mathrm{reaction}$. If any of them answers "no," the SSH should not be used without checking it against the full solution. The sixth step asks something else: whether the data would show the error (Chapter 8).

Table 17. Six steps to decide whether the SSH can be used in a mechanism.

	Question	How to answer it	If the answer is "no"
1	Who is the intermediate, and which steps consume it?	Write the rate equations and add the constants of the steps that consume I (Equation (7.1)).	Without a mechanism, there is no SSH.
2	Does I last much less than the time over which the reactants change?	Compare τ_I with τ_reaction (Equation (3.2)).	Intermediate too slow (Section 7.3): full system, slow step or total law (Section 7.5.3).
3	Does the intermediate hold little reactant?	Estimate [I] from the SSH itself and compare it with the reactants; for enzymes, $\varepsilon \ll 1$ (Equation (6.9)).	Too much intermediate (Section 7.4) or an equilibrium that builds up I: pre-equilibrium with mass balance if $k_2 \ll k_1 + k_{-1}$, else total law (Section 7.5.3).
4	Does the time window of interest start after the induction period?	Estimate t_ind, of the order of a few τ_I (Equation (7.2)).	For the earliest instants, use the full solution (Section 7.2).
5	Does a steady state exist for I?	Check that $\mathrm{d[I]}/\mathrm{d}t = 0$ has a positive solution and that I does not multiply itself (branched chains).	There is no SSH (Section 7.6).
6	Can the data show the error?	Are there points before the fast phase ends (I's initial mark)? Is it above about 3σ (σ: noise standard deviation) at the article's 13 times? Measure reactant and product or close the balance (Section 8.2).	The fit does not test the SSH; do not read elementary constants through the SSH formula.

Source: author's own work.

CHAPTER 8

The hypothesis against data

So far I have tested the steady-state hypothesis (SSH) almost entirely in theory and in simulation. In this chapter it meets data, and I ask three questions. First: in the enzyme assays already published, how large is the parameter ε of Segel & Slemrod (1989), which tells whether the SSH holds for an enzyme (Equation (6.9))? Second: when can a set of noisy points show that the SSH went wrong? Third: how can the SSH be tested on the bench, with cheap materials?

For the first question I use entries from a public database of enzyme kinetics (Section 8.1). For the second I use **synthetic data**: model curves with randomly drawn noise, which show what an assay could reveal if the model and the noise are right (Section 8.2). The third is a proposal for an experiment I have not done yet; what I say about its result is a model prediction (Section 8.3).

8.1 How large is ε in published assays?

In the derivation of the Michaelis-Menten equation (Section 6.1), I assumed there is far more substrate than enzyme. The criterion of Segel & Slemrod (1989) puts a number on this: the SSH holds when $\varepsilon = [\mathrm{E}]_0/([\mathrm{S}]_0 + K_\mathrm{M}) \ll 1$, where $[\mathrm{E}]_0$ and $[\mathrm{S}]_0$ are the initial concentrations of enzyme and substrate and K_M is the Michaelis constant (Equation (6.6)). The idea that the enzyme must be dilute is old. Straus & Goldstein (1943) already sorted an enzyme's behavior into "zones" by its concentration, and Cha (1970) studied the equation's errors when the enzyme is concentrated.

Some authors state, without showing data, that *in vitro* assays (done in a test tube, outside the cell) meet this condition with room to spare (SCHNELL; MAINI, 2003; CHOI; REMPALA; KIM, 2017), and Kim & Tyson (2020) point out that this is not always so. I found no survey that measured it in published assays, which does not prove that none exists. The closest was the analysis of 100 papers by Halling *et al.* (2018): 45 % of them do not even report the enzyme concentration. They measured whether the value is reported, not whether it meets the condition. I wanted to do that calculation.

I used SABIO-RK, a public database of enzyme kinetics, built by curators from published papers (WITTIG *et al.*, 2018). Each entry is a rate law taken from a paper, with its parameters (K_M, k_cat, v_max) and, when the paper reports them, the assay conditions, among them the enzyme concentration and the range of substrate concentrations. Here k_cat is the k_2 of Equation (6.1), the constant of the step that forms the product, and $v_\mathrm{max} = k_\mathrm{cat}[\mathrm{E}]_0$.

8.1.1 The sample

Each entry in the database has an identification number (ID), from 1 up to about 79,600. I drew 20,000 IDs (seed 2026) and downloaded the entries on 15 and 17 October 2026. I added a smaller, uniform sample, drawn earlier, with 734 IDs. In total I read 20,734 IDs, about a quarter of each ID range. Of these, 619 do not exist in the database, leaving 20,115 entries.

The sample covers all ID ranges almost evenly. Even so, I split the IDs into ranges of 4000 and gave each entry a weight equal to the inverse of the fraction of IDs I read in its range. This way, a range read a little more than the others does not count more in the result. The weights lie between 3.6 and 4.0, almost equal, and the three overall fractions, with and without them, differ by 0.1 percentage point or less. All fractions below use these weights. The 95 % confidence interval (CI: a range computed in a way that, repeated over many samples, contains the true value in 95 out of 100) comes from redrawing,

with replacement, the entries of each range 2000 times (the *bootstrap*).

8.1.2 What I computed

Many entries lack $[S]_0$, or give it as a range of values. So I computed

$$\varepsilon_{K_M} = \frac{[E]_0}{K_M} \geq \varepsilon, \tag{8.1}$$

which is an upper bound on the ε of Segel and Slemrod, because $[S]_0 \geq 0$. For K_M, this is the number that matters. The value of K_M is set by the points with small $[S]_0$, and there ε is close to ε_{K_M}. Srivastava *et al.* (2025) show that $[E]_0/K_M$ indicates the accuracy of the SSH, but conclude that its validity is better measured by $[E]_0/K_{VC}$, where $K_{VC} = k_2/k_1$ is the Van Slyke–Cullen constant. Since $K_M = (k_{-1} + k_2)/k_1$ is larger than K_{VC}, that ratio is larger than ε_{K_M}. The database does not give k_1, so I stay with ε_{K_M}; by their criterion, the fractions below are lower bounds.

From each entry I took the values of K_M (one per substrate) and the enzyme concentration, which the database records as that of the species "Enzyme", in the role of catalyst. It may come in mol/L, as mass per volume (mg/mL) or as activity (U/mL). I used the one in mol/L and, when it comes as a range, the smallest value. I converted the mass ones only when the entry gives the enzyme's molar mass and does not mention total protein (extracts, membranes). Even so, I kept these in a separate count, because the mass may include protein that is not the enzyme. Activity cannot be converted without knowing k_{cat} in that assay.

Of the 14,416 entries with a K_M, only 1850 give both K_M and the enzyme in mol/L. They yield 2007 K_M–$[E]_0$ pairs, from 741 papers. I set 39 of these pairs aside from the calculations, without deleting them from the file. In 26, the enzyme comes as a range that starts at zero, and the smallest positive value was not recorded. The other 13 come from a single paper. In 5 of its entries, the enzyme range, 0.25 mmol/L to 4 mmol/L, is identical to the substrate range of the same entry, and the other 8 repeat that range: most likely a copying error in the database. That leaves 1968 pairs, from 738 papers. Another 881 pairs have the enzyme converted from mass. In 2756 pairs the database lists the enzyme without a value, and in 1413 it is given as mass, without the molar mass.

8.1.3 Result

Figure 44 shows the distribution of ε_{K_M}. The typical value (the weighted median) is 9.4×10^{-4}: in most assays the enzyme really is present in small amounts. But the tail is long. I estimate that 24.1 % of the pairs have $\varepsilon_{K_M} > 0.01$ (CI 22.2 to 26.1 %), 9.0 % have $\varepsilon_{K_M} > 0.1$ (CI 7.6 to 10.4 %) and 2.5 % have $\varepsilon_{K_M} > 1$ (CI 1.8 to 3.4 %).

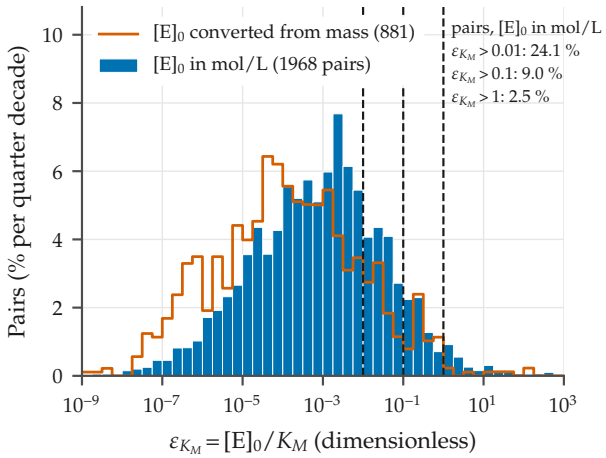


Figure 44. Distribution of $\varepsilon_{K_M} = [E]_0/K_M$ over the K_M–$[E]_0$ pairs of a sample of SABIO-RK. Bars: 1968 pairs with the enzyme in mol/L. Line: 881 pairs with the enzyme converted from mass to mol/L using the molar mass given in the entry. Each pair is weighted by the inverse of the fraction of IDs read in its ID range (weights from 3.6 to 4.0). Dashed lines mark 0.01, 0.1 and 1; the corner shows the estimated fraction of the mol/L pairs to the right of each.

Source: SABIO-RK data (WITTIG *et al.*, 2018), downloaded on 15 and 17 October 2026 (sample of 20,115 entries); analysis: `codigo/auditoria_sabiork.py`.

Many entries come from the same paper and repeat the same $[E]_0$. If I draw whole papers instead of entries, the CIs widen to 20.8–27.6 %, 6.8–11.4 % and 1.5–3.8 %. With the 39 set-aside pairs, the fractions would be 25.1 %, 9.9 % and 2.5 %; with a single copy of each repeated pair, 24.1 %, 9.1 % and 2.5 %; with only the 1904 pairs in which K_M is that of a substrate, 23.7 %, 8.2 % and 2.1 %. For the pairs with the enzyme converted from mass, they are 15.3 %, 6.3 % and

0.9 %. In 533 pairs the enzyme comes as a range, and I used the smallest value; with the largest, the three fractions would rise to 28.4 %, 11.5 % and 3.3 %.

Figure 45 splits the pairs by enzyme class. The EC number is the number the Enzyme Commission gives to each enzyme, and its first digit tells the type of reaction. Transferases (EC 2) have the largest fraction above 0.1: 13.2 % (CI 10.1 to 16.5 %). In hydrolases and isomerases this fraction is close to 6 %. Almost all pairs with the enzyme in mol/L (1938) come from *in vitro* assays. Another 29 come from 2 papers that the database marks as *in silico* (done on a computer), and 79 % of them exceed 0.01; 1 pair has no assay type recorded. In the publication decades with at least 30 pairs (from the 1970s to the 2020s), the fraction above 0.01 rises over time: 6.2 % in the 1970s, 18.8 % in the 1990s, 29.5 % in the 2010s and 39.1 % in the 2020s. The CIs of the decades with few pairs are wide, but those of the 1990s (14.5 to 23.5 %) and the 2010s (24.7 to 34.7 %) do not touch. I did not look into the cause.

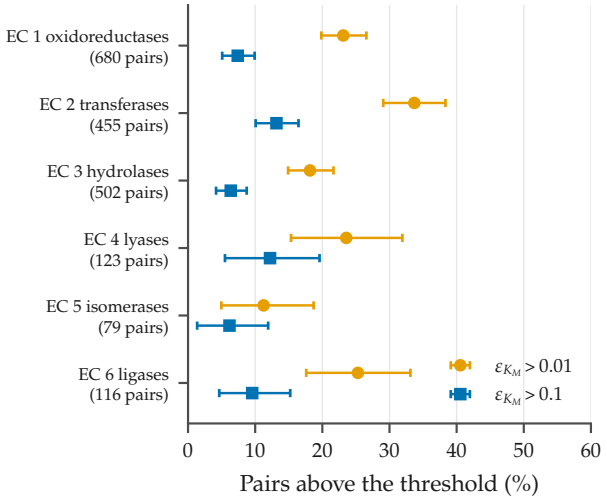


Figure 45. Estimated fraction of K_M–$[E]_0$ pairs (enzyme in mol/L) with $\varepsilon_{K_M} > 0.01$ (circles) and $\varepsilon_{K_M} > 0.1$ (squares), by enzyme class (first digit of the EC number). Bars are 95 % confidence intervals from the stratified *bootstrap* (redrawing within each ID range). Class EC 7, with 4 pairs, is left out.

Source: SABIO-RK data (WITTIG *et al.*, 2018), downloaded on 15 and 17 October 2026 (sample of 20,115 entries); analysis: `codigo/auditoria_sabiork.py`.

In 1563 pairs the entry also gives the $[S]_0$ range, and ε itself can be computed. At the largest $[S]_0$ of each assay, only 1.9 % of the pairs exceed $\varepsilon = 0.1$, and 0.7 % exceed 1. This agrees with what Schnell & Maini (2003) state: in most *in vitro* assays, the Segel and Slemrod condition holds with room to spare. But K_M is set by the points with small $[S]_0$. At the smallest positive $[S]_0$, recorded in 1110 pairs, 6.6 % of them exceed $\varepsilon = 0.1$, against 9.4 % by ε_{K_M} in the same pairs. So "the SSH holds in the assay" and "the published K_M is accurate" are different questions.

8.1.4 What a large ε does to the measured K_M

It is already known, from simulation, that the fitted K_M comes out wrong when $[E]_0$ is not small compared with K_M (STROBERG; SCHNELL, 2016; CHOI; REMPALA; KIM, 2017; YUN; HAN, 2020; WANG *et al.*, 2025). I do not present my error curves as a new result: I use them only to translate the values I found in the database into errors.

I simulated the full mechanism, $E + S \rightleftharpoons C \longrightarrow E + P$, where C is the complex (the ES of Equation (6.1)) and $[E] = [E]_0 - [C]$. On it I repeated the most common way of measuring K_M. For each $[S]_0$ in a grid (8 values from $0.2K_M$ to $5K_M$), the initial rate is the slope of the line fitted to the readings of [P] until 10 % of the substrate has turned into product. Then I fitted Equation (6.8) to the 8 points. Taking K_M as the unit of concentration and $1/k_2$ as the unit of time, the result depends only on ε_{K_M} and on $r = k_2/(k_{-1} + k_2)$. The base case uses $r = 0.5$, and I also tested other values of r, other grids and other reading windows (`codigo/vies_km_epsilon.py`). Even with $\varepsilon_{K_M} \to 0$, reading up to 10 % conversion already raises the fitted K_M by 5.4 %, because the substrate drops during the reading. So I call the *error caused by the enzyme* the relative difference between the fitted K_M and the one the same procedure gives with $\varepsilon_{K_M} = 10^{-6}$.

To isolate the simplest effect, I used the **total quasi-steady-state approximation** (tQSSA) of Borghans, Boer & Segel (1996). It is the idea of the total law of Chapter 7 (Section 7.5.3) carried over to enzymes: the substrate that counts is the total, free plus bound in the complex. Tzafriri

(2003) showed that it holds, at least roughly, for any enzyme and substrate concentrations. At the start of the reaction, almost nothing has turned into product, and the free substrate is $[S]_0 - [C]$, not $[S]_0$. Putting this and $[E] = [E]_0 - [C]$ into the balance of Equation (6.3) and dividing by k_1,

$$([E]_0 - [C])\,([S]_0 - [C]) = K_M[C]. \qquad (8.2)$$

This is a quadratic equation in [C]. The smaller root is the one that holds, because [C] cannot exceed $[E]_0$ or $[S]_0$, and the rate is $k_2[C]$. Fitting Equation (6.8) to this rate gives the error that comes only from the substrate bound in the complex, without the transient and without substrate consumption during the reading. In every initial-rate variant I simulated, the K_M error stayed above this one up to $\varepsilon_{K_M} \approx 4$ (Figure 46). That is why I use the tQSSA as a lower bound.

Putting $[C] = [E]_0/2$ into Equation (8.2), the rate reaches half its maximum at $[S]_0 = K_M + [E]_0/2$. Part of the substrate added to the tube is bound to the enzyme, and more must be added to reach half saturation. The same calculation appears in the study of inhibitors that bind very tightly to the enzyme (MORRISON, 1969). So the fit returns an apparent K_M, $K_{M,ap}$, larger than the true one. On my grid, with small ε_{K_M},

$$K_{M,ap} \approx K_M\left(1 + 0.84\,\varepsilon_{K_M}\right); \qquad (8.3)$$

the coefficient depends on where the $[S]_0$ points lie.

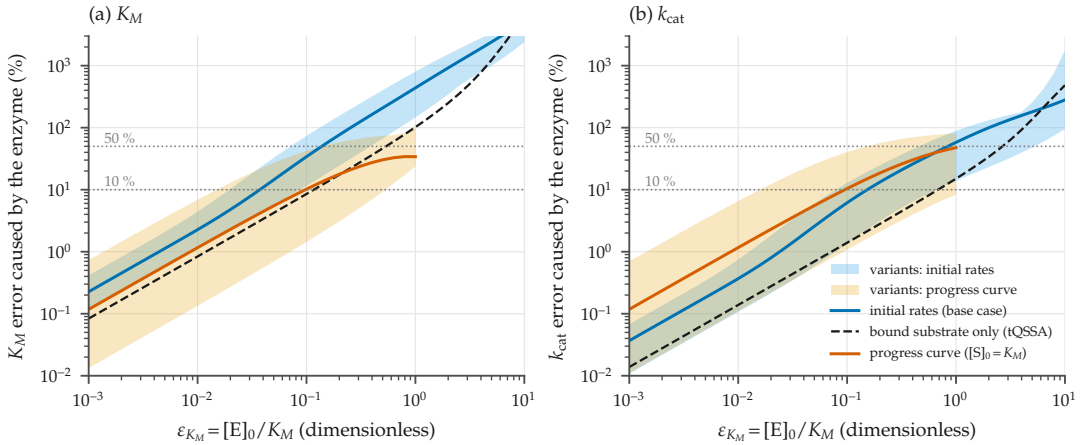


Figure 46. Error in K_M (a) and in k_{cat} (b) caused by the enzyme, in absolute value, against ε_{K_M}, from a simulation of the full mechanism and of the measurement procedure. Blue: initial rates read up to 10 % conversion and fitted with Equation (6.8) (base case: $r = 0.5$ and 8 values of $[S]_0$ from $0.2K_M$ to $5K_M$). Blue band: variants (other r, other grids and windows, readings after the transient). Dashed: substrate bound in the complex only (tQSSA, Equation (8.2)). Red: a single progress curve with $[S]_0 = K_M$; orange band: other r and $[S]_0$; both only up to $\varepsilon_{K_M} = 1$. With initial rates, K_M and k_{cat} come out larger than the true values; with the base-case progress curve, smaller. Dotted lines: 10 % and 50 %.

Source: simulation (`codigo/vies_km_epsilon.py`).

Figure 46 shows the result. With initial rates, k_{cat} is off by much less than K_M. By the tQSSA, the K_M error exceeds 10 % at $\varepsilon_{K_M} = 0.12$ and 50 % at $\varepsilon_{K_M} = 0.53$. If the readings start at $t = 0$, the transient (the time in which the complex forms) falls inside the reading window, and these thresholds drop to 0.037 and 0.14. If they start after the transient, they are 0.055 and 0.27. I also tested a single progress curve, that is, one reaction followed until near the end, with $[S]_0 = K_M$, fitted with the exact solution of the integrated Michaelis-Menten equation (SCHNELL; MENDOZA, 1997). There K_M comes out smaller than the true value: 10 % smaller at $\varepsilon_{K_M} = 0.1$ and 34 % smaller at $\varepsilon_{K_M} = 1$, and k_{cat} is off by as much as K_M or more: 48 % smaller at $\varepsilon_{K_M} = 1$. This is for the base case, $r = 0.5$. With k_{-1} larger than k_2 ($r = 0.1$), K_M comes out larger: 8.6 % above at $\varepsilon_{K_M} = 0.1$. The sign of the error depends on r. This agrees with Stroberg & Schnell (2016): with $[S]_0$ of the order of K_M, the progress curve returns the right values of K_M and v_{max} only if

the enzyme is below K_M.

Now I put the two parts together, under two assumptions: the recorded $[E]_0$ is that of the assay in which K_M was measured, and the fit was the ordinary Michaelis-Menten one, by initial rates. The published K_M is already the apparent one, $K_{M,ap}$. Since half saturation is at $K_M + [E]_0/2$, an apparent K_M that marks it gives $[E]_0/K_{M,ap} < 2$, whatever the $[S]_0$ grid. Only 1.7 % of the pairs (CI 1.1 to 2.4 %) go above 2. For them the calculation does not apply: the $[E]_0$ may come from another assay, or the authors may have discounted the bound enzyme.

If both assumptions hold, the K_M error caused by the enzyme exceeds 10 % in 5.3 to 9.3 % of the pairs and exceeds 50 % in 1.0 to 3.3 %, depending on the reading (the tQSSA gives the smallest values, and readings from $t = 0$ the largest). These are model estimates, not measurements. The CIs, the other variants and what changes when the $[S]_0$ grid is centered on the K_M of a pilot fit are in Appendix C.1.

In short, in most assays in the database the enzyme really is scarce, and the SSH is a good approximation. But about 9 in every 100 pairs with $[E]_0$ in mol/L have $\varepsilon_{K_M} > 0.1$, where the K_M error stops being small. These results hold for the entries that give $[E]_0$ in mol/L, about 14 % of the sample, not for all published K_M values: almost half of the papers do not even report the enzyme concentration (HALLING *et al.*, 2018), and those that do may not be typical labs. I did not check, paper by paper, whether the recorded $[E]_0$ is that of the K_M assay, nor look by hand at the pairs with $\varepsilon_{K_M} > 1$: some may come from experiments run on purpose with a lot of enzyme, such as those that follow only the start of the reaction. The counts and estimates are in `dados/sabiork_auditoria_resumo.json`; the application to the pairs, in `dados/simulacao/vies_km_epsilon_resumo.json`.

8.2 When the data show that the SSH went wrong

In Chapter 7 I compared the SSH with the exact solution, knowing all the constants. In an experiment it is the other way around: there are a few noisy points, and the constants are what one wants to find. The question changes. With how many points, at what times and with how much noise do the data show that the SSH went wrong? And, when they do not, is the constant that comes out of the fit right?

To answer, I generated thousands of synthetic data sets (`codigo/testabilidade_hee.py`). Each point is the exact value of the mechanism plus an error drawn from a normal distribution with standard deviation σ, as a fraction of $[A]_0$, independent from one point to the next, with a fixed seed. Since I know the right answer for each set, I can count how often the fit gets it right. All experimental designs cover the reaction up to 98.6 % of the way. The main one uses 13 times spaced like those of the December 2025 article data, with values rounded to 2 decimals, as there (RIBEIRO, 2025b). The others are 13 log-spaced times, closer together at the start, 50 evenly spaced times and 50 log-spaced times. In each set I fitted the candidate laws and chose by the lower AICc, the Akaike information criterion corrected for few points (AKAIKE, 1974; HURVICH; TSAI, 1989), the same as in Chapters 4 and 5. For [P] I compare two laws, the SSH law and the full solution. For [A], or [A] and [P] together, the law with a jump at the start also enters; it is the shape of the total law and of the pre-equilibrium. I say the data "show the SSH error" when AICc does not pick the SSH law. I used 200 sets per case, and 1000 to measure the false alarm rate.

8.2.1 What the product points do not tell

Start with the mechanism $A \rightleftharpoons I \to P$ of Section 7.5. For the product, the SSH (Equation (7.5)), the pre-equilibrium (Equation (7.6)) and the total law (Equation (7.9)) give the same shape,

$$[P] = [A]_0 \left(1 - e^{-kt}\right), \tag{8.4}$$

and only disagree about the value of k (1/s). The [P] points fix k, but do not say which of the three laws is right.

The k the fit returns lies very close to that of the total law, $k_1 k_2/(k_1 + k_{-1} + k_2)$. There is a simple reason. The area under the curve of $1 - [P]/[A]_0$

is the mean time a molecule of A takes to become P. In the exact solution (Equation (7.7)), this area is $(k_1 + k_{-1} + k_2)/(k_1 k_2)$, the inverse of the total-law k, and an exponential fitted to the points roughly keeps this area. Whoever reads this k through the SSH formula, knowing k_{-1} and k_2, gets a k_1 smaller than the true one (Figure 47a). Setting the SSH constant equal to the total-law constant gives

$$\frac{k_1^{\mathrm{SSH}}}{k_1} = \frac{1}{1+\kappa}, \qquad \kappa = \frac{k_1}{k_{-1}+k_2}, \tag{8.5}$$

where k_1^{SSH} is the k_1 computed from the SSH formula. The ratio κ is the one in the SSH criterion, the dashed line of Figure 42. With $\kappa = 0.11$, k_1 comes out 10 % too small (dotted line in Figure 48). In A ⟶ I ⟶ B, where $k_{-1} = 0$, the ratio becomes $\kappa = k_1/k_2$. On a grid of 361 points in the $(k_2/k_{-1}, K)$ plane, with $K = k_1/k_{-1}$, the 13 times of the article and $\sigma = 0.01$, the difference between the fitted k and the total-law k had a median of 0.065 % and a maximum of 7 %.

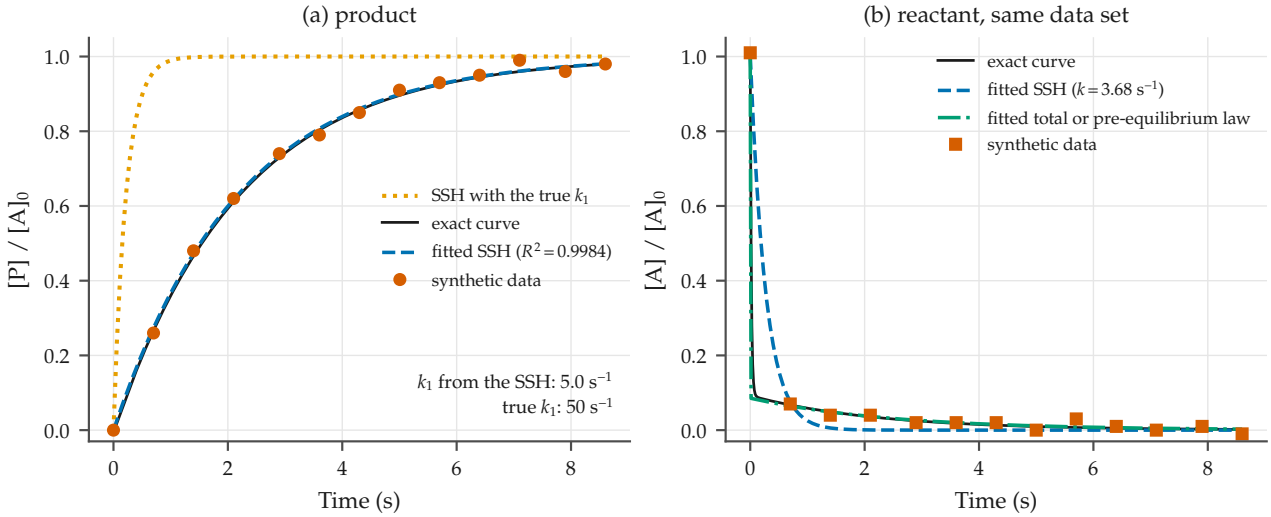


Figure 47. A case of high R^2 and a wrong constant. Synthetic data of the mechanism A ⇌ I ⟶ P with $k_1 = 50$/s, $k_{-1} = 5$/s and $k_2 = 0.5$/s ($\kappa = 9.1$), at the 13 times of the December 2025 article data, with Gaussian noise $\sigma = 0.01$ and 2 decimals; it is the first draw of the seed, not a chosen one. (a) Product: the fitted SSH law (dashed blue, $R^2 = 0.9984$) follows the points but, read through Equation (7.5), gives $k_1 = 5.0$/s. With the true k_1, the SSH would predict the dotted curve. (b) Reactant in the same data set: the fitted SSH (dashed blue) cannot follow the early drop; the law with an initial jump (green dash-dot, the shape of the total and pre-equilibrium laws) can.

Source: synthetic data (`codigo/testabilidade_hee.py`); times: data from the December 2025 article (RIBEIRO, 2025b); points in `dados/simulacao/testabilidade_exemplo_dados_sinteticos.csv`.

Figure 47 shows a single set, the first one the seed drew, not a chosen one. The mechanism has $k_1 = 50$/s, $k_{-1} = 5$/s and $k_2 = 0.5$/s ($\kappa = 9.1$), and the points have $\sigma = 0.01$ and 2 decimals. The SSH law fits [P] with $R^2 = 0.9984$ and $k = 0.457$/s, and AICc keeps it. Read through the SSH formula, this k gives $k_1 = 5.0$/s, ten times less than the true value; Equation (8.5) predicts 4.95/s. The reactant in the same data set tells another story. About 91 % of A turns into I within a few hundredths of a second, and [A] is already at 0.07 at the first point after zero. The SSH fitted to [A] needs $k = 3.68$/s, eight times the k of [P]. R^2 is still 0.994, but AICc prefers, by 16.8, the law with the jump at the start. After the jump, [A] starts at 0.086 in the fit, close to $1/(1+\kappa) = 0.099$; the exact value is 0.092.

Figure 48 extends the example to the whole $(k_2/k_{-1}, K)$ plane of A ⇌ I ⟶ P, with the article design and with 50 log-spaced points. In each column I read k_1 through the SSH formula from the k the SSH fits to the points at hand. With the article design, the k_1 read from [P] is off by more than 10 % at 182 of the 361 points, and 113

of them go unnoticed: AICc rejects the SSH in fewer than half of the sets, and the median R^2 stays above 0.998. The k_1 read from [A] alone is off by more than 10 % at 101 points, and 65 go unnoticed, nearly all with K above about 30: there A drops close to zero before the first point, and any fast exponential will do. With [A] and [P] from the same data set, fitted with the same constants, the SSH k_1 is off by more than 10 % at 169 points, and AICc rejects the SSH at all of them. With 50 log-spaced points and $\sigma = 0.001$, 42 of 192 points go unnoticed with [P] alone, 2 of 139 with [A] alone and none with both curves.

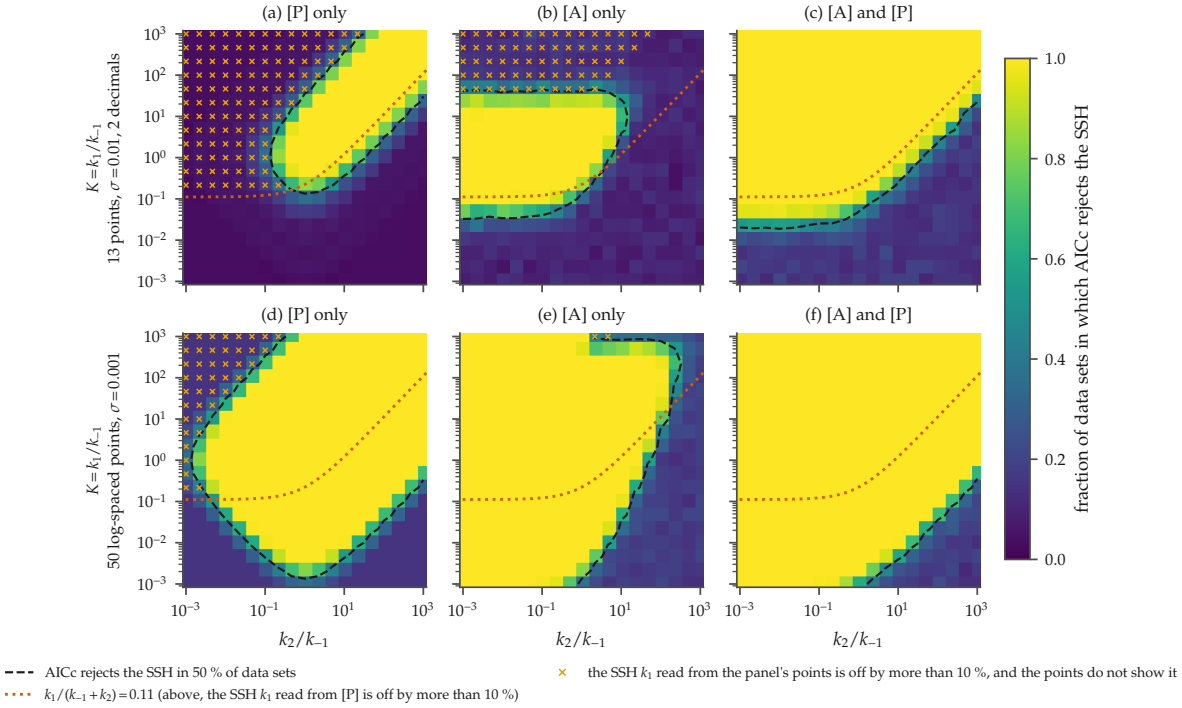


Figure 48. When the points reveal the SSH error in the mechanism A ⇌ I → P. The color is the fraction of synthetic data sets in which AICc rejects the SSH. Top row: 13 points at the times of the December 2025 article data, $\sigma = 0.01$, 2 decimals. Bottom row: 50 log-spaced points, $\sigma = 0.001$. Columns: (a, d) [P] only; (b, e) [A] only; (c, f) [A] and [P] from the same data set, fitted with the same constants. Dashed black: 50 %. Above the dotted line, $\kappa = 0.11$, the k_1 read from [P] through the SSH is off by more than 10 % (Equation (8.5)). The "x" marks show where the k_1 given by the SSH formula from that column's points is off by more than 10 % and AICc rejects the SSH in fewer than half of the data sets. With points drawn from the SSH law itself, AICc rejects it in 6 %, 10 % and 15 % of the data sets in the top three columns and in 6 %, 17 % and 17 % in the bottom ones. For [P], the curve depends only on the two constants of the exact solution; so the color of each point comes from A → I → B with the same ratio between them, checked by direct simulation at 8 points of each design (largest difference 0.06).

Source: simulation (`codigo/testabilidade_hee.py`); numbers in `dados/simulacao/testabilidade_aip.csv`.

The differences between the columns of the figure have a simple reason. In [P], the SSH error shows only through the fast phase, which in the pre-equilibrium regime is too small. In [A], it shows as the jump at the start. And the SSH requires the same k in both curves, with $[A] + [P] = [A]_0$; when I builds up, the two do not match. Comparing three laws has a price: with points drawn from the SSH law itself, AICc rejects it by mistake in 6 % of the sets with [P] alone, 10 % with [A] alone and 15 % with both curves (article design).

8.2.2 When the fast phase shows

When only the product is available, everything depends on the fast phase, the intermediate's mark at the start of the curve. In the mechanism A → I → B of Example 3.1, observing [B], this phase lasts about $1/k_{\text{fast}}$ and has height $1/(\rho - 1)$, as a fraction of $[A]_0$, with $\rho = k_{\text{fast}}/k_{\text{slow}}$,

where k_{fast} and k_{slow} are the larger and the smaller of the two constants. The [B] curve does not change when k_1 and k_2 swap places, a known ambiguity of series reactions (ALCOCK; BENTON; MOORE, 1970) (Section 7.5.1). So the chance that the points show the error is the same for k_2/k_1 and for k_1/k_2. With 13 points spaced like those of the article and $\sigma = 0.01$, AICc rejects the SSH in at least 90 % of the sets when $0.052 < k_2/k_1 < 19$, and in fewer than half when $k_2/k_1 < 0.030$ or $k_2/k_1 > 34$. On the $k_2 > k_1$ side this is enough: the SSH k_1 is off by more than 10 % only below $k_2/k_1 = 9.6$, and there detection already exceeds 90 %. The danger is on the other side, with the second step slow. With k_2/k_1 near 0.01, AICc rejects the SSH in 17 % of the sets, just above the false-alarm rate, which lies between 4 and 12 % in the four designs. The median R^2 is 0.9989, and the SSH returns a k_1 a hundred times smaller than the true one: in fact it returns k_2. In every set with $k_2/k_1 \geq 8.5$ or $k_2/k_1 \leq 0.12$, the SSH fit has $R^2 > 0.99$.

The experimental design matters as much as the noise. To compare designs before measuring, it is enough to compute the deviation D between the exact curve and the best SSH curve, without noise, at the design's times:

$$D^2 = \min_k \sum_{j=1}^{N} \left[y(t_j) - y_{\text{SSH}}(t_j; k) \right]^2 , \qquad (8.6)$$

where N is the number of points, $y(t_j)$ is the exact value at time t_j and $y_{\text{SSH}}(t_j; k)$ that of the SSH law with constant k. Up to a factor $1/N$, it is the quantity that Atkinson & Fedorov (1975) proposed to make as large as possible to tell two models apart; the idea of choosing the experiment for this is already in Box & Hill (1967). With 13 points, AICc rejects the SSH in half of the sets when $D \approx 1.8\,\sigma$ and in 90 % when $D \approx 3.2\,\sigma$; with 50 points, at $1.5\,\sigma$ and $2.8\,\sigma$. These thresholds are an approximate prediction, made with the noncentral F distribution, a formula from statistics that gives the expected fraction of detections without drawing data sets. For the designs without rounding, the predicted fraction was on average within 0.016 of the simulated one; with 2 decimals, the difference reached 0.13. With the same 13 points, log spacing increases D 3 to 4 times when $k_{\text{fast}}/k_{\text{slow}}$ exceeds 100.

This gives a short practical rule in three parts.

1. With the product alone, AICc notices the fast phase in half of the experiments when its height exceeds about $3\,\sigma$ with the article's 13 times, $1\,\sigma$ with 50 evenly spaced times or 13 log-spaced ones, and $0.4\,\sigma$ with 50 log-spaced times. For 90 %, about twice as much is needed. Points before $1/k_{\text{fast}}$ are worth more than extra points at the end.

2. If only the product is measured, the pre-equilibrium regime stays largely invisible: at 113 of 182 points with the article design, and still at 42 of 192 with 50 log-spaced points. One must also measure the reactant, or close the balance [A] + [P]. With both curves together, every k_1 error above 10 % showed up in both designs I tested.

3. A high R^2 does not test the SSH (SPIESS; NEUMEYER, 2010). In the cases the points do not reveal, the median R^2 stayed above 0.998.

8.2.3 The article data and enzymes

The December 2025 article data (RIBEIRO, 2025b) have 13 points in each case, one about every 0.7 s for ozone and every 0.7 min for ammonia, with 2 decimals, and the residuals of my fits to them are around 0.01 (0.014 for ozone and 0.010 for model M3 of ammonia). With $\sigma = 0.01$, a set like this reveals the SSH error at least 90 % of the time when the ratio between the fast and slow constants is below 19, less than half the time when it exceeds 34, and less than 20 % of the time when it exceeds 60. These rates assume noise that is independent between points. The ozone residuals tend to alternate in sign (Durbin-Watson of 3.2, Section 4.4.1), and with noise like that the rates may differ, although this test has little power with 13 rounded points. For ammonia, the M3 fit has a ratio of 3.35 between its two constants, well inside this range, and AICc does separate M3 from M1, the simple exponential, by a wide margin: the difference is 31.9 (Section 5.5). This shows the delay, not its cause. The ozone

points are of a reactant, and the O atom adjusts within milliseconds: at the times of the points, the exact curve and the SSH curve coincide (Section 4.3). These points could not show an SSH error even if there were one. In Sections 4.4.1 and 5.5.1, I use synthetic data to design assays that would test the laws of these two cases.

Something similar happens with enzymes. I repeated the test with a Michaelis-Menten progress curve (13 points, $\sigma = 0.01$, $[S]_0 = K_M$ and fast binding, $k_{-1} = 99\,k_2$, that is, $r \approx 0.01$), varying the Segel and Slemrod ε (Equation (6.9)). The integrated form of the SSH fits the curve with a median R^2 of 0.999 at every ε. Even so, it returns, in median, a K_M larger than the true value, as in the $r = 0.1$ variant above: 1.28 times the true value at $\varepsilon = 0.1$, 2.1 times at 0.3 and 6.0 times at 0.9, and a k_{cat} 1.09, 1.33 and 2.2 times the true value. The total law, which is the tQSSA of Equation (8.2) with the total substrate remaining in place of $[S]_0$, follows the full system to within 0.002 of $[S]_0$. But AICc prefers it in only 30 to 55 % of the sets: a single curve does not decide between the two laws. And the right law is not enough either. At $\varepsilon = 0.01$, 90 % of the sets give a total-law K_M between 0.79 and 1.30 times the true value; at $\varepsilon = 0.3$, between 0.26 and 2.1. With a lot of enzyme, K_M and k_{cat} are tied to each other, and a single curve cannot separate them. This agrees with Stroberg & Schnell (2016), for whom the SSH holding is not enough to estimate K_M and v_{max} from a progress curve, and with Choi, Rempala & Kim (2017), who found the same bias, corrected it with the total law and proposed combining curves made with little and with much enzyme.

What I did in this section is not a new method. Choosing the experiment to tell models apart goes back to Box & Hill (1967) and Atkinson & Fedorov (1975), and the SSH bias in enzymes has already been measured (STROBERG; SCHNELL, 2016; CHOI; REMPALA; KIM, 2017). What I add is the same test applied to the two general-chemistry mechanisms of Chapter 7, with maps in the plane of the constants, Equation (8.5) checked by simulation and a rule with numbers for designing the experiment.

8.3 A proposed bench test

The tests of the SSH in this work are simulations or fits to points that already existed. What is missing is an experiment in which I measure the intermediate itself. Here I propose one that can be done with a pharmacy glucose meter, sugar and an enzyme, invertase. I have not done it yet: what follows is the plan and what the model predicts.

8.3.1 The glucose the strip does not see yet

Invertase breaks sucrose (S), ordinary table sugar, into glucose and fructose (Section 6.1). The glucose comes out as alpha glucose (α) and only later becomes a mixture of alpha and beta glucose (β). This has been known since 1890 (O'SULLIVAN; TOMPSON, 1890 apud HUDSON, 1908) and also appears in Michaelis and Menten (MICHAELIS; MENTEN, 1913; JOHNSON; GOODY, 2011) and in Keilin & Hartree (1952a). These two forms, the **anomers**, differ only in the position of the OH group on carbon 1 of the ring. The interconversion between them, **mutarotation**, is first order in both directions, and I call k_m (1/min) the sum of the two constants. It is slow: at 25 °C, in slightly acidic water, $1/k_m$ is over 40 min (NELSON; BEEGLE, 1919). At equilibrium, about 64 % of the glucose is β (KEILIN; HARTREE, 1952a).

Hudson (1908) and Michaelis and Menten stopped each sample with an alkali, which kills the enzyme and completes the mutarotation, before measuring (MICHAELIS; MENTEN, 1913; JOHNSON; GOODY, 2011). Here I do the opposite: I want to measure the delay.

For that I use a glucose meter, the device people with diabetes use to measure their blood glucose. It reads glucose with an enzyme held on a disposable strip, and some of these enzymes tell the anomers apart. Keilin & Hartree (1952a) measured that glucose oxidase oxidizes α at only 0.64 % of the rate at which it oxidizes β, and already warned that glucose released from sucrose may be oxidized with a delay. Some preparations of this enzyme, however, carry a catalyst of mutarotation (KEILIN; HARTREE, 1952b). Perles & Volpe (2008), Perles & Volpe (2009) followed mu-

tarotation with a glucose meter that uses glucose dehydrogenase strips. The chemistry is known; what I propose is to use it as a test of the SSH.

The meter comes calibrated for glucose at equilibrium. A strip that reads only β therefore shows

$$R = \frac{[\beta]}{f_\beta}, \tag{8.7}$$

where R is the reading, $[\beta]$ is the concentration of β and f_β is the fraction of β at equilibrium. The mechanism has two steps:

$$\mathrm{S} \xrightarrow{\text{invertase}} \alpha + \text{fructose}, \tag{8.8}$$

$$\alpha \underset{k_{\beta\alpha}}{\overset{k_{\alpha\beta}}{\rightleftharpoons}} \beta, \tag{8.9}$$

with $k_m = k_{\alpha\beta} + k_{\beta\alpha}$ and $f_\beta = k_{\alpha\beta}/k_m$. I call $G = [\alpha] + [\beta] = [\mathrm{S}]_0 - [\mathrm{S}]$ the total glucose; fructose has the same concentration. The first step follows Michaelis-Menten (Equation (6.8)), slowed down by the glucose and fructose already formed, which compete with sucrose for the place on the enzyme:

$$v = \frac{v_{\max}[\mathrm{S}]}{K_\mathrm{M}\,(1 + G/K_F + G/K_G) + [\mathrm{S}]}, \tag{8.10}$$

with $K_\mathrm{M} = 16.7\,\mathrm{mmol.L^{-1}}$, $K_F = 58.8\,\mathrm{mmol.L^{-1}}$ (fructose) and $K_G = 91\,\mathrm{mmol.L^{-1}}$ (glucose), the constants of Michaelis and Menten (JOHNSON; GOODY, 2011). The second step gives

$$\begin{aligned} \frac{\mathrm{d}[\alpha]}{\mathrm{d}t} &= v - k_{\alpha\beta}[\alpha] + k_{\beta\alpha}[\beta], \\ \frac{\mathrm{d}[\beta]}{\mathrm{d}t} &= k_{\alpha\beta}[\alpha] - k_{\beta\alpha}[\beta]. \end{aligned} \tag{8.11}$$

I call $J = G - R$ the glucose the strip does not see yet. From Equation (8.7), $J = [\alpha] - (k_{\beta\alpha}/k_{\alpha\beta})[\beta]$. Differentiating and using Equation (8.11) gives, with no approximation at all,

$$\begin{aligned} \frac{\mathrm{d}J}{\mathrm{d}t} &= v - k_m J, \\ \frac{\mathrm{d}R}{\mathrm{d}t} &= k_m J. \end{aligned} \tag{8.12}$$

This is the A → I → B mechanism of Example 3.1: sucrose plays A, J plays I and the reading plays B, and f_β drops out. The role of k_2 goes to k_m, and that of k_1, approximately, to $k_\mathrm{ap} = v_{\max}/(K_\mathrm{M} + [\mathrm{S}]_0)$, the initial rate of invertase divided by $[\mathrm{S}]_0$. The ratio k_m/k_ap plays k_2/k_1, and it can be changed with the amount of enzyme.

The SSH for J gives

$$J_\mathrm{SSH} = \frac{v}{k_m}. \tag{8.13}$$

By the SSH rate law, $\mathrm{d}R/\mathrm{d}t = k_m J_\mathrm{SSH} = v$: the strip would read the total glucose, $R = G$, with no delay, as in Equation (3.16). Counting J_SSH in the mass balance, the reading becomes $R = G - v/k_m$. The delay of the full solution already has a general formula: in a sequence of enzyme reactions, the transition time of each intermediate is its steady-state concentration divided by the flux (EASTERBY, 1981), which here gives $1/k_m$.

The strong point of the experiment is that J can be measured. At each time, a drop of the mixture would go straight onto the strip and give R. Another sample would go into a small tube with sodium carbonate, an alkali like that of Michaelis and Menten, and would be read only half an hour later, with the mutarotation over: that one would give G. The difference between the two readings is the intermediate. In the sources I read, I did not find this reading written as an A → I → B that is exact in the I → B step, nor used to test the SSH. But the calculation is short, and I do not claim priority.

8.3.2 What the model predicts

To predict the curves of Figure 49, I fitted an Arrhenius line to the eight k_m values of Nelson & Beegle (1919), from 0.15 °C to 37 °C, in unbuffered water. It gives $E_\mathrm{a} = 69.4\,\mathrm{kJ/mol}$ (standard error of 0.9 kJ/mol) and, at 25 °C, $1/k_m = 42\,\mathrm{min}$. The values of Hudson (1907) and of Keilin & Hartree (1952a) in water fall within 6 % of the line, and those of Perles & Volpe (2009), with a glucose meter, 16 % and 18 % below it. But buffers speed up the interconversion: in $0.2\,\mathrm{mol.L^{-1}}$ phosphate it is 4.6 times faster than in water (KEILIN; HARTREE, 1952a). That is why the first step is to measure k_m in the buffer of the experiment itself. The numbers in this section are in `dados/simulacao/invertase_glicosimetro.json`.

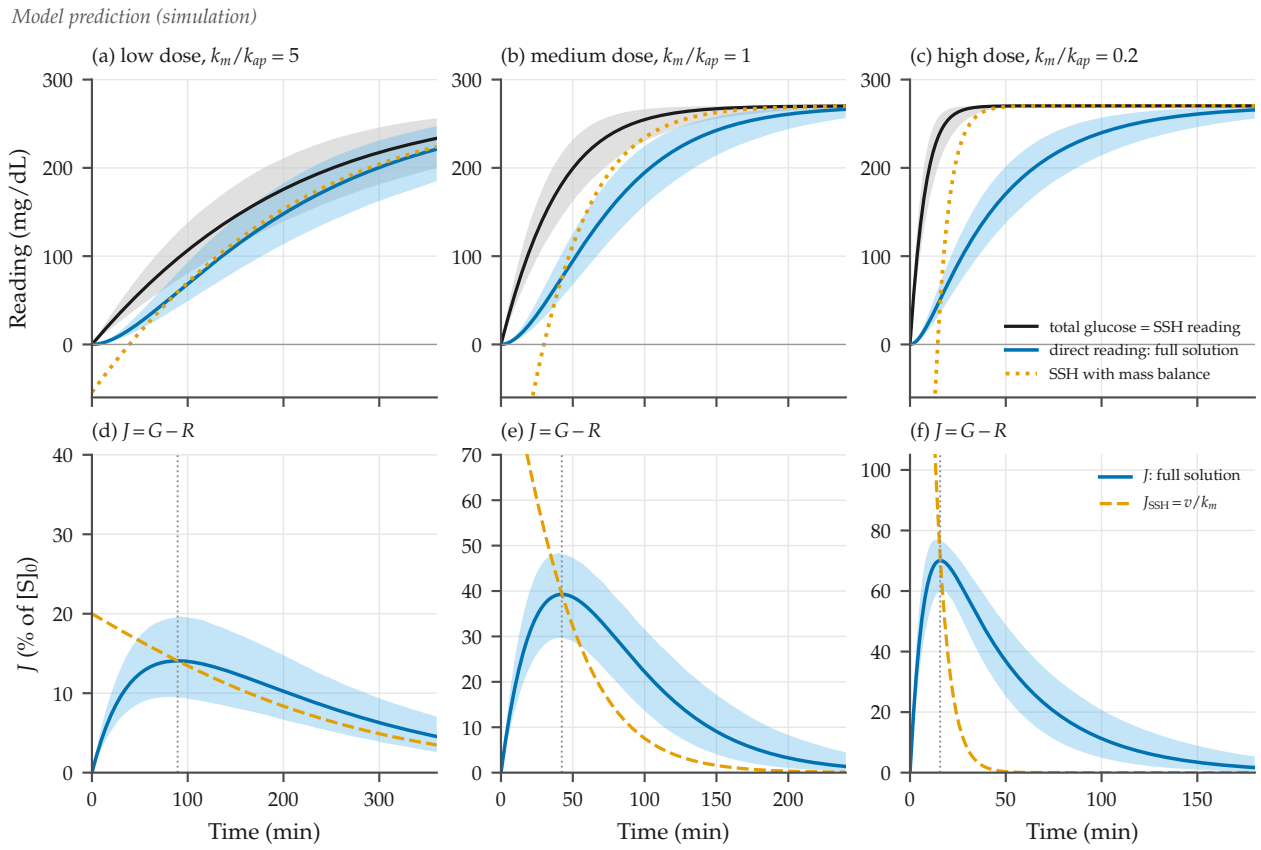


Figure 49. Model prediction for the invertase experiment at 25 °C, pH 4.7 and 15 mmol.L^{-1} sucrose, at three enzyme doses (k_m/k_{ap} = 5, 1 and 0.2). (a)–(c) Total glucose, read in the stopped aliquots (black), which is also the reading predicted by the SSH; direct strip reading from the full solution (blue); SSH with the mass balance, $R = G - v/k_m$ (dotted). (d)–(f) Glucose the strip does not see yet, $J = G - R$, as a percentage of $[S]_0$: full solution (blue) and SSH, $J = v/k_m$ (dashed); the vertical line marks the end of the induction period. Bands: 5th and 95th percentiles of 2000 draws on a log scale, with spreads I chose: 20 % in k_m (from the spread in the literature and buffer catalysis), 30 % in K_M, 50 % in K_F and K_G and 20 % in the initial rate.

Source: simulation (`codigo/invertase_glicosimetro.py`), with the invertase constants of Michaelis and Menten (JOHNSON; GOODY, 2011) and the mutarotation of Nelson & Beegle (1919).

Figure 49 shows the prediction with invertase for 15 mmol.L^{-1} sucrose (270 mg/dL of glucose at the end, in the unit the meter shows), at 25 °C and pH 4.7, at three enzyme doses. At the medium dose ($k_m/k_{ap} = 1$), the total glucose reaches half at 27 min, and the direct reading only gets there 40 min later. J rises to 39 % of $[S]_0$ (106 mg/dL) at 42 min. This is the largest SSH error in the reading (Equation (7.3)), close to the 37 % that Table 6 gives for $k_2 = k_1$. At the low dose ($k_m/k_{ap} = 5$), J reaches 14 %, close to the 13.4 % of the first-order case; at the high dose ($k_m/k_{ap} = 0.2$), 70 %. Halfway up, the direct reading lags 32 to 45 min behind the total glucose, close to $1/k_m$, as in Section 7.2. Even at the 5th percentile of the bands, the maximum J at the medium dose is above 30 % of $[S]_0$.

The SSH for the intermediate itself, $J \approx v/k_m$, can also be tested (Figure 49d–f). At the low dose, J/J_{SSH} rises from 1.13 to 1.28 between $1/k_m$ and $5/k_m$ after the induction period and tends to 1.36, the $k_2/(k_2 - k_1)$ of Figure 41. Here k_1 is the invertase constant at the end of the reaction, when the glucose and fructose already formed slow it down: from Equation (8.10), $k_1 = v_{max}/[K_M(1 + [S]_0/K_F + [S]_0/K_G)] = 1.34\,k_{ap}$. At the medium dose, the ratio goes from 2.3 to 7.1 between $1/k_m$ and $3/k_m$ after the induction period and keeps growing, as in Section 7.3 for $k_2 \le k_1$.

This first test would take an afternoon and needs no invertase: dissolve dextrose, ordinary crystalline glucose, which is α (PERLES; VOLPE, 2009), and read the solution from time to time.

With $v = 0$, all the glucose starts as J, and Equation (8.12) gives $R = G\left(1 - \mathrm{e}^{-k_m t}\right)$ for a strip that reads only β. A glucose oxidase strip should show 12 % of the final value at 5 min (from 9 to 16 %: 5th and 95th percentiles, with k_m uncertain by about 20 %) and half of it at 29 min (Figure 50a). A strip that reads both anomers would start at the final value. The difference is far beyond the ±15 % that the ISO 15197 standard allows for glucose meters (ISO, 2013), and would decide whether my meter will do.

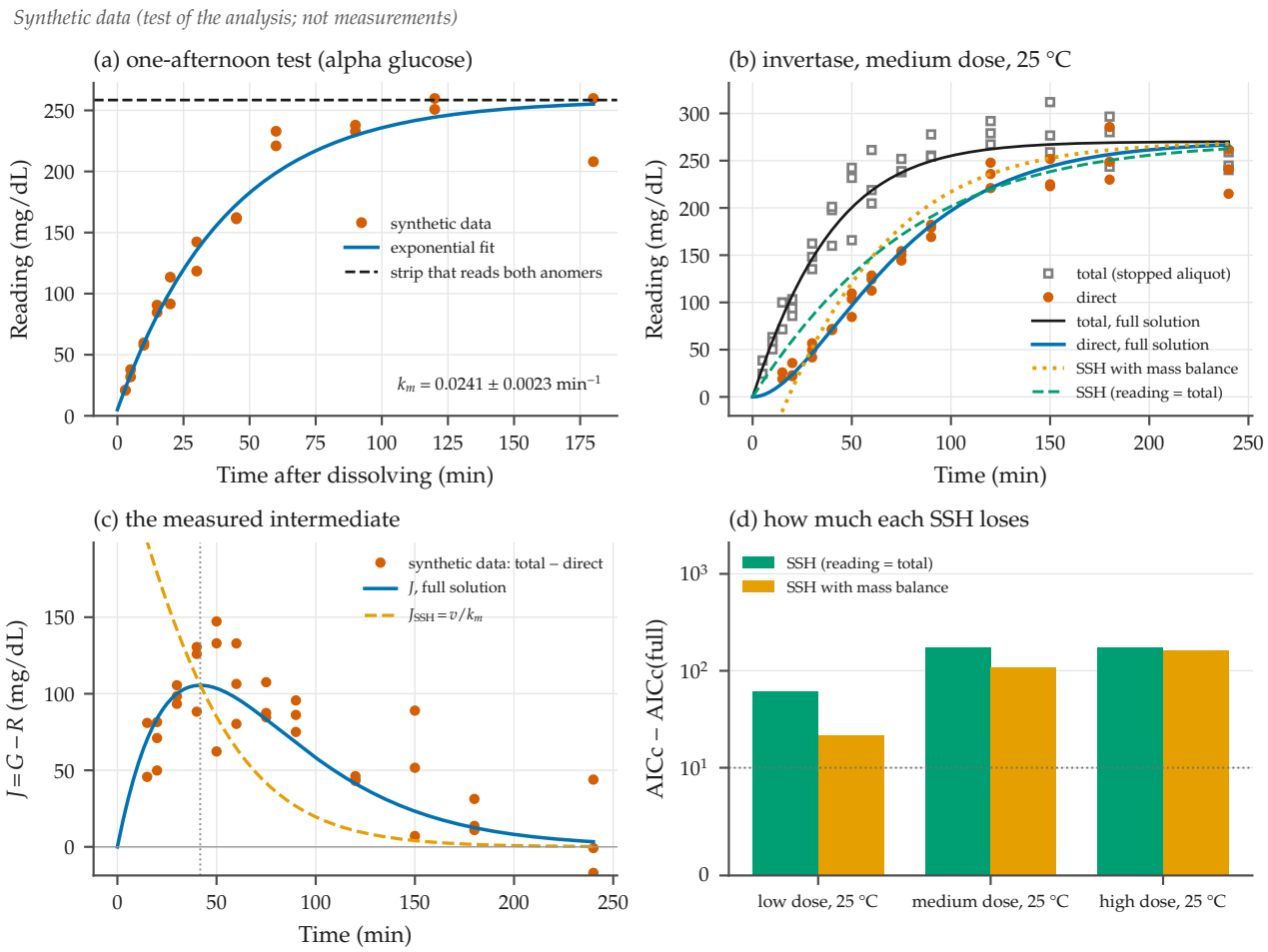


Figure 50. Test of the analysis program with synthetic data (model with noise at the ISO 15197 limit and a fixed seed; not measurements). (a) One-afternoon test: readings after dissolving alpha glucose, with the exponential fit. (b) Medium dose, three replicates: direct (circles) and total (squares) readings, converted with the calibration, with the totals multiplied by 1.125 (carbonate dilution); lines: fitted full solution and the two forms of the SSH. (c) J, total reading minus direct reading at the same time, with J from the full solution and from the SSH. (d) How far the AICc of each form of the SSH lies above that of the full solution, at the three doses; the dotted line marks 10.

Source: synthetic data (`codigo/invertase_glicosimetro.py`); fits: `codigo/analisar_glicosimetro.py`; numbers in `dados/simulacao/glicosimetro_ajuste_sintetico.json`.

The program that will analyze the readings is already written (`codigo/analisar_glicosimetro.py`). It fits the full solution and the two forms of the SSH ($R = G$ and $R = G - v/k_m$) and compares them by AICc. I tested it with synthetic data, with the largest noise ISO 15197 allows (Figure 50). It returned k_m between 0.88 and 1.02 times the true value, always within two standard errors, and picked the full solution at all three doses, with the AICc of the plain SSH ($R = G$) 61 to 175 higher and that of the SSH with the mass balance 22 to 162 higher.

In 100 synthetic data sets per case, like those of Figure 50, a single run at the medium dose,

with direct and total readings, was enough for AICc to pick the full solution in all of them, with k_m between 0.90 and 1.15 times the true value in 90 % of them. But the total readings are essential. The B curve in Equation (3.17) does not change if k_1 and k_2 swap places (Section 7.5.1), and almost the same happens here: with direct readings only (three runs), the fit mixes up the enzyme with mutarotation, and k_m came out wrong by more than a factor of 2 in 26 %, 22 % and 65 % of the sets at the low, medium and high doses. With the totals, this did not happen.

Temperature is the other knob. With the dose readjusted for the same initial rate, $1/k_m$ is 111 min at 15 °C and 17 min at 35 °C, and the maximum J falls from 59 % to 23 % of $[S]_0$.

There is a limit. Where the SSH does well, J is small. At the low dose and 35 °C, $k_m/k_{ap} = 12$, and J is only 4 to 8 % above v/k_m after the induction period (12 % at the end). But then the maximum J is 18 mg/dL, the size of the tolerance below 100 mg/dL, which is ±15 mg/dL (ISO, 2013). The glucose meter would show the SSH failing with room to spare; showing the SSH succeeding would take many replicates.

8.3.3 The protocol

1. *Materials.* A glucose meter used only for the experiment, which must not be the meter of a person with diabetes, and strips from a single lot; the strip's enzyme is on the package insert. Sugar, dextrose (glucose syrup will not do: it is already at equilibrium), yeast invertase, white vinegar, sodium bicarbonate and sodium carbonate, distilled water. A 0.01 g balance, syringes, small tubes with caps, a thermometer, a timer, a polystyrene box with water at 25 °C as a bath, safety glasses and gloves.

2. *Solutions.* Acetate buffer, 10 times concentrated: 30 mL of vinegar with 4 % acidity (20 mmol of acetic acid) and 0.84 g of bicarbonate (10 mmol), made up to 100 mL after the gas is gone; the pH is close to 4.7. Sucrose 150 mmol.L^{-1}: 5.13 g in 100 mL. Standards of 0, 5, 10, 15 and 20 mmol.L^{-1} glucose in the buffer, made the day before. Sodium carbonate 0.5 mol.L^{-1}: 5.30 g of the anhydrous salt in 100 mL.

3. *One-afternoon test.* Dissolve 0.30 g of dextrose monohydrate in 100 mL of buffer at 25 °C (15 mmol.L^{-1}) and read at 12 times, from 1 to 180 min, and on the next day, in two replicates; with the standards in duplicate, that is 36 strips. The strip will do if the 5 min reading is less than half of the next-day reading. Above 85 %, it reads both anomers, and another brand is needed. Between 50 and 85 %, I do not move on to invertase (Section 8.3.4).

4. *Run.* Mix 2.0 mL of sucrose, 2.0 mL of concentrated buffer and 15 mL of water in the bath and add 1.0 mL of diluted enzyme (20 mL with 15 mmol.L^{-1} sucrose). At each time, touch the strip to a drop of the mixture (direct reading, R) and put 0.40 mL into a small tube with 0.05 mL of carbonate, which is read after at least 30 min (times 1.125, it gives G). In a pilot run, adjust the enzyme dilution to the medium dose: at 20 min, the small tube should show about 95 mg/dL (107 mg/dL in the mixture) and the direct reading 23 mg/dL. The low dose is one fifth of the medium one, and the high dose five times it. At the medium dose, the run has 15 times, up to 240 min, plus the next day: 32 strips, plus 10 for the standards, read directly and with carbonate. Controls: sucrose without enzyme, enzyme without sucrose and a small tube read again on the next day.

5. *Safety.* Work under a teacher's supervision, in a laboratory. The carbonate is alkaline and irritates eyes and skin: wear glasses and gloves. Do not breathe in enzyme powder, which can cause allergies. Nothing in the experiment is to be eaten or drunk. With no blood and no lancet, the used strips go in the ordinary trash.

Some things can go wrong. The meter is made for blood, so the standards are made in the buffer itself and go through the same carbonate as the aliquots. The strip, however, was made for a pH

near 7, and the aliquot with carbonate is near pH 11. If the standards with carbonate read LO or well below the standards read directly, I add 0.10 mL of vinegar to each tube after the 30 min. The pH goes back to about 5, and the factor goes from 1.125 to 1.375. By then the alpha and beta mixture is already at equilibrium, and lowering the pH does not take it out of it. If I use this step, it applies to the standards and the aliquots together. A commercial invertase may have constants different from those of Michaelis and Menten. And I found no data on the selectivity of newer strips: that of the strips of Perles & Volpe (2009) rests on the curve they measured with their own meter. That is why the one-afternoon test comes before everything else. This experiment is my proposal for future work.

8.3.4 Commitments before the test

Before seeing any reading, I fix here what I will test and how I will decide. This is the idea of preregistration, used in many fields of science: the rules are written down first, so that the analysis does not bend to the result. Since the test has not been done, the values below are predictions of the model (`dados/simulacao/invertase_glicosimetro.json`).

1. *Hypothesis.* The glucose the strip does not see yet, $J = G - R$, follows the exact mechanism of Equation (8.12), and the SSH for it, $J_{\mathrm{SSH}} = v/k_m$ (Equation (8.13)), is off by the amount the model predicts.

2. *Condition to go on.* In the one-afternoon test, a glucose oxidase strip should show 12 % of the final value at 5 min (9 to 16 %; Section 8.3.2). I move on to invertase only if the 5 min reading is less than half of the next-day reading.

3. *Main outcome.* The maximum J of the full solution fitted to the medium-dose readings, at 25 °C, as a percentage of $[\mathrm{S}]_0$. The model predicts 39 % (30 to 49 %, 5th and 95th percentiles), at 42 min (33 to 53 min). Secondary outcome: J/J_{SSH} one time $1/k_m$ after the end of the induction period, which the model predicts to be 2.3; I did not compute a range for it.

4. *Decision.* The prediction counts as confirmed if both things hold: the AICc picks the full solution, with both SSH forms ($R = G$ and $R = G - v/k_m$) at least 10 above it, and the maximum J falls between 30 and 49 % of $[\mathrm{S}]_0$. Any other result counts as a refutation of the model with these constants, and I report it that way.

5. *Size.* In the Monte Carlo of the test's power, one medium-dose run, with direct and total readings, was enough for the AICc to pick the full solution in all 100 synthetic sets, with k_m between 0.90 and 1.15 times the true value in 90 % of them; with three runs, between 0.94 and 1.08. I do three medium-dose runs and decide with all three together: 32 strips per run and 10 for the standards per day, plus the 36 of the one-afternoon test. The low and high doses stay out of the decision.

6. *Exclusions.* A reading outside the meter's range (LO or HI on the screen) does not enter the fit. A strip that gives an error message is discarded, and the reading is repeated with another strip, noting the new time. Every exclusion is written down, with the reason.

7. *What does not change.* After seeing the data, I do not change the model, the three forms compared, the Michaelis and Menten constants K_{M}, K_F and K_G, the free fit of k_m, the AICc threshold of 10, the range of 30 to 49 % or the exclusion rules. The analysis is that of `codigo/analisar_glicosimetro.py` as it is today. Any analysis I come up with later goes into the text as an extra, separate from the decision.

9 CHAPTER Conclusion

9.1 Answer to the question

The question of this work, set out in Chapter 1, is when the steady-state hypothesis (SSH), also called the steady-state approximation, is a good approximation, how much it misses, and whether data can show that error, studied in detail in the breakdown of ozone by light and in the synthesis of ammonia. I split that question into four parts.

(a) The criterion is to compare two time scales: the intermediate's lifetime, τ_{I}, and the time over which the reactants change, τ_{reaction}. The SSH holds well when $\tau_{\mathrm{I}} \ll \tau_{\mathrm{reaction}}$ (Equation (3.2), Chapter 3). In the simplest mechanism with one intermediate, $\mathrm{A} \rightarrow \mathrm{I} \rightarrow \mathrm{B}$, with constants k_1 and k_2 for the two steps, that same ratio is the relative error left in [I] after the initial phase (Section 3.3.3). With $k_2 = 10k_1$, the largest error in [B] is 7.7 %, and with $k_2 = k_1$ it climbs to 37 % (Table 6, Chapter 7).

When the first step is reversible, $\mathrm{A} \rightleftharpoons \mathrm{I} \rightarrow \mathrm{P}$, with constant k_{-1} for the return of I to A, the SSH criterion becomes $k_1 \ll k_{-1} + k_2$: I must disappear much faster than it forms. The pre-equilibrium criterion is $k_2 \ll k_1 + k_{-1}$, and not just $k_2 \ll k_{-1}$, as Rae & Berberan-Santos (2004) had already pointed out (Section 7.5). Each criterion is the mirror of the other: swapping k_1 and k_2 leaves the exact product curve unchanged (this is the slow-fast ambiguity (SCHRÖDER *et al.*, 2016)) and takes the SSH law into the pre-equilibrium one (Section 7.5.1). Where $k_1/(k_{-1} + k_2) < 0.1$, the SSH is off by less than 8 % of $[\mathrm{A}]_0$ in [P] (Section 7.5.2). In a band around the diagonal $k_1 = k_2$, which starts at $k_1 = k_2 \approx 0.4\,k_{-1}$, neither works: with $k_1 = k_2 = 10\,k_{-1}$, both are off by 34 %. There the total law remains (Equation (7.9)), the SSH applied without neglecting I in the balance. It contains the SSH and the pre-equilibrium as limiting cases and is off by at most 14 % of $[\mathrm{A}]_0$ (Section 7.5.3).

But the criterion is enough without being required: the SSH can hold even when the criterion is not met. Inside the ozone cell, for example, the steady-state concentration of O, $[\mathrm{O}]_{\mathrm{ss}} = J/k_4$, where J is the photolysis frequency of $\mathrm{O_3}$ and k_4 the constant of $\mathrm{O} + \mathrm{O_3} \longrightarrow 2\,\mathrm{O_2}$, stays constant even as the lifetime of O, τ_{O}, grows while the $\mathrm{O_3}$ runs out, and the SSH for O still holds, with only the small deviation inherited from the start (Section 4.6).

(b) In ozone, inside a cell lit by constant light, the SSH's relative error in $[\mathrm{O_3}]$ is a fixed fraction of $J\tau_{\mathrm{O}}(0)$, where $\tau_{\mathrm{O}}(0)$ is the lifetime of O at the start, and comes to 0.015 %, 0.16 %, or 1.6 %, depending on the initial absorbance (Figure 21, Chapter 4). In the stratosphere, with the full Chapman mechanism, that error falls, at steady state, to about 1.4×10^{-10} at 25 km and 1.4×10^{-9} at 30 km (Figure 28). The calculated ozone comes out about three times larger than the observed value, but the SSH is not the cause of that gap, which comes from the missing catalytic cycles in the pure Chapman mechanism (Section 4.5). In ammonia, the intermediates are species trapped on the surface of the iron catalyst. The simple model in Section 5.4 uses illustrative values of the ratio a/b between the coefficients of $\mathrm{N_2}$ adsorption and of hydrogenation, 0.1, 1, and 10, which give fractions of the surface covered by N, θ_{N}, of 0.084, 0.382, and 0.730 (Example 5.1, Chapter 5). The SSH is far off only while the surface fills up. With fixed gas, at time $3/k_{\mathrm{hid}}$, its relative error in θ_{N} is at most 2.9 %; in the closed vessel, after $5\tau_{\mathrm{N}}$ (five lifetimes of N*) and with $\theta_{\mathrm{N}} > 0.01$, at most 5.5 % (Figure 34). A simulated reactor using the published Dyson and Simon rate law shows that, at start-up, the gas composition reaches 95 % of its final value between 11.4 s and 11.5 s (Section 5.6).

(c) In ozone, the 13 data points from the December 2025 article (RIBEIRO, 2025b) follow an

exponential decay, $e^{-k_{obs}t}$, with k_{obs} = 0.497/s (standard error 0.008/s), with R^2 = 0.998 on the linear scale of the data themselves and 0.979 for the straight line fitted to their logarithm (Table 9, Chapter 4). That is the first-order shape the SSH gives for the two-step mechanism. A curve of a single reactant or a single product, however, is not enough to test the SSH (Section 8.2). If the ozone in the cell had an absorbance above about 0.01 over 1 cm (the article does not give this concentration), the spacing between the points, about 0.7 s, is much larger than τ_O; then the exact solution and the SSH solution give practically the same curve, and what the points pin down is the constant k_{obs} (Section 4.4).

In ammonia, the 13 data points from the same article agree with three of the four models I fitted, with R^2 values of 0.998, 0.999, and 0.9994 (Table 14, Chapter 5). All three describe the delay at the start of the rise well, each with its own cause: a dead time, two steps in series, or a sigmoid. By AICc, the Akaike criterion that weighs the quality of the fit against the number of parameters, the best is the Weibull model, and what separates the three is the small residuals in the middle of the curve (Section 5.5). In synthetic data with independent noise, if the true curve were the two-step one, a Weibull advantage as large as the one the points show would appear in only 5 of 300 sets (Section 5.5.1). The plateau the points reach, in a normalized quantity, is the steady state of the reactor, which is not the same as the SSH for the surface-bound species. Those species follow the SSH during the rise as well, as long as they adjust much faster than the gas changes and the surface holds little matter compared with the gas (Section 7.4).

(d) The data show the SSH error only under certain conditions. In the mechanism A $\rightleftharpoons$ I $\rightarrow$ P, the SSH, the pre-equilibrium, and the total law give the product the same exponential shape, and the [P] points pin down only its constant (Equation (8.4)). That is why a curve of a single reactant or a single product can have $R^2 > 0.99$ and still give, through the SSH formula, a constant off by a factor of 10. In an example with synthetic data, the SSH fits [P] with R^2 = 0.998, and the k_1 read through its formula comes out at 5.0/s, against 50/s in the mechanism that generated the points. With 13 points spaced like those of the article and noise of 0.01, the k_1 read from [P] is off by more than 10 % at 182 of the 361 points of a grid of constants, and 113 of these errors go unnoticed by AICc. With the reactant and the product of the same set fitted together, every error in k_1 above 10 % showed up in both designs I tested. To test the SSH, then, one must measure both, or close the mass balance. With only the product and these 13 times, the fast phase at the start must exceed about 3σ, three times the noise, to show up in half of the experiments (Section 8.2).

The ozone points of the article could not show an SSH error even if there were one, because the O atom adjusts within milliseconds. In the cell assay with synthetic data, the SSH error in k_{obs}, from 0.04 % to 0.3 %, stays below the uncertainty of the slope, about 0.5 %, and the assay does not see it (Section 4.4.1). The ammonia points, on the other hand, do show the delay at the start of the rise. In the fit with two steps in series (M3), the ratio between the two constants is 3.35, well inside the range in which 13 points with this noise show the fast phase, and AICc separates M3 from the simple exponential by 31.9 (Section 8.2). This shows the delay, but not where it comes from, nor that the SSH fails for the surface species.

In the enzyme assays already published, the SSH usually holds, judging by a sample of SABIO-RK data (WITTIG *et al.*, 2018). In that sample, with 1968 pairs of Michaelis constant, K_M, and initial enzyme concentration, $[E]_0$, from 738 papers, the typical value of $\varepsilon_{K_M} = [E]_0/K_M$ is 9.4×10^{-4}. In the pairs whose entry gives the range of substrate concentration, $[S]_0$, only 1.9 % exceed 0.1 in the Segel and Slemrod parameter, $\varepsilon = [E]_0/([S]_0 + K_M)$, computed at the largest $[S]_0$ of the assay. But about 9 in every 100 pairs have $\varepsilon_{K_M} > 0.1$, the range in which the error of the fitted K_M stops being small (Section 8.1). "The SSH holds in the assay" and "the published K_M is accurate" are, then, different questions. These numbers hold for the entries that give the enzyme concentration in mol/L, about 14 % of the sample, and not for all published K_M values.

9.2 What was already known and what this work adds

Table 18. What was already known and what this work adds, result by result.

What was already known	What this work adds
SSH and pre-equilibrium in A ⇌ I → P. The comparison has appeared for decades in chemistry teaching journals (PYUN, 1971; GELLENE, 1995; PEREZ-BENITO, 2017). The correct pre-equilibrium criterion is in Rae & Berberan-Santos (2004), who credit the analysis to Pyun (1971), and the swap of k_1 and k_2 that leaves [P] unchanged is the slow-fast ambiguity (SCHRÖDER *et al.*, 2016). The slow root and the condition $q \ll 1$ are in Eilertsen, Schnell & Walcher (2024), who also note that the two time scales become equal when $k_{-1} \approx 0$ and $k_1 \approx k_2$.	The three-panel map of the error in [P], with the SSH map equal to the mirrored pre-equilibrium map (Figure 42); the band around the diagonal $k_1 = k_2$ where neither works; the measure q (Equation (7.10)) and the error of the total law as a single curve in q, with a ceiling of 0.14 of $[A]_0$ (Figure 43).
Total law. Adding the intermediate to the reactant is the total quasi-steady-state approximation for enzymes (BORGHANS; BOER; SEGEL, 1996), which holds at least approximately for any concentrations (TZAFRIRI, 2003).	Only its application to the general chemistry mechanism (Equation (7.9)) and the map of its error in the same plane (Figure 42c). The idea is not mine.
When the data show the error. Choosing the experiment to tell models apart goes back to Box & Hill (1967) and Atkinson & Fedorov (1975). For enzymes, the SSH holding is not enough to estimate K_M and v_{max} (STROBERG; SCHNELL, 2016), and its bias, seen in simulations, is corrected by the total law (CHOI; REMPALA; KIM, 2017). A high R^2 does not judge a nonlinear model (SPIESS; NEUMEYER, 2010).	The maps of when the points show the SSH error in the two general chemistry mechanisms; the result that, with [A] and [P] fitted together, every error in k_1 above 10 % showed up in both designs I tested; the rule that, with the 13 times of the paper, the fast phase must exceed about 3σ to show up in half of the sets; the application to the 13 + 13 points of the December 2025 paper (RIBEIRO, 2025b) (Section 8.2).
ε in published assays. The enzyme must be dilute (STRAUS; GOLDSTEIN, 1943; SEGEL; SLEMROD, 1989), and the fitted K_M is off when it is not (CHA, 1970; WANG *et al.*, 2025). Some authors state, without data, that assays meet the condition (SCHNELL; MAINI, 2003), and others note that this is not always so (KIM; TYSON, 2020). Halling *et al.* (2018) counted what papers leave out.	The count of ε_{K_M} in a sample of SABIO-RK data (WITTIG *et al.*, 2018) (Section 8.1). I did not find a similar survey, which does not prove that none exists.
Bench. Glucose comes out of sucrose as alpha, and glucose oxidase reads beta much faster; the same authors already warned that this glucose may be oxidized with a delay (KEILIN; HARTREE, 1952a). The glucose meter follows mutarotation (PERLES; VOLPE, 2008; PERLES; VOLPE, 2009).	Using mutarotation as the intermediate of an A → I → B that is exact in the I → B step, with k_m/k_{ap} set by the enzyme dose; the model's prediction, with ranges; the analysis program, tested only on synthetic data (Section 8.3). I did not find this reading in the sources I read. The test has not been done yet.
Tests with synthetic data. Designing the experiment to tell rival models apart (BOX; HILL, 1967; ATKINSON; FEDOROV, 1975).	The tests designed from the 13 + 13 points: the ozone one would pin down the slope of k_{obs} against J to about 0.5 % (Section 4.4.1); the ammonia one, with several flow rates, would tell which of the two steps is the gas exchange (Section 5.5.1).

Source: author's own work.

Many of the ideas in this work were already in the literature. What I did was put those pieces together in calculations, simulations and a proposed test. Table 18 separates, for each result, what I found in the sources I read and what I added; I did not do a complete search, so I do

not claim priority.

9.3 Next steps

From here, I would go down four paths. For three of them, an assay with synthetic data or the proposal in Chapter 8 already shows what the experiment would look like.

The first is to carry out the ozone photolysis in a cell, measuring the lamp's photon flux separately, and compare k_{obs} with $2J$ directly (Section 4.4). The assay with synthetic data in Section 4.4.1 shows the design: J measured by actinometry (measuring the light with a reference chemical reaction) at several light intensities, a background gas near 1 atm, so that $O(^1D)$ is deactivated, with J fitted by a law that already accounts for the O_2 formed, a lamp without the 185 nm mercury line (or a filter) and, after that, a series with O_2 in the gas. If the model and the noise are right, it would pin the slope b of k_{obs} against J, which the mechanism predicts to be 2, to about 0.5 % ($b = 2.001$ in the synthetic data).

The second is to extend the surface model of Example 5.1: let the other trapped species share the sites with N*, keep the reverse of the N_2 adsorption, treat each hydrogenation as a separate step, and use adsorption and hydrogenation constants measured on iron in place of the illustrative a/b ratios (Section 5.4). For this path I did not build a synthetic assay: it needs those constants first.

The third is to track the start-up of a reactor, changing one condition at a time, such as the flow rate or the temperature, and see which parameter of the models in Table 14 follows the change (Section 5.5). The assay with synthetic data in Section 5.5.1 shows that, under the hypotheses I tested, four flow rates with 3 replicates of 40 points each would tell whether the gas exchange is the slow or the fast one of the two steps in series. Their order would stay unanswered, because the curve does not change when the two constants swap places.

The fourth is to measure an intermediate on the bench. With invertase, sugar and a glucose meter, the glucose the strip does not see yet follows, with no approximation, the equation of the intermediate I in the A → I → B mechanism: it forms at the rate of invertase and disappears with the mutarotation constant, k_m. Only the first step is approximate, because invertase is not first order. The amount of enzyme sets the ratio k_m/k_{ap}, which plays the role of k_2/k_1 (Section 8.3). A first test, one afternoon long and with no enzyme, would tell whether my meter will do: according to the model, a glucose oxidase strip should show 12 % of the final value at 5 min. At the medium enzyme dose, the model predicts that this hidden glucose reaches 39 % of the initial sucrose, an SSH error easy to see on the meter. The program that will analyze the readings is ready, but it has only been tested with synthetic data.

In all four, the SSH keeps the role it had in this work: turning a multi-step mechanism into a rate law that can be written and understood without solving a system of differential equations every time. It played that same role for Bodenstein, Chapman, and Underhill in 1913, when there was still no computer to solve that system (Section 3.1). Today there is one, and the SSH law is still useful for another reason: it shows in a single formula how the rate depends on each concentration and each constant, which the table of numbers that comes out of the computer does not show.

When I started this work, I thought the SSH was just a shortcut to dodge the math. Now I know when I can use it and how much it misses, and that is what made me really like chemical kinetics.

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Glossary

The terms below are in alphabetical order. Each one points to the place in the body of the text where it is defined.

Absorbance (Abs**)** Measures how much light a sample takes out of a beam. The larger it is, the less light gets through to the other side. For ozone in a cell, $\text{Abs} = \sigma \ell [\text{O}_3]/\ln 10$ (see Section 4.2.1).

Absorption cross section (σ) The effective area a molecule offers for absorbing a photon (see Section 4.2.1).

Actinometry (chemical) Measuring light by the amount of product of a reaction with a known quantum yield (see Section 4.4.1).

Activation energy (E_a) The energy barrier the reactants must clear; measures how much k changes with temperature (see Section 2.5).

Active site The place on the surface of a catalyst, or of an enzyme, where the reaction happens (see Section 2.2.3).

Adsorption and desorption Adsorption is a molecule binding to the surface of a catalyst; desorption is it leaving, back into the gas (see Section 5.3).

AICc A number that compares fits with different parameter counts, penalizing the ones with more parameters. It is only valid for comparing models fitted to the same points and computed on the same scale (see Section 4.4).

Akaike weight The relative chance that a model is the best among those compared, given what the points show; the weights of all the models add up to 1 (see Section 5.5).

Anomers The two ring forms of glucose, alpha (α) and beta (β), which differ only in the position of the OH group on carbon 1 of the ring (see Section 8.3.1).

Arrhenius equation $k = A\,\mathrm{e}^{-E_a/(RT)}$ (see Section 2.5).

Autocatalysis A reaction in which the product itself speeds up its own formation, as in $\text{A} + \text{P} \longrightarrow 2\,\text{P}$ (see Section 2.2.3).

Average rate The change in concentration divided by the length of the interval, $\Delta[\text{X}]/\Delta t$ (see Section 2.1.1).

Bootstrap A way to find the confidence interval of a result: draw again, with replacement, from the data one has (or from the residuals of the fit), many times, and redo the calculation for each draw (see Sections 4.4 and 8.1).

Catalyst A substance that opens a path with a lower barrier for the same reaction and comes out unchanged at the end, without changing the energy of reactants or products (see Section 2.6).

Chain reaction A mechanism in which a reactive intermediate is remade on every turn, many times over, before the chain breaks (see Section 3.1).

Chapman cycle The four reactions that explain the ozone layer: photolysis of O_2 and of O_3, formation of O_3, and $\text{O} + \text{O}_3 \longrightarrow 2\,\text{O}_2$ (see Section 4.2).

Coefficient of determination (R^2) A number from 0 to 1 that says what fraction of the spread of the points around their mean the fitted curve follows. Close to 1, it says the shape works, not that the model is the right one. To compare models, I use the AICc (see Sections 2.4.3 and 4.4).

Collision theory The reaction happens when two molecules collide with enough energy and the right orientation (see Section 2.6).

Confidence interval (CI) A range computed in a way that, repeated over many samples, contains the true value in a fixed fraction of them: in 95 out of 100, for a 95 % CI (see Sections 4.4 and 8.1).

Conversion (per pass) Conversion is the fraction of the input reactant that reacts in one pass through the reactor (see Section 5.1).

Curve fitting (least squares) Choosing the parameters of a model that keep the residuals small, the difference between each point and the curve (see Section 4.4).

Dimensional analysis Checking that both sides of an equation have the same unit (see Section 2.2.5).

Dissociative adsorption Adsorption in which the molecule breaks apart as it binds to the surface, as in $N_2 \rightarrow 2\,N*$ (see Section 5.3).

EC number The number the Enzyme Commission gives to each enzyme. Its first digit tells the type of reaction, such as EC 2 for the transferases (see Section 8.1).

Elementary step A reaction that happens in a single event; only there can one be certain that the orders equal the stoichiometric coefficients (see Section 2.7).

Enzyme A biological catalyst E that binds the substrate, forms the complex ES, and returns the product without being consumed (see Section 6.1).

Enzyme-substrate complex (ES) The intermediate formed when the substrate binds to the enzyme, $E + S \rightleftharpoons ES$ (see Section 6.1).

Epsilon of K_M (ε_{K_M}) $\varepsilon_{K_M} = [E]_0/K_M$, an upper bound on the ε of Segel and Slemrod, used when $[S]_0$ is missing (see Section 8.1).

Equilibrium constant (K) In a reversible elementary step, the ratio of the forward and reverse rate constants, $K = k_1/k_{-1}$ (see Section 2.7).

Exothermic and endothermic reaction An exothermic reaction releases heat, $\Delta H < 0$; an endothermic one absorbs heat, $\Delta H > 0$ (see Section 2.6).

Factor-of-2 rule (Q_{10}) Rule of thumb: near room temperature, many reactions run about twice as fast for every 10 K rise (see Section 2.5.3).

False alarm When a criterion, such as AICc, rejects the right model on points drawn from that same model (see Section 5.5.1).

Fast phase The intermediate's mark at the start of the curve. In the mechanism $A \rightarrow I \rightarrow B$, it lasts about $1/k_{\text{fast}}$, where k_{fast} is the larger of the two constants (see Sections 7.8 and 8.2).

Free radical A species with an unpaired electron, generally very reactive (see Example 2.4).

Gibbs energy of activation ($\Delta G^{\ddagger}$) The Gibbs energy difference between the transition state and the reactants (see Section 2.6).

Glucose meter The device people with diabetes use to measure their blood glucose. It reads glucose with an enzyme held on a disposable strip (see Section 8.3).

Half-life ($t_{1/2}$) The time for the concentration to fall by half; for first order, $t_{1/2} = \ln 2/k$ (see Section 2.3).

In vitro Done in a test tube, outside the cell (see Section 8.1).

Induction period The early stretch in which the intermediate is still building up and the SSH does not yet hold (see Section 7.2).

Instantaneous rate The slope of the tangent line to the curve [X] against t at one instant, $\mathrm{d}[X]/\mathrm{d}t$ (see Section 2.1.2).

Integrated rate law A formula that gives concentration as a function of time, such as $[A] = [A]_0\, e^{-kt}$ (see Section 2.3).

Intermediate (reaction intermediate) A species that forms in one step and is consumed in another; some can be isolated, but the most reactive ones can only be detected, because they vanish almost as fast as they form (see Section 3.2).

Invertase An enzyme that breaks sucrose, ordinary table sugar, into glucose and fructose (see Section 6.1).

IUPAC The International Union of Pure and Applied Chemistry, which standardizes names and definitions in chemistry (see Section 2.1.3).

Isolation method Putting one reactant in large excess to measure the order in the other reactant alone (see Section 2.4.2).

Le Châtelier's principle A system at equilibrium, when disturbed, shifts in the direction that lessens the disturbance (see Section 5.2).

Maximum rate (v_{max}) The rate when all the enzyme is bound to the substrate, $v_{max} = k_2[E]_0$ (see Section 6.1).

Maxwell-Boltzmann distribution How the energies of the molecules of a gas are spread out at a given temperature (see Section 2.6).

Median The value that leaves half of the data below it and half above (see Section 8.1).

Method of initial rates Measuring the rate at the start of several runs, changing one concentration at a time (see Section 2.4.1).

Michaelis constant (K_M) $K_M = (k_{-1} + k_2)/k_1$: the substrate concentration at which the rate reaches half its maximum, when there is far more substrate than enzyme (see Section 6.1).

Molecularity The number of particles that come together in an elementary step (see Section 2.7).

Mutarotation The slow interconversion between the two forms of glucose, the anomers α and β. It is first order in both directions (see Sections 6.1 and 8.3.1).

Noncentral F distribution A formula from statistics that gives the expected fraction of detections of a test that compares two models, without drawing data sets (see Section 8.2).

Numerical simulation Solving the full system of rate equations on a computer, without approximation (see Section 3.5).

Odd oxygen The sum $[O] + [O_3]$. The reactions that simply trade one O for one O_3 do not change this sum; it is created only by O_2 photolysis and destroyed only by the $O + O_3$ reaction (see Section 4.5).

Pellet effectiveness factor (η) The ratio between the actual reaction rate in the catalyst, with the resistance the gas meets on its way into the pores, and the rate there would be without that resistance. It is dimensionless and lies between 0 and 1 (see Section 5.4).

Photolysis The breaking of a molecule by light absorption, as in $O_3 + h\nu \longrightarrow O_2 + O$ (see Section 4.1).

Photolysis frequency (J) $J = \sigma\Phi F$: the rate constant of a photolysis (see Section 4.2.1).

Plug flow A reactor regime in which all the gas moves forward at the same pace, with no back-mixing (see Section 5.6).

Pre-equilibrium (fast equilibrium) An approximation in which a reversible step stays practically at equilibrium, because the pair equilibrates much faster than the intermediate goes on to products. In the mechanism $A \rightleftharpoons I \rightarrow P$, the condition is $k_2 \ll k_1 + k_{-1}$, and not just $k_2 \ll k_{-1}$ (see Sections 3.3.2 and 7.5).

Pre-exponential factor (A) The number that multiplies the exponential in the Arrhenius equation, with the same unit as k (see Section 2.5).

Progress curve The product or substrate concentration against time in a single reaction, followed until near the end (see Section 8.1).

Promoter A substance that improves the catalyst's performance without being the main catalyst itself (see Section 5.3).

Pseudo-first order When a reactant in excess barely changes concentration, and the rate law starts to behave as first order in the other reactant (see Section 2.4.2).

Quantum yield The fraction of absorbed photons that actually breaks the molecule (see Section 4.2.1).

Quasi-steady state A more accurate name for the SSH. In general, [I] does not stay constant. It slowly tracks the falling reactants (see Section 3.2).

Rate constant (k) The number that multiplies the concentrations in the rate law. It changes with temperature and stays the same when the concentrations change (see Section 2.2.1).

Rate law An equation relating the rate to the concentrations, such as $v = k[\mathrm{A}]^n[\mathrm{B}]^m$ (see Section 2.2.1).

Rate-determining step The slowest step in a mechanism, the one that sets the rate of the whole reaction (see Section 2.7.2).

Reaction coordinate The axis that describes the progress of the bonds breaking and forming, on an energy plot (see Section 2.6).

Reaction mechanism The sequence of elementary steps that, added together, give the overall reaction (see Section 2.7).

Reaction order The exponent of a concentration in the rate law (see Section 2.2.1).

Reaction rate (v) The rate of consumption or of formation divided by the stoichiometric coefficient. It is the same for every species when the volume is constant and no intermediate builds up (see Sections 2.1.3 and 3.5).

Reactor steady state In a reactor with continuous flow, the regime in which all the concentrations become independent of time. It is not the SSH, which is about the intermediate (see Sections 5.6 and 7.1).

Reduced mass (μ) $\mu = m_\mathrm{A} m_\mathrm{B}/(m_\mathrm{A} + m_\mathrm{B})$: the mass that matters in the relative motion of two molecules (see Section 2.6).

Residual The difference between a point and the value the fitted model gives at the same instant (see Section 2.4.3).

Reversible and irreversible reaction A reversible reaction runs both ways; an irreversible one runs only one way under the conditions of the problem (see Section 2.3).

SABIO-RK A public database of enzyme kinetics, built by curators from published papers. Each entry is a rate law taken from a paper, with its parameters and, when the paper reports them, the assay conditions (see Sections 7.4 and 8.1).

Sampling weight How much each entry counts in an estimated fraction: the inverse of the fraction of IDs I read in its range (see Section 8.1).

Saturation The situation in which every site of an enzyme, or of a surface, is occupied and the rate stops rising (see Section 2.2.3).

Segel and Slemrod parameter (ε) $\varepsilon = [\mathrm{E}]_0/([\mathrm{S}]_0 + K_\mathrm{M})$. When $\varepsilon \ll 1$, there is little enzyme, and the SSH and the replacement of [S] by $[\mathrm{S}]_0$ hold (see Section 6.1).

Site-to-molecule ratio (ζ) In a closed vessel, the ratio between the number of surface sites and the number of N_2 molecules at the start (see Section 5.3).

Slow-fast ambiguity From the product curve alone, one cannot tell which of two steps in series is the slow one, because swapping k_1 and k_2 leaves that curve unchanged (see Sections 5.5 and 7.5.1).

Standard error How much a fitted parameter would typically vary if the test were repeated many times with the same noise (see Section 4.4).

Steady-state hypothesis (SSH) The approximation of setting $\mathrm{d[I]}/\mathrm{d}t \approx 0$ for an intermediate I. In the mechanism $\mathrm{A} \rightleftharpoons \mathrm{I} \rightarrow \mathrm{P}$, it holds when $k_1 \ll k_{-1} + k_2$ (see Sections 3.2 and 7.5).

Steric factor ($\mathcal{P}$) $\mathcal{P} = k_{\text{experimental}}/k_{\text{theoretical}}$: the fraction of energetic collisions with the right orientation (see Section 2.6).

Stiff system A system of rate equations with constants very different from one another. A simple integration method would need steps the size of the lifetime of the fastest intermediate. That is why special methods are used, such as Radau (see Section 7.7).

Stoichiometric coefficient The number that multiplies each species in the balanced chemical equation, negative for a reactant and positive for a product (see Section 2.1.3).

Stratosphere The layer of the atmosphere, above the troposphere, where most of the ozone lives (see Chapter 4).

Substrate The molecule S that the enzyme acts on (see Section 6.1).

Surface coverage (θ) The fraction of the surface's sites occupied by one species, a number from 0 to 1 (see Section 5.3).

Surface equilibrium constant (K_{N}) In ammonia synthesis, measures how much N* the surface would hold in equilibrium with the NH_3 and H_2 in the gas, $\theta_{\mathrm{N}}/\theta_* = K_{\mathrm{N}} P_{\mathrm{NH_3}}/P_{\mathrm{H_2}}^{3/2}$ (see Section 5.3).

Synthetic data Model curves plus noise drawn by code, with a fixed seed (the draw comes out the same every time the code runs). They show what an assay could reveal if the model and the noise are right, and they are never measurements (see Section 4.4.1).

Third body (M) Any molecule that carries away the excess energy of a recombination, as in $\mathrm{O} + \mathrm{O_2} + \mathrm{M} \longrightarrow \mathrm{O_3} + \mathrm{M}$ (see Section 4.2.1).

Time scale (characteristic time, τ) $\tau_{\mathrm{I}} = 1/\sum_j k'_j$: the average time a molecule of I lasts before being consumed (see Section 3.2).

Total law In the mechanism $\mathrm{A} \rightleftharpoons \mathrm{I} \rightarrow \mathrm{P}$, the SSH applied without neglecting I in the balance, $v = k_1 k_2([\mathrm{A}]_0 - [\mathrm{P}])/(k_1 + k_{-1} + k_2)$. It contains the SSH and the pre-equilibrium as limiting cases (see Section 7.5.3).

Total quasi-steady-state approximation (tQSSA) The SSH applied to the total substrate, free plus bound in the enzyme-substrate complex, as in Equation (8.2). It is the idea of the total law carried over to enzymes (see Section 7.5.3).

Transition state (activated complex) The highest-energy configuration along the reaction path; it lasts the time of one molecular vibration and cannot be isolated (see Section 2.6).

Appendix

Full derivations · Code · Tables of constants and points

APPENDIX

A Full derivations

This appendix shows, step by step, the algebra that the body of the text only cites or summarizes. I redid all of it in *sympy*, the Python library for symbolic mathematics. The script is at `codigo/conferir_deducoes_apendice_a.py`, and the output, with each of the 110 checks marked "OK", is in `dados/conferencias/deducoes_apendice_a.txt`.

A.1 Integrated rate laws of order 0, 1, 2 and general, half-life, $1/n$-life time and fraction reacted

For a reaction A $\longrightarrow$ products with rate law $v = -\mathrm{d}[\mathrm{A}]/\mathrm{d}t = k[\mathrm{A}]^{\alpha}$, each order α gives a different curve of [A] against time. Separating the variables, $\mathrm{d}[\mathrm{A}]/[\mathrm{A}]^{\alpha} = -k\,\mathrm{d}t$, and integrating from $t = 0$ (when $[\mathrm{A}] = [\mathrm{A}]_0$) to t, orders 0, 1, and 2 give

$$[\mathrm{A}] = [\mathrm{A}]_0 - kt, \qquad (\alpha = 0,\ k \text{ in mol L}^{-1}\,\mathrm{s}^{-1}), \tag{A.1}$$

$$[\mathrm{A}] = [\mathrm{A}]_0\,\mathrm{e}^{-kt}, \qquad (\alpha = 1,\ k \text{ in s}^{-1}), \tag{A.2}$$

$$\frac{1}{[\mathrm{A}]} = \frac{1}{[\mathrm{A}]_0} + kt, \qquad (\alpha = 2,\ k \text{ in L mol}^{-1}\,\mathrm{s}^{-1}). \tag{A.3}$$

In order 0 there is a limit on t, $0 \le t \le [\mathrm{A}]_0/k$, because [A] cannot go negative: the reaction stops once the reactant runs out. In order 1, the ln that shows up in the middle of the integration reads $\ln([\mathrm{A}]/c^{\circ})$, with $c^{\circ} = 1\,\mathrm{mol/L}$. The fraction already reacted is $f = 1 - [\mathrm{A}]/[\mathrm{A}]_0 = 1 - \mathrm{e}^{-kt}$, which does not depend on $[\mathrm{A}]_0$.

Setting $[\mathrm{A}] = [\mathrm{A}]_0/2$ in each of these,

$$t_{1/2} = \frac{[\mathrm{A}]_0}{2k} \quad (\text{order 0}), \qquad t_{1/2} = \frac{\ln 2}{k} \quad (\text{order 1}), \qquad t_{1/2} = \frac{1}{k[\mathrm{A}]_0} \quad (\text{order 2}). \tag{A.4}$$

The order-1 half-life does not depend on $[\mathrm{A}]_0$. That is why radioactive decay, which follows this law, has a fixed half-life no matter how many atoms the sample has. In order 2 it falls as $[\mathrm{A}]_0$ rises, that is, the more concentrated the reactant, the faster it disappears in proportion, because the reaction depends on $[\mathrm{A}]^2$.

General order α (the first version of this work (RIBEIRO, 2025a) had a flipped sign here; I fixed it). For any order $\alpha \neq 1$, the same algebra gives

$$\int_{[\mathrm{A}]_0}^{[\mathrm{A}]} [\mathrm{A}]'^{-\alpha}\,\mathrm{d}[\mathrm{A}]' = -k\int_0^t \mathrm{d}t' \quad\Longrightarrow\quad \frac{[\mathrm{A}]^{1-\alpha} - [\mathrm{A}]_0^{1-\alpha}}{1-\alpha} = -kt. \tag{A.5}$$

Multiplying both sides by $-(1-\alpha) = (\alpha - 1)$,

$$[\mathrm{A}]^{1-\alpha} = [\mathrm{A}]_0^{1-\alpha} + (\alpha - 1)\,k\,t \qquad (\alpha \neq 1). \tag{A.6}$$

Note the factor $(\alpha - 1)$ multiplying kt. This is the form that recovers $[\mathrm{A}] = [\mathrm{A}]_0 - kt$ at $\alpha = 0$ and $1/[\mathrm{A}] = 1/[\mathrm{A}]_0 + kt$ at $\alpha = 2$ (the script checks both substitutions). With the sign flipped, as in the July 2025 version, a term is left over that does not vanish in those cases, and that is how the error shows up in the algebra.

Setting $[\mathrm{A}] = [\mathrm{A}]_0/2$ in Equation (A.6) and solving for t,

$$t_{1/2} = \frac{2^{\alpha-1} - 1}{(\alpha - 1)\, k\, [\mathrm{A}]_0^{\alpha-1}} \qquad (\alpha \neq 1). \tag{A.7}$$

Again, $\alpha = 0$ gives $[\mathrm{A}]_0/(2k)$, $\alpha = 2$ gives $1/(k[\mathrm{A}]_0)$, and the limit $\alpha \to 1$ gives $\ln 2/k$: the same three half-lives seen above. With the sign flipped, this formula gives a negative half-life at $\alpha = 2$.

The $1/n$**-life time** $t_{1/n}$ (for [A] to fall to $[\mathrm{A}]_0/n$) follows the same way:

$$t_{1/n} = \frac{n^{\alpha-1} - 1}{(\alpha - 1)\, k\, [\mathrm{A}]_0^{\alpha-1}} \qquad (\alpha \neq 1). \tag{A.8}$$

With $n = 2$ this returns to Equation (A.7), and the limit $\alpha \to 1$ gives $\ln n/k$. This is Eq. (67) of the first version of this work. Of that version's general-order-α formulas, Eqs. (61) to (63), (67) and (68), only (67) was already right. The other four had the sign flipped, and those are the ones I fixed here. Eqs. (64) to (66), for orders 0, 1 and 2, were right.

A.2 A + B with different initial concentrations

For the reaction A + B ⟶ products, second order, with $v = k[\mathrm{A}][\mathrm{B}]$ and $[\mathrm{A}]_0 \neq [\mathrm{B}]_0$, it is easier to track how much A has already reacted, $x = [\mathrm{A}]_0 - [\mathrm{A}] = [\mathrm{B}]_0 - [\mathrm{B}]$ (the same amount of A and of B disappears, mole for mole):

$$\frac{\mathrm{d}x}{\mathrm{d}t} = k\,([\mathrm{A}]_0 - x)\,([\mathrm{B}]_0 - x)\,. \tag{A.9}$$

To integrate, the product $([\mathrm{A}]_0 - x)([\mathrm{B}]_0 - x)$ moves to the left-hand side, and its inverse splits into partial fractions:

$$\frac{1}{([\mathrm{A}]_0 - x)([\mathrm{B}]_0 - x)} = \frac{1}{[\mathrm{A}]_0 - [\mathrm{B}]_0}\left[\frac{1}{[\mathrm{B}]_0 - x} - \frac{1}{[\mathrm{A}]_0 - x}\right]. \tag{A.10}$$

Integrating from 0 to x and from 0 to t gives directly

$$\ln \frac{[\mathrm{B}]_0([\mathrm{A}]_0 - x)}{[\mathrm{A}]_0([\mathrm{B}]_0 - x)} = ([\mathrm{A}]_0 - [\mathrm{B}]_0)\, kt. \tag{A.11}$$

Replacing $[\mathrm{A}]_0 - x$ with [A] and $[\mathrm{B}]_0 - x$ with [B],

$$\ln \frac{[\mathrm{B}]_0\,[\mathrm{A}]}{[\mathrm{A}]_0\,[\mathrm{B}]} = ([\mathrm{A}]_0 - [\mathrm{B}]_0)\, kt. \tag{A.12}$$

At $t = 0$ the logarithm is $\ln([\mathrm{B}]_0[\mathrm{A}]_0/([\mathrm{A}]_0[\mathrm{B}]_0)) = \ln 1 = 0$, as it should. If $[\mathrm{A}]_0 = [\mathrm{B}]_0$, Equation (A.12) does not hold (the denominator vanishes), and one returns to Equation (A.3) with [A] = [B].

A.3 First-order reversible reaction

Example 9 of the first version of this work treats the equilibrium $\mathrm{A} \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} \mathrm{B}$. Here I start with only A ($[\mathrm{A}]_0$; [B] = 0 at $t = 0$). Since $[\mathrm{A}] + [\mathrm{B}] = [\mathrm{A}]_0$ at all times, the rate equation for A becomes a function of [A] alone:

$$\frac{\mathrm{d}[\mathrm{A}]}{\mathrm{d}t} = -k_1[\mathrm{A}] + k_{-1}\,([\mathrm{A}]_0 - [\mathrm{A}]) = -\,(k_1 + k_{-1})\,[\mathrm{A}] + k_{-1}[\mathrm{A}]_0. \tag{A.13}$$

It is a linear equation with a constant term, and the solution can be checked by direct substitution. As $t \to \infty$ the exponential vanishes, and the ratio of B to A settles at the equilibrium constant K_{eq}. The

gap between [A] and that final value falls as $e^{-t/\tau}$, with characteristic time τ:

$$[A] = \frac{[A]_0}{k_1 + k_{-1}} \left[k_{-1} + k_1 e^{-(k_1+k_{-1})t}\right], \tag{A.14}$$

$$\left.\frac{[B]}{[A]}\right|_{t\to\infty} = \frac{k_1}{k_{-1}} = K_{eq}, \qquad \tau = \frac{1}{k_1 + k_{-1}}. \tag{A.15}$$

The reaction relaxes to equilibrium with a constant that is the *sum* of the two rate constants, $k_1 + k_{-1}$. In the July 2025 version the indices appeared as "forward" and "reverse" without warning. Here I use k_1 and k_{-1}.

A.4 The exact solution of A → I → B, and the case $k_1 = k_2$

Section 3.4 cites the result of this mechanism, $A \xrightarrow{k_1} I \xrightarrow{k_2} B$, in Equations (3.15) and (3.17). The full derivation follows. Starting with only A, the rate equations are

$$\begin{aligned} \frac{d[A]}{dt} &= -k_1[A], \\ \frac{d[I]}{dt} &= k_1[A] - k_2[I], \\ \frac{d[B]}{dt} &= k_2[I]. \end{aligned} \tag{A.16}$$

Step 1. The first equation is order 1 (Equation (A.2)): $[A] = [A]_0 e^{-k_1 t}$.

Step 2. Carrying this into the equation for I and moving the term with I to the left, $d[I]/dt + k_2[I] = k_1[A]_0 e^{-k_1 t}$.

Step 3. This linear equation is solved by multiplying both sides by an **integrating factor**, $e^{k_2 t}$: by the product rule, the derivative of $[I] e^{k_2 t}$ is exactly the left-hand side times that factor. If $k_1 \neq k_2$, integrating from 0 to t (with $[I] = 0$ at $t = 0$),

$$\frac{d}{dt}\left([I] e^{k_2 t}\right) = k_1[A]_0 e^{(k_2-k_1)t} \quad \Longrightarrow \quad [I] = \frac{k_1[A]_0}{k_2 - k_1}\left(e^{-k_1 t} - e^{-k_2 t}\right). \tag{A.17}$$

Then, using the mass balance $[B] = [A]_0 - [A] - [I]$,

$$[B] = [A]_0\left[1 - \frac{k_2 e^{-k_1 t} - k_1 e^{-k_2 t}}{k_2 - k_1}\right]. \tag{A.18}$$

The case $k_1 = k_2 = k$. Here the exponent $(k_2 - k_1)t$ from step 3 (Equation (A.17)) vanishes, and the right-hand side of the integrating-factor equation becomes constant, $k[A]_0$. The integral of a constant is linear in t, and

$$[I] e^{kt} = k[A]_0 t \quad \Longrightarrow \quad [I] = k[A]_0 t e^{-kt}, \qquad [B] = [A]_0\left[1 - (1 + kt) e^{-kt}\right]. \tag{A.19}$$

This is the same result one gets by taking $k_2 \to k_1$ in Equations (A.17) and (A.18) (the script checks both routes, direct and by limit, and they agree).

The maximum of I. Differentiating Equation (A.17) and setting it to zero, $k_1 e^{-k_1 t} = k_2 e^{-k_2 t}$. Taking the logarithm,

$$t_{ind} = \frac{\ln(k_2/k_1)}{k_2 - k_1}, \qquad \frac{[I]_{max}}{[A]_0} = \left(\frac{k_1}{k_2}\right)^{k_2/(k_2-k_1)}. \tag{A.20}$$

The steady-state hypothesis (SSH; Equation (3.15)) replaces the equation for I with $d[I]/dt \approx 0$, which gives $[I]_{SSH} = (k_1/k_2)[A]$ and $[B]_{SSH} = [A]_0 - [A]$. Subtracting Equation (A.18) from that last formula,

$$[B]_{SSH} - [B] = [I]. \tag{A.21}$$

That is, the SSH's absolute error in [B] is exactly what is still stored in [I]. This is why the largest error occurs at t_{ind}, and equals $[\text{I}]_{\max}$ (Equation (A.20)). Dividing Equation (A.17) by $[\text{I}]_{\text{SSH}}$, after the initial period (when $t \gg 1/(k_2 - k_1)$) the ratio $[\text{I}]/[\text{I}]_{\text{SSH}}$ tends to $k_2/(k_2 - k_1)$. This means that the relative error in [I], and also in the rate $k_2[\text{I}]$, tends to $k_1/(k_2 - k_1)$ of the SSH value, or k_1/k_2 of the exact value: for example, 11 % and 10 % when $k_2 = 10k_1$.

A.5 The algebra of the two-step mechanism

Example 3.2, in Section 3.5, uses the mechanism $\text{A} + \text{B} \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} \text{I} + \text{D}$, $\text{I} + \text{B} \xrightarrow{k_2} \text{C} + \text{D}$ (the same "ladder" as Section 3.3, with overall reaction $\text{A} + 2\,\text{B} \longrightarrow \text{C} + 2\,\text{D}$) to show that the SSH contains, as particular cases, both the rate-determining step and the pre-equilibrium. The algebra behind Equation (3.10) goes like this.

The rate of change of the intermediate I adds what the first step forms, in both directions, to what the second step consumes. By the SSH that derivative is approximately zero. Factoring [I] out of the terms that consume it gives $[\text{I}]_{\text{ss}}$:

$$\frac{\text{d}[\text{I}]}{\text{d}t} = k_1[\text{A}][\text{B}] - k_{-1}[\text{I}][\text{D}] - k_2[\text{I}][\text{B}] \approx 0 \quad \Longrightarrow \quad [\text{I}]_{\text{ss}} = \frac{k_1[\text{A}][\text{B}]}{k_{-1}[\text{D}] + k_2[\text{B}]}. \tag{A.22}$$

The reaction rate is that of the second step, $v = k_2[\text{I}]_{\text{ss}}[\text{B}]$:

$$v = \frac{k_1 k_2\,[\text{A}][\text{B}]^2}{k_{-1}[\text{D}] + k_2[\text{B}]}. \tag{A.23}$$

The denominator of this fraction holds the two possible fates of the intermediate: going back to the reactants ($k_{-1}[\text{D}]$) or moving on to the products ($k_2[\text{B}]$). These are what decide which simple case shows up. If $k_2[\text{B}] \gg k_{-1}[\text{D}]$ (I almost always moves on to C), the term $k_{-1}[\text{D}]$ drops out of the denominator and $v \approx k_1[\text{A}][\text{B}]$, that is, step 1 is the rate-determining one. If $k_{-1}[\text{D}] \gg k_2[\text{B}]$ (I almost always returns to the reactants), the term $k_2[\text{B}]$ drops out and $v \approx (k_1k_2/k_{-1})[\text{A}][\text{B}]^2/[\text{D}]$, which is the result for step 1 in pre-equilibrium with step 2 rate-determining. Equation (A.23) does not pick one case, because it holds in both limits and throughout the interval between them. That is what Example 20 and Observations 8 and 9 in the first version of this work were meant to show. With the SSH there is no need to know beforehand which step is the slow one.

A.6 Pre-equilibrium with the second step also reversible

The first version of this work had two sections with the same title, "Non-elementary reaction with the intermediate in the rate-determining step (pre-equilibrium)". The first is already in Section 3.3.2, with step 1 fast and reversible and step 2 slow and irreversible. Here is the second one, in which step 2, despite being the slower one, is also treated as reversible and close to its own equilibrium, $\text{A} + \text{B} \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} \text{I} + \text{D}$, $\text{I} + \text{B} \underset{k_{-2}}{\overset{k_2}{\rightleftharpoons}} \text{C} + \text{D}$.

Each step near equilibrium gives a constant. Multiplying the two, the [I] in one cancels the one in the other:

$$K_1 = \frac{[\text{I}][\text{D}]}{[\text{A}][\text{B}]}, \qquad K_2 = \frac{[\text{C}][\text{D}]}{[\text{I}][\text{B}]}, \tag{A.24}$$

$$K_1K_2 = \frac{[\text{I}][\text{D}]}{[\text{A}][\text{B}]} \cdot \frac{[\text{C}][\text{D}]}{[\text{I}][\text{B}]} = \frac{[\text{C}][\text{D}]^2}{[\text{A}][\text{B}]^2} = K_{\text{eq}}. \tag{A.25}$$

The equilibrium constant of the overall reaction, $\text{A} + 2\,\text{B} \rightleftharpoons \text{C} + 2\,\text{D}$, is the product of the two step constants: $K_{\text{eq}} = K_1K_2$. Whenever a mechanism is a sequence of steps in pre-equilibrium, the overall

constant is assembled this way, by multiplying the step constants, without solving any differential equation.

A.7 Michaelis-Menten and Briggs-Haldane

Section 6.1 presents the Michaelis-Menten equation via the SSH, which is the derivation by Briggs and Haldane (1925). Here are the two derivations side by side, to show where they part ways. The mechanism is $\mathrm{E} + \mathrm{S} \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} \mathrm{ES} \xrightarrow{k_2} \mathrm{E} + \mathrm{P}$, with $[\mathrm{E}]_0 = [\mathrm{E}] + [\mathrm{ES}]$ (free enzyme plus enzyme bound to substrate).

Michaelis and Menten (1913): pre-equilibrium. They assumed the first step is always at equilibrium, which gives the dissociation constant $K_S = k_{-1}/k_1 = [\mathrm{E}][\mathrm{S}]/[\mathrm{ES}]$. Substituting $[\mathrm{E}] = [\mathrm{E}]_0 - [\mathrm{ES}]$ and solving for [ES],

$$[\mathrm{ES}] = \frac{[\mathrm{E}]_0[\mathrm{S}]}{K_S + [\mathrm{S}]} \quad \Longrightarrow \quad v = k_2[\mathrm{ES}] = \frac{v_{\max}[\mathrm{S}]}{K_S + [\mathrm{S}]}, \qquad v_{\max} = k_2[\mathrm{E}]_0. \tag{A.26}$$

Briggs and Haldane (1925): the SSH. Instead of assuming equilibrium, they assumed only that $\mathrm{d}[\mathrm{ES}]/\mathrm{d}t \approx 0$:

$$k_1\left([\mathrm{E}]_0 - [\mathrm{ES}]\right)[\mathrm{S}] - k_{-1}[\mathrm{ES}] - k_2[\mathrm{ES}] \approx 0 \quad \Longrightarrow \quad [\mathrm{ES}] = \frac{[\mathrm{E}]_0[\mathrm{S}]}{K_\mathrm{M} + [\mathrm{S}]}, \tag{A.27}$$

with $K_\mathrm{M} = (k_{-1} + k_2)/k_1$. The rate law has the same form as Equation (A.26), with K_M in place of K_S, and the two constants are linked by $K_\mathrm{M} = K_S + k_2/k_1$. The pre-equilibrium is the particular case of the SSH in which $k_2 \ll k_{-1}$, and then $K_\mathrm{M} \to K_S$. When k_2 is not much smaller than k_{-1}, K_M stops measuring only the enzyme's affinity for the substrate, because it also carries the rate of product formation.

A.8 The ammonia surface

Section 5.4 presents the simple coverage model, $a(1-\theta_\mathrm{N})^2 = b\theta_\mathrm{N}$ (Equation (5.12)), and the form near equilibrium (Equation (5.13)). Here is the algebra behind each.

The physical root. The equation $a(1-\theta)^2 = b\theta$ is quadratic in θ, and its roots are

$$\theta_\pm = \frac{(2a + b) \pm \sqrt{b^2 + 4ab}}{2a}. \tag{A.28}$$

The root with the + sign does not work, because $\theta_+ - 1 = \left(b + \sqrt{b^2 + 4ab}\right)/(2a)$ is positive whenever a and b are positive; so $\theta_+ > 1$, and a coverage cannot exceed 1. What is left is the root with the − sign. With $r = a/b$,

$$\theta_\mathrm{N} = \frac{2r + 1 - \sqrt{1 + 4r}}{2r}. \tag{A.29}$$

For $r \to 0$ (fast desorption compared with adsorption), $\theta_\mathrm{N} \approx r$, that is, the coverage grows in a straight line with a/b. For $r \to \infty$ (adsorption much faster), $1 - \theta_\mathrm{N} \approx 1/\sqrt{r} \to 0$, and the surface becomes almost entirely covered with N*. The constant governing the return to the steady state after a small disturbance is $2a(1-\theta_\mathrm{N}) + b$; substituting θ_N, it equals $\sqrt{b^2 + 4ab}$, and the characteristic time is the inverse of that.

Temkin and Pyzhev. Their empirical rate law is

$$v = k_+[\mathrm{N}_2]\left(\frac{[\mathrm{H}_2]^3}{[\mathrm{NH}_3]^2}\right)^\alpha - k_-\left(\frac{[\mathrm{NH}_3]^2}{[\mathrm{H}_2]^3}\right)^{1-\alpha}, \qquad 0 < \alpha < 1. \tag{A.30}$$

It passes a simple test. At equilibrium, $v = 0$, and this gives k_+/k_- equal to the equilibrium constant for any α between 0 and 1:

$$v = 0 \quad \Longrightarrow \quad \frac{k_+}{k_-} = \frac{[NH_3]^2}{[N_2][H_2]^3} \qquad \text{for any } \alpha. \tag{A.31}$$

That is why the Temkin-Pyzhev equation, a purely empirical law, is consistent with thermodynamics, and stood for decades even before a widely accepted surface mechanism existed.

A.9 Mars and van Krevelen

Section 6.2 uses the Mars and van Krevelen (1954) redox mechanism for SO_3: SO_2 takes an oxygen from the oxidized catalyst, and O_2 gives that oxygen back afterward. With θ_{ox} the fraction of catalyst sites that are oxidized,

$$SO_2 + O* \longrightarrow SO_3 + * \qquad (k_{red}), \tag{A.32}$$

$$O_2 + 2* \longrightarrow 2\,O* \qquad (k_{ox}). \tag{A.33}$$

The first step consumes one oxidized site, and the second restores two. As in Section 6.2, I take the reoxidation rate proportional to the fraction of empty sites, $1 - \theta_{ox}$, and not to its square, which is what the rule for an elementary step (Equation (2.25)) would give for two sites. This is an assumption of the model. By the SSH in θ_{ox},

$$2k_{ox}[O_2]\,(1 - \theta_{ox}) - k_{red}[SO_2]\,\theta_{ox} \approx 0 \quad \Longrightarrow \quad \theta_{ox} = \frac{2k_{ox}[O_2]}{k_{red}[SO_2] + 2k_{ox}[O_2]}. \tag{A.34}$$

With the rate of SO_3 formation given by $v = k_{red}[SO_2]\,\theta_{ox}$,

$$v = \frac{2\,k_{red}k_{ox}\,[SO_2][O_2]}{k_{red}[SO_2] + 2k_{ox}[O_2]}. \tag{A.35}$$

There are two simple limits. If reoxidation is fast ($k_{ox}[O_2] \gg k_{red}[SO_2]$), $v \to k_{red}[SO_2]$, zero order in O_2, because the catalyst almost never sits reduced waiting for oxygen. If reduction is the fast one, $v \to 2k_{ox}[O_2]$. Both are legitimate cases of the same formula.

A.10 H_2 + Br_2

Example 2.4 cites the rate law of the reaction $H_2 + Br_2 \longrightarrow 2\,HBr$ without deriving it. It comes from a chain mechanism, with H and Br radicals as intermediates (HERZFELD, 1919):

$$Br_2 \longrightarrow 2\,Br \qquad (k_i), \tag{A.36}$$

$$Br + H_2 \longrightarrow HBr + H \qquad (k_1), \tag{A.37}$$

$$H + Br_2 \longrightarrow HBr + Br \qquad (k_2), \tag{A.38}$$

$$H + HBr \longrightarrow H_2 + Br \qquad (k_{-1}), \tag{A.39}$$

$$2\,Br \longrightarrow Br_2 \qquad (k_t). \tag{A.40}$$

The second and third steps form HBr; the fourth consumes it. The SSH in H gives $[H]_{ss}$:

$$k_1[Br][H_2] - k_2[H][Br_2] - k_{-1}[H][HBr] \approx 0 \quad \Longrightarrow \quad [H]_{ss} = \frac{k_1[Br][H_2]}{k_2[Br_2] + k_{-1}[HBr]}. \tag{A.41}$$

Adding the SSH equations for Br and for H, the H terms cancel, because they appear with opposite signs in the two, leaving only the balance between initiation and termination, $2k_i[\mathrm{Br_2}] = 2k_t[\mathrm{Br}]^2$:

$$[\mathrm{Br}]_{\mathrm{ss}} = \sqrt{\frac{k_i[\mathrm{Br_2}]}{k_t}}. \tag{A.42}$$

The quantity $\mathrm{d[HBr]}/\mathrm{d}t$ adds what steps 2 and 3 produce minus what step 4 consumes: $\mathrm{d[HBr]}/\mathrm{d}t = k_1[\mathrm{Br}][\mathrm{H_2}] + k_2[\mathrm{H}][\mathrm{Br_2}] - k_{-1}[\mathrm{H}][\mathrm{HBr}]$. Using Equation (A.41) to replace $k_2[\mathrm{Br_2}] + k_{-1}[\mathrm{HBr}]$ with $k_1[\mathrm{Br}][\mathrm{H_2}]/[\mathrm{H}]$ in the denominator, simplifying, substituting Equation (A.42), and dividing everything by $[\mathrm{Br_2}]$,

$$\frac{\mathrm{d[HBr]}}{\mathrm{d}t} = \frac{2\,k_1 k_2\,[\mathrm{Br}][\mathrm{H_2}][\mathrm{Br_2}]}{k_2[\mathrm{Br_2}] + k_{-1}[\mathrm{HBr}]} = \frac{k\,[\mathrm{H_2}][\mathrm{Br_2}]^{1/2}}{1 + k'\,[\mathrm{HBr}]/[\mathrm{Br_2}]} = \frac{k\,[\mathrm{H_2}][\mathrm{Br_2}]^{3/2}}{[\mathrm{Br_2}] + k'[\mathrm{HBr}]}, \tag{A.43}$$

$$k = 2k_1\sqrt{\frac{k_i}{k_t}}, \qquad k' = \frac{k_{-1}}{k_2}. \tag{A.44}$$

Equation (A.43) gives $\mathrm{d[HBr]}/\mathrm{d}t$. Since HBr has coefficient 2 in the overall equation, the reaction rate is $v = (1/2)\,\mathrm{d[HBr]}/\mathrm{d}t$. Written for v, the same law becomes

$$v = k_1\sqrt{\frac{k_i}{k_t}}\,\frac{[\mathrm{H_2}][\mathrm{Br_2}]^{1/2}}{1 + k'\,[\mathrm{HBr}]/[\mathrm{Br_2}]}, \tag{A.45}$$

with $k = k_1\sqrt{k_i/k_t}$, half the constant in Equation (A.44). It is the same form as Equation (6.30) in Chapter 6, where $k_i, k_1, k_2, k_{-1}, k_t$ appear renamed k_1, k_2, k_3, k_4, k_5, and also the form in Example 2.4, which gives $\mathrm{d[HBr]}/\mathrm{d}t$ and states there that $v = (1/2)\,\mathrm{d[HBr]}/\mathrm{d}t$. The reaction is order 1 in $\mathrm{H_2}$ and $1/2$ in $\mathrm{Br_2}$ at the start, and the product enters later to slow the reaction down. Bodenstein & Lind (1907) measured the rate law; Herzfeld (1919) proposed the chain that explains it.

A.11 Collision theory, the Eyring equation, and Arrhenius from two points

Collisions. Section 2.6 presents the rate constant of an elementary bimolecular reaction in the form $k = \mathcal{P}\,\sigma_{\mathrm{col}}\,\bar{u}_{\mathrm{rel}}\,N_{\mathrm{A}}\,\mathrm{e}^{-E_0/RT}$, with $\bar{u}_{\mathrm{rel}} = (8k_{\mathrm{B}}T/\pi\mu)^{1/2}$, where $\sigma_{\mathrm{col}} = \pi d_{\mathrm{AB}}^2$ is the collision cross section (d_{AB} is the sum of the radii), μ is the reduced mass, E_0 is the minimum collision energy needed to react, and $\mathcal{P}$ is the steric factor (the fraction of collisions with the right orientation). The Arrhenius activation energy, $E_{\mathrm{a}} = -R\,\mathrm{d}\ln k/\mathrm{d}(1/T)$, comes from writing $\ln k = \text{const.} + (1/2)\ln T - E_0/(RT)$ and differentiating with respect to $1/T$:

$$k = \mathcal{P}\,\sigma_{\mathrm{col}}\left(\frac{8k_{\mathrm{B}}T}{\pi\mu}\right)^{1/2} N_{\mathrm{A}}\,\mathrm{e}^{-E_0/RT}, \tag{A.46}$$

$$E_{\mathrm{a}} = E_0 + \frac{1}{2}RT. \tag{A.47}$$

The measured activation energy, therefore, is not the minimum collision energy. It comes out a bit larger, by $RT/2$ (about 1.2 kJ/mol at 298 K), because of the $\sqrt{T}$ in the mean relative speed $\bar{u}_{\mathrm{rel}}$.

Eyring. The transition-state theory equation is $k = \kappa\,(k_{\mathrm{B}}T/h)\,\mathrm{e}^{-\Delta G^{\ddagger}/RT}$, with $\Delta G^{\ddagger} = \Delta H^{\ddagger} - T\Delta S^{\ddagger}$ and κ the transmission coefficient (close to 1). Differentiating $\ln k$ with respect to $1/T$ the same way, $E_{\mathrm{a}} = \Delta H^{\ddagger} + RT$ for a unimolecular reaction or one in solution. In the gas phase, for a bimolecular reaction, the constant written in concentration units gains a factor RT/p° (where $p^{\circ} = 1\,\mathrm{bar}$ is the standard pressure), which comes from the change of standard state (from pressure to concentration), and the same calculation gives $E_{\mathrm{a}} = \Delta H^{\ddagger} + 2RT$. The gap between E_{a} and the barrier $\Delta H^{\ddagger}$ is again of order RT.

Arrhenius from two points. With k measured at two temperatures, k_1 at T_1 and k_2 at T_2, writing $k = A\,\mathrm{e}^{-E_a/RT}$ at both points and dividing cancels the factor A:

$$\ln\frac{k_2}{k_1} = -\frac{E_a}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right) \quad\Longrightarrow\quad E_a = -\frac{R\ln(k_2/k_1)}{1/T_2 - 1/T_1}. \tag{A.48}$$

This is the same calculation as in Example 2.6, for temperature instead of concentration.

A.12 Units of the rate constant

For a rate law $v = k[\mathrm{A}]^n$, the unit of k follows from solving $k = v/[\mathrm{A}]^n$. Since v has units of concentration over time,

$$[k] = \frac{[\text{concentration}]}{[\text{time}]\cdot[\text{concentration}]^n} = [\text{concentration}]^{1-n}\,[\text{time}]^{-1}. \tag{A.49}$$

Table 19 lists the result for $n = 0$, 1, 2, and 3, in SI (mol m^{-3}, s) and in the practical unit (mol L^{-1}, s).

Table 19. Unit of k according to the reaction order n (Equation (A.49)).

n	Practical unit	SI unit
0	mol/(L s)	$10^3 \times$ mol/(m^3 s)
1	1/s	1/s
2	L/(mol s)	$10^{-3} \times$ m^3/(mol s)
3	L^2/(mol^2 s)	$10^{-6} \times$ m^6/(mol^2 s)

Source: author's own work, from Equation (A.49).

Two useful conversions. For an ideal gas, $P = cRT$, and $k_p = k_c\,(RT)^{1-n}$ links the same constant written in pressure and in concentration. In atmospheric chemistry, a second-order constant in cm^3 molecule^{-1} s^{-1} becomes L/(mol s) on multiplying by $N_A/1000$.

A case that needs care: Eq. (25) of the July 2025 version. It gives the k of the $H_2 + Br_2$ rate law (Equation (A.43)) without the $[H_2]$ in the denominator, which changes the unit. Solving for k correctly,

$$[k] = \frac{[v]\,([\mathrm{Br_2}] + k'\,[\mathrm{HBr}])}{[\mathrm{H_2}]\,[\mathrm{Br_2}]^{3/2}} = (\text{mol L}^{-1})^{-1/2}\,\text{s}^{-1} = (\text{mol m}^{-3})^{-1/2}\,\text{s}^{-1}\ (\text{SI}). \tag{A.50}$$

Without the $[H_2]$ in the denominator the calculation gives another unit, a wrong one, and that is how I noticed that the factor was missing from the printed equation.

A.13 Chapman and the general form with O_2

Section 4.3 derives the SSH in the two-step mechanism for photolysis in a cell, and Section 4.5 extends that to Chapman's mechanism, with four reactions. Below is the algebra common to both versions.

Two steps. With $\mathrm{O_3} \xrightarrow{J} \mathrm{O_2} + \mathrm{O}$ and $\mathrm{O} + \mathrm{O_3} \xrightarrow{k_4} 2\,\mathrm{O_2}$, the SSH in O ($J[\mathrm{O_3}] - k_4[\mathrm{O}][\mathrm{O_3}] \approx 0$) gives $[\mathrm{O}]_{ss} = J/k_4$. In the equation for O_3, $\mathrm{d}[\mathrm{O_3}]/\mathrm{d}t = -J[\mathrm{O_3}] - k_4[\mathrm{O}][\mathrm{O_3}]$, the two terms become equal and their sum gives $-2J[\mathrm{O_3}]$ (Equation (4.11)).

With O_2 present. Adding $\mathrm{O} + \mathrm{O_2} + \mathrm{M} \xrightarrow{k_2} \mathrm{O_3} + \mathrm{M}$, which reforms ozone, the SSH in O gives $[\mathrm{O}]_{ss}$. Substituting into the equation for O_3 and combining terms (the numerator becomes $2k_4[\mathrm{O_3}]$), one

gets Equation (4.13) of Chapter 4:

$$[\mathrm{O}]_{\mathrm{ss}} = \frac{J[\mathrm{O_3}]}{k_4[\mathrm{O_3}] + k_2[\mathrm{O_2}][\mathrm{M}]}, \tag{A.51}$$

$$\frac{\mathrm{d}[\mathrm{O_3}]}{\mathrm{d}t} = -k_{\mathrm{obs}}[\mathrm{O_3}], \qquad k_{\mathrm{obs}} = \frac{2Jk_4[\mathrm{O_3}]}{k_4[\mathrm{O_3}] + k_2[\mathrm{O_2}][\mathrm{M}]}. \tag{A.52}$$

Without $\mathrm{O_2}$ ($[\mathrm{O_2}] \to 0$), this fraction goes to $2J$.

Chapman. With Chapman's four reactions (Equations (4.1) to (4.4)), odd oxygen $[\mathrm{Ox}] = [\mathrm{O}] + [\mathrm{O_3}]$ is created only by $\mathrm{O_2}$ photolysis and destroyed only by the $\mathrm{O} + \mathrm{O_3}$ reaction, because the terms with k_2 and J_3 cancel when the two rate equations are added. The SSH in O, dropping the terms that are much smaller than the others, gives $[\mathrm{O}]_{\mathrm{ss}}/[\mathrm{O_3}] = J_3/(k_2[\mathrm{O_2}][\mathrm{M}])$. Equating production and loss of $[\mathrm{Ox}]$, $2J_1[\mathrm{O_2}] = 2k_4[\mathrm{O}]_{\mathrm{ss}}[\mathrm{O_3}]$, and substituting $[\mathrm{O}]_{\mathrm{ss}}$, gives Equation (4.21) of Chapter 4, a steady-state concentration rather than a decay law:

$$[\mathrm{O_3}]_{\mathrm{ss}} = [\mathrm{O_2}]\sqrt{\frac{J_1 k_2[\mathrm{M}]}{J_3 k_4}}. \tag{A.53}$$

A.14 The total law: why the error depends only on q

Section 7.5.3 says three things about the total law (Equation (7.9)) in the mechanism $\mathrm{A} \rightleftharpoons \mathrm{I} \to \mathrm{P}$: its largest error in [P] depends only on q (Equation (7.10)), it is about $q/4$ when q is small, and it never exceeds 0.140 of $[\mathrm{A}]_0$. Here is the calculation. I measure concentrations as a fraction of $[\mathrm{A}]_0$, that is, $[\mathrm{A}]_0 = 1$, and start with $[\mathrm{I}]_0 = [\mathrm{P}]_0 = 0$.

Step 1: the two curves. The exact [P] is Equation (7.7), with λ_+ and λ_- the roots of Equation (7.8). In a quadratic equation, the sum of the roots is minus the coefficient of the λ term, and the product is the constant term: $\lambda_+ + \lambda_- = k_1 + k_{-1} + k_2$ and $\lambda_+\lambda_- = k_1 k_2$. The total law is first order in $1 - [\mathrm{P}]$, and it integrates like Equation (A.2):

$$[\mathrm{P}]_{\mathrm{tot}} = 1 - \mathrm{e}^{-kt}, \qquad k = \frac{k_1 k_2}{k_1 + k_{-1} + k_2} = \frac{\lambda_+\lambda_-}{\lambda_+ + \lambda_-}. \tag{A.54}$$

Step 2: change of variables. I define the ratio of the roots and a dimensionless time,

$$r = \frac{\lambda_-}{\lambda_+} \quad (0 < r \le 1), \qquad u = \lambda_- t. \tag{A.55}$$

Then $\lambda_+ t = u/r$ and $kt = u/(1 + r)$, and the difference between the two curves becomes

$$D = [\mathrm{P}]_{\mathrm{tot}} - [\mathrm{P}] = \frac{\mathrm{e}^{-u} - r\,\mathrm{e}^{-u/r}}{1 - r} - \mathrm{e}^{-u/(1+r)}. \tag{A.56}$$

The three constants are gone, and only r and u are left. The largest error over the reaction, $F = \max_t |D|$, is the maximum of Equation (A.56) over u, so it depends only on r. What is left is to link r to q. Dividing top and bottom by λ_+^2,

$$q = \frac{4\lambda_+\lambda_-}{(\lambda_+ + \lambda_-)^2} = \frac{4r}{(1 + r)^2}, \qquad \frac{\mathrm{d}q}{\mathrm{d}r} = \frac{4(1 - r)}{(1 + r)^3}. \tag{A.57}$$

The derivative is positive for $r < 1$. So q always grows with r, each q matches a single r, and F depends only on q. The case $r = 1$ (a double root, $q = 1$) only happens with $k_{-1} = 0$ and $k_1 = k_2$, because $(k_1 + k_{-1} + k_2)^2 - 4k_1k_2 = (k_1 - k_2)^2 + k_{-1}(k_{-1} + 2k_1 + 2k_2)$.

Step 3: small q. With $r \ll 1$, the fast root λ_+ is much larger than the slow one. After a few $1/\lambda_+$ (the fast transient), $e^{-u/r}$ becomes negligible. Expanding what is left of Equation (A.56) in r (I redid this in sympy),

$$D \approx r\,(1-u)\,e^{-u} + O(r^2). \tag{A.58}$$

Right after the transient, u is still almost 0, and $D \approx r$. Then D decreases, changes sign at $u = 1$ and has a minimum of $-r/e^2 \approx -0.14\,r$ at $u = 2$, much smaller in size. At the start, $t = 0$, Equation (A.56) gives $D = 0$. In between, while $e^{-u/r}$ dies out, D rises from 0 to close to r. So the largest error is $F \approx r$, and from Equation (A.57), $q \approx 4r$ when r is small:

$$F \approx r \approx \frac{q}{4} \qquad (q \ll 1). \tag{A.59}$$

This error is the debt of the induction period, which the total law does not describe. At the start, the exact [P] barely leaves zero, because I is still building up, while the total law already starts at rate k. After the transient, the slow part of Equation (7.7) is $1 - e^{-\lambda_-(t-t_a)}$, the same curve delayed by $t_a = -\ln(1-r)/\lambda_- \approx 1/\lambda_+$. The debt is about $k/\lambda_+ = r/(1+r) \approx r$. In the simulation (`codigo/validade_simetria.py`), $F/(q/4)$ is 0.99999 at $q = 10^{-6}$, 0.996 at 0.001, 0.976 at 0.01 and 0.871 at 0.1 (`dados/simulacao/validade_simetria.json`). The script recomputes the maximum of Equation (A.56) at these four values of q, and all four match. With larger q, the next terms in r matter, and F falls below $q/4$.

Step 4: $q = 1$. With $k_{-1} = 0$ and $k_1 = k_2 = \lambda$, the mechanism is A $\to$ I $\to$ P with equal constants, and the exact [P] is the one in Equation (A.19) (Appendix A.4), $[\mathrm{P}] = 1 - (1+\lambda t)\,e^{-\lambda t}$. In the total law, $k = \lambda^2/(2\lambda) = \lambda/2$, and $[\mathrm{P}]_{\mathrm{tot}} = 1 - e^{-\lambda t/2}$. With $x = \lambda t$ (the same as taking $r \to 1$ in Equation (A.56)),

$$D = (1+x)\,e^{-x} - e^{-x/2}, \qquad \frac{dD}{dx} = e^{-x}\left(\frac{e^{x/2}}{2} - x\right) = 0 \quad \Longleftrightarrow \quad e^{x/2} = 2x. \tag{A.60}$$

The equation $e^{x/2} = 2x$ has x both inside and outside the exponential and has no closed-form solution; I solved it numerically in sympy (the `nsolve` function). It has two roots: $x = 0.7148$, the maximum, and $x = 4.307$, a minimum. At the maximum, replacing $e^{-x/2}$ by $2x\,e^{-x}$, $D = (1-x)\,e^{-x} = 0.1395$; at the minimum, $D = -0.0446$. So $F = 0.1395$, at $\lambda t = 0.7148$, or $kt = 0.3574$. The simulation gives the same 0.13954 and 0.3574 in the limit $q = 1$ (`validade_simetria.json`).

Step 5: F grows with q. I did not prove this for every q. I checked it on the 400-point curve in `dados/simulacao/validade_lei_total_curva.csv`, from $q = 10^{-6}$ to 1: F rises at every point, from 2.5×10^{-7} to 0.1395. That is why 0.1395 is the largest error of the total law in this mechanism, and it only happens at $q = 1$. The slow root close to k and the condition $q \ll 1$ are in Eilertsen, Schnell & Walcher (2024); the calculation here is the one for the error in [P].

B APPENDIX Code

Section 1.4 explains, in half a page, the five sources of every number in this work: verified literature, the data from the December 2025 article (RIBEIRO, 2025b), the SABIO-RK data (WITTIG *et al.*, 2018), simulation, and synthetic data. This appendix is the practical side of that: it says which script generates each figure and each table, with what command, from what file, and to what file. All the code is in `codigo/`, it is plain Python, and it depends on nothing outside the standard library and the packages named in Appendix B.2.

B.1 Figures and tables generated by script

The five tables in this section list, for each script, the figures and tables it produces, the command that runs it, what it reads, and what it writes. A figure marked "author's own drawing" comes from no calculation at all: it is a qualitative diagram, with no computed number. The rest come from simulation (numerical integration or curve fitting), from synthetic data (a model curve with drawn noise), from the SABIO-RK data, or from a literature file. The tables of Chapters 4, 5 and 7 that come from no script are listed too, with a dash in the script column.

Every figure script accepts `--idioma en`. In that case the figure comes out in English, in `figuras/en/`, and the files in `dados/` come out identical to the Portuguese run. Table 20 starts with the scripts of Chapters 1 to 3, 6 and 7.

Table 20. Scripts in `codigo/` and the figures and tables they generate (part 1 of 5: chapters 1, 2, 3, 6, and 7).

Figure(s)/table	Script and command	Inputs	Outputs
Figure 1	`figura_wilhelmy.py` `python codigo/figura_wilhelmy.py`	no file; sketch and illustrative curve, values chosen for the drawing in the script itself	`figuras/intro_wilhelmy.pdf|png`
Figures 2 to 15 and Table 4	`figuras_fundamentos.py` `python codigo/figuras_fundamentos.py`	no file; illustrative values and, in Figure 7b, constants from `codigo/constantes.py` (Burkholder *et al.*, 2019)	`figuras/fund_*.pdf|.png` (14 files); `dados/simulacao/fund_contas.json`, `fund_q10.csv`, `fund_maxwell_boltzmann.csv`
Figures 16, 17 and 19	`figuras_ssh.py` `python codigo/figuras_ssh.py`	no file; illustrative constants (k_1, k_{-1}, k_2, $[A]_0$, $[B]_0$) set inside the script itself	`figuras/ssh_perfil_duas_etapas.pdf|png`, `ssh_fabrica_invertida.pdf|png`, `ssh_exemplo20.pdf|png`; `dados/simulacao/ssh_exemplo20.csv`, `.json`
Figures 18 and 40 to 42 and Tables 6 and 15	`validade_ssh.py` `python codigo/validade_ssh.py`	no file; illustrative ratios k_2/k_1 and $K = k_1/k_{-1}$; constants from `codigo/constantes.py` only in the cross-check of the ozone time scales (item 5 of the script; the number used in Chapter 4 is the one from `ozonio.py`)	`figuras/validade_AIB_concentracoes.pdf|png`, `validade_inducao.pdf|png`, `validade_AIB_erro.pdf|png`, `validade_mapa_ssh_preeq.pdf|png`; nine files in `dados/simulacao/` (`validade_*.csv|json`)

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Table 20. Scripts and what they generate, part 1 of 5: chapters 1, 2, 3, 6, and 7 (continued)

Figure(s)/table	Script and command	Inputs	Outputs
Table 34 and Figure 55 (Appendix C.1)	`varredura_validade.py` `python codigo/ varredura_validade.py`	no file; imports `validade_ssh.py` without changing it (100 k_2/k_1 ratios, against 7, and a 20×20 grid in k_2/k_{-1} and K, coarser than the 61×61 one of the map)	`figuras/varredura_validade_erro.pdf|png`, `varredura_validade_classificacao.pdf|png`; `dados/simulacao/varredura_validade_aib.csv`, `_preeq.csv`, `_resumo.json`
Figure 43 and Table 16 (Sections 7.5.1 and 7.5.3)	`validade_simetria.py` `python codigo/ validade_simetria.py`	`dados/simulacao/validade_mapa.csv` and `validade_mapa_lei_total.csv`, written by `validade_ssh.py` (run it first)	`figuras/validade_lei_total.pdf|png`; `dados/simulacao/validade_lei_total_curva.csv`, `validade_pontos_referencia.csv`, `validade_simetria.json`
Table 17	—	no file; author's own work, built from the results of Chapter 7	—
Figure 39	`figuras_outras.py` `python codigo/ figuras_outras.py`	no file; $v_{max} = 1$ mmol/(L s), $K_M = 0.5$ mmol/L (illustrative, the same as in the first version of this work)	`figuras/outras_michaelis_menten.pdf|png` (no CSV: it is a closed-form expression)

Source: author's own work.

Table 21 gives the scripts of Chapter 4, the ozone one: the constants, the simulation and the two sweeps built on `ozonio.py`, the fit to the ozone data, and the synthetic data of the assay.

Table 21. Scripts in `codigo/` and the figures and tables they generate (part 2 of 5: chapter 4).

Figure(s)/table	Script and command	Inputs	Outputs
Table 8 (the table is written in the chapter itself; the values are in `constantes.py`)	`gerar_constantes_literatura.py` `python codigo/gerar_constantes_literatura.py` `conferir_constantes.py` `python codigo/ conferir_constantes.py`	`codigo/constantes.py`, with the source and page of each constant (Burkholder *et al.*, 2019; Hodges *et al.*, 2019; NOAA, 1976; Jacob, 1999)	`dados/constantes_literatura.csv`, which also feeds the tables of Appendix C; `dados/conferencias/conferir_constantes.txt`
J_1 and J_3 in Table 8 and in Table 32	`ler_jacob_fig10_5.py` and `ler_jacob_fig10_5_altitudes.py` (run separately)	local copies of chapter 10 and of p. 168 of Jacob (1999), which stay out of the repository	only the terminal and `dados/conferencias/jacob1999_fig10_5_altitudes.txt`; the values read are in `constantes.py` and `varredura_ozonio_altitude.py`
Figures 20 to 22 and 26 to 29 and Table 10	`ozonio.py` `python codigo/ ozonio.py`	`codigo/constantes.py` (Burkholder *et al.*, 2019; NOAA, 1976; Jacob, 1999)	seven figures `figuras/ozonio_*.pdf|png`; eight files in `dados/simulacao/` (`ozonio_*.csv|json`)
Table 32 and Figure 53 (Appendix C.1)	`varredura_ozonio_altitude.py` `python codigo/ varredura_ozonio_altitude.py`	`codigo/constantes.py`; J_1, J_3 at 25 and 30 km from the vector reading of Jacob (1999) and a new, pixel-based reading of the same figure at the other altitudes (`codigo/ler_jacob_fig10_5_altitudes.py`)	`figuras/varredura_ozonio_altitude.pdf|png`, `_sensibilidade.pdf|png`; `dados/simulacao/varredura_ozonio_altitude.csv`, `.json`

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Table 21. Scripts and what they generate, part 2 of 5: chapter 4 (continued)

Figure(s)/table	Script and command	Inputs	Outputs
Table 31 and Figure 52 (Appendix C.1)	`varredura_ozonio_celula.py` `python codigo/varredura_ozonio_celula.py`	no file; imports `ozonio.py` without changing it; J and absorbance illustrative (same convention as `ozonio.py`), k_2, k_4 from `codigo/constantes.py`	`figuras/varredura_ozonio_celula.pdf|png`; `dados/simulacao/varredura_ozonio_celula.csv`, `.json`
Figure 23 and Table 9	`ajuste_ozonio.py` `python codigo/ajuste_ozonio.py`	`dados/chemis3_ozonio.csv` (data from the December 2025 article), `dados/chemis3_modelo_ozonio.csv` (curve published in the article)	`figuras/ozonio_ajuste.pdf|png`; `dados/simulacao/ozonio_ajuste.json`, `.csv`, `ozonio_fluxo_fotons.json`
Figures 24 and 25 (Section 4.4.1)	`dados_sinteticos.py` `python codigo/dados_sinteticos.py`	`dados/chemis3_ozonio.csv` (data from the December 2025 article, which give J_{ref} and the noise); imports `ozonio.py`, `ajuste_ozonio.py`, and `constantes.py` unchanged	`figuras/sintetico_ozonio_J.pdf|png`, `sintetico_ozonio_o2.pdf|png`; `dados/simulacao/dados_sinteticos_ozonio.csv`, `_ozonio_ajustes.csv`, `dados_sinteticos.json` (synthetic data)

Source: author's own work.

Table 22 gives those of Chapter 5, the ammonia one: the reactor, the theory, the surface and its sweep, the fit to the ammonia data, and the synthetic data of the assay.

Table 22. Scripts in `codigo/` and the figures and tables they generate (part 3 of 5: chapter 5).

Figure(s)/table	Script and command	Inputs	Outputs
Figures 30 to 32 and 38	`haber.py` `python codigo/haber.py`	`codigo/constantes.py` (K_a from Gillespie and Beattie, 1930; Dyson and Simon rate law, 1968)	four figures `figuras/amonia_*.pdf|png`; six files in `dados/simulacao/` (`amonia_*.csv|json`)
Table 13 (the values of K_a and φ_i) and the algebra of the theory in Chapter 5	`deducoes_amonia.py` `python codigo/deducoes_amonia.py`	`codigo/constantes.py` (K_a: Gillespie and Beattie, 1930; fugacity: Cheema and Krewer, 2018; Dyson and Simon, 1968); checks `amonia_equilibrio.csv` from `haber.py` (run it first)	`dados/simulacao/amonia_teoria.json`; `dados/conferencias/deducoes_amonia.txt`
Figures 33 and 34	`amonia_superficie.py` `python codigo/amonia_superficie.py`	no file; illustrative parameters (a/b, ε), chosen by me and declared in the script	`figuras/amonia_mecanismo_esquema.pdf|png`, `amonia_cobertura_ssh.pdf|png`; `dados/simulacao/amonia_cobertura_ssh.csv`, `amonia_superficie.json`
Table 30 and Figure 51 (Appendix C.1)	`varredura_amonia.py` `python codigo/varredura_amonia.py`	no file; imports `amonia_superficie.py` and `haber.py` without changing them; surface parameters illustrative, reactor law from Dyson & Simon (1968)	`figuras/varredura_amonia_superficie.pdf|png`, `_reator.pdf|png`; `dados/simulacao/varredura_amonia_superficie.csv`, `_reator.csv`, `_resumo.json`
Figure 35 and Table 14	`ajuste_amonia.py` `python codigo/ajuste_amonia.py`	`dados/chemis3_amonia.csv` (data from the December 2025 article), `dados/chemis3_modelo_amonia.csv` (curve published in the article)	`figuras/amonia_ajuste.pdf|png`; `dados/simulacao/amonia_ajuste.json`, `_curvas.csv`, `_residuos.csv`

continued on next page

Table 22. Scripts and what they generate, part 3 of 5: chapter 5 (continued)

Figure(s)/table	Script and command	Inputs	Outputs
Figures 36 and 37 (Section 5.5.1)	`dados_sinteticos.py` (the same run as in Table 21)	`dados/chemis3_amonia.csv` (data from the December 2025 article, which give the parameters and the noise); imports `ajuste_amonia.py` and `haber.py` without changing them	`figuras/sintetico_amonia_aicc.pdf|png`, `sintetico_amonia_vazao.pdf|png`; `dados/simulacao/dados_sinteticos_amonia.csv`, `_amonia_ajustes.csv`, `_amonia_confusao.csv`, and the same `dados_sinteticos.json` (synthetic data)
Tables 11 and 12	—	no file; author's own work (Table 11) and literature (Table 12)	—

Source: author's own work.

Table 23 gives those of Chapter 8: the SABIO-RK download and audit, the simulation of the K_M error, the testability of the SSH, and the bench proposal.

Table 23. Scripts in `codigo/` and the figures and tables they generate (part 4 of 5: chapter 8).

Figure(s)/table	Script and command	Inputs	Outputs
the SABIO-RK entries of Section 8.1	`baixar_sabiork.py` `python codigo/baixar_sabiork.py --n 20000 --semente 2026 --processos 3` (optional; run separately)	the SABIO-RK server, over the internet	`dados/sabiork_cache/` (one JSON per entry, kept out of the repository because of its size); `dados/sabiork_download.json`
Figures 44 and 45	`auditoria_sabiork.py` `python codigo/auditoria_sabiork.py`	`dados/sabiork_cache/kl_*.json` and `dados/sabiork_download.json` (SABIO-RK data); without the cache, the CSV files the script itself wrote before	`figuras/sabiork_epsilon_distribuicao.pdf|png`, `sabiork_epsilon_classe_ec.pdf|png`; `dados/sabiork_entradas.csv`, `sabiork_auditoria.csv`, `_resumo.json`, `_tabela.csv`
Figure 46	`vies_km_epsilon.py` `python codigo/vies_km_epsilon.py`	`dados/sabiork_entradas.csv` and `sabiork_auditoria.csv`, written by `auditoria_sabiork.py` (run it first)	`figuras/vies_km_epsilon.pdf|png`; `dados/simulacao/vies_km_epsilon.csv`, `_resumo.json`
Figures 47 and 48	`testabilidade_hee.py` `python codigo/testabilidade_hee.py`	`dados/chemis3_ozonio.csv` (only the 13 times); imports `validade_ssh.py` without changing it	six figures `figuras/testabilidade_*.pdf|png` (two in the text); `dados/simulacao/testabilidade_aib.csv`, `_aip.csv`, `_mm.csv`, `_exemplo_dados_sinteticos.csv` (synthetic data), `_resumo.json`
Figure 49 (Section 8.3)	`invertase_glicosimetro.py` `python codigo/invertase_glicosimetro.py`	no data file; literature constants written in the script, each with its source (Johnson & Goody (2011), Nelson & Beegle (1919), and others); R from `constantes.py`	three figures `figuras/invertase_*.pdf|png` (one in the text); six CSV files `dados/simulacao/invertase_*.csv`, among them `invertase_dados_sinteticos.csv` (synthetic data), and `invertase_glicosimetro.json`

continued on next page

Table 23. Scripts and what they generate, part 4 of 5: chapter 8 (continued)

Figure(s)/table	Script and command	Inputs	Outputs
Figure 50	`analisar_glicosimetro.py` `python codigo/analisar_glicosimetro.py`	`dados/simulacao/invertase_dados_sinteticos.csv` (synthetic data; with `--dados`, the bench spreadsheet); imports `invertase_glicosimetro.py`	`figuras/glicosimetro_ajuste_sintetico.pdf|png`; `dados/simulacao/glicosimetro_ajuste_sintetico.csv`, `.json`

Source: author's own work.

Table 24 closes the list with the scripts of the appendices: the check of the derivations in Appendix A and, in Appendix C, the bootstrap of the two fits, the constants tables, and the general data table. It also says which files the two tables with the data from the December 2025 article come from. The constants and data tables are described in more detail in Appendix C.

Table 24. Scripts in `codigo/` and the figures and tables they generate (part 5 of 5: Appendices A and C).

Figure(s)/table	Script and command	Inputs	Outputs
the derivations of Appendix A	`conferir_deducoes_apendice_a.py` `python codigo/conferir_deducoes_apendice_a.py`	no file; the formulas of the appendix, written in *sympy* in the script itself	`dados/conferencias/deducoes_apendice_a.txt` (110 checks, all "OK")
Table 33 and Figure 54 (Appendix C.1)	`bootstrap_ajustes.py` `python codigo/bootstrap_ajustes.py`	`dados/chemis3_ozonio.csv`, `chemis3_amonia.csv`; imports `ajuste_ozonio.py` and `ajuste_amonia.py` without changing them	`figuras/varredura_bootstrap.pdf|png`; `dados/simulacao/bootstrap_ajustes.csv`, `.json`
Tables 25 and 26	—	`dados/chemis3_ozonio.csv` and `dados/chemis3_amonia.csv` (data from the December 2025 article); `chemis3_modelo_ozonio.csv` and `chemis3_modelo_amonia.csv` (curve published in the article)	`tese_v2/tab_dados_chemis3.tex`
Tables 27 to 29	`tabelas_apendice_c.py` `python codigo/tabelas_apendice_c.py`	`dados/constantes_literatura.csv`	`tese_v2/tab_apc_constantes.tex` and, with `--idioma en`, its twin in `tese_v2_en/`
Table 36	`tabela_geral_dados.py` `python codigo/tabela_geral_dados.py`	every small CSV/JSON in `dados/` and `dados/simulacao/` (including those of the sweeps, of the synthetic data, and of SABIO-RK) and the constants selection from `tabelas_apendice_c.py`	`dados/tabela_geral_dados.csv`, `_resumo.md`; `tese_v2/tab_geral_dados.tex` and its twin in `tese_v2_en/`

Source: author's own work.

B.2 Environment

I ran the code for this work on Python 3.13.16, with numpy 2.5.3, scipy 1.18.1, sympy 1.14.0, matplotlib 3.11.2, and pandas 3.0.5. Some scripts also use PyMuPDF 1.28.2, to read PDF files, and

OpenCV (`opencv-python-headless` 4.10.0.84), only for the check against the page image (`extrair_pontos_chemis3.py` and `verificar_extracao_chemis3.py`); without it, the first still runs, but `dados/chemis3_resumo_extracao.json` comes out without that check and gets a different SHA-256. Pillow 12.3.0, which comes with matplotlib, reads Jacob's page in `ler_jacob_fig10_5_altitudes.py`. The SABIO-RK download uses the `curl` program. I compiled the monograph with TeX Live 2023 (Debian) and `latexmk` 4.83, class `abntex2`. I checked these versions on the same machine where I ran the code and compiled the monograph (the OpenCV one comes from the installation record), with commands such as `python3 -c "import numpy; print(numpy.__version__)"`. A reproduction on another machine, with similar versions, should give the same numbers in the text, because no script relies on a feature from a recent version of any of these libraries; only the last digits of some fits may change (Appendix B.5).

B.3 What each script does

Appendix B.1 already states which figure or table each script generates; here the focus is on each one's method. The list covers the 46 scripts in `codigo/`: the two modules the others import, the ones that generate figures, data, and tables, in chapter order, and, at the end, the tools that only serve to build and check the monograph.

`constantes.py` Keeps, in one place, every physical, kinetic, and thermodynamic constant that the ozone and ammonia chapters use, each with its source, table or page, and unit, plus short functions that simply apply the source's formula (for example, $k(T)$ for the $O + O_3$ reaction). Other scripts import from here instead of typing the number again. Run on its own, it just prints a summary; imported, it reads and writes nothing.

`estilo.py` Defines the single style used by every figure in the monograph: color palette (Okabe and Ito, readable for colorblind readers), sizes, axes with a decimal point in the English figures (a decimal comma in the Portuguese ones), and the functions `aplicar_estilo`, `salvar_figura`, and `ler_idioma` that every figure script calls. Sets the rule that the data from the December 2025 article appear as filled points and the curve published in the article as a dashed line.

`figura_wilhelmy.py` Draws the only figure of the introduction (Figure 1): the sketch of a polarimeter, like the one Wilhelmy used in 1850, and the qualitative curve of the rotation angle falling, crossing zero and ending negative as sucrose turns into invert sugar. It is the author's own drawing: the values in the curve were chosen for the picture and are neither measured rotations nor rate constants, which is why the axes carry no numbers.

`figuras_fundamentos.py` Generates the 14 figures of the kinetics fundamentals chapter (Chapter 2): from the secant lines of average rate to the biscuit factory. Each panel uses illustrative values, except the Arrhenius plot with O, NO, and OH against ozone, which uses the literature (Burkholder *et al.*, 2019). Also records the calculations cited in the text (`fund_contas.json`) and the factor-2 table (Table 4).

`fund_metodo_grafico.py` Checks, with an exact first-order reaction, the claim that "at low conversion, any plot looks like a line": it generates 11 evenly spaced points up to 20%, 75%, and 90% conversion and computes the R^2 of a line fitted to [A] against t (the wrong plot for order 1). The three R^2 values cited in Chapter 2 come from the JSON this script writes; there is no figure.

`figuras_ssh.py` Generates the three figures of the steady-state hypothesis chapter: the two-step energy profile, the inverted factory (the same analogy from chapter 2, with the oven slower

than the dough machine), and the simulation of Example 3.2, which integrates the full system and compares the exact [I] with the SSH expression.

`validade_ssh.py` The main script of Chapter 7: it solves $A \longrightarrow I \longrightarrow B$ and $A \rightleftharpoons I \longrightarrow P$ both by the exact solution and by the SSH, measures the hypothesis's error for several ratios k_2/k_1 and K, finds the induction period, and builds the validity map (SSH versus pre-equilibrium). It also checks the Segel and Slemrod (1989) criterion for Michaelis-Menten and redoes the ozone time scales as a cross-check on `ozonio.py`.

`varredura_validade.py` Simulation sweep on top of `validade_ssh.py` (Appendix C.1): imports that script's functions without changing them and runs the SSH error in $A \rightarrow I \rightarrow B$ over 100 log-spaced ratios k_2/k_1 (against the seven already tabulated) and the SSH/pre-equilibrium map over a 20×20 grid, finer than the three-row table in Chapter 7. Checks every result against the old files in `dados/simulacao/` and, at ten points, against an independent `solve_ivp` integration.

`validade_simetria.py` Checks the swap of k_1 and k_2 in the mechanism $A \rightleftharpoons I \rightarrow P$ (Section 7.5.1): it shows that [P] does not change and that the pre-equilibrium map is the SSH map mirrored. It computes the error of the total law as a function of q, the criteria of both approximations and the band where neither works, and checks 14 reference points with `solve_ivp`. It exits with an error if any check fails.

`figuras_outras.py` Generates the only figure of the other-applications chapter: the Michaelis-Menten curve with the same illustrative $v_{\max}$ and K_M as the figure in the first version of this work, marking the asymptote, the point $[S] = K_M$, and the initial line.

`gerar_constantes_literatura.py` Writes `dados/constantes_literatura.csv`, one row per constant in `constantes.py`, with value, unit, range of validity, source, page, address, and access date. Before writing, it checks some values against the module itself and stops with an error if they disagree.

`conferir_constantes.py` Checks `constantes.py` against the tables of the sources: the 1976 standard atmosphere against Tables I and II, the Gillespie and Beattie K_a against JANAF, the JPL constants against the IUPAC ones and the JPL uncertainty factors, plus order-of-magnitude calculations. It writes `dados/conferencias/conferir_constantes.txt`.

`ler_jacob_fig10_5.py` **and** `ler_jacob_fig10_5_altitudes.py` Read J_1 and J_3 from Figure 10-5 of Jacob (1999): the first from the vector paths of the chapter PDF (the values at 25 and 30 km), the second from the pixels of an image of the page (20 to 36 km). They need those copies, which stay out of the repository for copyright reasons; the values they read are already in `constantes.py`, in `varredura_ozonio_altitude.py`, and in `dados/conferencias/`.

`ozonio.py` Solves the ozone photolysis: the two-step cell with constant light, the same cell with the ozone-reforming reaction added, and the full Chapman mechanism in the stratosphere (with and without the chlorine cycle), always comparing the full solution (`solve_ivp`) with the steady-state hypothesis in [O]. Records the time scales τ_O, τ_{O_3}, and τ_{Ox}, and checks atom conservation with two integrators.

`varredura_ozonio_altitude.py` Simulation sweep on top of `ozonio.py`: runs the full Chapman mechanism from 20 to 36 km, with J_1, J_3 from the vector reading at 25 and 30 km and a new, pixel-based reading of the same Jacob (1999) Figure 10-5 at the other altitudes (up to 31% different from the vector reading, noted in Figure 53's caption). Then it runs two sensitivity

checks that are not the real profile, to separate the effect of each variable: one with T, $[M]$ varying and J fixed, another with J scaled by a factor and T, $[M]$ fixed.

`varredura_ozonio_celula.py` Another simulation sweep on top of `ozonio.py`: runs the two-step illuminated cell, with and without O_2, over 30 values of J, 8 of absorbance, and 38 of O_2 fraction (illustrative, as in `ozonio.py`), fits an effective k_{obs} to each simulated curve, and compares it with $2J$ and with the general form with O_2 (Equation (4.13)).

`ajuste_ozonio.py` Fits an exponential decay to the 13 ozone data points from the December 2025 article, with `curve_fit`: three models (one parameter, two parameters, a line in $\ln y$), a 95% confidence interval, R^2, AIC, AICc, and a check of the residuals. Ends by estimating what photon flux matches the fitted k and comparing that flux with the one from published mercury lamps.

`haber.py` Simulates a simple ammonia synthesis reactor: the equilibrium fraction of NH_3 against T and P, the per-pass conversion in an isothermal fixed bed, and a dynamic start-up model, with the Temkin and Pyzhev equation in the form given by Dyson and Simon (1968). Also runs four sanity checks (for example, that the rate is exactly zero at the equilibrium composition).

`deducoes_amonia.py` Checks in *sympy* the algebra of the ammonia theory (Langmuir isotherms, the simple coverage model, the near-equilibrium form, and the Temkin-Pyzhev and Dyson and Simon laws) and computes the numbers that Chapter 5 takes from literature formulas. It writes `dados/simulacao/amonia_teoria.json` and `dados/conferencias/deducoes_amonia.txt`.

`amonia_superficie.py` Draws the diagram of the six steps on the iron surface (Ertl's 2008 mechanism) and solves the steady-state hypothesis for the nitrogen coverage θ_N, with fixed gas composition and with the gas slowly running out in a closed vessel. All parameters are illustrative and I chose them myself: the model is there to show the idea and does not measure any real catalyst.

`varredura_amonia.py` Simulation sweep on top of `amonia_superficie.py` and `haber.py`: solves the surface's SSH over a grid of 40 ratios a/b and four ratios ζ (illustrative parameters), and the Dyson-Simon reactor's conversion over a grid of 21 temperatures and 11 pressures, for the two flow examples already used in `haber.py`; marks, without dropping, the cells outside the published fitting range of the rate law (149-309 atm; 603-768 K, Nadiri *et al.* (2024)).

`ajuste_amonia.py` Fits five models to the 13 ammonia data points from the same article (the article's own curve, a simple exponential decay, the same with a dead time, a product of two steps in series, and a Weibull), with the same kind of confidence interval, R^2, AICc, BIC, and Durbin-Watson as the ozone script, plus a rounding-sensitivity check by Monte Carlo with a fixed seed.

`bootstrap_ajustes.py` Non-parametric bootstrap, a simulation sweep on top of `ajuste_ozonio.py` and `ajuste_amonia.py` (Appendix C.1): two variants (residual bootstrap and case resampling), 5000 replicates each, for the ozone and ammonia fit parameters; counts, replicate by replicate, which of the ammonia's five models wins by AICc, and runs a sensitivity check restricted to the rounding of the data. Checks two calculations against the values already recorded by the two fitting scripts.

`dados_sinteticos.py` Builds, by simulation, the sets of points that a well-designed assay would measure for ozone and for ammonia, with the parameters and the noise taken from the 13 + 13 points of the December 2025 article, and runs on them the analyses that would test the laws (Sections 4.4.1 and 5.5.1). For ozone: one series with five values of J and another with four

O_2 fractions, with 40 points per curve and 3 replicates. For ammonia: four flow rates, three hypotheses about what changes with the flow rate, and the fraction of times the AICc picks the true model in 300 sets per case. Everything it writes is synthetic data (a model curve with Gaussian noise drawn with a fixed seed), and no point is a measurement.

`baixar_sabiork.py` Downloads, over the internet, a random sample of entries from the public SABIO-RK database (WITTIG *et al.*, 2018): one JSON per entry in `dados/sabiork_cache/`, which stays out of the repository because of its size, and the download record in `dados/sabiork_download.json`. It waits 0.3 s between requests, so as not to overload the server, and does not download again what it already has. The sample used in Chapter 8 was downloaded on 15 and 17 October 2026.

`auditoria_sabiork.py` For each K_M–$[E]_0$ pair from the same SABIO-RK entry, it computes $\varepsilon_{K_M} = [E]_0/K_M$, which is an upper bound on the parameter of Segel & Slemrod (1989), and estimates the fraction of pairs above 0.01, 0.1, and 1, with a 95% bootstrap confidence interval, overall, by EC class, by experiment type, and by decade (Section 8.1). Each entry gets a weight according to its ID range, the inverse of the fraction read from that range; since the sample covers all ranges almost evenly, the weights are almost equal. It converts the enzyme concentration to mol/L and leaves out, and counts, the pairs whose unit does not allow the conversion. Without `dados/sabiork_cache/`, it redoes the calculations from the CSV files it wrote before.

`vies_km_epsilon.py` Simulates the full Michaelis-Menten mechanism, with the enzyme mass balance, and repeats on the simulated data the usual measurement procedure (initial rates on a grid of $[S]_0$, or a progress curve), to see how much K_M and k_{cat} miss as a function of $[E]_0/K_M$. It applies these curves to the SABIO-RK pairs from `auditoria_sabiork.py`, with the same estimator and the same bootstrap. It redoes the calculation with the $[S]_0$ grid centered on the apparent K_M (a pilot fit, then the grid rebuilt until it is centered on the K_M it gives) and with the half-saturation limit, $[E]_0/K_{M,ap} < 2$, and applies both readings to the SABIO-RK pairs.

`testabilidade_hee.py` Asks when a set of points can show that the SSH has failed (Section 8.2). It generates synthetic data (the exact curve of the mechanism with Gaussian noise drawn with a fixed seed) in four assay designs, among them the 13 times of the December 2025 article, fits the SSH law and the full model to each set, and counts in what fraction of the sets the AICc rejects the SSH, with 200 replicates per case. It does this for $A \to I \to B$, for $A \rightleftharpoons I \to P$ (looking at the reactant, the product, or both), and for Michaelis-Menten, and it also records the example of a high R^2 with the wrong constant. In the English run, it reuses the calculation already recorded.

`invertase_glicosimetro.py` Prepares the bench proposal of Section 8.3, which I have not carried out yet. Invertase splits sucrose and releases alpha glucose, which only later turns into the alpha–beta mixture (mutarotation); a glucose-meter strip that reads only the beta form shows the glucose that has already turned beta, and the difference between the total glucose and the reading follows exactly the $I \to B$ step of the $A \to I \to B$ mechanism; the $A \to I$ step is the Michaelis-Menten rate of invertase. The script fits the mutarotation constant against temperature (Arrhenius on the eight values of Nelson & Beegle (1919)), predicts the readings at three enzyme doses and at three temperatures, writes synthetic data in the format of the bench spreadsheet, and computes, by Monte Carlo with a fixed seed, how often the AICc would recognize the full solution. Nothing it writes is a measurement: the curves are model predictions.

`analisar_glicosimetro.py` Analyzes the glucose-meter readings. It is ready to take the bench spreadsheet; with no arguments, it runs on the synthetic data from `invertase_glicosimetro.py`, which serve as a test, because the parameters it returns must match the ones

used to generate the data. It calibrates the meter with standards, fits the one-afternoon test (dissolved alpha glucose), and fits three models to the readings with invertase, chosen by AICc: the full solution, the SSH with the mass balance, and the plain SSH.

`conferir_deducoes_apendice_a.py` Redoes the derivations of Appendix A in *sympy* and checks each one against the formula in the text. It writes `dados/conferencias/deducoes_apendice_a.txt`, with the 110 checks marked "OK".

`tabelas_apendice_c.py` Builds the three constants tables of Appendix C directly from `dados/constantes_literatura.csv`, without retyping a single number: a converter reads each expression from the CSV with the `ast` module and rebuilds the formula in LaTeX. Running it again changes nothing, because the script writes no date and uses no random number, and it does not depend on any system order.

`tabela_geral_dados.py` Merges, in long format (one row per value), every numerical value the final monograph uses: the 13+13 data points from the December 2025 article, the simulation CSV and JSON files (including the sweeps', Appendix C.1), the synthetic data, the summaries of the SABIO-RK audit, and the literature-constants selection from `tabelas_apendice_c.py`, each value with the label of its source. Writes `dados/tabela_geral_dados.csv` (the full list) and the summary table `tab_geral_dados.tex` (Table 36), in Portuguese and in English.

Data from the December 2025 article Five scripts take care of the `dados/chemis3_*` files: one writes them, together with the two point tables of Appendix C (Tables 25 and 26), and four check those files and run calculations on the 13 + 13 points, such as R^2 under several definitions, compared with what the article reports. What each one does is in `dados/INVENTARIO.md`.

`reproduzir_tudo.py` Runs, in the right order, every script that writes outputs, and records the SHA-256 of each output (Appendix B.5).

`compilar_capitulo.py` The tool that compiles one chapter on its own, without waiting for the full monograph: it drops the requested file into a test document with the `abntex2` class and the shared preamble, runs `latexmk`, and summarizes the errors and warnings from the log. The PDF and the log go to `compilacao_teste/`, outside the repository. A reference to another chapter's label shows up as "??" in that test when the full monograph has not been compiled yet, and that is expected.

`auditar_diagramacao.py` Reads the PDF of the monograph, or of a chapter test, and points out a column that ends well above the foot of the page while the text goes on in the next one, and a figure or table far from the page where it is first cited.

`concordancia_cref.py` Fixes the article before `\cref` in the Portuguese text: the package writes "Seção", "Figura", "Tabela", and "Equação", which are feminine in Portuguese, so "no" before one of those labels becomes "na".

`conferir_revisao.py` Compares a chapter's version before and after a style edit and lists every number, citation key, label, cross-reference, and displayed-equation content that changed from one side to the other. That way one can see whether a style edit changed a fact by accident.

`normalizar_texto_pdf.py` A support script, unrelated to any figure or table in the monograph. It rejoins letters and accent marks in the text of the PDFs of my earlier work, which `pdftotext` returns separated.

`renderizar_paginas_pdf.py` Another support script, also unrelated to any figure or table: it saves images of the pages of the PDFs of my 2025 work, for reference.

B.4 Key excerpts

This section shows, line by line, three excerpts of the code: the integration of the full system with `solve_ivp`, in two scripts (the steady-state hypothesis one and the ozone one), and the ozone curve fit. The complete code, with comments and docstrings, is in `codigo/`.

B.4.1 Integration of the full system

In `validade_ssh.py`, the function `aib_numerica` writes the three differential equations of the mechanism $\mathrm{A} \longrightarrow \mathrm{I} \longrightarrow \mathrm{B}$ (with $k_1 = 1$ and $k_2 = r$) and calls `solve_ivp` to check the exact analytical solution used in the rest of the script:

```
def aib_numerica(tau, r):
    """Mesma coisa por integração numérica (k1 = 1, k2 = r), para conferir."""
    def f(t, y):
        return [-y[0], y[0] - r * y[1], r * y[1]]
    s = solve_ivp(f, (0.0, float(tau[-1])), [1.0, 0.0, 0.0], method="Radau",
                  t_eval=tau, jac=lambda t, y: [[-1, 0, 0], [1, -r, 0], [0, r, 0]], **TOL)
    return s.y
```

In `ozonio.py`, the function `celula_rhs` writes a similar system for photolysis in a cell ($\mathrm{O_3} \xrightarrow{\mathrm{J}} \mathrm{O_2} + \mathrm{O}$, $\mathrm{O} + \mathrm{O_3} \xrightarrow{\mathrm{k_4}} 2\,\mathrm{O_2}$). This system enters `solve_ivp` via `integrar`, called inside `simular_celula`; the literal call is in the helper function `integrar`, lines 111–116 of the same file:

```
def celula_rhs(J, k4):
    def f(t, y):
        o3, o = y
        return [-J * o3 - k4 * o * o3, J * o3 - k4 * o * o3]

    def jac(t, y):
        o3, o = y
        return [[-J - k4 * o, -k4 * o3], [J - k4 * o, -k4 * o3]]
    return f, jac


def simular_celula():
    J, k4 = J_CELULA, K4_CELULA
    o_ee = J / k4
    casos = []
    for A in ABSORBANCIAS:
        o3_0 = o3_inicial(A)
        f, jac = celula_rhs(J, k4)
        atol = [o3_0 * 1e-16, o_ee * 1e-16]
        s = integrar(f, [o3_0, 0.0], T_EVAL_CELULA[-1], T_EVAL_CELULA, atol, jac=jac)
```

B.4.2 Curve fitting

In `ajuste_ozonio.py`, the function `ajuste_nlin` calls `curve_fit` to fit each of the three models to the 13 ozone data points, and the function `ic95` uses Student's t distribution for the 95% confidence interval (CI) of each parameter:

```
def ajuste_nlin(f, t, y, p0):
    p, cov = curve_fit(f, t, y, p0=p0, maxfev=100000) # absolute_sigma=False: s2 dos residuos
    ep = np.sqrt(np.diag(cov))
    return p, cov, ep


```

```
def ic95(valor, ep, gl):
    tc = stats.t.ppf(0.975, gl)
    return tc, (valor - tc * ep, valor + tc * ep)
```

B.5 How to reproduce everything

A single command redoes every number, figure, and table this work generates by code, starting from the project root:

```
python3 codigo/reproduzir_tudo.py
```

This script calls, in this order, 32 of the scripts from Appendices B.1 and B.3 that write a file to `figuras/`, `dados/`, or `tese_v2/`. First come the two that write and check the files with the data from the December 2025 article, the constants file, and the checks that make no figure (`gerar_constantes_literatura`, `conferir_constantes`, `conferir_deducoes_apendice_a`). Then come the figures of the introduction and of the fundamentals, steady-state hypothesis, and other-applications chapters (`figura_wilhelmy`, `figuras_fundamentos`, `fund_metodo_grafico`, `figuras_ssh`, `figuras_outras`). After those come the simulations, fits, and sweeps, each sweep right after the script it depends on (`validade_ssh`, `varredura_validade`, `validade_simetria`, `testabilidade_hee`, `ozonio`, `varredura_ozonio_altitude`, `varredura_ozonio_celula`, `ajuste_ozonio`, `haber`, `deducoes_amonia`, `amonia_superficie`, `varredura_amonia`, `ajuste_amonia`, `bootstrap_ajustes`), the synthetic data and the bench proposal (`dados_sinteticos`, `invertase_glicosimetro`, `analisar_glicosimetro`), and the SABIO-RK audit (`auditoria_sabiork`, `vies_km_epsilon`). Finally, the tables of Appendix C, the audit of the article data, and the general data table (`tabelas_apendice_c`, `auditoria_chemis3`, `tabela_geral_dados`, which depends on every CSV and JSON generated before it and so runs last). Each of these scripts runs twice, in Portuguese and in English, whenever it accepts `--idioma`. No step uses the internet: without the downloaded SABIO-RK entries, which stay out of the repository because of their size, `auditoria_sabiork` redoes the calculations from the CSV files in `dados/`. At the end, the script writes the SHA-256 of every output to `dados/sha256_saidas.txt`.

On my machine, the whole run takes about 45 minutes. The slowest scripts are `vies_km_epsilon` (about 14 minutes), `testabilidade_hee` (about 6), and `invertase_glicosimetro` (about 4); in the English run, all three reuse the calculation already recorded. Since none of the scripts draws a random number without a fixed seed, running `reproduzir_tudo.py` again on the same machine gives the same files, byte for byte (with OpenCV installed). On another machine, the last digits of some fits may change (in one value of the fitted curve of the ammonia model M3, the eighth significant digit changed), without changing any number in the text.

Left out of that list are the SABIO-RK download (`baixar_sabiork.py`), which is optional, uses the internet, and is slow; the two readers of Jacob's figure, which need the copies of the book; and the two independent checks of the article data, `verificar_extracao_chemis3.py` (which needs OpenCV) and `verificar_contas_chemis3.py` (which uses the result of the first); both write to `dados/verificacao/`. Since the SABIO-RK database changes over time, a new download may not give the same sample as on 15 and 17 October 2026.

APPENDIX C

Tables of constants and data points

This appendix gathers, in one place, the numbers the ozone and ammonia chapters use. First come the data from the December 2025 article (RIBEIRO, 2025b). Then come the physical, kinetic and thermodynamic constants from published sources, the simulation sweeps (Appendix C.1) and, at the end, a general table with every data set I used (Appendix C.2).

Table 25 gives the 13 ozone data points from the December 2025 article, and Table 26 the 13 ammonia data points. Next to each point is the value of the curve published in the article at the same time. The times have one decimal place and the values two, so the rounding reaches ±0.05 in time and ±0.005 in value. That is the smallest error each point carries. The points scatter more than that around the fitted curves (residual standard deviation of 0.014 for ozone and 0.009 for ammonia, with M4), and it is this scatter that sets the width of the confidence intervals.

Table 25. Ozone data from the December 2025 article: "normalized ozone concentration," a quantity with no unit.

Point	t (s)	Normalized $[O_3]$	Article curve	Point − curve
1	0.0	0.99	1.000	−0.010
2	0.7	0.72	0.705	0.015
3	1.4	0.48	0.497	−0.017
4	2.1	0.38	0.350	0.030
5	2.9	0.22	0.235	−0.015
6	3.6	0.15	0.165	−0.015
7	4.3	0.13	0.117	0.013
8	5.0	0.08	0.082	−0.002
9	5.7	0.06	0.058	0.002
10	6.4	0.05	0.041	0.009
11	7.1	0.02	0.029	−0.009
12	7.9	0.03	0.019	0.011
13	8.6	0.01	0.014	−0.004

The normalized $[O_3]$ values have two decimal places, so each one carries a rounding of up to ±0.005; the times have one decimal place (±0.05 s). The "article curve" column gives the value of the curve published in the article at each time.

Source: data from the December 2025 article (RIBEIRO, 2025b); files `dados/chemis3_ozonio.csv` (data) and `dados/chemis3_modelo_ozonio.csv` (article curve).

Table 26. Ammonia data from the December 2025 article: "reactor efficiency," a quantity with no unit. The article gives time in minutes; the column in seconds is the same time multiplied by 60.

Point	t (min)	t (s)	Efficiency	Article curve	Point − curve
1	0.0	0	0.00	0.000	0.000
2	0.7	42	0.20	0.343	−0.143
3	1.4	84	0.45	0.568	−0.118

continued on next page

Table 26. Ammonia data from the December 2025 article (continued)

Point	t (min)	t (s)	Efficiency	Article curve	Point − curve
4	2.1	126	0.62	0.716	−0.096
5	2.9	174	0.77	0.824	−0.054
6	3.6	216	0.88	0.885	−0.005
7	4.3	258	0.92	0.924	−0.004
8	5.0	300	0.95	0.950	0.000
9	5.7	342	0.96	0.967	−0.007
10	6.4	384	0.98	0.978	0.002
11	7.1	426	0.99	0.986	0.004
12	7.9	474	0.99	0.991	−0.001
13	8.6	516	1.00	0.994	0.006

The efficiency values have two decimal places, so each one carries a rounding of up to ±0.005; the times have one decimal place (±0.05 min). The "article curve" column gives the value of the curve published in the article at each time.

Source: data from the December 2025 article (RIBEIRO, 2025b); files `dados/chemis3_amonia.csv` (data) and `dados/chemis3_modelo_amonia.csv` (article curve).

The three constants tables come from a single file, `dados/constantes_literatura.csv`. A script builds them straight from it (`codigo/tabelas_apendice_c.py`, described in Appendix B), with no number typed by hand, so what appears here always matches the spreadsheet. Table 27 gives the fundamental physical constants; Table 28, those for ozone kinetics and photochemistry; and Table 29, those for ammonia synthesis. Each row carries its source, so anyone who wants to check a number used in some chapter knows where it came from.

Table 27. Fundamental physical constants (CODATA 2018).

Quantity	Value or expression	Unit and range of validity	Source
h (Planck constant)	$6.62607015 \times 10^{-34}$	J s; exact (SI 2019)	(TIESINGA *et al.*, 2021)
k_B (Boltzmann constant)	1.380649×10^{-23}	J/K; exact	(TIESINGA *et al.*, 2021)
R (gas constant, $R = N_A k_B$)	8.314462618	J/(mol K); exact (repeating decimal)	(TIESINGA *et al.*, 2021)
N_A (Avogadro constant)	$6.02214076 \times 10^{23}$	1/mol; exact	(TIESINGA *et al.*, 2021)

Source: literature: Tiesinga *et al.* (2021). Values and expressions identical to `dados/constantes_literatura.csv` (generated by `codigo/tabelas_apendice_c.py`).

Table 28. Literature constants for ozone kinetics and photochemistry, used in the fundamentals and ozone chapters.

Quantity	Value or expression	Unit and range of validity	Source
k_2: $O + O_2 + M \longrightarrow O_3 + M$ (M = air)	$6.1 \times 10^{-34}\,(T/298)^{-2.4}$	cm^6 molecule^{-2} s^{-1}; data 200-268 K; note A1: consistent extrapolation down to 100 K	(BURKHOLDER *et al.*, 2019)
k_4: $O + O_3 \longrightarrow 2\,O_2$	$8.0 \times 10^{-12}\,e^{-2060/T}$	cm^3 molecule^{-1} s^{-1}; data 220-409 K	(BURKHOLDER *et al.*, 2019)
k_5: $Cl + O_3 \longrightarrow ClO + O_2$	$2.3 \times 10^{-11}\,e^{-200/T}$	cm^3 molecule^{-1} s^{-1}; data 184-1350 K	(BURKHOLDER *et al.*, 2019)
k_6: $O + ClO \longrightarrow Cl + O_2$	$2.8 \times 10^{-11}\,e^{85/T}$	cm^3 molecule^{-1} s^{-1}; data 220-1250 K	(BURKHOLDER *et al.*, 2019)
$NO + O_3 \longrightarrow NO_2 + O_2$	$3.0 \times 10^{-12}\,e^{-1500/T}$	cm^3 molecule^{-1} s^{-1}; data 195-443 K	(BURKHOLDER *et al.*, 2019)

continued on next page

Table 28. Literature constants for ozone kinetics (continued)

Quantity	Value or expression	Unit and range of validity	Source
$OH + O_3 \longrightarrow HO_2 + O_2$	$1.7 \times 10^{-12}\ e^{-940/T}$	cm^3 molecule^{-1} s^{-1}; data 190-357 K	(BURKHOLDER *et al.*, 2019)
σ of O_3 at 253.65 nm (consensus value)	$(1.1329 \pm 0.0035) \times 10^{-17}$ (u); U95 = 0.0069×10^{-17}	cm^2 molecule^{-1}; referenced to 296 K	(HODGES *et al.*, 2019)
σ of O_3 at 253.65 nm (JPL, cross-check)	$(1.141 \pm 0.009) \times 10^{-17}$	cm^2 molecule^{-1}; room temperature (295-298 K)	(BURKHOLDER *et al.*, 2019)
Φ of $O(^1D)$ in the photolysis of O_3, 220–305 nm	0.90	dimensionless; 220-305 nm, independent of T	(BURKHOLDER *et al.*, 2019; ATKINSON *et al.*, 2004)
Φ of $O(^3P)$ in the photolysis of O_3, 220–305 nm	0.10	dimensionless; 220-305 nm	(ATKINSON *et al.*, 2004)
volume fraction of O_2 in dry air	0.209476	dimensionless; 0-80 km (homogeneous mixing)	(NOAA; NASA; USAF, 1976)
T, P, ρ of air at 25 km (1976 standard atmosphere)	221.552; 2549.2; 4.0084×10^{-2}	K; Pa; kg/m^3; mid-latitude annual averages (model)	(NOAA; NASA; USAF, 1976)
[M] (number density of air) at 25 km	8.3341×10^{23} (8.3341×10^{17} cm^{-3})	$1/m^3$; same as above	(NOAA; NASA; USAF, 1976)
T, P, ρ of air at 30 km (1976 standard atmosphere)	226.509; 1197.0; 1.8410×10^{-2}	K; Pa; kg/m^3; mid-latitude annual averages (model)	(NOAA; NASA; USAF, 1976)
[M] (number density of air) at 30 km	3.8278×10^{23} (3.8278×10^{17} cm^{-3})	$1/m^3$; same as above	(NOAA; NASA; USAF, 1976)
J_1 (O_2) and J_3 (O_3) at 25 and 30 km (read from a figure)	J_{O_2}: 3.4×10^{-12} (25 km), 2.5×10^{-11} (30 km); J_{O_3}: 5.2×10^{-4} (25 km), 7.7×10^{-4} (30 km)	1/s; order of magnitude (hand-drawn figure)	(JACOB, 1999)
total chlorine entering the stratosphere (ceiling of Cl_y)	3660 (1993, peak); 3240 (2020)	ppt (10^{-12} mol/mol); global average	(SALAWITCH *et al.*, 2023)

Source: literature: Burkholder *et al.* (2019), Hodges *et al.* (2019), Atkinson *et al.* (2004), NOAA; NASA; USAF (1976), Jacob (1999), Salawitch *et al.* (2023). Values and expressions identical to `dados/constantes_literatura.csv` (generated by `codigo/tabelas_apendice_c.py`).

Table 29. Literature constants used in the ammonia synthesis chapter.

Quantity	Value or expression	Unit and range of validity	Source
$\log_{10} K_a$ of $\frac{1}{2}N_2 + \frac{3}{2}H_2 \rightleftharpoons NH_3$	$-2.691122 \log_{10}(T) - 5.519265 \times 10^{-5}\ T + 1.848863 \times 10^{-7}\ T^2 + 2001.6/T + 2.6899$	K_a in 1/atm (T in K); data from 325 to 952 degrees C (598-1225 K), 10-1000 atm	(GILLESPIE; BEATTIE, 1930)
k_0 (pre-exponential factor of k_-, Dyson and Simon)	8.849×10^{14}	kmol/(m^3 h) (NH_3, per m^3 of bed); industrial iron, 150-300 atm, 640-770 K	(DYSON; SIMON, 1968)
E (activation energy of k_-, Dyson and Simon)	40,765 cal/mol = 170.56 kJ/mol	J/mol; same as above	(DYSON; SIMON, 1968)
α (exponent of the Temkin equation)	0.5 (literature: 0.5 to 0.75)	dimensionless	(YANCY-CABALLERO; BIEGLER; GUIRARDELLO, 2015; CHEEMA; KREWER, 2018)
effectiveness factor of the catalyst pellet	0.3 to 0.5	dimensionless; industrial catalyst	(YANCY-CABALLERO; BIEGLER; GUIRARDELLO, 2015)
φ_{N_2} (fugacity coefficient)	$0.93431737 + 0.3101804 \times 10^{-3}\ T + 0.295895 \times 10^{-3}\ P - 0.270729 \times 10^{-6}\ T^2 + 0.4775207 \times 10^{-6}\ P^2$	dimensionless (T in K, P in atm); synthesis conditions (600-800 K, 150-300 atm); not checked outside that range	(CHEEMA; KREWER, 2018)
φ_{H_2} (fugacity coefficient)	chained exponentials in T and P; full formula and values computed in Table 13	dimensionless (T in K, P in atm); Shaw and Wones: 0-1000 degrees C, up to 3000 atm (title of the article)	(CHEEMA; KREWER, 2018; SHAW; WONES, 1964)

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Table 29. Literature constants for the ammonia chapter (continued)

Quantity	Value or expression	Unit and range of validity	Source
φ_{NH_3} (fugacity coefficient)	$0.1438996 + 0.2028538 \times 10^{-2}\,T - 0.4487672 \times 10^{-3}\,P - 0.1142945 \times 10^{-5}\,T^2 + 0.2761216 \times 10^{-6}\,P^2$	dimensionless (T in K, P in atm); synthesis conditions; not checked outside that range	(CHEEMA; KREWER, 2018)
$\Delta_r H^\circ_{298}$ of $N_2 + 3\,H_2 \longrightarrow 2\,NH_3$	−91.88 (JANAF: −91.796)	kJ per mol of N_2; 298.15 K	(LINSTROM; MALLARD, 2026; CHASE JR., 1998)
$\Delta_r H^\circ$ of the same reaction at 700 K	−105.236	kJ per mol of N_2; 700 K	(CHASE JR., 1998)
industrial synthesis pressure	10 MPa to 25 MPa (100 to 250 bar)	iron catalyst	(EUROPEAN COMMISSION, 2006)
industrial synthesis temperature	350 to 550	°C; iron catalyst	(EUROPEAN COMMISSION, 2006)
per-pass conversion	0.20 to 0.30	dimensionless	(EUROPEAN COMMISSION, 2006)
usual operating envelope (P, T, H_2:N_2, inerts, conversion)	150-300 bar; 623-773 K; H_2:N_2 from 2:1 to 3:1; inerts 0-15 mol%; conversion 25-35%	—	(CHEEMA; KREWER, 2018)

Source: literature: Gillespie & Beattie (1930), Dyson & Simon (1968), Yancy-Caballero, Biegler & Guirardello (2015), Cheema & Krewer (2018), Shaw & Wones (1964), Linstrom & Mallard (2026), Chase Jr. (1998), European Commission (2006). Values and expressions identical to `dados/constantes_literatura.csv` (generated by `codigo/tabelas_apendice_c.py`).

C.1 Simulation sweeps

After closing the chapters, I went back to the same numerical models and ran each one over a range of conditions far wider than the few points each chapter shows. Everything in this section is simulation. The numbers come from new scripts in `codigo/`, which reuse the functions the chapters already use without changing them, and they are recorded, point by point, in `dados/simulacao/`.

Ammonia: surface and reactor over more conditions

I ran the surface model over a grid of 40 ratios a/b and four ratios ζ between sites and N_2 molecules, and the Dyson-Simon reactor over a temperature (350 to 550 °C) and pressure (100 to 300 atm) grid larger than the one in Chapter 5. All the surface parameters remain illustrative, as in the chapter. In that reactor grid, 222 of the 462 cells (48%) fall outside the range with published fitting data for the Dyson-Simon law (149-309 atm; 603-768 K, Nadiri *et al.* (2024)). I left those cells marked in the table and the figure, rather than hiding them, because there the law is being used outside the range in which it was measured. Table 30 gives a sample of the reactor grid.

Table 30. Per-pass and equilibrium N_2 conversion over a sample of the temperature and pressure grid in which I ran the reactor of Jinasena, Lie & Glemmestad (2018) (example A), Dyson-Simon rate law, $\eta = 0.4$.

T (°C)	P (atm)	$X_{equilibrium}$ (%)	Per-pass X (%)	X/X_{eq}	Within range
350	100	50.9	5.6	0.11	no
350	200	64.2	11.7	0.18	yes
400	200	51.4	19.3	0.38	yes
450	100	23.6	14.5	0.62	no
450	160	33.4	22.0	0.66	yes
450	200	38.5	26.5	0.69	yes
450	260	44.7	32.5	0.73	yes
450	300	48.2	36.2	0.75	yes

continued on next page

Table 30. Per-pass N_2 conversion over a temperature and pressure grid (continued)

T (°C)	P (atm)	$X_{equilibrium}$ (%)	Per-pass X (%)	X/X_{eq}	Within range
500	200	27.2	26.3	0.97	no
550	200	18.3	18.3	1.00	no
550	300	26.8	26.8	1.00	no

Source: simulation (`codigo/varredura_amonia.py`, on top of `codigo/haber.py`; rate law of Dyson & Simon (1968); flow, volume and composition from Jinasena, Lie & Glemmestad (2018)). "Within range": 149-309 atm and 603-768 K, the range with published fitting data for the rate law (Nadiri *et al.* (2024)); outside it the law is being extrapolated. Numbers in `dados/simulacao/varredura_amonia_reator.csv`.

Within that range, in example A, the largest per-pass conversion was 37.8 % (470 °C, 300 atm) and the smallest 9.3 % (350 °C, 160 atm). In example B (Cheema & Krewer (2018)), they were 29.8 % (490 °C, 300 atm) and 4.1 % (350 °C, 160 atm). Figure 51 shows the whole grid, for both flow examples. On the surface, the SSH error grows with a/b and with ζ, and the largest of the whole grid (0.143 in θ_N) shows up where the reaction is fastest and the gas scarcest.

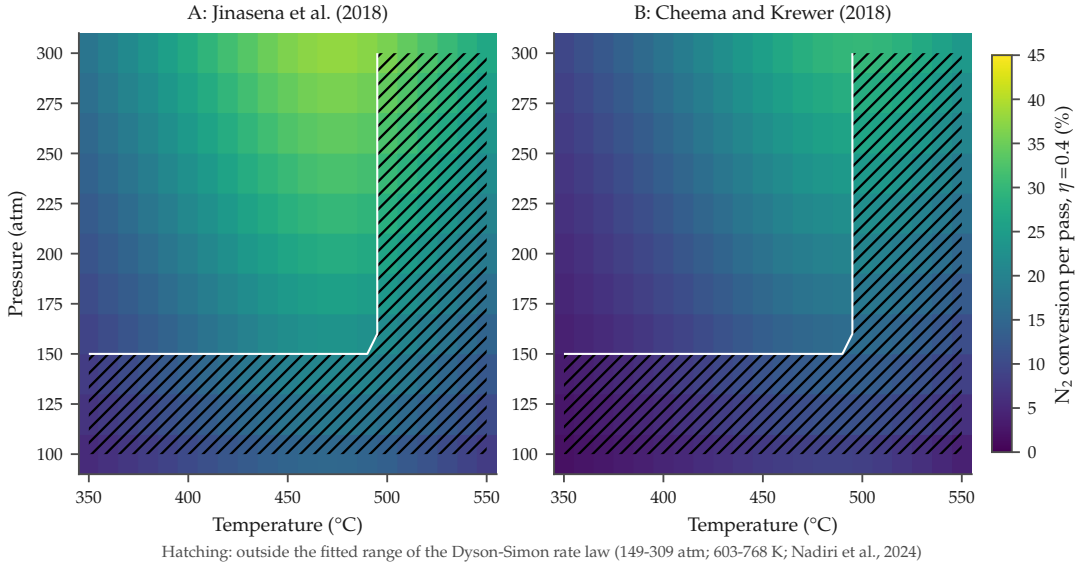


Figure 51. N_2 conversion per pass ($\eta = 0.4$) over a temperature-pressure grid, Dyson-Simon rate law, for the two flow examples of Jinasena, Lie & Glemmestad (2018) and Cheema & Krewer (2018). Hatching marks cells outside the published fitting range of the rate law (149-309 atm; 603-768 K; Nadiri *et al.* (2024)). In those cells the law is being extrapolated beyond the published data.

Source: simulation (`codigo/varredura_amonia.py`, on top of `codigo/haber.py`; rate law of Dyson & Simon (1968); flow, volume and composition from Jinasena, Lie & Glemmestad (2018) and Cheema & Krewer (2018)); validity range: Nadiri *et al.* (2024).

Illuminated cell over more J, absorbance and O_2 fraction

I ran the illuminated-cell model (Sections 4.3 and 4.3.1) over a much wider range of J, absorbance and O_2 fraction than the points already shown in the chapter. All three parameters remain illustrative, as in the chapter. Effective k_{obs} only matches $2J$ while the O atom's lifetime stays small compared with ozone's decay time: it drops from 99.999 % of $2J$ at $J = 10^{-3}$ s^{-1} to 89.55 % at $J = 10$ s^{-1} (absorbance 0.1). Table 31 gives a sample of the three sweeps.

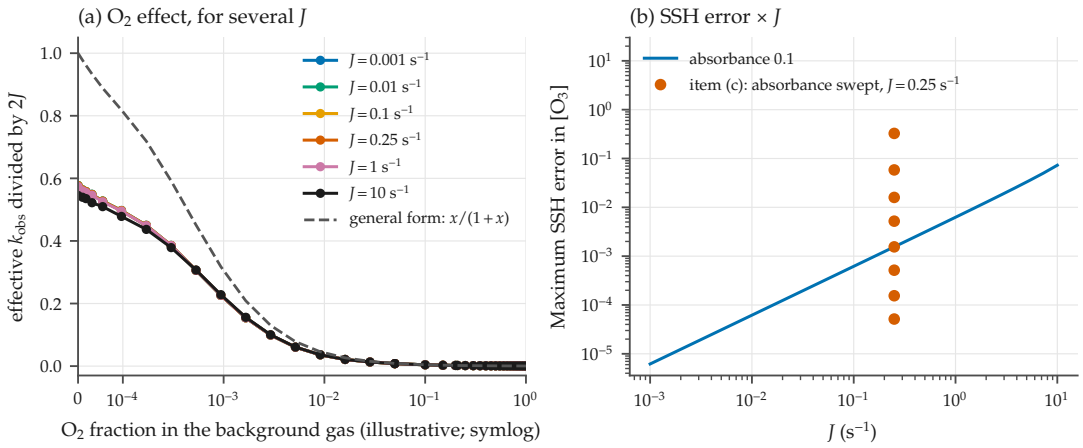


Figure 52. Sweep of the illuminated-cell model (Sections 4.3 and 4.3.1). (a) Effective k_{obs}, fitted to the simulated curve, divided by $2J$, against the O_2 fraction in the background gas (illustrative), for six values of J (10^{-3} to 10 s^{-1}, absorbance 0.1, $P = 1$ atm). The horizontal axis is on a symlog scale, because the transition happens in a narrow band near zero. The dashed curve is the general form $k_{obs} = 2J\,x/(1+x)$ computed with the initial concentrations (Equation (4.13)). The six curves nearly coincide, because that ratio does not depend on J. (b) Maximum SSH error in $[O_3]$ against J, from 10^{-3} to 10 s^{-1}, with absorbance fixed at 0.1 (line); the circles are the eight absorbance values swept with $J = 0.25\ s^{-1}$ fixed.

Source: simulation (`codigo/varredura_ozonio_celula.py`), with k_2, k_4 from Burkholder *et al.* (2019) and σ from Hodges *et al.* (2019); numbers in `dados/simulacao/varredura_ozonio_celula.csv`.

Table 31. Sweep of the illuminated-cell model (two-step mechanism, with and without O_2): effective k_{obs} compared with $2J$, and maximum SSH error in $[O_3]$. Effective k_{obs}: k of an exponential fitted to the simulated curve while $[O_3]$ stays above 10 % (series a and c) or 50 % (series b) of its initial value. Series b is at $P = 1$ atm. There the curve with no O_2 at the start is not exponential ($R^2 = 0.97$), and k depends on the window. The table shows 13 of the 266 cases of the sweep. Absorbance and J are illustrative, as in Figure 21; the full grid is in `dados/simulacao/varredura_ozonio_celula.csv`.

Sweep	Parameter	effective k_{obs} (s^{-1})	effective $k_{obs}/2J$	max. SSH error in $[O_3]$
(a) J, absorbance 0.1	$J = 0.001\ s^{-1}$	0.0020000	0.99999	6.19×10^{-6}
(a) J, absorbance 0.1	$J = 0.00924\ s^{-1}$	0.018471	0.99988	5.71×10^{-5}
(a) J, absorbance 0.1	$J = 1.083\ s^{-1}$	2.1357	0.98636	6.78×10^{-3}
(a) J, absorbance 0.1	$J = 10\ s^{-1}$	17.91	0.89550	7.28×10^{-2}
(c) absorbance, $J = 0.25\ s^{-1}$	0.001	0.40032	0.80064	0.327
(c) absorbance, $J = 0.25\ s^{-1}$	0.01	0.48475	0.96951	1.60×10^{-2}
(c) absorbance, $J = 0.25\ s^{-1}$	0.1	0.49839	0.99679	1.55×10^{-3}
(c) absorbance, $J = 0.25\ s^{-1}$	1	0.49984	0.99968	1.55×10^{-4}
(b) O_2 fraction, $J = 0.25\ s^{-1}$, 1 atm	0	0.28819	0.57637	1.42×10^{-3}
(b) O_2 fraction, $J = 0.25\ s^{-1}$, 1 atm	9.39×10^{-4}	0.11312	0.22623	2.44×10^{-4}
(b) O_2 fraction, $J = 0.25\ s^{-1}$, 1 atm	9.10×10^{-3}	0.017983	0.035966	6.43×10^{-5}
(b) O_2 fraction, $J = 0.25\ s^{-1}$, 1 atm	0.209476 (air)	0.0010172	0.0020344	3.21×10^{-6}
(b) O_2 fraction, $J = 0.25\ s^{-1}$, 1 atm	1	2.1746×10^{-4}	4.349×10^{-4}	6.75×10^{-7}

Source: simulation (`codigo/varredura_ozonio_celula.py`), with k_4 and k_2 from Burkholder *et al.* (2019); full numbers in `dados/simulacao/varredura_ozonio_celula.csv`.

The most striking case is that of 1 atm. Even starting with no O_2 at all, at 1 atm the very O_2 that photolysis produces already slows the ozone's fall, and effective k_{obs} drops to 58 % of $2J$ before any O_2 comes from outside (with the fit only until half the ozone is used up; Figure 52). In the window of

the 13 points, from 0 to 8.6 s, the same case gives 0.37 (Section 4.4.1). The difference comes only from the window, because the curve is no longer exponential.

Chapman mechanism over more altitudes

Chapter 4 only has J_1 and J_3 (the photolysis frequencies) with a published source at 25 and 30 km, read from Figure 10-5 of Jacob (1999). To run the same Chapman mechanism from 20 to 36 km, I made a new, pixel-based reading of the same figure at the missing altitudes. At the two points where the two readings can be compared, this new reading differs from the vector reading by up to 16 % (25 km) and 31 % (30 km) in J_1, and by under 3 % in J_3. That is why Table 32 states, at each altitude, which J comes from the vector reading and which comes from this new reading, and that the result inherits the same uncertainty.

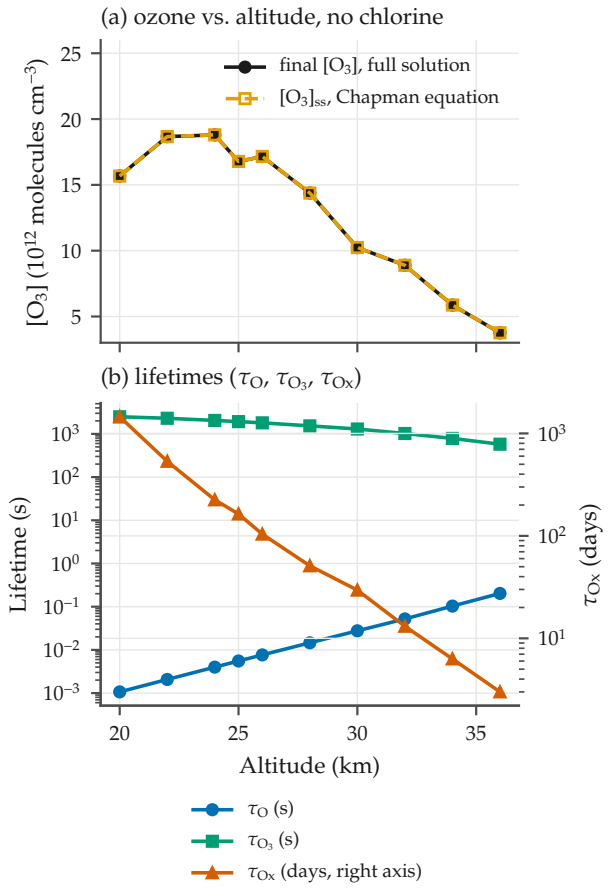


Figure 53. (a) Final $[O_3]$ (full solution) and $[O_3]_{ss}$ (Chapman equation) against altitude, 20 to 36 km, no chlorine. (b) Lifetimes τ_O, τ_{O_3} (left axis, seconds) and τ_{Ox} (right axis, days), same altitude range. T and $[M]$ from the 1976 standard atmosphere; J_1, J_3 read from Jacob's (1999) Figure 10-5, with vector reading at 25 and 30 km and new reading at the other 8 altitudes, which differs by up to 31 % from the vector reading in J_1 (Table 32).

Source: simulation (`codigo/varredura_ozonio_altitude.py`); numbers in `dados/simulacao/varredura_ozonio_altitude.csv`.

With these values, the steady-state $[O_3]$ peaks near 22-24 km (1.880×10^{13} cm^{-3} at 24 km), within the 20 to 30 km range of the observed peak of the ozone layer (JACOB, 1999, p. 168), but higher than the Chapman curve drawn in Jacob's own book, which already exceeds observed ozone by a factor of 2 or more (JACOB, 1999, p. 169). Figure 53 shows the same ozone profile and the lifetimes against altitude, with the same warning about where each J comes from. I also ran a sensitivity test, which separates the effect of T and $[M]$ from that of J and therefore does not represent the real profile: with

J fixed at its 25 km value, steady-state ozone at 20 km is 4 times larger than in the sweep with J varying with altitude. So a good part of the real profile's flattening comes from J falling with altitude, and only part of it from the air becoming thinner.

Table 32. Chapman mechanism as a function of altitude, 20 to 36 km, no chlorine. T and $[M]$ from the 1976 standard atmosphere; J_1 and J_3 read from Jacob's (1999) Figure 10-5, with vector reading at 25 and 30 km and new reading at the other 8 altitudes, which differs by up to 31 % from the vector reading in J_1. $[O_3]_{ss}$: Chapman equation. τ_O, τ_{O_3}: lifetimes of the O atom and the O_3 molecule. τ_{Ox}: lifetime of odd oxygen ($O + O_3$).

z (km)	T (K)	$[O_3]_{ss}$ (cm^{-3})	τ_O (s)	τ_{O_3} (s)	τ_{Ox} (days)
20	216.7	1.567×10^{13}	1.066×10^{-3}	2491	1454
22	218.6	1.867×10^{13}	2.067×10^{-3}	2300	534.1
24	220.6	1.880×10^{13}	3.991×10^{-3}	2048	224.8
25	221.6	1.678×10^{13}	5.531×10^{-3}	1923	163.6
26	222.5	1.716×10^{13}	7.654×10^{-3}	1802	104.0
28	224.5	1.438×10^{13}	1.459×10^{-2}	1534	51.01
30	226.5	1.024×10^{13}	2.764×10^{-2}	1298	29.55
32	228.5	8.893×10^{12}	5.206×10^{-2}	1021	13.12
34	233.7	5.870×10^{12}	1.033×10^{-1}	785.7	6.298
36	239.3	3.761×10^{12}	2.027×10^{-1}	577.2	3.001

Source: simulation (`codigo/varredura_ozonio_altitude.py`); T and $[M]$ from NOAA; NASA; USAF (1976); k_2, k_4 from Burkholder *et al.* (2019); J_1, J_3 at 25 and 30 km from Jacob (1999) (vector reading), at the other altitudes a new reading (same figure, `codigo/ler_jacob_fig10_5_altitudes.py`); numbers in `dados/simulacao/varredura_ozonio_altitude.csv`.

Bootstrap of the ozone and ammonia fits

I ran a non-parametric bootstrap (5000 replicates, two variants) of the fits to the 13+13 ozone and ammonia data points from the December 2025 article. Figure 54 shows the replicates of k for ozone and of τ for ammonia, and Table 33 compares the intervals.

Table 33. Non-parametric bootstrap (5000 replicates, fixed seed) of the fits to the ozone and ammonia data of the December 2025 article. Student's 95% CI: value ± t · standard error. For ozone it is the one in Table 9. For ammonia, Table 14 gives the profile-likelihood CI, which is slightly different (τ of M1 [1.84; 2.24]; k'_b of M3 [1.71; 2.77]). Here I use Student's for both series, so both are compared with the bootstrap alike. Bootstrap 95% CI: 2.5 and 97.5 percentiles of 5000 residual-bootstrap replicates and of 5000 case-resampling replicates. For ammonia, the residual-bootstrap replicates come from M4's curve and residuals, so for M1, M2 and M3 that column is not a CI of those models.

Series, model	Parameter	Fitted value	95% CI Student	95% CI residual bootstrap	95% CI case resampling
Ozone, M1	k (s^{-1})	0.4973	[0.4788; 0.5158]	[0.4798; 0.5128]	[0.4757; 0.5172]
Ozone, M2	A	0.9964	[0.9682; 1.0247]	[0.9743; 1.0229]	[0.9484; 1.0612]
Ozone, M2	k (s^{-1})	0.4956	[0.4718; 0.5193]	[0.4752; 0.5168]	[0.4705; 0.5299]
Ammonia, M1	τ (min)	2.0348	[1.8274; 2.2422]	[1.9899; 2.0637]	[1.8142; 2.2201]
Ammonia, M2	t_0 (min)	0.3551	[0.2811; 0.4291]	[0.3006; 0.3781]	[0.3159; 0.6990]
Ammonia, M2	τ (min)	1.6810	[1.5763; 1.7857]	[1.6342; 1.7398]	[1.4349; 1.7721]
Ammonia, M3	k'_a (min^{-1})	0.6480	[0.5957; 0.7003]	[0.6127; 0.7039]	[0.5997; 0.7764]
Ammonia, M3	k'_b (min^{-1})	2.1729	[1.6867; 2.6591]	[1.7797; 2.6452]	[1.3554; 2.6380]
Ammonia, M4	τ_W (min)	2.1271	[2.0844; 2.1698]	[2.0850; 2.1617]	[2.0748; 2.1700]
Ammonia, M4	γ	1.3073	[1.2567; 1.3579]	[1.2601; 1.3532]	[1.2356; 1.3575]

Source: data from the December 2025 article (RIBEIRO, 2025b); bootstrap: simulation (`codigo/bootstrap_ajustes.py`); numbers in `dados/simulacao/bootstrap_ajustes.csv`.

For ozone, both bootstrap variants stay close to the Student's-t 95% confidence interval (CI) of Table 9, [0.4788; 0.5158]: they give [0.4798; 0.5128] with the residuals and [0.4757; 0.5172] with case resampling. For ammonia, I generated the replicates from the curve and residuals of M4, the model that wins by AICc (Table 14). So the middle column, for M1, M2 and M3, is not a CI of those models. It shows how much the parameter of a wrong model swings when the data come from M4, with noise of 0.009. The CI for τ of M1 comes out narrow ([1.9899; 2.0637], against [1.8142; 2.2201] with case resampling) because that noise is about four times smaller than the scatter of the points around M1's own curve (about 0.04). These intervals measure the uncertainty of the fit to the data. Varying each point only within the rounding of the data (±0.05 in time, ±0.005 in value) gives, for ozone, $k \in [0.4905; 0.5041]$, a much narrower range than either interval above.

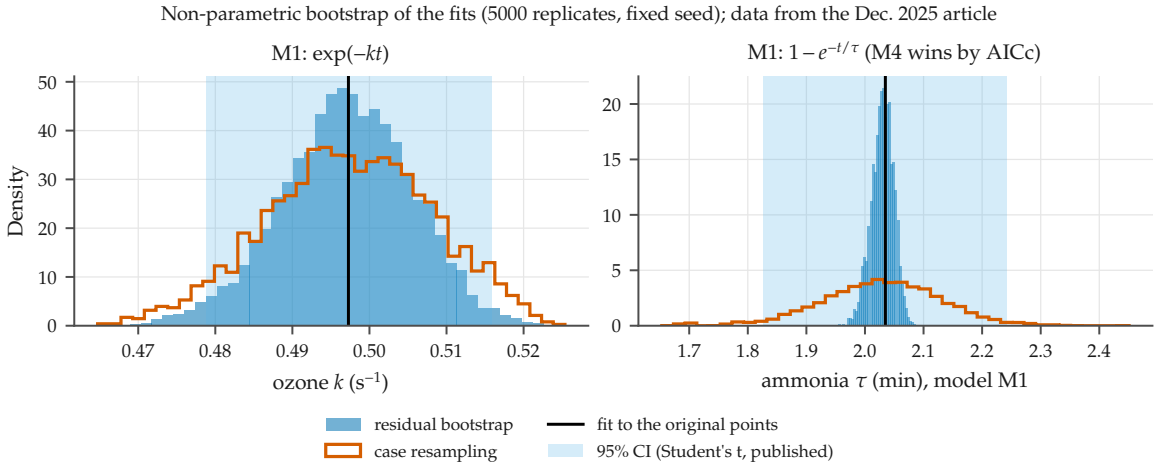


Figure 54. Histograms of 5000 residual-bootstrap replicates (filled) and 5000 case-resampling replicates (outline), for k of the ozone exponential fit (M1, left) and τ of the ammonia $1 - e^{-t/\tau}$ fit (M1, right). The black line marks the fit to the 13 data points; the light-blue band marks the Student's-t 95% CI (for ammonia, the asymptotic one, not the profile CI of Chapter 5). For ammonia, the model that wins by AICc on these data is M4 (Weibull), and the panel shows M1 because it is the model whose parameter is named τ. The width shown here is the uncertainty of the fit to the 13 data points.

Source: data from the December 2025 article (RIBEIRO, 2025b); bootstrap: simulation (`codigo/bootstrap_ajustes.py`); numbers in `dados/simulacao/bootstrap_ajustes.json`.

SSH validity over more ratios and a finer grid

I computed the SSH error in the A → I → B mechanism over 100 log-spaced ratios k_2/k_1, from 1 to 1000. Table 34 shows ten of these ratios, other than the seven that Chapter 3 already tabulates.

Table 34. Sweep of the SSH error in the A → I → B mechanism, k_2/k_1 from 1 to 1000 in 100 log-spaced values (ten grid values, other than the seven already tabulated in Table 6). Column 2: largest difference between SSH [B] and exact [B], as a fraction of $[A]_0$ (= $[I]_{max}/[A]_0$). Column 3: the time of that maximum error, in units of $1/k_1$. Column 4: $(k_1/k_2 - \text{exact value})/(k_1/k_2)$, how far the simple ratio k_1/k_2 overshoots the exact value of column 2, as a fraction of k_1/k_2 itself.

k_2/k_1	$\max([B]_{SSH} - [B])/[A]_0$	$t_{\max\text{ error}}$ $(1/k_1)$	error of the k_1/k_2 approximation
1.52	0.294	0.805	55.3 %
3.05	0.190	0.544	41.9 %
7.06	0.103	0.323	27.6 %
15.2	0.0543	0.192	17.4 %
20.1	0.0425	0.157	14.5 %
32.8	0.0274	0.110	10.4 %

continued on next page

Table 34. Sweep of the SSH error in the A → I → B mechanism (continued)

k_2/k_1	$\max([B]_{SSH} - [B])/[A]_0$	$t_{\max\,\text{error}}$ $(1/k_1)$	error of the k_1/k_2 approximation
70.6	0.0133	0.0612	5.94 %
152	0.00636	0.0333	3.27 %
200.9	0.00485	0.0265	2.62 %
705.5	0.00140	0.00931	0.93 %

Source: simulation (`codigo/varredura_validade.py`); numbers in `dados/simulacao/varredura_validade_aib.csv`.

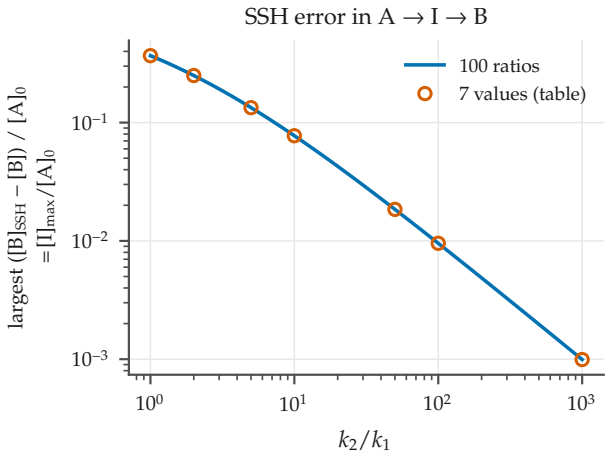


Figure 55. Largest SSH error in [B], as a fraction of $[A]_0$, against k_2/k_1 from 1 to 1000, in 100 log-spaced values (log-log scale). The circles mark the seven values already tabulated in Table 6.

Source: simulation (`codigo/varredura_validade.py`); numbers in `dados/simulacao/varredura_validade_aib.csv`.

Figure 55 shows all 100 ratios together. The maximum error falls off continuously between the seven points already tabulated in Chapter 3, for example 4.25 % at $k_2/k_1 = 20.1$ and 0.48 % at $k_2/k_1 = 200.9$. I also swept a 20×20 grid in the A ⇌ I → P mechanism (Chapter 7). With the 10 % error threshold, 142 of the 400 points (35.5 %) have only the SSH working, 142 only the pre-equilibrium, 67 (17 %) both together, and 49 (12 %) neither. That last region is a band around the diagonal $k_1 = k_2$, where neither step is much faster than the other. On this grid, with a step of 2.07, the first point on the diagonal where neither works is $k_1 = k_2 = 0.70\,k_{-1}$. The point before it, $0.34\,k_{-1}$, still has an error of 8.7 %. This agrees with the $0.4\,k_{-1}$ threshold of Chapter 7. For the same reason, the band the grid shows, from $k_1/k_2 = 0.23$ to 4.3, is narrower than the one from 0.14 to 7.3 in that chapter. The fractions for "only the SSH" and "only the pre-equilibrium" are equal (142 points each) because of the $k_1 \leftrightarrow k_2$ symmetry (Section 7.5.1). The errors come from the same functions as in Chapter 7, which I checked with `solve_ivp` at five points of that chapter's map (largest difference 1.1×10^{-13} of $[A]_0$; `dados/simulacao/validade_resumo.json`).

K_M error caused by the enzyme in the SABIO-RK pairs

This is the full calculation that Section 8.1 summarizes. It assumes that the recorded $[E]_0$ is that of the assay in which K_M was measured and that the fit was the ordinary Michaelis-Menten one, by initial rates. I compare each pair by $[E]_0/K_{M,ap}$, because the published K_M is already the apparent one.

With the $[S]_0$ grid tied to the true K_M (8 values from $0.2K_M$ to $5K_M$), $[E]_0/K_{M,ap}$ never goes above 0.57 by the tQSSA, 0.34 with readings after the transient and 0.19 with readings from $t = 0$: this is the ceiling of each reading. But one may center the grid on the K_M of a pilot fit. I redid the calculation

that way, until the grid was centered on the K_M it gives. The ratio then always grew with ε_{K_M} and, at the largest value I simulated ($\varepsilon_{K_M} = 1000$), reached 1.30 (from $t = 0$), 1.31 (after the transient) and 1.38 (tQSSA), still rising. An apparent K_M that marks half saturation cannot go above 2, whatever the grid.

By the tQSSA, 3.4 % of the pairs lie above the fixed-grid ceiling: 1.2 % between that ceiling and 1.38, 0.5 % (9 pairs) between 1.38 and 2, and 1.7 % above 2. From $t = 0$, the share between the fixed-grid ceiling and 1.30 is 4.4 %. The pairs between the fixed-grid ceiling and the largest value of the centered grid have two readings: with the fixed grid, they cannot come from an ordinary fit; with the centered one, they can, with a K_M error of 88 % or more. Without reading each paper, I cannot tell which holds.

Table 35 gives, for each reading and variant, the share of pairs that lie below the ceiling and have a K_M error caused by the enzyme above 10 % and above 50 %. For the 10 % cut, the main readings go from 5.3 to 9.3 % of the pairs and the other variants from 6.4 to 13.2 %; for the 50 % cut, from 1.0 to 3.3 % and from 1.4 to 5.9 %. With the centered grid, the main readings become 6.5–13.2 % and 2.1–6.6 %. These are model estimates, not measurements.

Table 35. Estimated share of the 1968 SABIO-RK K_M–$[E]_0$ pairs (enzyme in mol/L, with the sample weights) with a K_M error caused by the enzyme above 10 % and above 50 %, among those below the largest $[E]_0/K_{M,ap}$ of each reading. Ceiling: largest simulated $[E]_0/K_{M,ap}$; with an asterisk, the ratio was still rising at the largest simulated ε_{K_M}, and the value is not a ceiling. In brackets, the 95 % CI of the stratified *bootstrap*. The variants start from the readings from $t = 0$.

	Grid tied to K_M			Grid centered on $K_{M,ap}$		
Reading	Ceiling	> 10 % (%)	> 50 % (%)	Ceiling	> 10 % (%)	> 50 % (%)
tQSSA	0.57	5.3 [4.3; 6.4]	1.0 [0.6; 1.4]	1.38*	6.5 [5.4; 7.7]	2.1 [1.5; 2.8]
From $t = 0$	0.19	9.3 [8.0; 10.7]	3.3 [2.5; 4.1]	1.30*	13.2 [11.7; 14.7]	6.6 [5.5; 7.8]
After the transient	0.34	8.6 [7.4; 9.9]	2.7 [2.0; 3.4]	1.31*	10.3 [9.0; 11.8]	3.9 [3.0; 4.8]
$r = 0.1$	0.43	6.4 [5.3; 7.5]	1.4 [0.8; 2.0]	1.31*	8.2 [7.0; 9.5]	2.9 [2.1; 3.7]
$r = 0.9$	0.16*	12.3 [10.9; 13.8]	5.3 [4.2; 6.3]	1.30*	16.9 [15.2; 18.6]	9.3 [8.0; 10.6]
Grid 0.1 to $10K_M$	0.28	10.0 [8.6; 11.4]	3.6 [2.8; 4.5]	1.51*	12.9 [11.4; 14.5]	6.4 [5.3; 7.6]
Grid 1 to $20K_M$	0.43	7.8 [6.5; 9.0]	3.1 [2.4; 4.0]	2.29*	10.1 [8.8; 11.6]	4.8 [3.8; 5.8]
20 % window	0.28	6.9 [5.8; 8.1]	2.2 [1.6; 2.9]	1.23*	9.3 [8.1; 10.6]	4.3 [3.4; 5.3]
5 % window	0.15*	13.2 [11.8; 14.9]	5.9 [4.9; 7.0]	1.33*	18.0 [16.3; 19.7]	9.8 [8.5; 11.2]

Source: SABIO-RK data (WITTIG *et al.*, 2018); simulation (`codigo/vies_km_epsilon.py`); numbers in `dados/simulacao/vies_km_epsilon_resumo.json`.

C.2 General data table

I gathered in Table 36 every numerical value I used in this research: where each set comes from (data from the December 2025 article, SABIO-RK data, simulation, synthetic data or literature), how many values it has and where it appears in the text. The full list, value by value, is in `dados/tabela_geral_dados.csv`, generated by `codigo/tabela_geral_dados.py`.

Table 36. General table of data: every dataset used in this monograph, with the number of values, where it is cited, and its source file. Full values, one row per value, in `dados/tabela_geral_dados.csv`.

Dataset	Origin	N	Cited in	File[a]
December 2025 article: ozone data	article data (13)	13	Figure 23; Table 25	`chemis3_ozonio.csv`
December 2025 article: ammonia data	article data (13)	13	Figure 35; Table 26	`chemis3_amonia.csv`
SSH in A→I→B: maximum error as a function of k2/k1	simulation (98)	98	Table 6; Figure 41	`validade_AIB_erro.csv`
SSH in A→I→B: slow regime (k2/k1 < 1)	simulation (16)	16	Chapter 7 (slow regime, k2/k1<1)	`validade_AIB_lento.csv`

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Table 36. General data table (continued)

Dataset	Origin	N	Cited in	File[a]
Pre-equilibrium: maximum error of the SSH and of the pre-equilibrium	simulation (50)	50	Table 15	`validade_preeq_K.csv`
Segel's approximation (enzyme-substrate)	simulation (20)	20	Chapter 7 (Segel's approximation)	`validade_segel.csv`
The Q10 rule (rate factor per 10 °C)	simulation (44)	44	Section 2.5.3	`fund_q10.csv`
Exponential fit to the 13 ozone points: points vs. models	simulation (91); article data (26)	117	Figure 23	`ozonio_ajuste.csv`
Residuals of the 5 models fitted to the ammonia points	simulation (78); article data (13)	91	Figure 35	`amonia_ajuste_residuos.csv`
Chapter 3 example (A + B ⇌ I + D, I + B → C + D) solved on the computer	simulation (10)	10	Figure 19	`ssh_exemplo20.json`
Fit to ozone: parameters, CI and AICc of the models	simulation (172)	172	Table 9	`ozonio_ajuste.json`
Ozone in the cell: photon-flux estimate	simulation (45)	45	Section 4.4 (photon flux, exercise)	`ozonio_fluxo_fotons.json`
Ozone: numerical verification (cell and stratosphere)	simulation (86)	86	Chapter 4 (numerical cross-check)	`ozonio_verificacao.json`
Ozone in the stratosphere: time scales and the chlorine cycle	simulation (177)	177	Table 10; figs. in ch. 4	`ozonio_escalas.json`
Fit to ammonia: 5 models, AICc, Akaike weight, DW	simulation (356)	356	Table 14	`amonia_ajuste.json`
Ammonia: per-pass conversion, start-up and time scales of the simulated reactor	simulation (122)	122	Figure 31; Figure 32; Figure 38	`amonia_reator.json`
Ammonia coverage on the surface (simple model, illustrative parameters)	simulation (52)	52	Figure 34	`amonia_superficie.json`
Ammonia: theory (Ka, fugacity, example 5.1, equilibrium cross-check)	simulation (83)	83	Table 13; Figure 30	`amonia_teoria.json`
Ammonia: sanity tests of the simulated reactor	simulation (288)	288	verification of codigo/haber.py	`amonia_sanidade.json`
Fundamentals: secant and tangent of the decay example	simulation (30)	30	Section 2.1 (decay-example calculations)	`fund_contas.json`
Fundamentals: R^2 of the wrong first-order plot	simulation (9)	9	Section 2.4.3 (R^2 at low conversion)	`fund_metodo_grafico.json`
SSH validity: numerical cross-checks and induction summary	simulation (92)	92	Table 17	`validade_resumo.json`
Sweep: SSH validity (more k2/k1 ratios and a fine pre-equilibrium grid)	simulation (21)	21	Appendix C.1; Table 34	`varredura_validade_resumo.json`
Sweep: Chapman mechanism as a function of altitude (20-40 km)	simulation (870)	870	Appendix C.1; Table 32	`varredura_ozonio_altitude.json`
Sweep: illuminated cell over more J, absorbance and O2 fraction	simulation (101)	101	Appendix C.1; Table 31	`varredura_ozonio_celula.json`
Sweep: ammonia – surface and reactor over more conditions	simulation (34)	34	Appendix C.1; Table 30	`varredura_amonia_resumo.json`
Sweep: bootstrap of the ozone and ammonia fits	simulation (155)	155	Appendix C.1; Table 33	`bootstrap_ajustes.json`
Symmetry k1 ↔ k2, criteria and total law in A⇌I→P	simulation (352)	352	Section 7.5.1; Figure 43; Table 16	`validade_simetria.json` and 1 more
Synthetic data: assays that would test the ozone and ammonia laws	synthetic data (1900); simulation (73)	1973	Section 4.4.1; Section 5.5.1	`dados_sinteticos.json` and 3 more
SABIO-RK: [E]0/KM and Segel and Slemrod's ε in published assays	SABIO-RK data (978)	978	Section 8.1; Figure 44; Figure 45	`sabiork_auditoria_resumo.json` and 2 more
Error of KM and kcat against [E]0/KM, applied to the SABIO-RK pairs	simulation (1435)	1435	Figure 46	`vies_km_epsilon_resumo.json`
SSH testability: detecting the error in synthetic data sets	simulation (1123); synthetic data (63)	1186	Section 8.2; Figure 47; Figure 48	`testabilidade_resumo.json` and 1 more
Bench proposal (invertase and glucose meter): model predictions	simulation (713); literature (32); synthetic data (12)	757	Section 8.3; Figure 49	`invertase_glicosimetro.json` and 2 more
Bench proposal: analysis of the synthetic glucose-meter data	synthetic data (274); literature (3)	277	Figure 50	`glicosimetro_ajuste_sintetico.json` and 1 more

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Table 36. General data table (continued)

Dataset	Origin	N	Cited in	File[a]
Fundamental physical constants (CODATA 2018)	literature (4)	4	Table 27	`constantes_literatura.csv`
Literature constants for ozone kinetics and photochemistry	literature (17)	17	Table 28	`constantes_literatura.csv`
Literature constants for the ammonia synthesis	literature (14)	14	Table 29	`constantes_literatura.csv`

Source: author's own compilation from `dados/tabela_geral_dados.csv` (generated by `codigo/tabela_geral_dados.py`). [a]Full path in the `arquivo_de_origem` column of that CSV ("and n more": the set has values from n other files); simulation and synthetic-data files are under `dados/simulacao/`, and the article data, the SABIO-RK data, and the literature constants under `dados/`.